Tunneling and Risk Management

A Geometric Framework for Hidden Transitions, Barrier Permeability, and Institutional Risk

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Preface

This book develops a geometric interpretation of risk. Its central concern is not only the probability of adverse outcomes, but the structure that makes those outcomes reachable. The research programme investigates states, landscapes, barriers, hidden pathways, and the conditions under which a system may enter an adverse region without following the expected sequence of deterioration. It asks where a risk-bearing system is located, how it can move, which protections oppose that movement, and why apparently remote outcomes can sometimes become accessible with surprising speed.

Probability remains indispensable. Institutions need estimates of default, loss, failure, disruption, sanction, and contagion. Such estimates support pricing, provisioning, capital allocation, insurance, control design, regulation, and intervention. Nothing in this book diminishes that role. The argument is instead that a reported probability often compresses the mechanism that produced it. Two borrowers may have the same probability of default while facing different refinancing paths. Two operating systems may have similar incident frequencies while possessing radically different dependencies and recovery barriers. Two institutions may appear individually resilient while their shared assets, funding sources, and responses create a fragile collective configuration. A geometric representation seeks to recover this lost structure.

The book therefore treats risk as movement through a state space. A system occupies a configuration described by economically, operationally, legally, or institutionally meaningful coordinates. Those configurations form a manifold of feasible states rather than an unrestricted spreadsheet of variables. A metric determines which movements are consequential. Drift, covariance, jumps, feedback, and intervention govern mobility. Stable, fragile, adverse, and recovery regions give the space decision meaning. Protective structures appear as barriers with height, width, continuity, reliability, capacity, and direction. Candidate pathways connect the current state to possible outcomes. This language permits probability to be located within an explicit architecture of accessibility.

Quantum tunneling provides the book's most provocative source of mathematical inspiration. In physical systems, tunneling describes nonzero transmission through a finite barrier that classical mechanics would treat as forbidden. The analogy is attractive because risk systems also contain nominal protections, distributed uncertainty, interacting pathways, and transitions that may not resemble ordinary over-the-barrier deterioration. Yet the analogy must be handled with care. Borrowers, institutions, contracts, and computer systems are not asserted to be literal quantum objects. Complex amplitudes, phase, interference, action, and transmission are introduced as modelling constructs whose legitimacy depends on interpretation, estimation, validation, and comparison with credible classical alternatives.

This qualification is central to the research programme. A surprising event is not automatically tunneling. Gradual drift may have been missed. A jump may have occurred. The state vector may omit a decisive variable. A barrier may have eroded, moved, or contained a hidden gap. A regime may have changed. A model may simply be wrong. The burden of proof therefore rests with the tunneling extension. It earns a place only when it identifies stable structure, improves explanation or prediction, clarifies intervention, and survives tests against nonlinear, path-dependent, network, and regime-switching classical models.

The intended audience includes risk researchers, quantitative analysts, economists, credit professionals, operational and cyber specialists, lawyers, regulators, model validators, and institutional decision-makers. Some readers will approach the subject through geometry and stochastic processes; others will begin with practical problems of liquidity, controls, evidence, infrastructure, or governance. The chapters are designed to support both routes. Mathematical objects are introduced with their institutional interpretation, while empirical and governance chapters translate the formal framework into data requirements, measurements, stress tests, limits, ownership, and intervention authority.

The book proceeds from concept to implementation. The early chapters establish state spaces, metrics, covariance geometry, barriers, and the mathematics of tunneling. The middle chapters distinguish physical theory from a disciplined risk translation and apply the framework to credit, operational, cyber, legal, regulatory, conduct, institutional, and systemic settings. The later chapters focus on empirical detection, measurement, stress testing, and governance. The appendices consolidate notation, propose an implementation blueprint, and present worked calculations. This sequence reflects a deliberate principle: sophisticated formalism should follow clear state definition, credible classical modelling, and measurable protection rather than precede them.

Several practical commitments run throughout the text. Data must be timestamp-correct so that analysis uses only information available at the decision date. Barriers must correspond to protections that can be tested for availability and reliability. Pathways must be observable or reconstructable rather than invented retrospectively. Phase must be grounded in timing, sequencing, persistence, and synchronization evidence. Action and transmission must be calibrated and accompanied by uncertainty. Composite measures must preserve their components. Interventions must have owners, authority, capacity, and response windows. Model performance must be assessed out of sample and against strong benchmarks.

The framework is intentionally pluralistic. Geometry may be constructed structurally, statistically, through machine learning, or through a governed hybrid. Pathways may be estimated with survival models, process mining, causal graphs, network methods, temporal point processes, simulation, or expert evidence. Artificial intelligence may assist with document extraction, scenario development, pattern discovery, and reporting, but it does not remove the need for source grounding, validation, auditability, and human accountability. The relevant question is never whether a method appears advanced. It is whether the method reveals a reproducible mechanism and improves a consequential decision.

This book should consequently be read as a research programme rather than a declaration that every risk problem requires tunneling mathematics. Many problems are explained adequately by conventional probability, stochastic processes, network contagion, threshold models, or scenario analysis. In those cases, the simpler model should prevail. The geometric and amplitude-based extensions become useful when location, direction, barriers, pathway interaction, and timing contain decision-relevant information that an aggregate probability conceals.

The governing ambition is modest in statement but demanding in execution: to make the routes to failure as visible as the estimated likelihood of failure. If successful, the framework helps analysts ask better questions, detect fragility earlier, distinguish nominal from effective protection, and choose interventions that alter the structure of accessibility. The reader is invited to challenge every analogy, test every proposed mechanism, and retain only those elements that withstand evidence. That discipline is not a limitation of the framework. It is the condition under which a geometric account of risk can become useful.

How to Read This Book

This book is organized as a cumulative research journey, but it can also be read selectively. Readers seeking the full argument should proceed in sequence. The opening chapters introduce the geometric language of states, manifolds, metrics, covariance, barriers, pathways, and accessibility. The mathematical chapters then develop the physical foundations of tunneling and explain how concepts such as amplitude, phase, interference, action, and transmission may be translated into risk modelling. Later chapters apply the framework to credit, operational, cyber, legal, regulatory, institutional, and systemic risk before turning to empirical detection, measurement, stress testing, and governance.

Readers primarily interested in practical implementation may begin with the conceptual chapters and then move directly to the chapters on detecting hidden pathways, measuring tunneling risk, stress testing, and governing barrier permeability. These chapters show how the theoretical objects become data structures, indicators, scenarios, limits, ownership arrangements, and intervention decisions. The appendices provide a consolidated notation guide, an empirical implementation blueprint, and worked examples that can be used alongside the main text.

The equations should be read structurally rather than as claims that every application requires the same mathematical specification. Each formal object answers a particular question. The state vector identifies the system's relevant condition. The metric defines meaningful distance. The law of motion describes ordinary movement. The barrier represents measurable protection. Path action compares transition routes. Phase represents timing and alignment. Transmission expresses effective accessibility. When a simpler classical model answers the practical question adequately, that model should remain the benchmark.

Readers should pay particular attention to the distinction between physical theory and risk interpretation. The book does not claim that institutions, borrowers, contracts, or technological systems are literal quantum objects. The tunneling framework is a disciplined modelling extension whose value depends on observable definitions, timestamp-correct evidence, comparison with classical alternatives, out-of-sample validation, uncertainty reporting, and actionable intervention insight.

Each chapter develops a central proposition and concludes by connecting its results to the next stage of the framework. The synthesis sections can therefore be used for rapid review, while the detailed sections provide derivations, assumptions, examples, and governance implications. Cross-references and figures are intended to help readers move between conceptual, mathematical, empirical, and institutional layers without losing the overall architecture.

The most productive reading stance is critical rather than reverential. Ask what is observed, what is estimated, which assumptions create the geometry, whether the barrier is real and measurable, whether proposed pathways are reproducible, and whether the tunneling extension improves a decision. The framework should be retained where it reveals structure that probability alone conceals and rejected where it adds complexity without evidence.

Each chapter is maintained as an independent LaTeX file and can be drafted, revised, compiled, or replaced without disturbing the rest of the manuscript.

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Chapter 1

From Probability to Geometry

1.1 The Point of Departure

Risk management has traditionally been expressed through the language of probability. We estimate the probability that a borrower will default, the likelihood that an operational process will fail, the frequency with which a cyber incident may occur, or the chance that a legal claim will generate a material loss. These estimates are indispensable. They support pricing, provisioning, capital allocation, insurance, regulation, control design, and institutional decision-making.

Yet a probability reported as a single number provides only a partial description of risk.

A scalar probability tells us how likely an outcome is under a specified model. By itself, it does not explain how that outcome becomes reachable. It does not identify where the system is currently located, what adverse states surround it, which barriers separate the present condition from those states, or what forces govern movement from one configuration to another. A sufficiently rich probabilistic model can contain this structure; the point is that the reported probability alone does not reveal it.

The present book begins from the proposition that risk can be understood not only as uncertainty over outcomes, but also as movement through a structured space.

This requires a change in emphasis. Conventional risk analysis often begins with the adverse event: default, operational failure, litigation, insolvency, regulatory sanction, or systemic collapse. It then attempts to estimate the probability of that event. The geometric perspective begins earlier. It asks what the system is, where it is located, how it evolves, what barriers protect it, and which paths connect the present state to adverse regions.

The central object of analysis is therefore not merely the event. Schematically, it is the combination

$$\text{State} + \text{Geometry} + \text{Dynamics} + \text{Barriers} + \text{Transition Mechanisms}. \tag{1.1}$$

This leads to the foundational proposition of the book:

> Risk is not only the probability of arriving at an adverse state. It is also the geometry and dynamics that make that state reachable.

The chapter develops this proposition in three stages. First, it introduces state spaces, metrics, dynamics, reachability, and barriers. Second, it reviews the physical mathematics of quantum tunneling. Third, it separates the literal physical theory from a tunneling-inspired risk model and states the assumptions required for that translation.

1.1.1 What Probability Reveals—and What It Conceals

The distinction is not merely presentational. Two systems with identical forecast probabilities can require opposite interventions because their paths, buffers, and response times differ. A probability estimate becomes more decision-useful when analysts can decompose it into the mechanisms that generate accessibility. This decomposition permits management to distinguish persistent deterioration from concentrated vulnerability, select monitoring indicators that correspond to the dominant route, and test whether a proposed control changes the path itself or merely changes the reported probability. Geometry therefore supplements probabilistic ranking with an account of causation, timing, and intervention leverage.

Probability is powerful because it compresses uncertainty into a manageable quantity. A probability estimate allows institutions to compare exposures, rank risks, calculate expected losses, and allocate resources. It can summarise historical experience and support decisions under uncertainty.

However, compression removes structure.

Consider two borrowers with the same one-year probability of default. A conventional model may treat them as broadly comparable. Yet one borrower may be deteriorating gradually through a familiar sequence of declining revenues, weakening liquidity, and rising leverage. The other may appear financially stable but depend on one fragile source of refinancing.

The first borrower follows a visible path toward default. The second may be separated from default by an apparently robust barrier that contains one narrow and highly consequential weakness. The same estimated probability can therefore conceal radically different transition mechanisms.

The problem becomes even clearer in operational risk. Two institutions may report similar frequencies of technology incidents. One may suffer repeated but contained disruptions caused by known process weaknesses. The other may experience few incidents but rely on a highly concentrated infrastructure in which one hidden dependency could produce catastrophic failure. Historical frequency may suggest that the second institution is safer, although its underlying geometry is more fragile.

Legal risk provides another example. A contract may appear to provide strong protection and historical litigation rates may be low. Yet a particular combination of ambiguous language, inconsistent conduct, regulatory reinterpretation, and jurisdictional conflict may create a pathway toward liability that has little historical precedent. The exposure cannot be understood merely as a low-probability event. It is a transition made possible by the structure of the legal landscape.

Probability answers the question

> How likely is the adverse outcome?

A geometric and dynamic model asks a broader set of questions:

> Where is the system now? How does it move? What forces act upon it? What barriers separate it from danger? Which paths lead toward failure? Are there hidden routes through those barriers?

The geometric perspective does not reject probability. It places probability inside a richer structural model.

1.2 State Space, Distance, and Geometry

The first building block is the state space. A state is a sufficiently informative description of the system at a given time; its coordinates should contain the variables needed to model subsequent evolution at the chosen level of abstraction.

Let a risk-bearing system be represented by a state vector

$$\mathbf{x}_t \in \mathcal{M}, \tag{1.2}$$

where $\mathcal{M}$ is the state space and $\mathbf{x}_t$ describes the system at time t. When $\mathcal{M}$ has the smooth local structure required for differential calculus, we treat it as a manifold.

For a borrower, the coordinates of $\mathbf{x}_t$ may include leverage, liquidity, profitability, debt maturity, collateral quality, market access, and management behaviour. For an operational system, they may include system load, process complexity, control effectiveness, staff capacity, data quality, third-party dependence, and recovery capability. For a legal position, they may include contractual protection, documentary evidence, conduct, precedent, regulatory posture, jurisdiction, and enforcement risk.

Each point on $\mathcal{M}$ corresponds to one possible configuration of the system at the selected level of description.

Some regions of the manifold represent stability. Others represent fragility, distress, failure, or recovery. These regions need not be separated by simple straight lines. They may overlap, curve, expand, contract, or change shape under stress.

A borrower may remain solvent while entering a region in which liquidity has become highly sensitive to market confidence. An institution may remain operational while moving toward a configuration in which redundancy is low and recovery times are increasing. A contract may remain formally enforceable while conduct and precedent gradually weaken its practical protection.

The state-space representation therefore replaces a binary distinction between success and failure with a continuum of configurations.

Yet the state space alone is not sufficient.

A manifold merely tells us where the system could be. It does not tell us how the system moves. Without a law of motion, the representation is static. A point placed on a surface has no trajectory, no direction, no momentum, and no notion of reachability.

A complete risk geometry must therefore be dynamical.

1.2.1 Distance and the Geometry of the Manifold

Risk distance must also be interpreted relative to the decision horizon and the institution's capacity to respond. A state that is economically distant over one week may be operationally close over one year, especially when financing maturities, contractual deadlines, or control degradation create predictable movement. Conversely, a numerically nearby adverse region may remain practically remote when strong restoring forces, independent controls, and rapid intervention capacity oppose the transition. The metric should therefore be calibrated against observed transitions and not selected solely for mathematical convenience. Validation should examine whether shorter measured distances are associated with earlier adverse events, stronger warning signals, and greater intervention urgency after controlling for ordinary probability forecasts.

The manifold must contain a meaningful notion of distance.

Let the local distance between nearby states be defined by

$$\mathrm{d}s^2 = \mathrm{d}\mathbf{x}^\mathsf{T}\mathbf{g}(\mathbf{x})\mathrm{d}\mathbf{x}, \tag{1.3}$$

where $\mathbf{g}(\mathbf{x})$ is a symmetric positive-definite metric tensor. Positive definiteness ensures that every non-zero local displacement has positive squared length.

This expression means that not every numerical change has the same risk significance. A one-unit change in liquidity may matter more than a one-unit change in administrative expenses. A small deterioration in legal documentation may be more consequential in one jurisdiction than in another. A modest increase in operational complexity may be harmless when controls are independent but dangerous when they share a common dependency.

The metric determines which changes are large, which are small, and which directions are costly or easy to traverse.

This geometry may be flat or curved. In a flat state space, the relationship among variables is relatively stable. In a curved space, the meaning of a movement depends on where the system is located. A decline in liquidity may be manageable for a low-leverage borrower but critical for a highly leveraged one. A control failure may be minor under normal operating conditions but severe during a period of system congestion.

Distance is therefore state-dependent.

Two adverse regions may lie at the same numerical distance from the current state but differ greatly in practical accessibility. One may lie in the direction of the system's natural drift. The other may require movement against strong stabilizing forces.

Geometry identifies the terrain. Dynamics determine how the system travels through it.

1.3 Dynamics, Potential Landscapes, and Directional Mobility

The general evolution of the state may be written as

$$\mathrm{d}\mathbf{x}_t = \mathbf{b}(\mathbf{x}_t, t)\,\mathrm{d}t + \mathbf{\Sigma}(\mathbf{x}_t, t)\,\mathrm{d}\mathbf{W}_t + \mathrm{d}\mathbf{J}_t. \tag{1.4}$$

This equation contains three forms of movement.

The first term,

$$\mathbf{b}(\mathbf{x}_t, t)\,\mathrm{d}t, \tag{1.5}$$

is the deterministic drift. It represents the direction in which the system tends to move under its current conditions. A borrower with weak cash generation and rising refinancing costs may drift toward distress. An institution with increasing complexity and deteriorating controls may drift toward operational fragility. A legal position may drift toward liability as conduct becomes increasingly inconsistent with contractual protections.

The second term,

$$\mathbf{\Sigma}(\mathbf{x}_t, t)\,\mathrm{d}\mathbf{W}_t, \tag{1.6}$$

represents continuous stochastic variation, where $\mathbf{W}_t$ is a vector Brownian motion. It captures market shocks, behavioural fluctuations, operating uncertainty, information noise, and other disturbances. The matrix $\mathbf{\Sigma}$ determines both the local magnitude and the directional structure of the diffusion.

The third term,

$$\mathrm{d}\mathbf{J}_t, \tag{1.7}$$

represents discontinuous jumps. These may include the sudden withdrawal of funding, a cyberattack, an adverse court ruling, a regulatory intervention, the discovery of fraud, or the failure of a critical supplier.

This formulation is deliberately general. It does not assume that all risk systems behave like conservative physical systems. Financial, operational, and legal systems are typically open, adaptive, path-dependent, and subject to intervention. Their laws of motion include feedback, learning, incentives, institutional constraints, and exogenous shocks.

The state at time t is therefore not merely a location. It is a location embedded in an evolving process.

1.3.1 Potential Landscapes and Gradient Dynamics

The potential should not be interpreted as a directly observed physical energy. It is a modelling device that summarizes the relative stability of configurations and the directional forces implied by the chosen risk model. Different empirical constructions may produce different landscapes, including statistical loss surfaces, structural stress functions, or expert-calibrated resilience scores. A credible specification must state which variables determine the potential, how its scale is identified, and whether its gradients predict observed movement. Sensitivity analysis is essential because local minima, ridges, and saddle points may shift when the sample, regime, or institutional structure changes. The landscape is useful only when those features correspond to economically meaningful persistence, resistance, and transition behaviour.

In some settings, it is useful to introduce a potential function

$$V = V(\mathbf{x}, t). \tag{1.8}$$

The potential defines a landscape of valleys, slopes, ridges, and barriers. Stable configurations lie near local minima. Unstable configurations lie near peaks or ridges.

The simplest deterministic dynamics are

$$\frac{\mathrm{d}\mathbf{x}_t}{\mathrm{d}t} = -\nabla V(\mathbf{x}_t, t). \tag{1.9}$$

The system moves in the direction of decreasing potential.

A financially stable borrower may occupy a basin supported by adequate capital, predictable cash flow, and diversified funding. If it is displaced slightly, stabilizing mechanisms tend to return it toward the basin. An operational system may occupy a stable region because controls, redundancy, and recovery procedures correct small disturbances.

Under these dynamics, adverse transitions occur when the system moves out of one basin and into another.

The potential function is not necessarily a physical energy. In risk management, it is better interpreted as a structural measure of stability, resistance, or transition difficulty. It summarises how strongly the system is drawn toward particular regions.

The landscape may also change through time. Capital may be consumed, controls may deteriorate, regulations may change, market confidence may collapse, or new technological vulnerabilities may emerge.

A system can therefore remain at approximately the same observed state while the landscape beneath it changes.

A borrower may retain the same leverage ratio while access to refinancing disappears. Its location appears unchanged, but the barrier separating it from default has become lower or narrower.

Risk management must monitor both the movement of the state and the deformation of the landscape.

1.3.2 Variance as Directional Mobility

This interpretation emphasizes that variance is informative only in relation to direction and location. High variance along a well-buffered coordinate may have little effect on adverse accessibility, while modest variance aligned with a weak boundary may materially increase transition risk. Analysts should therefore examine eigenvectors, projected covariance, and state-dependent correlations rather than relying exclusively on total volatility. Empirical validation can compare whether adverse-direction variance predicts barrier contact, covenant breach, control exhaustion, or other intermediate events. The resulting measure is not a replacement for variance-based risk statistics; it explains how those statistics interact with the geometry and why identical aggregate volatility can imply different transition possibilities across systems.

Variance is normally treated as a statistical measure of dispersion. Geometrically, conditional variance describes the local spread of possible future states around the current location.

A system with low conditional variance has a tightly concentrated short-horizon distribution. A system with high conditional variance can reach a larger neighbourhood over the same horizon, subject to its drift and constraints.

Yet uncertainty is rarely equal in every direction.

A borrower may be highly sensitive to interest rates but relatively insensitive to small changes in administrative expenses. An operational system may absorb changes in demand but remain vulnerable to the failure of one specialised provider. A legal position may be stable under ordinary interpretation but highly sensitive to a shift in regulatory policy.

For the diffusion in Equation (1.4), the instantaneous covariance rate may be written as

$$\mathbf{C}(\mathbf{x}, t) = \mathbf{\Sigma}(\mathbf{x}, t)\mathbf{\Sigma}^{\mathsf{T}}(\mathbf{x}, t). \tag{1.10}$$

This matrix defines a local uncertainty geometry. Instead of a circular cloud around the present state, the system may occupy an elongated ellipsoid. Its principal axes identify the directions of greatest and least mobility.

If

$$\mathbf{C}\mathbf{v}_i = \lambda_i \mathbf{v}_i, \tag{1.11}$$

then the eigenvectors $\mathbf{v}_i$ identify the principal directions of local uncertainty and the eigenvalues λ_i give the corresponding variance rates.

This is important because the magnitude of uncertainty may be moderate while its direction is dangerous.

A system may appear stable under aggregate volatility measures while remaining highly mobile along one narrow corridor leading toward an adverse region.

The relevant question is therefore not only

> How uncertain is the state?

It is also

> In which direction does uncertainty allow the system to move?

1.4 Hamiltonian Structure, Paths, and Reachability

The language of Hamiltonian dynamics offers a useful conceptual extension.

In a conservative physical system, the Hamiltonian is written as

$$H(\mathbf{q}, \mathbf{p}) = T(\mathbf{p}) + V(\mathbf{q}), \tag{1.12}$$

where $\mathbf{q}$ denotes position, $\mathbf{p}$ denotes momentum, T is kinetic energy, and V is potential energy. The equations of motion are

$$\dot{\mathbf{q}} = \frac{\partial H}{\partial \mathbf{p}}, \tag{1.13}$$

$$\dot{\mathbf{p}} = -\frac{\partial H}{\partial \mathbf{q}}. \tag{1.14}$$

This framework is useful because it distinguishes the system's current position from its direction and speed of movement.

Two institutions may occupy similar observed states while possessing very different momentum. One may be stabilizing. The other may be deteriorating rapidly. Two borrowers may have the same leverage and liquidity today, but one may be experiencing accelerating deposit withdrawals or refinancing pressure.

Momentum may represent accumulated directional forces such as accelerating losses, declining confidence, deposit outflows, growing litigation exposure, control deterioration, increasing operational complexity, or contagion from connected institutions.

However, a purely Hamiltonian representation is too restrictive for most risk systems. These systems are not closed and do not conserve energy. They contain friction, dissipation, control interventions, learning, information arrival, and irreversible losses.

A dissipative stochastic analogue may be written as

$$\mathrm{d}\mathbf{x}_t = \left([\mathbf{J}(\mathbf{x}_t) - \mathbf{R}(\mathbf{x}_t)]\nabla H(\mathbf{x}_t) + \mathbf{u}_t\right)\mathrm{d}t + \mathbf{\Sigma}(\mathbf{x}_t)\,\mathrm{d}\mathbf{W}_t, \tag{1.15}$$

where $\mathbf{J}$ is typically skew-symmetric and generates Hamiltonian-like motion, $\mathbf{R}$ is typically positive semidefinite and represents dissipation, $\mathbf{u}_t$ represents external intervention, and the stochastic term represents uncertainty. This is a modelling template, not a claim that institutional systems literally obey Hamiltonian mechanics.

Hamiltonian language should therefore be treated as an important special case and as a bridge to quantum mechanics, not as the universal law governing every risk system.

1.4.1 Paths, Reachability, and Action

Reachability must be specified with respect to both the admissible dynamics and the available time. A path that exists mathematically may be irrelevant if it requires impossible sequencing, unavailable resources, or more time than the decision horizon permits. Conversely, a path that appears improbable under average conditions may become dominant when constraints, correlations, or institutional responses change. Action provides a disciplined way to rank these alternatives, but its components require empirical interpretation. The analyst should identify which terms represent stress, resistance, duration, or control consumption and test whether lower-action paths are observed more frequently. This makes path analysis an auditable extension of conventional scenario design rather than a purely geometric illustration.

Once the law of motion is specified, we can study possible trajectories.

Starting from an initial state $\mathbf{x}_0$, the dynamics generate a family of possible paths. A particular realized path may be denoted by

$$\gamma = \{\mathbf{x}_t : 0 \leq t \leq T\}. \tag{1.16}$$

Let $A \subset \mathcal{M}$ denote an adverse region. The central question is whether there exists an admissible path and a time T such that

$$\mathbf{x}_T \in A. \tag{1.17}$$

If such a path exists, the adverse state is reachable.

Within a specified model, reachability logically precedes a positive transition probability: a genuinely unreachable state must have probability zero. The qualification "within a specified model" is essential, because a state may appear unreachable only because the model omits a relevant path or mechanism.

The relevant path is not always the geometrically shortest one.

A trajectory may be evaluated through an action functional

$$\mathcal{A}[\gamma] = \int_0^T L\left(\mathbf{x}_t, \dot{\mathbf{x}}_t, t\right) \,\mathrm{d}t, \tag{1.18}$$

where L is a specified transition-cost function. In applications, the choice of L must be derived from the dynamics or justified empirically; otherwise "minimum action" is only a metaphor.

A path aligned with the system's natural drift may have low action even if it is geometrically long. A short path that requires movement against strong stabilizing forces may have high action.

The most relevant route is therefore often the path of least resistance or minimum action:

$$\gamma^* = \arg\min_{\gamma} \mathcal{A}[\gamma]. \tag{1.19}$$

This distinction is central to geometric risk analysis. Distance alone is insufficient. We must understand the forces, constraints, and transition costs associated with movement.

1.5 Barriers and Adverse Transition Mechanisms

Risk management is largely concerned with barriers.

Capital protects creditors from losses. Collateral protects lenders from borrower deterioration. Liquidity reserves protect institutions from funding disruptions. Contracts protect parties from opportunistic behaviour. Internal controls protect organisations from fraud, error, and unauthorised action. Cybersecurity systems protect data and infrastructure.

The geometric perspective interprets these safeguards as barriers separating stable and adverse regions.

A barrier is not simply present or absent. It has several properties.

Barrier height represents the severity of deterioration required to cross it. A strongly capitalised borrower may need to experience substantial losses before default. A well-documented contract may require an aggressive interpretation before liability emerges.

Barrier width represents the persistence or number of transitions required to complete the movement. A borrower may survive a sharp but temporary shock while failing under a smaller but prolonged one. An institution may survive one isolated control failure but not a sustained sequence of failures.

Barrier shape reflects the fact that protection may be strong in one direction and weak in another. Capital may protect against credit losses but not against a sudden liquidity freeze. A contract may be robust in one jurisdiction and fragile in another.

Barrier continuity reflects whether the protective structure contains gaps. An institution may possess strong controls in most areas while remaining exposed through one narrow dependency.

Barrier permeability describes the degree to which the barrier permits effective transmission. This is not identical to barrier height. A barrier may be high but thin, or low but wide. The combination matters.

The relevant question is therefore not merely

> Does a control exist?

It is

> What is the geometry of the barrier, how does it evolve, and where is it permeable?

1.5.1 Three Mechanisms of Adverse Transition

These mechanisms should be treated as competing explanations rather than labels assigned after an event. Before invoking tunneling, the analyst should test whether ordinary drift, diffusion, a jump, a moving boundary, or a visible control gap accounts for the transition. Each explanation implies different evidence and different intervention. Drift suggests gradual remediation, shock-driven escape suggests loss absorption and contingency capacity, while a hidden sub-barrier pathway suggests redesigning dependencies, sequencing, or coupling. Maintaining this diagnostic order reduces the risk that the tunneling language becomes a metaphor for surprise. It also creates falsifiable comparisons in which the extended model must outperform clearly specified classical alternatives.

The dynamic framework allows us to distinguish three mechanisms through which a system may enter an adverse region.

Gradual Drift

The state may move progressively toward the boundary through ordinary dynamics. This is classical deterioration.

A borrower experiences falling revenues, weakening liquidity, rising leverage, and eventual default. The path is visible and continuous.

Shock-Driven Escape

A sufficiently large disturbance may push the system over the barrier. This is a classical rare-event transition.

A borrower may suffer an extreme market shock. An operational system may be overwhelmed by an unusually severe disruption. The transition is improbable but follows the ordinary logic of barrier crossing.

Tunneling

The system may effectively transition through a barrier that the ordinary dynamics treat as prohibitive.

The distinction can be summarised as

$$\begin{aligned} \text{Drift} &\longrightarrow \text{movement along the landscape,} && (1.20)\\ \text{Shock-driven escape} &\longrightarrow \text{movement over the barrier,} && (1.21)\\ \text{Tunneling} &\longrightarrow \text{effective transmission through the barrier.} && (1.22) \end{aligned}$$

Tunneling is therefore not merely another name for tail risk.

A tail event is defined by low probability. A tunneling event is defined by the structure of the transition.

A rare event may occur through an ordinary path. In this book, a *tunneling-inspired* event is one in which an adverse region is connected to the present state by a mechanism that the baseline transition model treats as inaccessible, negligible, or absent. The term describes a mathematical and structural analogy; it does not imply literal quantum behaviour in institutions.

To avoid a category error, the discussion that follows keeps three levels distinct: the established physical theory of quantum tunneling, the abstract mathematics of barriers and transmission, and the proposed use of that mathematics in risk modelling. Results at the first level do not automatically validate claims at the third.

1.5.2 What, Precisely, Tunnels?

The answer must remain model-relative and operationally interpretable. What crosses the barrier is not a literal borrower, institution, contract, or computer system. It is probability weight, state amplitude, or another formally defined representation of possible configurations and transition routes. The modeller must specify how that representation is estimated, normalized, and connected to observable outcomes. Without this bridge, the language of transmission cannot support prediction or intervention. The practical test is whether changes in the represented weight precede adverse events, respond coherently to barrier-strengthening actions, and add stable information beyond classical state probabilities and pathway interaction terms.

The word *tunneling* can produce a misleading image. It may suggest that a classical particle follows a hidden trajectory through a solid wall, as if it drilled a microscopic passage from one side to the other.

That is not the quantum-mechanical description.

In quantum mechanics, the primitive object is not a particle with a perfectly defined position and trajectory. The system is represented by a state vector

$$|\psi(t)\rangle \in \mathcal{H}, \tag{1.23}$$

where $\mathcal{H}$ is a Hilbert space.

The state vector contains the information required to calculate the probabilities of possible measurement outcomes.

In the position representation, the same state is described by the wave function

$$\psi(x,t) = \langle x \mid \psi(t)\rangle. \tag{1.24}$$

The quantity $\psi(x,t)$ is generally complex-valued. It is not itself a probability. According to the Born rule, the probability density for a position measurement at x and time t is

$$\rho(x,t) = |\psi(x,t)|^2. \tag{1.25}$$

The state is normalised so that

$$\int_{-\infty}^{\infty} |\psi(x,t)|^2 \, \mathrm{d}x = 1. \tag{1.26}$$

The wave function assigns probability amplitudes across configuration space. It need not be concentrated near a single classical location and may have non-zero amplitude in several spatial regions at the same time.

What tunnels is consequently not a classical point following a concealed path. What penetrates the barrier is the quantum state itself. More precisely, the wave function extends into the classically forbidden region and, when the barrier has finite width, can retain a non-zero amplitude on its far side.

This distinction is fundamental for the risk-management analogy. A tunneling interpretation should not begin by imagining an institution as a small object unexpectedly jumping through a wall. It should begin by asking whether the representation of the institution's possible configurations has a non-zero extension into states that the classical risk model treats as inaccessible.

1.6 Quantum Evolution and the Rectangular Barrier

The quantum state does not evolve arbitrarily. Its law of motion is the time-dependent Schrödinger equation:

$$\mathrm{i}\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = \widehat{H}|\psi(t)\rangle, \tag{1.27}$$

where i is the imaginary unit, $\hbar$ is the reduced Planck constant, and $\widehat{H}$ is the Hamiltonian operator.

For a single non-relativistic particle moving in one dimension, the Hamiltonian is commonly written as

$$\widehat{H} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x), \tag{1.28}$$

where m is the particle's mass and $V(x)$ is the potential-energy function.

Substituting Equation (1.28) into Equation (1.27) gives the position-space form

$$\mathrm{i}\hbar\frac{\partial\psi(x,t)}{\partial t} = \left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)\right]\psi(x,t). \tag{1.29}$$

This equation performs the role that the law of motion performed in the preceding geometric discussion. The potential describes the landscape, while the Hamiltonian determines the evolution of the state over that landscape.

There is, however, an important difference from classical mechanics. In the classical Hamiltonian formulation, the system follows a trajectory through phase space. In quantum mechanics, the Hamiltonian evolves a state vector. The result is not generally one unique path but an evolving distribution of probability amplitudes.

For a Hamiltonian that does not depend explicitly on time, the formal solution is

$$|\psi(t)\rangle = \exp\left(-\frac{\mathrm{i}}{\hbar}\widehat{H}t\right)|\psi(0)\rangle. \tag{1.30}$$

The time-evolution operator is therefore

$$\widehat{U}(t) = \exp\left(-\frac{\mathrm{i}}{\hbar}\widehat{H}t\right). \tag{1.31}$$

Quantum tunneling therefore does have a law of motion. The phenomenon does not arise because the system ceases to obey dynamics. It arises because the relevant dynamics govern a wave function rather than a classical point particle.

1.6.1 The Rectangular Barrier

For a time-independent potential, solutions may be separated into spatial and temporal components:

$$\psi(x,t) = \phi(x)\exp\left(-\frac{\mathrm{i}Et}{\hbar}\right), \tag{1.32}$$

where E is an energy eigenvalue.

Substitution into the time-dependent Schrödinger equation yields the time-independent equation

$$\widehat{H}\phi(x) = E\phi(x), \tag{1.33}$$

or, explicitly,

$$-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2\phi(x)}{\mathrm{d}x^2} + V(x)\phi(x) = E\phi(x). \tag{1.34}$$

Consider a rectangular potential barrier of height V_0 and width a:

$$V(x) = \begin{cases} 0, & x < 0, \\ V_0, & 0 \le x \le a, \\ 0, & x > a. \end{cases} \tag{1.35}$$

Suppose that the particle has energy

$$0 < E < V_0. \tag{1.36}$$

Classically, energy conservation would require the kinetic energy inside the barrier to be

$$K = E - V_0 < 0. \tag{1.37}$$

Because classical kinetic energy cannot be negative, the region $0 \le x \le a$ is classically forbidden for a particle with $E < V_0$.

Quantum mechanically, however, the Schrödinger equation still has a valid solution inside that region.

Before the barrier, the general stationary solution is

$$\phi_{\mathrm{I}}(x) = A\exp(\mathrm{i}kx) + B\exp(-\mathrm{i}kx), \qquad x < 0, \tag{1.38}$$

where

$$k = \frac{\sqrt{2mE}}{\hbar}. \tag{1.39}$$

The first term represents an incident wave travelling toward the barrier. The second represents a reflected wave travelling away from it.

Inside the barrier, the Schrödinger equation becomes

$$\frac{\mathrm{d}^2\phi_{\mathrm{II}}(x)}{\mathrm{d}x^2} = \kappa^2\phi_{\mathrm{II}}(x), \tag{1.40}$$

where

$$\kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar}. \tag{1.41}$$

The solution is

$$\phi_{\mathrm{II}}(x) = C\exp(\kappa x) + D\exp(-\kappa x), \qquad 0 \le x \le a. \tag{1.42}$$

The solution no longer oscillates as it does in a classically allowed region. Instead, it contains exponentially increasing and decreasing components. After the boundary conditions are imposed, the physically relevant result is that the amplitude decays through the barrier but does not generally become exactly zero.

Beyond the barrier, the transmitted solution is

$$\phi_{\mathrm{III}}(x) = F\exp(\mathrm{i}kx), \qquad x > a. \tag{1.43}$$

There is therefore a non-zero transmitted amplitude F on the far side of the barrier.

For this finite potential with constant mass, the wave function and its first derivative must be continuous at both boundaries:

$$\phi_{\mathrm{I}}(0) = \phi_{\mathrm{II}}(0), \tag{1.44}$$

$$\left.\frac{\mathrm{d}\phi_{\mathrm{I}}}{\mathrm{d}x}\right|_{x=0} = \left.\frac{\mathrm{d}\phi_{\mathrm{II}}}{\mathrm{d}x}\right|_{x=0}, \tag{1.45}$$

$$\phi_{\mathrm{II}}(a) = \phi_{\mathrm{III}}(a), \tag{1.46}$$

$$\left.\frac{\mathrm{d}\phi_{\mathrm{II}}}{\mathrm{d}x}\right|_{x=a} = \left.\frac{\mathrm{d}\phi_{\mathrm{III}}}{\mathrm{d}x}\right|_{x=a}. \tag{1.47}$$

These matching conditions connect the incident, reflected, decaying, and transmitted components of the state. The barrier does not divide the wave function into unrelated pieces. It modifies one globally defined solution.

This is the mathematical core of tunneling.

1.7 Transmission and Barrier Scaling

The presence of amplitude beyond the barrier is not by itself the complete description of transmission. Quantum mechanics also defines a probability current:

$$j(x,t) = \frac{\hbar}{2mi}\left[\psi^*(x,t)\frac{\partial\psi(x,t)}{\partial x} - \psi(x,t)\frac{\partial\psi^*(x,t)}{\partial x}\right]. \tag{1.48}$$

Equivalently,

$$j(x,t) = \frac{\hbar}{m}\,\mathrm{Im}\left[\psi^*(x,t)\frac{\partial\psi(x,t)}{\partial x}\right]. \tag{1.49}$$

The probability density and probability current satisfy the continuity equation

$$\frac{\partial\rho(x,t)}{\partial t} + \frac{\partial j(x,t)}{\partial x} = 0. \tag{1.50}$$

This equation expresses conservation of total probability.

The transmission coefficient is defined as the ratio of transmitted probability current to incident probability current:

$$T = \frac{j_{\text{transmitted}}}{j_{\text{incident}}}. \tag{1.51}$$

Similarly, the reflection coefficient is

$$R = \frac{|j_{\text{reflected}}|}{j_{\text{incident}}}. \tag{1.52}$$

For a lossless one-dimensional barrier,

$$R + T = 1. \tag{1.53}$$

Tunneling is therefore not simply the observation that the wave function is mathematically non-zero in a forbidden region. It is the existence of a non-zero transmitted current beyond the barrier.

1.7.1 Barrier Height, Width, and Exponential Suppression

Together, these quantities determine whether protection is merely visible in form or genuinely resistant across the full transition route.

Height and width describe different dimensions of protection and should be measured separately. Height represents the intensity of deterioration, shock, or control failure required at a particular state. Width represents the duration, number of stages, institutional distance, or persistence needed to complete the adverse transition. A barrier may be high but narrow when one severe event can defeat it quickly, or low but wide when many modest conditions must remain aligned for an extended period. These alternatives can produce similar headline protection scores while implying different vulnerabilities and intervention strategies.

The exponential relationship makes measurement discipline especially important. Small errors in estimated height, width, resistance, or scale can create large changes in the calculated transmission coefficient. The model should therefore report uncertainty intervals, parameter sensitivity, and alternative barrier definitions rather than a single precise number. Historical episodes, stress tests, and intervention outcomes can help determine whether estimated transmission responds realistically when protection strengthens or weakens. If the exponential form produces unstable results or cannot be calibrated against observations, a simpler monotone accessibility measure may be preferable. The theoretical form is valuable because it highlights nonlinear sensitivity, but empirical use requires conservative estimation and transparent validation.

This distinction should remain visible in every committee report and model review.

For the rectangular barrier above, with $E < V_0$ and equal potentials on both sides, the exact transmission coefficient is

$$T = \left[1 + \frac{V_0^2 \sinh^2(\kappa a)}{4E(V_0 - E)}\right]^{-1}. \tag{1.54}$$

When the barrier is sufficiently high or wide so that

$$\kappa a \gg 1, \tag{1.55}$$

the dominant behaviour is approximately

$$T \propto \exp(-2\kappa a). \tag{1.56}$$

Using Equation (1.41), this becomes

$$T \propto \exp\left[-\frac{2a}{\hbar}\sqrt{2m(V_0 - E)}\right]. \tag{1.57}$$

This expression provides the key intuition for the risk-management framework.

Transmission decreases when the barrier becomes wider, when the barrier becomes higher relative to the system's energy, or when the effective inertia of the system increases.

The dependence is exponential rather than linear. A modest change in effective barrier width or height may therefore produce a large change in transmission.

The essential insight is that a barrier's effectiveness cannot be summarised by height alone. A high but narrow barrier may be more permeable than a somewhat lower but substantially wider barrier.

1.7.2 Tunneling Through a General Barrier

General barriers are rarely rectangular or stationary. Their resistance may vary across state dimensions, depend on direction, and change as institutions react. Liquidity support may be strong until several entities draw simultaneously; collateral may retain value until market depth disappears; a legal defence may weaken as evidence or precedent evolves. The action integral accommodates this variation by accumulating local resistance along each candidate route. It therefore directs attention toward the complete path rather than one average barrier score.

In implementation, the integral will usually be approximated on a graph, sequence model, discretized state space, or simulated trajectory set. Each edge or segment should carry an interpretable resistance based on observed protection, transition difficulty, and time. Competing routes can then be ranked by total action, with uncertainty propagated from their component estimates. The most important validation question is whether low-action routes resemble the mechanisms observed before real adverse events. If they do not, the apparent mathematical precision has no operational value. A general-barrier model succeeds when it identifies actionable weak routes earlier and more reliably than conventional scenarios.

That comparison must be repeated whenever the system or regime materially changes. It should also be documented with the parameter ranges, alternative paths, and evidence used to support the selected resistance profile.

The rectangular barrier is analytically useful, but real barriers need not have vertical walls or constant height.

For a smooth potential $V(x)$, let x_1 and x_2 denote the classical turning points satisfying

$$V(x_1) = V(x_2) = E. \tag{1.58}$$

Between these points,

$$V(x) > E, \qquad x_1 < x < x_2. \tag{1.59}$$

Under the semiclassical Wentzel–Kramers–Brillouin (WKB) approximation, the leading exponential contribution to the transmission coefficient is

$$T_{\mathrm{WKB}} \approx \exp\left[-2\int_{x_1}^{x_2} \kappa(x)\,\mathrm{d}x\right], \tag{1.60}$$

where

$$\kappa(x) = \frac{\sqrt{2m[V(x)-E]}}{\hbar}. \tag{1.61}$$

Equivalently,

$$T_{\mathrm{WKB}} \approx \exp\left[-\frac{2}{\hbar}\int_{x_1}^{x_2} \sqrt{2m[V(x)-E]}\,\mathrm{d}x\right]. \tag{1.62}$$

Define the under-barrier action

$$S_{\mathrm{B}} = \int_{x_1}^{x_2} \sqrt{2m[V(x) - E]}\,\mathrm{d}x. \tag{1.63}$$

Then

$$T_{\mathrm{WKB}} \approx \exp\left(-\frac{2S_{\mathrm{B}}}{\hbar}\right). \tag{1.64}$$

The action S_{B} combines barrier height, width, and shape into one quantity. It is not merely the maximum height of the barrier that governs transmission. The entire forbidden region contributes to the exponent.

This offers a more precise interpretation of barrier geometry. The relevant object is the integrated resistance encountered across the transition region.

1.8 Quantum Tunneling and Classical Barrier Crossing

The distinction between classical escape and quantum tunneling must remain explicit.

In classical dynamics, a system crosses a barrier when it acquires enough energy to pass over it:

$$E_{\text{effective}} > V_{\text{barrier}}. \tag{1.65}$$

In quantum tunneling, transmission occurs even when

$$E < V_{\text{barrier}}. \tag{1.66}$$

The system does not acquire the classical energy required to climb over the barrier. Instead, the wave function penetrates the forbidden region and produces non-zero transmitted probability current on the other side.

This distinction matters for the book. A very large financial shock that overwhelms capital is analogous to classical barrier crossing. It is not, by itself, evidence of tunneling.

A tunneling-inspired risk model becomes relevant when the adverse transition cannot be adequately represented as ordinary movement over the barrier under the baseline dynamics, but a well-defined alternative pathway still produces transmission. This criterion concerns model structure, not evidence that a macroscopic institution behaves quantum mechanically.

1.9 Risk-State Amplitudes and Effective Operators

A disciplined translation into risk management now becomes possible.

Let the possible configurations of a risk-bearing system form a state space $\mathcal{M}$. In the classical model, the system occupies a state

$$\mathbf{x}_t \in \mathcal{M}, \tag{1.67}$$

and evolves according to Equation (1.4).

In a tunneling-inspired representation, we may instead associate the system with a state amplitude over configurations:

$$\Psi(\mathbf{x}, t). \tag{1.68}$$

If Ψ is normalised with respect to a chosen measure μ on $\mathcal{M}$, the corresponding state density is

$$\rho_{\mathrm{R}}(\mathbf{x}, t) = |\Psi(\mathbf{x}, t)|^2. \tag{1.69}$$

At this stage, Ψ should not be interpreted as a literal physical wave function. It is a modelling construct representing latent configurations and the strength with which the current institutional condition is associated with different regions of the state space. The choice of measure, normalization, and empirical interpretation must be specified before ρ_{R} can be used quantitatively.

Suppose that $S \subset \mathcal{M}$ is a stable region, $A \subset \mathcal{M}$ is an adverse region, and $B \subset \mathcal{M}$ is a barrier separating them.

The baseline model may imply that ordinary trajectories cannot enter A without crossing the top of B.

A tunneling-inspired model asks how much state weight lies in the adverse region:

$$P_A(t) = \int_A |\Psi(\mathbf{x}, t)|^2 \, \mathrm{d}\mu(\mathbf{x}), \tag{1.70}$$

where $\mathrm{d}\mu(\mathbf{x})$ is the chosen volume measure on the state manifold. Mere positivity of $P_A(t)$ is usually too weak to be operationally meaningful, because many smooth distributions have non-zero tails. Applications should therefore estimate its magnitude, its change through time, or whether it exceeds a decision-relevant threshold.

The important idea is not that an institution is physically present in several states in the quantum-mechanical sense. The idea is that the observed institutional state may be an incomplete projection of a richer configuration.

Latent dependencies, unobserved exposures, behavioural responses, legal interpretations, and network couplings may already connect the system to adverse regions.

The visible state may lie safely before the barrier while the broader configuration has a non-negligible extension into or beyond it.

1.9.1 An Effective Risk-Tunneling Operator

The operator is effective in the modelling sense: it summarizes the assumed evolution of a risk representation at the chosen scale. Its terms should be linked to measurable mobility, resistance, coupling, and time dependence, and competing specifications should be compared out of sample. No institutional conclusion follows merely from writing an operator with the form of a Hamiltonian. Its value depends on whether it generates calibrated state weights, credible transition paths, and intervention sensitivities that remain stable across regimes and reference classes.

For research purposes, the analogy may be extended by postulating an effective evolution equation

$$\mathrm{i}\eta \frac{\partial}{\partial t}|\Psi(t)\rangle = \widehat{\mathcal{H}}_{\mathrm{R}}|\Psi(t)\rangle, \tag{1.71}$$

where $|\Psi(t)\rangle$ is an abstract risk-state vector, $\widehat{\mathcal{H}}_{\mathrm{R}}$ is an effective risk-transition operator, and η is a scaling parameter governing the relationship between state dispersion and barrier penetration.

In local Euclidean coordinates, an illustrative effective operator could take the form

$$\widehat{\mathcal{H}}_{\mathrm{R}} = -\frac{\eta^2}{2}\nabla\cdot\left[\mathbf{M}^{-1}(\mathbf{x})\nabla\right] + V_{\mathrm{R}}(\mathbf{x},t), \tag{1.72}$$

where $V_{\mathrm{R}}(\mathbf{x},t)$ is the risk-potential landscape and $\mathbf{M}(\mathbf{x})$ is a positive-definite effective resistance or inertia matrix.

The differential term represents the tendency of the state amplitude to spread across nearby configurations.

This equation is not an empirically validated law of institutional behaviour. It is a formal research hypothesis whose purpose is to identify which components of barrier transmission might be translated into measurable risk quantities. Moreover, the factor i generates norm-preserving evolution only under suitable operator conditions, whereas real institutions are open and dissipative. Empirical applications may therefore require a non-unitary evolution equation, such as a master equation or diffusion generator, instead.

Under a semiclassical approximation, and after choosing units so that S_{R}/η is dimensionless, an effective risk-transmission coefficient might have the form

$$T_{\mathrm{R}} \approx \exp\left[-\frac{2}{\eta}S_{\mathrm{R}}\right], \tag{1.73}$$

where the risk-barrier action is

$$S_{\mathrm{R}} = \int_{\gamma_{\mathrm{B}}} \sqrt{2\left[V_{\mathrm{R}}(\mathbf{x},t) - E_{\mathrm{R}}\right]}\,\mathrm{d}s_{\mathbf{M}}. \tag{1.74}$$

Here, γ_{B} is a path through the barrier region, E_{R} is the system's effective capacity to absorb stress, and $\mathrm{d}s_{\mathbf{M}}$ is distance measured using the effective resistance geometry. The integrand is real along portions of the path for which $V_{\mathrm{R}}(\mathbf{x},t) > E_{\mathrm{R}}$.

The dominant tunneling path would be the path minimizing the barrier action:

$$\gamma_{\mathrm{B}}^{*} = \arg\min_{\gamma_{\mathrm{B}}} S_{\mathrm{R}}[\gamma_{\mathrm{B}}]. \tag{1.75}$$

The corresponding approximation would be

$$T_{\mathrm{R}} \approx \exp\left[-\frac{2}{\eta} S_{\mathrm{R}}\left[\gamma_{\mathrm{B}}^{*}\right]\right]. \tag{1.76}$$

This formulation gives mathematical content to the idea of a hidden pathway, conditional on the specified potential, metric, and scaling. The relevant pathway is not necessarily the shortest path in ordinary coordinates; it is the path of minimum integrated barrier resistance under the model.

1.9.2 The Risk-Management Interpretation

Translation requires a documented mapping from every mathematical object to an empirical quantity and a decision use. Barrier height might correspond to liquidity capacity, collateral resilience, control strength, or evidentiary protection; width might correspond to time, stages, or independent safeguards. Phase must be grounded in timing or synchronization data rather than assigned narratively. The resulting outputs should be expressed in conventional governance terms such as probability, expected loss, limit usage, warning lead time, and intervention effect. This discipline preserves the explanatory benefit of the framework while preventing metaphor from substituting for measurement.

The translation from quantum mechanics to risk management can be summarised as follows.

Table 1.1: Translation from quantum concepts to risk-management interpretations.

Quantum concept	**Risk-management interpretation**
Quantum state	Structured representation of latent system configurations
Wave function	Amplitude assigned to possible configurations
Probability density	Relative weight associated with regions of the risk state space
Hamiltonian	Operator governing state evolution and coupling
Potential landscape	Structural landscape of stability, vulnerability, and resistance
Classically allowed region	States reachable through ordinary modelled dynamics
Classically forbidden region	States treated as inaccessible under the conventional transition model
Barrier height	Severity of deterioration or stress required for transition
Barrier width	Persistence, institutional distance, or number of reinforcing transitions required
Wave-function decay	Declining but non-zero latent exposure through the protective structure
Transmission coefficient	Intensity of effective transition into the adverse region
Under-barrier action	Integrated resistance along a possible transition pathway
Turning points	Boundaries at which ordinary dynamics cease to support the transition
Probability current	Directional flow of state weight across a boundary

This translation also clarifies what tunneling is not.

Tunneling is not simply a large shock, a rare observation, an unexpected event, a jump process, an omitted variable, or a failure of forecasting.

Any of these may occur without tunneling.

A risk transition should be described as tunneling only when three conditions are satisfied:

1. A clearly specified baseline model classifies the adverse region as inaccessible or prohibitively difficult to reach through ordinary barrier crossing.

   Explanation. The baseline determines what the word "forbidden" means and prevents the tunneling claim from becoming circular. It should specify the state variables, transition law, shocks, jumps, constraints, and decision horizon before the extended model is introduced. Analysts must demonstrate that ordinary drift, diffusion, regime change, or visible threshold crossing produces negligible or prohibitively costly access under this benchmark. The conclusion should be supported by simulation, historical calibration, or structural analysis rather than intuition. If the baseline can already reach the adverse region with realistic probability, the event is classical within that model and no tunneling interpretation is required.

2. A protective barrier can be meaningfully identified between the current and adverse regions.

   Explanation. The barrier must correspond to observable or defensible protection, not merely to a line drawn around the adverse state. Relevant components may include liquidity, collateral, covenants, capital, legal rights, controls, redundancy, response capacity, or policy support. The model should measure their height, width, continuity, reliability, independence, and activation time. It should also identify gaps and shared dependencies that would make the protection only nominal. Evidence that the barrier changes outcomes under stress or intervention is especially important. Without a measurable protective structure, surprising transmission is more plausibly explained by misspecification, omitted variables, or ordinary exposure than by a sub-barrier mechanism.

3. An omitted coupling, latent pathway, or distributed-state representation generates effective transmission without requiring the ordinary over-the-barrier mechanism.

   Explanation. The extended mechanism must identify how adverse accessibility arises despite the benchmark barrier. A coupling may connect variables or institutions that the baseline treats as independent; a latent pathway may sequence individually modest weaknesses; a distributed representation may assign non-zero weight to configurations excluded by a point-state description. Whatever form is chosen, it must yield testable implications for timing, pathway concentration, interference, or intervention response. The mechanism should improve out-of-sample prediction or explanation after comparison with nonlinear interactions and jump models. Otherwise, the added mathematical language increases complexity without establishing that genuine sub-barrier transmission has been detected.

These conditions are necessary but not sufficient. A useful tunneling-inspired model must also define observable or estimable quantities, produce testable implications, and improve on simpler explanations such as model misspecification, omitted-variable bias, or an ordinary jump process.

1.10 The Minimal Model, Analytical Order, and Synthesis

A complete geometric model of risk requires at least seven elements:

1. A state manifold $\mathcal{M}$.

   Explanation. The manifold defines the feasible configurations of the system and the structural relationships among its variables. It is more than a data matrix because accounting identities, contractual rules, physical capacities, and institutional dependencies exclude many numerical combinations. Construction may begin with explicit constraints, observed state clouds, simulation, or learned embeddings, but the resulting representation must preserve economically meaningful neighbourhoods. Analysts should document its boundaries, dimensions, reference population, and regime dependence. A poorly specified manifold can create artificial proximity or conceal important routes. Validation therefore asks whether nearby states exhibit similar dynamics, barrier conditions, and outcomes in data not used to construct the representation.

2. A state vector $\mathbf{x}_t$ locating the system on the manifold.

   Explanation. The state vector contains the smallest sufficiently informative collection of variables needed to model future transitions at the chosen horizon. Its coordinates should represent capacity, obligations, protection, behaviour, dependencies, and access to resources rather than every available field. Variables must be timestamp-correct, consistently transformed, and interpretable in direction. If relevant memory is omitted, the state may require augmentation with lagged or cumulative variables. The vector should also carry data-quality information because missing or stale observations affect confidence. A useful state definition supports both prediction and intervention by showing which dimensions are changing and which changes can be influenced by management or policy.

3. A metric $\mathbf{g}(\mathbf{x})$ defining meaningful distance.

   Explanation. The metric determines which movements are consequential and how changes in different coordinates combine. Euclidean distance after standardization may be an initial benchmark, but it assumes equal, state-independent importance. A richer metric can reflect covariance, transition cost, causal sensitivity, or learned local geometry. It should be positive definite, stable enough for monitoring, and calibrated against observed transitions. Analysts must explain why movement in one direction is effectively longer or shorter than movement in another. Validation examines whether metric distance improves discrimination, lead time, or pathway reconstruction beyond raw variable differences. If it does not, the simpler geometry should remain the operational choice.

4. A potential or risk landscape $V(\mathbf{x}, t)$.

   Explanation. The landscape represents relative stability, vulnerability, or resistance across the state space. Local minima may correspond to persistent operating regimes, while ridges and saddle regions may separate stable and adverse configurations. The function can be structural, statistical, expert-calibrated, or hybrid, but its scale and interpretation must be explicit. Because institutions and environments change, the landscape may be time-dependent and should be versioned. Analysts should test whether its gradients correspond to observed drift and whether its high-resistance regions actually delay transitions. A landscape that merely reproduces the outcome labels without explaining movement provides visualization, but not an empirically credible transition model.

5. A law of motion governing the evolution of the system.

   Explanation. The law of motion specifies how the state changes through drift, diffusion, jumps, feedback, and intervention. It forms the classical benchmark against which unusual accessibility is assessed. Estimation should respect time ordering, censoring, regimes, and state dependence, while validation should test both one-step forecasts and longer transition paths. The model may be a stochastic differential equation, hazard system, state-space model, agent-based simulation, or another appropriate process. Its assumptions about continuity and shocks are particularly important because a jump can mimic apparent tunneling. A credible implementation therefore compares alternative dynamics and reports where the data cannot reliably distinguish among them.

6. Stable, fragile, and adverse regions separated by barriers.

   Explanation. Region definitions translate the continuous state space into operational conditions. Stable states perform within tolerance and retain usable protection. Fragile states remain functional but depend increasingly on depleted buffers, concentrated resources, exceptions, or delayed intervention. Adverse states must correspond to observable events such as default, outage, sanction, or systemic impairment. Boundaries should be defined before evaluation and tested for sensitivity because moving a threshold can alter measured distance and transmission. Multiple adverse regions may be needed when legal, financial, operational, and reputational outcomes differ. Governance should understand both the formal boundary and the earlier fragile zone where preventive action remains feasible.

7. A transition mechanism capable of generating transmission through an apparently prohibitive barrier.

   Explanation. The transition mechanism connects the geometry to observable adverse outcomes. It may consist of interacting pathways, hidden dependencies, phase alignment, minimum-action routes, or another structure that produces non-zero accessibility without ordinary barrier crossing. The mechanism should specify magnitudes, timing variables, and how separate routes combine. It must also respond coherently to intervention: strengthening the relevant barrier or disrupting the sequence should reduce estimated transmission. Comparison with classical interaction, contagion, and jump models is mandatory. The mechanism is justified only when it adds stable, explainable, and actionable value rather than fitting a complicated retrospective narrative to events already known.

The state alone is not the model. The surface alone is not the model. The potential alone is not the model.

The complete framework is

$$\boxed{\begin{aligned}\text{Risk Model} = \text{Geometry} + \text{Dynamics}\\ + \text{Barriers} + \text{Transition Mechanism.}\end{aligned}} \tag{1.77}$$

Only after these elements have been specified does it become meaningful to estimate transition probabilities.

1.10.1 A Revised Order for Risk Analysis

The revised order is intended as a disciplined workflow rather than an inflexible sequence. In practice, state selection, probability estimation, barrier construction, and pathway discovery will inform one another. Initial models may reveal omitted variables; stress tests may expose missing barriers; intervention analysis may show that an apparently predictive variable has no controllable mechanism. The sequence nevertheless imposes useful governance checkpoints. Analysts should not estimate sophisticated transmission measures before defining the event, state, benchmark dynamics, and protection structure on which those measures depend.

Each stage should produce an auditable output. State construction produces a data dictionary and feasibility rules. Geometry produces a metric and region definitions. Dynamics produce drift, covariance, jump, and regime estimates. Barrier analysis produces protection components, reliability adjustments, dependencies, and capacity measures. Path analysis produces ordered routes and action estimates. Only then should the model combine pathways, estimate transmission, and compare interventions. This ordering makes validation modular: a failure in one layer can be diagnosed without treating the entire model as an opaque score.

The workflow also clarifies decision accountability. Business experts help define feasible states and interventions; risk functions challenge assumptions and limits; model developers estimate the structural objects; validation tests stability and incremental value; governance bodies authorize actions. Probability remains central, but it becomes the output of a transparent chain whose geometry and mechanisms can be reviewed, stressed, and improved.

The geometric framework suggests a different sequence of analysis:

1. Define the state variables.

   Explanation. Begin with the decision and adverse event, then select variables that represent the mechanisms capable of changing that outcome. The state should include resources, obligations, protection, behaviour, dependencies, and relevant memory while remaining parsimonious enough to estimate and explain. Each variable needs a definition, source, availability timestamp, transformation, expected direction, and quality flag. Analysts should test whether omitted information produces path dependence or unstable residuals. This stage also identifies variables that management can influence, which later supports intervention analysis. A disciplined state definition prevents the modelling exercise from becoming an unrestricted search across convenient but poorly governed data.

2. Construct the relevant state space or manifold.

   Explanation. Organize the selected variables into the set of feasible configurations rather than assuming every combination is possible. Accounting constraints, legal rules, system capacities, and institutional processes should be encoded explicitly where feasible. Data-driven embeddings can reveal nonlinear structure, but they require stability checks, interpretability analysis, and out-of-sample validation. The manifold should preserve meaningful neighbourhoods and distinguish structural impossibility from merely rare observation. Analysts should inspect coverage across sectors, regimes, and outcome classes because sparse regions may create unreliable distances. The completed state space provides the coordinate system within which boundaries, barriers, paths, and movement can be defined consistently.

3. Specify the metric and the meaning of distance.

   Explanation. Decide how displacements in the state space should be weighted and combined. The metric may begin with standardized Euclidean distance, covariance-adjusted distance, transition cost, or a learned local measure. Its choice must reflect the intended interpretation: economic difference, dynamic accessibility, intervention effort, or another clearly stated concept. Analysts should verify positive definiteness, scaling stability, and sensitivity to regime changes. Empirical tests should determine whether shorter distances predict faster or more frequent transitions after controlling for conventional risk scores. Reporting should distinguish geometric proximity from probability because a nearby state may remain dynamically inaccessible when drift and protection oppose movement.

4. Identify stable, fragile, and adverse regions.

   Explanation. Translate operational definitions into regions of the state space using observable thresholds, labelled outcomes, structural conditions, or supervised boundaries. Stable and adverse states alone are insufficient because the fragile region captures systems that still function while protection is being consumed. Region definitions should reflect the decision horizon and materiality standard, and multiple adverse outcomes may require separate boundaries. Analysts should freeze definitions before testing to avoid using future knowledge. Sensitivity analysis can reveal whether conclusions depend on arbitrary thresholds. These regions create the destinations and warning zones required for distance measurement, barrier construction, and escalation rules.

5. Define the law of motion.

   Explanation. Specify the baseline process that moves the system through its state space. The process should include deterministic tendencies, stochastic variation, discontinuities, regimes, feedback, and management action to the extent supported by evidence. Estimation must use only information available at each date and should account for censoring and unequal observation frequency. Alternative specifications should be compared because excessive diffusion, missing jumps, or ignored feedback can create false evidence of hidden transmission. The chosen law of motion becomes the null model for accessibility. Its limitations should therefore be documented before any claim that an adverse state is classically difficult to reach.

6. Estimate drift, uncertainty, momentum, and jumps.

   Explanation. Decompose observed movement into directional tendency, local covariance, persistent trajectory effects, and discontinuous events. Drift shows where the system tends to move, while covariance reveals the directions in which uncertainty is expanding. Momentum or persistence can distinguish a temporary fluctuation from sustained deterioration. Jump estimates capture sudden funding withdrawals, incidents, rulings, or other discrete changes that might otherwise be mistaken for tunneling. These components should be estimated with uncertainty and stress-tested across regimes. Their projections toward the adverse boundary provide interpretable early-warning measures and establish whether classical dynamics already explain the apparent transition route.

7. Represent protective barriers.

   Explanation. Map real protections into height, width, continuity, reliability, independence, residual capacity, and activation time. Nominal amounts should be adjusted for availability under stress and for shared dependencies that reduce effective layering. Evidence may come from financial capacity, control tests, legal analysis, recovery exercises, or historical intervention outcomes. The barrier should cover the relevant path rather than exist only in aggregate. Analysts should identify visible gaps separately because movement through a gap is classical bypass, not tunneling. A governed barrier representation names owners, limits, testing schedules, and uncertainty so that weaknesses can be monitored and corrected.

8. Identify plausible transition paths.

   Explanation. Construct ordered sequences that connect the current state to each adverse region. Paths should arise from process evidence, historical episodes, causal analysis, graphs, simulations, and expert challenge rather than unrestricted storytelling. Each stage needs an observable indicator and a defensible relationship to the next stage. Competing paths should remain distinct when they involve different mechanisms or interventions. Analysts should record timing, prerequisites, crossed barrier components, and possible interruptions. Path definitions can then be validated against event histories and near misses. This step changes scenario analysis from a collection of shocks into a structured map of how adverse outcomes may unfold.

9. Search for hidden or low-action pathways.

   Explanation. Evaluate candidate routes for integrated resistance rather than focusing only on the largest visible weakness. A hidden path may pass through moderate vulnerabilities that become consequential when sequenced or synchronized. Graph algorithms, minimum-action methods, process mining, and constrained simulation can help rank routes, but the resistance assigned to each segment must remain interpretable. Analysts should test path stability under alternative parameters and determine whether the route was detectable before known events. A low-action path is operationally valuable when it identifies a specific barrier component, dependency, or timing relationship that can be altered before the adverse sequence completes.

10. Evaluate barrier permeability.

    Explanation. Combine barrier integrity, path action, directional mobility, timing, and transmission into a transparent assessment of adverse accessibility. Permeability should not be reduced to one opaque score; component levels, trends, and uncertainty must remain visible. The measure should rise when protection weakens, routes shorten, or harmful pathways align, and it should fall when validated interventions strengthen or interrupt the structure. Calibration against historical events and benign periods is necessary to control false warnings. Governance should connect permeability thresholds to investigation, escalation, and authorized action while recognizing that an elevated measure indicates accessibility, not certainty of failure.

11. Estimate transition probabilities.

    Explanation. Translate the structural model into probabilities that can support pricing, limits, capital, and intervention decisions. Pathway magnitudes should remain anchored in classical frequencies, hazards, or calibrated simulations, while any amplitude or interference extension must demonstrate incremental value. Probabilities require temporal and cross-entity validation, calibration plots, uncertainty intervals, and comparison with strong benchmarks. The model should distinguish total adverse probability from contributions associated with state deterioration, barrier weakness, pathways, and timing. This decomposition allows committees to understand why risk changed and prevents the final estimate from becoming detached from the geometry and evidence that produced it.

12. Design interventions that reshape the geometry or dynamics.

    Explanation. Compare actions according to how they raise barrier height, increase width, remove dependencies, close gaps, redirect drift, or change pathway timing. Examples include liquidity support, maturity extension, collateral enhancement, control separation, evidence preservation, and coordinated policy response. Intervention analysis should consider cost, capacity, authority, delivery time, side effects, and uncertainty. Counterfactual or causal methods are preferable where evidence permits, supplemented by stress testing and expert review. The selected action should target the dominant route and arrive before pathway completion. Post-intervention monitoring must verify that measured accessibility actually declined and update the model when the observed effect differs from expectation.

Probability appears late in this conceptual sequence, not because it is unimportant, but because the probability calculation should be grounded in a prior structural representation. In practice, model construction and probability estimation will often be iterative.

Probability can then be defined over states or trajectories in a structured landscape rather than reported only as an isolated forecast about an outcome.

1.10.2 Chapter Synthesis

The synthesis is deliberately structural. It does not argue that all risk processes require curved manifolds, Hamiltonians, or amplitude models. It establishes a hierarchy of questions that must be answered before such tools are justified. A simpler classical model remains preferable whenever it explains observed transitions, supports intervention, and validates reliably. The geometric and tunneling extensions earn their place only when they reveal stable pathway, barrier, or timing information that would otherwise remain hidden and when that information improves real risk decisions.

This chapter has established the conceptual foundation of the book.

In this framework, risk is represented as a dynamical geometry. A system occupies a state on a manifold, and that state evolves according to a specified law of motion. The geometry defines meaningful distance. The dynamics encode drift, volatility, momentum, and jumps. Potential landscapes identify stable regions and barriers. Model-relative reachability determines whether adverse states can be entered. A justified action functional distinguishes comparatively easy paths from difficult ones.

This framework also clarifies the role of tunneling.

Gradual deterioration moves the system along the landscape. Large shocks may carry it over a barrier. Tunneling represents effective transmission through a barrier that the ordinary model treats as prohibitive.

The quantum analogy adds a further conceptual layer. The system need not be represented only as a point following one trajectory; a model may instead assign state weight across possible configurations. The barrier attenuates that weight but need not terminate it. In the physical theory, this produces a non-zero transmitted current even when classical reasoning treats the region as forbidden. In risk management, the corresponding quantities remain modelling constructs that require independent empirical justification.

The central proposition can therefore be stated more precisely:

> Risk is determined not only by where the system is located, but by the geometry of the landscape, the law governing its motion, the structure of the barriers, and the mechanisms through which effective transmission into adverse regions can occur.

The chapters that follow will formalize each component. We will construct risk landscapes, examine variance and covariance as geometric objects, model barriers and protective structures, develop the mathematics of tunneling, and translate physical concepts into risk-management tools.

The guiding question for the book is no longer merely

> How likely is failure?

It is

> Given the geometry, the dynamics, and the barriers of the system, how can failure become reachable?

That is the point at which probability becomes geometry, and geometry becomes risk management.

Chapter 2

State Spaces and Risk Landscapes

2.1 The Purpose of the State-Space Representation

The first chapter argued that risk should be understood not only as uncertainty over outcomes, but as movement through a structured space of possible configurations. Chapter 2 develops that proposition more formally.

The central object is the *risk state space*: a mathematical representation of the configurations that a borrower, institution, contract, operational process, technological system, or interconnected network may occupy. The purpose of this representation is not merely to create an abstract diagram. It is to organize the variables of the system so that notions such as stability, fragility, distance, direction, reachability, barrier height, and transition become mathematically meaningful.

A conventional risk model often begins by defining an adverse event and estimating its probability. A state-space model begins by defining the system itself. It asks which variables are sufficient to describe its relevant condition, how those variables interact, which combinations correspond to stable or adverse states, and how the structure of the space changes over time.

The basic representation is

$$\mathbf{x} * t = \begin{bmatrix} x * 1, t \; x_{2,t} \vdots x_{n,t} \end{bmatrix} \in \mathcal{M}, \tag{2.1}$$

where $\mathbf{x}_t$ is the state vector of the system at time (t), and $\mathcal{M}$ is the manifold of admissible states.

The coordinates of $\mathbf{x}_t$ are not arbitrary data fields. They are intended to capture the dimensions along which the risk-bearing system can meaningfully change. The quality of the state-space model therefore depends critically on the choice of coordinates.

For a borrower, one possible representation is

$$\mathbf{x} * t^{\text{credit}} = \begin{bmatrix} \text{leverage}_t \\ \text{liquidity}_t \\ \text{cash-flow strength}_t \\ \text{collateral quality}_t \\ \text{refinancing access}_t \\ \text{market confidence}_t \end{bmatrix}. \tag{2.2}$$

For an operational system, the state may instead be

$$\mathbf{x} * t^{\text{operational}} = \begin{bmatrix} \text{system load}_t \\ \text{control effectiveness}_t \\ \text{staff capacity}_t \\ \text{dependency concentration}_t \\ \text{data integrity}_t \\ \text{recovery capability}_t \end{bmatrix}. \tag{2.3}$$

The state vector should be sufficiently rich to capture the principal mechanisms of transition, but sufficiently parsimonious to remain interpretable and estimable. A state space containing every observable variable may become too large to understand, while an excessively compressed representation may eliminate the very pathways the model is intended to reveal.

The construction of the state space is therefore an act of modelling judgment. It determines what the system is allowed to be, how it may differ from neighbouring states, and which dimensions of fragility remain visible.

2.2 From Coordinates to a Manifold

A list of variables creates a coordinate space, but not necessarily a meaningful manifold.

If the variables were independent and unrestricted, the state space could be represented as a subset of Euclidean space,

$$\mathcal{M} \subseteq \mathbb{R}^n. \tag{2.4}$$

In practice, however, risk variables are constrained by accounting identities, institutional rules, contractual relationships, regulatory limits, technological architecture, and behavioural dependencies. Not every numerical combination corresponds to a feasible state.

For example, leverage, liquidity, and collateral quality cannot vary entirely independently. A borrower with severe leverage may retain liquidity temporarily, but only under particular financing conditions. An operational system cannot simultaneously exhibit zero staff capacity and full manual recovery capability unless some external support mechanism exists. A legal position cannot be described by contractual language alone if conduct has altered the enforceability of the agreement.

The admissible states therefore form a structured subset of the ambient coordinate space. This subset may be curved, constrained, fragmented, or composed of several connected regions. It is more naturally represented as a manifold $\mathcal{M}$.

Locally, a manifold resembles ordinary Euclidean space. Around each state $\mathbf{x}$, there exists a coordinate chart

$$\varphi_\alpha : U_\alpha \longrightarrow \mathbb{R}^n, \tag{2.5}$$

where $U_\alpha \subseteq \mathcal{M}$ is a local neighbourhood.

The use of local charts matters because the most useful variables may change across regions. The coordinates that describe a healthy borrower may not be the most informative coordinates once liquidity stress becomes dominant. Similarly, a legal-risk model may shift from contractual variables to litigation and enforcement variables after a dispute has crystallized.

The manifold concept permits different local representations to coexist within one global structure.

This also allows the geometry to be nonlinear. The meaning of a change in one variable may depend on the level of another. A decline in liquidity has a different significance at low leverage than at high leverage. A weakening of one operational control may have limited effect when redundancies are intact but become critical when several dependencies are already concentrated.

The state space is therefore not merely a spreadsheet of variables. It is a structured set of feasible configurations whose local relationships may vary from one region to another.

2.3 What Is a State?

The word *state* should be used carefully. A state must contain enough information to determine, at least probabilistically, the system's future evolution.

In a Markovian representation,

$$\mathbb{P}\left(\mathbf{x} * t + \Delta t \mid \mathbf{x} * t, \mathbf{x} * t - \Delta t, \ldots\right) = \mathbb{P}\left(\mathbf{x} * t + \Delta t \mid \mathbf{x}_t\right). \tag{2.6}$$

The current state $\mathbf{x}_t$ contains all information required for the transition law.

Many risk systems are not naturally Markovian. Legal risk depends on historical conduct. Credit risk may depend on the path by which leverage was accumulated. Operational risk may depend on unresolved incidents, control fatigue, or accumulated technical debt. Institutional risk may depend on reputation, memory, and previous interventions.

When path dependence matters, the state vector must be enlarged. A history-dependent system may be rendered Markovian by including relevant memory variables:

$$\widetilde{\mathbf{x}} * t = \begin{bmatrix} \mathbf{x} * t \ \mathbf{m}_t \end{bmatrix}, \tag{2.7}$$

where $\mathbf{m}_t$ summarizes the relevant history.

The practical implication is important. A state is not simply the set of variables that happen to be observed at time (t). It is the smallest sufficiently informative representation of the system's condition for the purpose of modelling future transitions.

This distinction separates a descriptive dashboard from a genuine state-space model.

2.3.1 A Metric for Risk Distance

The choice of metric must be governed as carefully as the selection of state variables. A metric determines which differences the model treats as material, so it can amplify or suppress particular forms of fragility before any probability is calculated. Analysts should document its scale, reference population, estimation window, and state dependence. Competing metrics should be compared through transition prediction, neighbourhood stability, and intervention sensitivity. If a complicated metric does not improve those properties beyond standardized Euclidean or covariance-adjusted distance, the simpler measure should remain the operational benchmark. Distance is useful only when its meaning can be explained to decision-makers and reproduced under validation.

Once the state manifold has been defined, the next question is how to measure distance.

In ordinary Euclidean space, the distance between two states $\mathbf{x}$ and $\mathbf{y}$ is

$$d_{\mathrm{E}}(\mathbf{x}, \mathbf{y}) = \sqrt{(\mathbf{x}-\mathbf{y})^{\mathsf{T}}(\mathbf{x}-\mathbf{y})}. \tag{2.8}$$

This metric treats all directions equally after unit normalization. Such symmetry is rarely appropriate for risk.

A movement in liquidity may be more consequential than an equal standardized movement in administrative costs. A change in governing law may alter legal exposure more sharply than a similar numerical change in contract length. A small increase in system load may be innocuous under normal capacity but dangerous when the system is already near saturation.

A more general local distance is therefore defined by

$$\mathrm{d}s^2 = \mathrm{d}\mathbf{x}^{\mathsf{T}}\mathbf{g}(\mathbf{x}, t)\mathrm{d}\mathbf{x}, \tag{2.9}$$

where $\mathbf{g}(\mathbf{x}, t)$ is a positive-definite metric tensor.

The metric determines how infinitesimal changes are weighted. It can encode:

- the economic significance of each variable;

  Explanation. Economic significance determines how strongly a change in one coordinate should affect measured risk distance. Standardization alone cannot provide this meaning because variables with similar statistical variation may have very different consequences for solvency, continuity, enforceability, or recovery. The metric should therefore reflect materiality at the relevant horizon, using historical outcomes, structural relationships, stress tests, and expert evidence. Liquidity deterioration may deserve greater weight near a refinancing date, while documentation quality may become decisive after a dispute begins. The assigned importance should be reviewed across sectors and regimes rather than assumed universal. Analysts should also separate predictive importance from economic importance: a variable may forecast outcomes because it proxies for another mechanism without being a suitable intervention target. A defensible weighting explains why the coordinate matters, how its effect was estimated, and whether changing it alters adverse accessibility. This makes distance interpretable and prevents arbitrary scaling choices from silently determining the model's conclusions.

- the interaction among dimensions;

  Explanation. Risk variables rarely operate independently, so the metric must capture how simultaneous changes reinforce or offset one another. A decline in liquidity has a different meaning when leverage is low than when debt service is already concentrated. A control weakness may remain manageable while redundancy is intact but become critical when monitoring and recovery also deteriorate. Interaction terms or off-diagonal metric components represent these conditional effects. Their estimation should be constrained by sufficient data, structural logic, and stability tests because unrestricted interactions can overfit noise. Analysts should compare the resulting geometry with simpler diagonal metrics and classical nonlinear models. They should also identify whether interactions are symmetric, directional, or regime-dependent. If two dimensions share a common dependency, their apparent diversification may be overstated. A credible interaction structure improves neighbourhood quality and transition prediction while remaining explainable in operational terms. It should reveal which combinations create fragility and which coordinated interventions can restore effective distance from the adverse region.

- the cost of moving from one configuration to another;

  Explanation. Movement cost represents the resources, time, institutional changes, or improbable sequence required to transform one state into another. It is not necessarily a monetary cost. In credit risk it may reflect refinancing effort, asset sales, covenant negotiation, or sustained cash-flow deterioration. In operational risk it may reflect the number of controls that must fail and the duration of an attack sequence. A transition-cost metric should be grounded in observed pathways, process constraints, or calibrated simulations. Costs may be asymmetric: deterioration can occur quickly while recovery requires prolonged investment, approval, and evidence. They may also change under stress when market access, staffing, or legal options disappear. Analysts should document the unit of cost and avoid combining incomparable quantities without normalization. Validation should examine whether lower-cost state pairs exhibit more frequent or faster transitions. When movement cost has empirical meaning, geodesic distance becomes a practical measure of accessibility rather than a purely numerical separation between observations.

- the resistance created by controls or constraints;

  Explanation. Controls and constraints increase effective distance by making some directions harder to traverse. Capital requirements, liquidity buffers, contractual protections, access controls, approval stages, and recovery systems can all create resistance. The metric should use effective rather than nominal resistance, adjusting for availability, reliability, independence, capacity, and activation time. Several visible controls may provide little additional protection when they depend on the same data, vendor, infrastructure, or decision-maker. Resistance can also be directional: a covenant may restrict distributions but offer limited protection against a sudden market closure. Estimation may use control tests, historical intervention effects, legal analysis, and stress exercises. Analysts should distinguish genuine resistance from a visible gap; movement through an uncovered route is not difficult merely because other parts of the boundary are strong. A governed resistance measure identifies the owner and evidence for each component and predicts how proposed control changes alter path length, barrier action, and adverse accessibility.

- the state-dependent sensitivity of the system.

  Explanation. State dependence means that the consequence of a marginal change varies with the system's current location. A small rise in utilization may be harmless at normal capacity but destabilizing near saturation. The same collateral decline may have little effect when covenant headroom is large and a severe effect when lenders can accelerate. The metric should therefore permit local weights or curvature rather than imposing one global relationship. Estimation can use local regressions, regime models, supervised embeddings, or structural thresholds, but the resulting sensitivity must remain stable enough for monitoring. Analysts should map where sensitivities change sharply and determine whether those changes correspond to observed fragility or merely to sparse data. Uncertainty should rise in regions with limited coverage. State-dependent sensitivity also guides intervention timing because an action delivered before a high-curvature region may be far more effective than the same action delivered later. This feature turns geometry into a context-aware representation of risk rather than a fixed scoring formula.

The length of a path

$$\gamma : [0,1] \longrightarrow \mathcal{M} \tag{2.10}$$

is

$$\ell[\gamma] = \int_0^1 \sqrt{\dot{\gamma}(s)^\mathsf{T} \mathbf{g}(\gamma(s), t)\dot{\gamma}(s)}, \mathrm{d}s. \tag{2.11}$$

The geodesic distance between two states is the minimum path length:

$$d_{\mathcal{M}}(\mathbf{x}, \mathbf{y}) = \inf_{\gamma:\mathbf{x}\to\mathbf{y}} \ell[\gamma]. \tag{2.12}$$

This notion of distance is already richer than ordinary numerical proximity. It reflects the structure of the manifold and the effective difficulty of movement.

However, geodesic distance is still not identical to dynamic accessibility. A state may be geometrically close but dynamically difficult to reach because the law of motion points away from it. Conversely, a more distant state may be reached easily if it lies along the natural drift.

The metric describes the terrain. The transition law describes the preferred movement across it.

2.4 Curvature and State-Dependent Fragility

Curvature captures the idea that the geometry itself changes across the manifold.

In a flat state space, nearby trajectories remain approximately parallel. In a curved state space, initially similar trajectories may converge or diverge as they evolve.

This has an immediate risk interpretation. Two institutions may begin with nearly identical observable states but respond very differently to the same disturbance because they occupy a region of high structural curvature. A small difference in dependency concentration, governance quality, or refinancing access may be amplified by the geometry.

The metric tensor determines the Christoffel symbols

$$\Gamma^k * ij = \frac{1}{2} g^{k\ell} \left(\frac{\partial g * \ell j}{\partial x^i} + \frac{\partial g_{i\ell}}{\partial x^j} - \frac{\partial g_{ij}}{\partial x^\ell} \right), \tag{2.13}$$

which describe how coordinate directions change across the manifold.

A geodesic satisfies

$$\frac{\mathrm{d}^2 x^k}{\mathrm{d}s^2} + \Gamma^k_{ij} \frac{\mathrm{d}x^i}{\mathrm{d}s} \frac{\mathrm{d}x^j}{\mathrm{d}s} = 0. \tag{2.14}$$

The book will not require full differential-geometric machinery in every application. The conceptual importance of curvature is more immediate: risk sensitivity depends on location.

The same shock can have a minor effect in one region and a catastrophic effect in another. The same control can be effective in one institutional configuration and insufficient in another. The same contractual ambiguity can be harmless before a dispute but decisive after adverse conduct has accumulated.

Curvature is therefore one way of formalizing context dependence.

2.5 Risk Regions

The manifold must be partitioned into regions with distinct risk meanings.

Let

$$\mathcal{M} = S \cup F \cup A \cup R, \tag{2.15}$$

where:

- S is the stable region;

  Explanation. The stable region contains configurations in which the system performs within tolerance and retains sufficient capacity to absorb ordinary disturbances. Stability should not be defined only by the absence of failure. A borrower may be current on payments but unstable if it depends on one imminent refinancing event; a platform may be available but unstable if recovery capacity is exhausted. Stable-state criteria should therefore include buffer adequacy, control reliability, access to resources, and resilience under plausible shocks. The region may contain several basins corresponding to different sustainable operating models. Analysts should validate stability through persistence, low transition rates to fragility, and successful stress absorption. Boundaries need periodic review because environmental changes can make previously stable configurations vulnerable. Management reporting should distinguish deep stability from states located close to the fragile boundary. This permits preventive action before the system leaves the region and avoids interpreting temporary compliance with a threshold as evidence of durable resilience.

- F is the fragile or vulnerable region;

  Explanation. The fragile region contains systems that still function but depend increasingly on depleted buffers, favourable timing, exceptions, concentrated support, or delayed intervention. It is the most important early-warning zone because adverse outcomes have not yet occurred and meaningful options remain available. Fragility may appear as rising barrier utilization, narrowing covenant headroom, shared control dependencies, unstable market access, or sensitivity to small shocks. The region should be defined with observable indicators and validated against subsequent deterioration, recovery, and near misses. Analysts should avoid treating it as a single severity band; different fragile states may have distinct dominant pathways and require different actions. A system can also move within the fragile region while its headline probability remains unchanged. Monitoring should therefore include distance, direction, barrier integrity, and response time. Governance thresholds in this region should trigger investigation and preventive measures rather than waiting for a formal adverse boundary to be crossed. The region should therefore remain visible in routine governance reporting and escalation decisions.

- A is the adverse region;

  Explanation. The adverse region represents outcomes that meet a defined materiality standard, such as default, prolonged outage, regulatory sanction, unenforceability, or systemic impairment. Its definition must be operationally observable and fixed before model evaluation. Broad labels should be decomposed when different outcomes involve different pathways, losses, or intervention authorities. For example, distressed restructuring and legal bankruptcy may both be adverse in credit analysis but require separate boundaries and timing. The region can also be time-dependent if regulation, market structure, or service tolerances change. Analysts should document entry criteria, event dates, censoring rules, and recovery conditions. Sensitivity tests should show whether model conclusions depend on arbitrary thresholds. The adverse region is not merely the set of historically observed failures; simulated and expert-defined severe states may extend it when data are sparse. Clear definition ensures that distance, action, transmission, and probability estimates refer to the same governed outcome. This consistency is essential for validation, comparison, and accountable decision-making.

- R is the recovery region.

  Explanation. The recovery region contains configurations in which the system has left immediate adversity but has not necessarily returned to durable stability. Recovery can involve restored service with temporary controls, a borrower emerging from restructuring, remediation after a legal event, or a market supported by exceptional policy measures. These states often retain elevated vulnerability because resources are depleted, confidence is weak, and dependencies remain unresolved. Treating recovery as identical to stability can therefore understate relapse risk. The region should include criteria for restoration quality, residual capacity, remediation completion, and the withdrawal of emergency support. Analysts should model paths from adversity to recovery and from recovery back to fragility or stability, recognizing that they may differ from the original failure path. Recovery metrics help evaluate intervention effectiveness and determine when temporary measures can safely end. Governance should require evidence that barriers have been rebuilt rather than relying only on the disappearance of the adverse event.

These regions need not be disjoint in a simple empirical classification. Uncertainty, incomplete observation, and model ambiguity may create transitional zones. A state can exhibit features of both stability and fragility.

A risk region may be defined through a scalar classification function

$$h : \mathcal{M} \longrightarrow \mathbb{R}. \tag{2.16}$$

For example, an adverse region may be written as

$$A = \{\mathbf{x} \in \mathcal{M} : h(\mathbf{x}) \geq c\}, \tag{2.17}$$

where (c) is a critical threshold.

This threshold representation is useful but incomplete. Risk boundaries are often multidimensional and may not be captured by one linear score. A borrower may be safe under high leverage if liquidity and refinancing access are strong, but unsafe at the same leverage when market confidence collapses.

The boundary of the adverse region,

$$\partial A, \tag{2.18}$$

may therefore be curved, irregular, or time-dependent.

The distance to this boundary is a natural geometric risk indicator:

$$d_A(\mathbf{x}) = \inf_{\mathbf{y} \in A} d_{\mathcal{M}}(\mathbf{x}, \mathbf{y}). \tag{2.19}$$

Yet this measure should not be interpreted in isolation. A state may be distant from the adverse region under the metric but aligned with a rapid dynamic pathway toward it.

Risk classification must ultimately combine region, distance, and motion.

2.6 The Risk Landscape

The manifold identifies possible states. The risk landscape assigns structure to those states.

Let

$$V_{\mathrm{R}} : \mathcal{M} \times \mathbb{R}_{+} \longrightarrow \mathbb{R} \tag{2.20}$$

be a risk-potential or stability function.

The interpretation of V_{R} depends on the application. It may represent instability, transition resistance, effective cost, stress, or negative resilience.

Stable configurations may correspond to local minima:

$$\nabla V_{\mathrm{R}}(\mathbf{x}^{*}, t) = \mathbf{0}, \tag{2.21}$$

with

$$\nabla^{2} V_{\mathrm{R}}(\mathbf{x}^{*}, t) \succ \mathbf{0}. \tag{2.22}$$

The Hessian

$$\mathbf{H} * V(\mathbf{x}, t) = \nabla^{2} V * \mathrm{R}(\mathbf{x}, t) \tag{2.23}$$

describes local curvature of the potential landscape.

A steep basin implies strong restoring forces. Small disturbances are corrected rapidly. A shallow basin indicates weak stability. The system may wander widely even under moderate shocks.

A ridge or saddle point separates basins. The height and shape of that separating structure determine the difficulty of transition.

This allows risk management to distinguish several situations that may appear similar in a conventional score:

- a deeply stable state far from any boundary;

  Explanation. A deeply stable state lies well inside a resilient basin and retains substantial effective distance from fragile or adverse boundaries. Its buffers are not merely adequate under current conditions; they remain usable under a broad range of plausible shocks. Drift tends to restore the system toward the basin, controls are independent and reliable, and ordinary stochastic variation is unlikely to reach the boundary within the decision horizon. This does not imply zero risk. Large jumps, moving landscapes, or omitted dependencies can still create adverse outcomes, so monitoring remains necessary. The practical distinction is that intervention urgency is low and management can focus on maintaining resilience rather than immediate remediation. Validation should show persistent performance, low barrier utilization, and successful stress absorption. Deep stability also provides a useful reference class for measuring deterioration. Analysts should avoid assuming that distance alone proves resilience; the state must remain deep under alternative metrics, stressed geometries, and realistic changes in external conditions.

- a stable state in a shallow basin;

  Explanation. A shallow basin supports ordinary operation but provides limited restoring force when the system is disturbed. The state may satisfy all formal requirements while small shocks, correlations, or behavioural responses produce meaningful movement toward fragility. Examples include a borrower with acceptable ratios but thin liquidity, a system with controls that work under normal load, or a legal position that depends on unresolved interpretation. The distinction from deep stability is important because headline classifications can be identical even though warning lead times differ. Analysts should measure basin depth through stress response, local potential curvature, recovery speed, and residual barrier capacity. A shallow basin calls for closer monitoring and inexpensive preventive action, such as increasing flexibility or reducing concentration. Governance should not wait for a boundary breach, since the state can leave the basin quickly when the landscape moves. Scenario analysis should identify the smallest disturbances capable of dislodging the system. Those disturbances should also be linked to specific monitoring and response triggers.

- a state near a saddle point;

  Explanation. A saddle point is locally stable in some directions and unstable in others. A system near such a point may appear resilient when assessed along familiar variables while remaining highly sensitive to one less visible direction. For instance, profitability and capital may be stable while refinancing access deteriorates, or authentication may be strong while a shared recovery dependency weakens. Small directional changes can determine which basin the system enters next, making covariance orientation and drift especially important. Analysts should identify the unstable directions through local gradients, eigenvectors, scenario perturbations, and pathway analysis. Monitoring should focus on variables that distinguish movement toward stability from movement toward adversity rather than on aggregate distance alone. Interventions can be highly effective if they are applied before the state commits to an adverse route. The saddle-point interpretation therefore explains why apparently modest developments sometimes produce abrupt changes in trajectory and why timing matters as much as intervention size.

- a state in a narrow corridor between two barriers;

  Explanation. A narrow corridor restricts the set of feasible movements and can channel the system toward a small number of outcomes. The barriers on either side may represent contractual constraints, limited funding choices, technology dependencies, or policy rules. Although the state may be distant from an adverse boundary in ordinary coordinates, the lack of alternative routes reduces flexibility and makes local shocks more consequential. Analysts should examine corridor width, direction, exit points, and whether both barriers share dependencies. A corridor can also create path dependence because earlier decisions determine which options remain available later. Stress testing should vary the barriers themselves, not only the state, to determine whether the corridor narrows or closes. Management may improve resilience by creating optionality, extending maturities, adding independent recovery routes, or renegotiating constraints. The important warning is that limited degrees of freedom can increase adverse accessibility even when current performance appears satisfactory. Optionality should consequently be measured and governed as a resilience resource.

- a state located before a high but thin barrier.

  Explanation. A high but thin barrier provides strong resistance over a short distance. It may require one severe vulnerability, concentrated maturity, decisive legal fact, or common infrastructure failure to cross, but the transition can occur rapidly once that condition is met. Traditional scoring may emphasize the barrier's apparent strength and overlook its limited width. The system has little time for detection and response because few independent stages separate the current state from adversity. Analysts should measure both the severity required to breach the barrier and the duration or number of controls involved. Scenario design should include focused shocks capable of exploiting the narrow route. Appropriate interventions often increase width by adding independent stages, delaying execution, diversifying support, or creating recovery time. Governance should preauthorize rapid action because approval delays may exceed the available window. This configuration illustrates why nominal protection and practical resilience are not equivalent. Response readiness must therefore be tested against the short available intervention window.

The potential landscape converts a static map into a structural model of stability and resistance.

2.7 How Can the Landscape Be Constructed?

The risk landscape is not directly observed. It must be inferred.

There are several possible approaches.

Structural Construction

Structural construction begins with relationships that must hold because of accounting, contractual, technological, legal, or physical constraints. A corporate balance sheet, for example, cannot contain arbitrary combinations of assets, liabilities, cash flow, and leverage. A payment system cannot promise recovery times that are inconsistent with its infrastructure and staffing. A contract cannot be evaluated independently of governing law, required evidence, and the conduct that affects enforceability. These restrictions define feasible regions before statistical estimation begins.

The principal advantage of structural construction is interpretability. Every boundary can be traced to a rule, capacity, identity, or institutional mechanism. This makes the resulting manifold suitable for stress testing and counterfactual analysis because an intervention can be represented as a change in a known constraint. The disadvantage is incompleteness. Formal rules may omit behavioural adaptation, hidden dependencies, or empirical regularities that materially shape observed states. Structural models should therefore be challenged against data and near-miss evidence.

A practical workflow identifies hard constraints, soft constraints, and regime-dependent constraints separately. Hard constraints define impossible configurations. Soft constraints assign low feasibility to unusual but possible states. Regime-dependent constraints change under crisis, legal reinterpretation, market closure, or operational disruption. Each constraint should have an owner, evidence source, review frequency, and uncertainty assessment. The resulting state space should be tested for internal consistency and for coverage of historical observations. States that violate the structure require investigation: they may indicate data errors, missing variables, or a genuine structural change. Structural construction is credible when it narrows the feasible space without excluding real transitions that the model must explain.

The potential may be derived from known economic, contractual, operational, or institutional relationships.

In credit risk, the landscape may reflect cash-flow generation, leverage constraints, collateral, funding access, and covenant structure. In operational risk, it may reflect control redundancy, system architecture, dependency networks, and recovery capacity.

A structural model is interpretable but may omit important behavioural or nonlinear effects.

Data-Driven Construction

Data-driven construction estimates the shape of feasible states from repeated observations. Methods may include principal components, manifold learning, autoencoders, diffusion maps, clustering, graph embeddings, or supervised representations that preserve outcome-relevant variation. These approaches can reveal nonlinear relationships that are difficult to specify structurally, especially when many variables interact or when the effective dimension is much smaller than the number of observed fields.

The estimated geometry must nevertheless remain timestamp-correct and stable. Training data should be separated from validation and test periods, and transformations must use only information available at the observation date. Analysts should examine whether neighbourhoods persist across samples, whether latent dimensions have interpretable sensitivities, and whether the representation behaves coherently during stress. A model that rearranges the state space dramatically after minor data changes cannot support reliable distance or pathway monitoring.

Coverage is another critical concern. Rare adverse states may be poorly represented because the historical sample contains few events. The method may then compress fragile and stable observations together or place crisis states outside the learned support. Oversampling, simulation, expert scenarios, and supervised objectives can help, but each introduces assumptions that require disclosure. The representation should also preserve heterogeneous reference classes; a geometry learned across incompatible sectors or systems may create false neighbours. Data-driven construction is most useful when it improves out-of-sample transition prediction while retaining enough interpretability to identify which variables, barriers, and interventions explain movement through the learned space.

The landscape may be inferred from observed state transitions.

If the system is approximately governed by gradient dynamics,

$$d\mathbf{x} * t = -\nabla V * \mathrm{R}(\mathbf{x} * t), dt + \boldsymbol{\Sigma}(\mathbf{x} * t), d\mathbf{W}_t, \tag{2.24}$$

then observed drift may provide information about the gradient of the potential.

Under restrictive equilibrium conditions, a stationary density may have the approximate form

$$\rho_\infty(\mathbf{x}) \propto \exp\left[-\frac{V_{\mathrm{R}}(\mathbf{x})}{D}\right], \tag{2.25}$$

where (D) is an effective noise scale.

This relation suggests

$$V_{\mathrm{R}}(\mathbf{x}) = -D \log \rho_\infty(\mathbf{x}) + C, \tag{2.26}$$

where (C) is an arbitrary constant.

Such a construction must be used cautiously. Institutional systems may not be stationary, reversible, or near equilibrium. Nonetheless, it provides one possible bridge between observed densities and geometric landscapes.

2.7.1 Expert and Hybrid Construction

Expert construction is valuable when data are sparse, regimes change, or important protections are qualitative. Legal enforceability, sponsor support, control independence, political capacity, and recovery readiness often cannot be inferred reliably from historical frequencies alone. Subject-matter experts can identify feasible states, critical dependencies, moving boundaries, and scenarios that have not yet appeared in the sample. Their contribution should be elicited through structured definitions, documented evidence, and explicit uncertainty rather than informal scoring.

Hybrid construction combines this knowledge with empirical estimation. Structural constraints may define the admissible domain, data-driven models may estimate local geometry, and expert scenarios may extend coverage into severe but plausible regions. The components should remain separable so validation can determine which conclusions arise from observations and which arise from judgement. Sensitivity tests should vary expert weights, scenario assumptions, and structural rules to reveal whether the geometry depends excessively on one input.

Governance is essential because expert views can become anchored, inconsistent, or influenced by organizational incentives. Panels should include diverse functions, record disagreements, and revisit assumptions after events or interventions. Calibration exercises can compare expert assessments with realized transitions and with model-implied neighbourhoods. Hybrid models should also include abstention or confidence measures when neither data nor expertise provides strong support. The objective is not to average incompatible sources mechanically. It is to construct a coherent geometry in which empirical regularities, institutional constraints, and forward-looking scenarios reinforce one another while their respective limitations remain transparent.

Many important risks are data-poor. Legal-risk transitions, catastrophic operational failures, and institutional collapses may be too rare for purely statistical estimation.

In such cases, the landscape may be constructed through a combination of:

- expert judgment;

  Explanation. Expert judgment contributes information that is absent, sparse, or difficult to encode in historical data. Specialists may identify infeasible states, hidden dependencies, emerging legal interpretations, control weaknesses, and plausible pathways that have not yet produced recorded losses. Their input should be elicited through structured questions, explicit definitions, and evidence requirements rather than unrecorded intuition. Multiple experts should be used where possible to reveal disagreement and reduce individual anchoring. Judgments need confidence ranges, review dates, and clear ownership. Analysts should test how the geometry changes when expert assumptions are varied or removed. Historical calibration can compare earlier assessments with realized transitions and near misses. Expert judgment is most valuable when it extends empirical coverage without concealing uncertainty. It should not override contradictory evidence automatically, and model reports must distinguish estimated relationships from expert-imposed constraints. A governed process turns expertise into transparent, challengeable inputs rather than an invisible source of model precision.

- stress testing;

  Explanation. Stress testing explores how the state space, metric, barriers, and pathways behave under adverse but plausible conditions. It can reveal feasible configurations that are rare or absent in historical samples and show whether the learned geometry remains meaningful when correlations rise or resources become constrained. A geometric stress test should change more than individual variables; it may move boundaries, rotate covariance, weaken controls, activate pathways, and alter response timing. Scenarios should be calibrated from historical episodes, market distributions, structural models, and expert challenge. Analysts should recompute distance, barrier integrity, action, and transmission under each scenario and compare the results with conventional loss estimates. Stress tests also support validation by checking whether stronger shocks produce coherent structural deterioration. Their assumptions and endogenous responses must be documented. The purpose is not to predict one future, but to test whether the model represents accessibility and intervention capacity across a range of severe environments.

- scenario analysis;

  Explanation. Scenario analysis constructs coherent narratives that connect changes in the environment to ordered changes in the system. Unlike isolated sensitivity tests, a scenario specifies timing, dependencies, management responses, and the sequence through which conditions propagate. It is particularly useful for legal, operational, and systemic risks where historical frequencies are limited and the path matters. Each narrative should be translated into reviewable quantitative assumptions about states, boundaries, barriers, and pathways. Analysts should distinguish exogenous shocks from endogenous reactions and identify which assumptions are necessary for the adverse outcome. Alternative scenarios can explore different orderings or intervention times while holding shock severity constant. This helps isolate the importance of phase and sequencing. Scenario analysis should not be used to justify predetermined conclusions; it requires plausibility review, benchmark comparison, and uncertainty reporting. Its contribution to landscape construction is strongest when it identifies missing regions and mechanisms that can later be tested or monitored empirically.

- causal modelling;

  Explanation. Causal modelling helps distinguish variables that merely predict a state from mechanisms that actually change accessibility. Directed graphs, structural equations, natural experiments, and intervention studies can identify how resources, controls, behaviour, and external conditions influence one another. This matters because a predictive embedding may place states close together for reasons that do not support effective action. Causal assumptions should be explicit, including confounders, timing, and the conditions under which relationships remain valid. Analysts can use causal models to define feasible interventions, estimate barrier effects, and prevent paths that violate known ordering. Validation may include placebo tests, sensitivity to unmeasured confounding, and comparison with observed policy or management actions. Causality is rarely identified perfectly in complex risk systems, so uncertainty must remain visible. Its value lies in organizing evidence about mechanisms and improving counterfactual reasoning, not in converting every association into a definitive claim. A causal layer makes the geometry more useful for decision design.

- network structure;

  Explanation. Network structure represents relationships among institutions, systems, controls, assets, vendors, or legal obligations that cannot be captured by entity-level variables alone. Links may transmit losses, information, funding pressure, operational failure, or confidence. Network topology can constrain feasible states and create shortcuts through otherwise strong barriers. Important measures include concentration, centrality, community structure, shared dependencies, and the availability of alternative routes. The network should be timestamped because links and weights change, especially under stress. Analysts must also account for missing or uncertain edges rather than treating the observed graph as complete. Stress testing can remove critical nodes, increase correlations, or reweight exposures to examine how the manifold deforms. Network evidence is particularly valuable for identifying systemic corridors and common points of failure. It should be combined with balance-sheet and behavioural information so that connectivity is not mistaken for vulnerability without considering capacity, direction, and response. Directionality and substitution capacity must also remain explicit in the assessment.

- historical near-misses;

  Explanation. Near-misses contain information about pathways that approached an adverse boundary but were interrupted before material loss occurred. They can reveal weak barriers, critical timing, and effective interventions that outcome-only datasets overlook. Examples include covenant breaches followed by waivers, cyber intrusions contained before data loss, outages recovered within tolerance, or regulatory concerns resolved through remediation. Records should capture the sequence of events, detection time, controls used, and the reason the path stopped. Analysts can compare near-misses with completed events to identify which variables and interventions distinguish recovery from failure. Reporting biases require care because successful containment may be documented less consistently than losses. Near-miss evidence can also improve fragile-region definitions and provide negative examples for pathway validation. Its greatest value is prospective: it shows where the system has already tested its barriers and whether the same interruption would remain available under different timing, capacity, or simultaneous stress. That interruption mechanism should be retested periodically under realistic combined stress.

- synthetic simulation;

  Explanation. Synthetic simulation generates feasible states and transitions when observed data do not cover severe or novel conditions. Structural equations, agent-based models, Monte Carlo processes, digital twins, and controlled generative methods can explore combinations that are rare but plausible. Simulated states should obey accounting, physical, legal, and operational constraints; otherwise they distort the learned manifold. Parameters require calibration and uncertainty ranges, and outputs should be compared with historical episodes and expert expectations. Simulation is useful for testing barrier capacity, pathway competition, moving boundaries, and intervention timing. It can also create balanced samples for algorithm development, but synthetic observations must never be presented as independent empirical evidence. Analysts should report which conclusions depend on simulation assumptions and conduct sensitivity analysis across alternative models. The objective is to expand structural coverage and test mechanisms, not to manufacture precision. Well-governed simulation complements sparse data by exposing how the geometry might behave outside the observed sample.

- observed partial transitions.

  Explanation. Partial transitions are sequences in which the system moves through identifiable stages of a pathway without reaching the final adverse event. They include liquidity deterioration that stops before default, control compromise without material impact, escalating complaints without sanction, or asset sales that do not become a systemic fire sale. These observations provide direct evidence about intermediate states, transition timing, and barrier segments. Analysts should record where the sequence began, which stages occurred, what interrupted it, and whether the system returned to stability or remained fragile. Sequence and survival methods can use partial transitions without treating them as complete failures. They also help validate whether proposed paths reflect actual orderings rather than retrospective narratives. Censoring and reporting quality must be addressed because later stages may be unobserved. Partial-transition evidence strengthens landscape construction by populating the regions between ordinary operation and adversity and by showing which controls preserve distance in practice. Their interruption points should become explicit candidates for preventive intervention design.

The result should not be treated as a discovered physical surface. It is a disciplined representation of institutional knowledge about stability and transition resistance.

2.7.2 Moving States and Moving Landscapes

The distinction between state movement and landscape movement is central to early warning. Conventional dashboards often hold thresholds and relationships fixed, so risk appears to change only when observed variables deteriorate. In practice, refinancing markets close, legal interpretations evolve, controls become obsolete, and shared infrastructure becomes more concentrated. These changes alter the location of boundaries, the strength of barriers, or the meaning of distance even when the measured state remains nearly unchanged.

Empirical implementation should therefore estimate separate indicators for state displacement and structural deformation. State displacement records changes in the coordinates of the entity. Structural deformation records changes in metrics, feasible regions, transition dynamics, correlations, and protection. Analysts can compare a fixed-geometry distance with a current-geometry distance to determine whether danger is approaching because the entity moved or because the environment changed around it. This distinction supports better intervention: state deterioration may require remediation within the entity, while landscape movement may require market, legal, infrastructure, or policy action.

Moving landscapes also create model risk. A representation calibrated in normal conditions may become unreliable when correlations rise or constraints bind. Monitoring should include drift tests, regime indicators, boundary stability, and recalibration triggers. Historical replay and scenario analysis can show how earlier signals would have changed under alternative geometries. Governance should avoid continuous unreviewed refitting, however, because frequent parameter changes can make risk measures incomparable through time. A versioned approach balances responsiveness and interpretability by updating the geometry at controlled intervals while reporting the effects of each structural change.

A central distinction of the framework is the difference between movement of the system and deformation of the landscape.

The state evolves as

$$\mathbf{x} * t \longrightarrow \mathbf{x} * t + \Delta t, \tag{2.27}$$

while the landscape may evolve as

$$V_{\mathrm{R}}(\mathbf{x}, t) \longrightarrow V_{\mathrm{R}}(\mathbf{x}, t + \Delta t). \tag{2.28}$$

Risk may increase even when the observed state changes little.

A borrower may maintain similar leverage and profitability while the refinancing market closes. An institution may retain the same formal controls while a software vulnerability alters the path to cyber compromise. A contract may remain unchanged while judicial precedent transforms its effective interpretation.

Four distinct mechanisms should therefore be distinguished:

1. The state moves toward the adverse region.
2. The adverse region expands toward the state.
3. The barrier becomes lower or narrower.
4. The metric changes, making the adverse region effectively closer.

The last mechanism is particularly important. A change in network connectivity, legal interpretation, or market confidence may alter the effective meaning of distance without visibly changing the coordinates.

A dynamic risk geometry therefore requires

$$\mathcal{G} * t = (\mathcal{M} * t, \mathbf{g} * t, V * \mathrm{R}, t, S_t, F_t, A_t)\,, \tag{2.29}$$

where the manifold, metric, potential, and risk regions may all evolve.

2.8 Classical Density and Complex Amplitude

The state-space framework supports two distinct representations that must remain conceptually separate.

The first is a classical probability density

$$\rho(\mathbf{x}, t) \geq 0, \tag{2.30}$$

normalized according to

$$\int_{\mathcal{M}} \rho(\mathbf{x}, t), \mathrm{d}\mu(\mathbf{x}) = 1. \tag{2.31}$$

This density represents uncertainty about the system's state. It may evolve through a Fokker–Planck equation associated with the stochastic law of motion:

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot (\mathbf{b}\rho) + \frac{1}{2} \sum_{i,j} \frac{\partial^2}{\partial x_i \partial x_j} (C_{ij}\rho)\,. \tag{2.32}$$

This is the baseline classical representation.

The second possibility is a complex amplitude

$$\Psi(\mathbf{x}, t) \in L^2 (\mathcal{M}, \mathbb{C})\,, \tag{2.33}$$

from which observable state weight is derived as

$$\rho(\mathbf{x}, t) = |\Psi(\mathbf{x}, t)|^2. \tag{2.34}$$

The complex representation contains phase information that a classical density does not.

At this stage, Chapter 2 does not make the full conceptual transition to quantum-style interference. That transition belongs in Chapter 5. It is nevertheless important to establish that the manifold can support either a real-valued density or a complex amplitude.

The complex amplitude may be written in polar form as

$$\Psi(\mathbf{x}, t) = R(\mathbf{x}, t) \exp\left[\mathrm{i}\theta(\mathbf{x}, t)\right], \tag{2.35}$$

where (R) is the amplitude magnitude and θ is the phase.

The density depends only on the magnitude:

$$\rho(\mathbf{x}, t) = R^2(\mathbf{x}, t). \tag{2.36}$$

The phase becomes relevant when multiple transition channels are combined. It will later allow the book to represent constructive and destructive interference among pathways.

For now, the important distinction is:

$$\boxed{\begin{aligned} \text{Classical representation} &: \rho(\mathbf{x}, t), \\ \text{Amplitude representation} &: \Psi(\mathbf{x}, t) \in \mathbb{C}. \end{aligned}} \tag{2.37}$$

The first describes uncertainty over states. The second contains additional relational information that may affect how transition channels combine.

2.9 The Minimum Operational Risk Landscape

A risk landscape becomes operational only when its components are explicitly defined.

At minimum, the model requires:

1. a manifold of admissible states $\mathcal{M}$;
2. a state vector $\mathbf{x} * t$;
3. a coordinate system or collection of local charts;
4. a metric tensor $\mathbf{g}(\mathbf{x}, t)$;
5. stable, fragile, and adverse regions;
6. a potential or stability function $V * \mathrm{R}(\mathbf{x}, t)$;
7. a law of motion;
8. a rule governing deformation of the geometry;
9. a probability density or state amplitude over the manifold.

These elements can be summarized as

$$\boxed{\mathfrak{R} * t = (\mathcal{M}, \mathbf{x} * t, \mathbf{g} * t, V * \mathrm{R}, t, \mathcal{D} * t, \rho * t \text{ or} \Psi_t)}, \tag{2.38}$$

where $\mathcal{D}_t$ denotes the dynamical law.

This object provides the foundation for the chapters that follow.

Variance and covariance will describe directional mobility within the manifold. Barriers will be defined as structured regions of high transition resistance. Reachability and action will identify relevant paths. Tunneling will examine transmission through regions that ordinary dynamics classify as inaccessible.

2.10 Chapter Synthesis

This chapter has transformed the intuitive idea of a risk surface into a formal state-space architecture.

A risk-bearing system is represented by a state vector on a manifold of admissible configurations. The manifold is not merely a list of variables. It reflects institutional constraints, accounting relations, legal dependencies, technological architecture, and behavioural structure.

A metric defines meaningful distance. Curvature captures the fact that the importance of a movement depends on the region in which it occurs. Stable, fragile, adverse, and recovery states form structured regions whose boundaries may be nonlinear and time-dependent.

A risk potential assigns stability and resistance to the manifold. Local minima represent basins of attraction, while ridges and saddle regions separate alternative regimes. The landscape may be inferred from structural relationships, data, expert judgment, or hybrid methods.

The geometry itself may change. Risk can rise because the state moves, because the adverse region expands, because the barrier weakens, or because the metric changes and makes danger effectively closer.

Finally, the chapter has introduced two possible representations over the state manifold. A classical probability density captures uncertainty over configurations. A complex amplitude contains both magnitude and phase. The full significance of phase, superposition, and interference will be developed in Chapter 5.

The central proposition of Chapter 2 is therefore:

> Risk becomes geometrically tractable when the possible configurations of a system are organized into a dynamic manifold in which distance, stability, direction, and accessibility are explicitly defined.

Chapter 3 will build on this foundation by examining variance, covariance, directional mobility, and the geometry of movement.

Chapter 3

Variance, Covariance, and the Geometry of Movement

3.1 From Uncertainty to Mobility

The previous chapter constructed the risk state space as a manifold of admissible configurations. It introduced the state vector, the metric, the potential landscape, the distinction among stable and adverse regions, and the possibility that the geometry itself may change over time.

The present chapter asks a more specific question:

> Given the current state of the system, in which directions can it move?

This question requires a careful distinction among probability, covariance, diffusion, mobility, and geometry. In conventional probability theory, covariance is a statistical moment of a random variable or stochastic process. It measures the extent to which components of a random vector vary jointly. In the framework developed here, covariance retains that classical probabilistic origin. It is not introduced as a free-standing geometric primitive.

Once covariance is associated with a state manifold, however, it acquires a geometric interpretation. It defines the scale, orientation, and local anisotropy of stochastic movement. It determines whether uncertainty is evenly distributed across directions or concentrated along particular pathways.

The logical sequence is therefore

$$\boxed{\text{Probability Law} \longrightarrow \text{Covariance Tensor} \longrightarrow \text{Mobility Geometry.}} \tag{3.1}$$

The probability law is the underlying object. Covariance is its second central moment. The uncertainty ellipsoid is the geometric representation of that covariance.

This distinction matters because the book does not seek merely to rename statistical covariance as geometry. It seeks to place a probabilistic object within a geometric architecture in which its directional meaning becomes visible.

The central proposition of the chapter is:

Covariance originates in probability theory, but on a state manifold it becomes a tensor describing the local geometry of stochastic mobility.

3.1.1 The State Increment and Classical Covariance

Let the state of the system at time (t) be

$$\mathbf{x}_t \in \mathcal{M}, \tag{3.2}$$

where $\mathcal{M}$ is the state manifold.

Over a finite interval Δt, define the state increment

$$\Delta \mathbf{x} * t = \mathbf{x} * t + \Delta t - \mathbf{x}_t. \tag{3.3}$$

The conditional expected increment is

$$\boldsymbol{\mu} * t = \mathbb{E}\left[\Delta \mathbf{x} * t \mid \mathbf{x}_t\right]. \tag{3.4}$$

The conditional covariance matrix is then

$$\mathbf{C} * t = \mathrm{Cov}\left(\Delta \mathbf{x} * t \mid \mathbf{x}_t\right), \tag{3.5}$$

or explicitly,

$$\mathbf{C} * t = \mathbb{E}\left[\left(\Delta \mathbf{x} * t - \boldsymbol{\mu} * t\right)\left(\Delta \mathbf{x} * t - \boldsymbol{\mu} * t\right)^{\mathsf{T}}, \Big|, \mathbf{x} * t\right]. \tag{3.6}$$

This is the classical and canonical covariance of a random vector. Its diagonal elements are conditional variances, while its off-diagonal elements are conditional covariances.

The conditioning information is essential. An unconditional covariance matrix averages movement across states that may have very different dynamics, constraints, and risk meanings. Liquidity and leverage, for example, may display modest covariance during ordinary periods but move sharply together when refinancing access deteriorates. Operational load and control failure may appear weakly related until capacity is nearly exhausted. The local object should therefore be estimated over a horizon and information set that correspond to the decision being made. Analysts must state whether the increment is daily, monthly, or scenario-based; whether the estimate is historical, forward-looking, or model-implied; and whether observations are comparable across regimes.

Estimation also requires discipline when the state dimension is large relative to the sample. Sample covariance may become unstable, singular, or dominated by transient observations. Shrinkage, factor structure, robust estimators, Bayesian regularization, or rolling windows may improve stability, but each method changes the geometry that follows. The selected estimator should be challenged through out-of-sample transition prediction, eigenvalue stability, sensitivity to extreme observations, and comparison with simpler diagonal or low-rank alternatives. Missing data and asynchronous observations should be handled explicitly because artificial covariance can arise from interpolation or mismatched timestamps.

Finally, covariance must be interpreted as mobility rather than causation. A positive off-diagonal entry shows that two increments tend to move together after conditioning; it does not establish that one variable causes the other. Causal claims require additional structural evidence. This distinction prevents the uncertainty ellipsoid from being treated as a complete mechanism. It is a local probabilistic summary that becomes decision-useful when combined with drift, barriers, adverse directions, and the institutional processes capable of producing the observed co-movement.

For the (i)-th and (j)-th state variables,

$$C_{ij,t} = \mathbb{E}\left[\left(\Delta x_{i,t} - \mu_{i,t}\right)\left(\Delta x_{j,t} - \mu_{j,t}\right), |, \mathbf{x}_t\right]. \tag{3.7}$$

The corresponding correlation coefficient is

$$\rho_{ij,t} = \frac{C_{ij,t}}{\sqrt{C_{ii,t} C_{jj,t}}}. \tag{3.8}$$

Covariance therefore remains rooted in probability theory. It is meaningful because the state increment is modelled as a random vector.

The geometric interpretation arises only after this probabilistic object is placed in the tangent space of the manifold.

3.2 Continuous-Time Dynamics and the Covariance Rate

Suppose the system follows the stochastic differential equation

$$\mathrm{d}\mathbf{x} * t = \mathbf{b}(\mathbf{x} * t, t), \mathrm{d}t + \mathbf{\Sigma}(\mathbf{x} * t, t), \mathrm{d}\mathbf{W} * t, \tag{3.9}$$

where:

- $\mathbf{b}(\mathbf{x} * t, t)$ is the drift vector;
- $\mathbf{\Sigma}(\mathbf{x} * t, t)$ is the volatility or dispersion matrix;
- $\mathbf{W}_t$ is a vector-valued Wiener process.

Conditional on the current state,

$$\mathbb{E}\left[\mathrm{d}\mathbf{x} * t \mid \mathbf{x} * t\right] = \mathbf{b}(\mathbf{x}_t, t), \mathrm{d}t. \tag{3.10}$$

The covariance of the infinitesimal increment is

$$\mathrm{Cov}\left(\mathrm{d}\mathbf{x} * t \mid \mathbf{x} * t\right) = \mathbf{\Sigma}(\mathbf{x} * t, t)\mathbf{\Sigma}^{\mathsf{T}}(\mathbf{x} * t, t), \mathrm{d}t. \tag{3.11}$$

Define the instantaneous covariance rate

$$\mathbf{A}(\mathbf{x} * t, t) = \mathbf{\Sigma}(\mathbf{x} * t, t)\mathbf{\Sigma}^{\mathsf{T}}(\mathbf{x}_t, t). \tag{3.12}$$

Then

$$\operatorname{Cov}\left(\mathrm{d}\mathbf{x} * t \mid \mathbf{x} * t\right) = \mathbf{A}(\mathbf{x}_t, t), \mathrm{d}t. \tag{3.13}$$

Some conventions define the diffusion tensor as

$$\mathbf{D}(\mathbf{x} * t, t) = \frac{1}{2}\mathbf{A}(\mathbf{x} * t, t) = \frac{1}{2}\mathbf{\Sigma}\mathbf{\Sigma}^{\mathsf{T}}. \tag{3.14}$$

Under this convention,

$$\operatorname{Cov}\left(\mathrm{d}\mathbf{x} * t \mid \mathbf{x} * t\right) = 2\mathbf{D}(\mathbf{x}_t, t), \mathrm{d}t. \tag{3.15}$$

This terminology must remain precise:

$$\boxed{\begin{gathered}\text{Covariance of increments} = \operatorname{Cov}\left(\mathrm{d}\mathbf{x} * t \mid \mathbf{x} * t\right),\\ \text{Covariance rate} = \mathbf{A} = \mathbf{\Sigma}\mathbf{\Sigma}^{\mathsf{T}},\\ \text{Diffusion tensor} = \mathbf{D} = \frac{1}{2}\mathbf{A}.\end{gathered}} \tag{3.16}$$

The three objects are closely related, but they are not identical.

From Covariance to a Mobility Tensor

Once the covariance rate is defined at each state, it can be interpreted geometrically.

At a point $\mathbf{x} \in \mathcal{M}$, an infinitesimal state change belongs to the tangent space

$$\mathrm{d}\mathbf{x} \in T_{\mathbf{x}}\mathcal{M}. \tag{3.17}$$

The covariance rate may then be viewed as a symmetric positive-semidefinite rank-two tensor,

$$\mathbf{A}(\mathbf{x}, t) \in T_{\mathbf{x}}\mathcal{M} \otimes T_{\mathbf{x}}\mathcal{M}. \tag{3.18}$$

It describes how strongly random disturbances generate movement in different tangent directions.

In this sense, the covariance rate becomes a local mobility tensor. It identifies directions in which the system is highly mobile and directions in which movement is constrained.

The term *mobility tensor* is useful because it emphasizes the geometric interpretation without obscuring the probabilistic origin. When the object is derived from a stochastic process, the mobility tensor is the geometric interpretation of its covariance or diffusion structure.

There may also be situations in which a mobility tensor is specified directly from structural knowledge rather than estimated from an empirical probability law. For example, legal experts may identify pathways through which a dispute can evolve, or operational experts may identify network directions through which failures can propagate.

In such cases, one may define a structural tensor

$$\mathbf{K}(\mathbf{x}, t), \tag{3.19}$$

representing feasible or institutionally meaningful mobility.

However, $\mathbf{K}$ should not be called covariance unless it is the second central moment of a probability distribution or stochastic process.

The terminology should therefore be:

- **covariance tensor**, when derived from a probability law;
- **diffusion tensor**, when it appears in a stochastic evolution equation;
- **mobility tensor**, when emphasizing geometric movement;
- **structural mobility tensor**, when defined from non-probabilistic institutional structure.

This distinction prevents probability, movement, and geometry from being conflated.

Coordinate Transformations and Tensorial Meaning

A genuine geometric object must transform consistently under a change of coordinates.

Suppose the system is represented in coordinates $\mathbf{x}$, and a new coordinate system is introduced through

$$\mathbf{y} = \mathbf{f}(\mathbf{x}). \tag{3.20}$$

Let the Jacobian matrix be

$$\mathbf{J}_f(\mathbf{x}) = \frac{\partial \mathbf{f}}{\partial \mathbf{x}}. \tag{3.21}$$

To first order,

$$\mathrm{d}\mathbf{y} = \mathbf{J}_f \mathrm{d}\mathbf{x}. \tag{3.22}$$

The covariance transforms as

$$\mathbf{A} * \mathbf{y} = \mathbf{J} * f\mathbf{A} * \mathbf{x}\mathbf{J} * f^{\mathsf{T}}. \tag{3.23}$$

This transformation law is crucial. It means that the covariance structure is not simply a table of numerical correlations attached to one arbitrary coordinate system. It is a geometric tensor whose representation changes consistently when the coordinates change.

The same local mobility can therefore be described using leverage and liquidity ratios, transformed risk scores, principal components, or nonlinear local coordinates, provided the tensor is transformed correctly.

This tensorial interpretation is what permits covariance to enter the geometry of the risk manifold without losing its probabilistic foundation.

3.3 The Uncertainty Ellipsoid

If local state increments are approximately Gaussian, contours of equal conditional density satisfy

$$(\Delta\mathbf{x} - \boldsymbol{\mu})^{\mathsf{T}}\,\mathbf{C}^{-1}\,(\Delta\mathbf{x} - \boldsymbol{\mu}) = c, \tag{3.24}$$

where (c>0) is a constant.

This equation defines an ellipsoid centred at the expected increment $\boldsymbol{\mu}$.

Let the eigendecomposition of the covariance matrix be

$$\mathbf{C} = \mathbf{Q}\boldsymbol{\Lambda}\mathbf{Q}^{\mathsf{T}}, \tag{3.25}$$

where

$$\boldsymbol{\Lambda} = \operatorname{diag}\left(\lambda_1, \lambda_2, \ldots, \lambda_n\right), \tag{3.26}$$

and the columns of $\mathbf{Q}$ are the eigenvectors

$$\mathbf{v} * 1, \mathbf{v} * 2, \ldots, \mathbf{v}_n. \tag{3.27}$$

Each eigenvector identifies a principal direction of stochastic mobility. Each eigenvalue measures the variance along that direction.

If

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n, \tag{3.28}$$

then $\mathbf{v}_1$ is the dominant direction of uncertainty.

The semi-axis lengths of the ellipsoid are proportional to

$$\sqrt{\lambda_1}, \sqrt{\lambda_2}, \ldots, \sqrt{\lambda_n}. \tag{3.29}$$

A spherical cloud corresponds to approximately equal eigenvalues. An elongated ellipsoid indicates mobility concentrated along one or a few directions.

The uncertainty ellipsoid is therefore not a new probabilistic object. It is the geometric visualization of a classical covariance matrix.

Aggregate Volatility and Directional Vulnerability

Several scalar quantities summarize total variance. The trace is

$$\mathrm{tr}(\mathbf{C}) = \sum_{i=1}^{n} \lambda_i, \tag{3.30}$$

while the determinant is

$$\det(\mathbf{C}) = \prod_{i=1}^{n} \lambda_i. \tag{3.31}$$

The trace measures total variance. The determinant is related to the volume of the uncertainty ellipsoid.

These measures can conceal directional fragility.

Consider

$$\mathbf{\Lambda}_1 = \mathrm{diag}\,(1, 1, 1)\,, \tag{3.32}$$

and

$$\mathbf{\Lambda}_2 = \mathrm{diag}\,(2.8, 0.1, 0.1)\,. \tag{3.33}$$

Both have trace equal to (3), but their geometries differ sharply.

The first system has uncertainty distributed evenly across directions. The second has almost all mobility concentrated along one dominant axis.

If that axis points toward an adverse boundary, the second system may be much more fragile despite having the same total variance.

Let $\mathbf{n}_A(\mathbf{x})$ denote a unit vector pointing toward the nearest adverse boundary. The variance projected in that direction is

$$\sigma_A^2(\mathbf{x}, t) = \mathbf{n} * A^{\mathsf{T}} \mathbf{C}(\mathbf{x}, t) \mathbf{n} * A. \tag{3.34}$$

This quantity measures stochastic mobility specifically toward danger.

The relevant proposition is therefore:

> Total volatility is less informative than volatility projected onto pathways leading toward adverse states.

3.4 Covariance and the Risk Metric Are Different Objects

The covariance tensor and the metric tensor must not be conflated.

The risk metric

$$\mathbf{g}(\mathbf{x}, t) \tag{3.35}$$

defines the meaning of distance:

$$\mathrm{d}s^2 = \mathrm{d}\mathbf{x}^\mathsf{T}\mathbf{g}(\mathbf{x}, t)\mathrm{d}\mathbf{x}. \tag{3.36}$$

The covariance rate

$$\mathbf{A}(\mathbf{x}, t) \tag{3.37}$$

defines the directions and intensity of stochastic movement.

The distinction can be summarized as

$$\boxed{\mathbf{g} = \text{geometry of distance}, \qquad \mathbf{A} = \text{geometry of stochastic mobility}.} \tag{3.38}$$

The metric says how consequential a displacement is. Covariance says how likely random dynamics are to generate displacement in a particular direction.

A direction may have high variance but low structural significance. Another direction may have low variance but extreme institutional importance.

This is why covariance should not automatically be used as the risk metric.

The Mahalanobis Metric and Its Limits

If the covariance matrix is positive definite, its inverse defines the Mahalanobis distance:

$$\mathrm{d}s_\mathbf{C}^2 = \mathrm{d}\mathbf{x}^\mathsf{T}\mathbf{C}^{-1}\mathrm{d}\mathbf{x}. \tag{3.39}$$

For two finite states,

$$d_\mathbf{C}^2\left(\mathbf{x}, \mathbf{y}\right) = \left(\mathbf{x} - \mathbf{y}\right)^\mathsf{T} \mathbf{C}^{-1} \left(\mathbf{x} - \mathbf{y}\right). \tag{3.40}$$

This geometry measures distance relative to typical variability. Movement along a high-variance direction is treated as comparatively short. Movement along a low-variance direction is treated as comparatively long.

The Mahalanobis metric is useful for anomaly detection, statistical similarity, and information geometry.

It is not necessarily the same as the structural risk metric.

A statistically common movement may be economically dangerous. A rare movement may be institutionally inconsequential. Therefore,

$$\mathbf{g} \neq \mathbf{C}^{-1} \tag{3.41}$$

in general.

The book should preserve three distinct notions:

- statistical distance, often induced by $\mathbf{C}^{-1}$;
- structural risk distance, defined by $\mathbf{g}$;
- stochastic mobility, defined by $\mathbf{A}$ or $\mathbf{D}$.

The three may be related, but none should be identified with another without justification.

3.5 Combining the Metric and the Covariance

Although the metric and covariance are distinct, their interaction is central.

The expected squared geometric displacement generated by the diffusion is

$$\mathbb{E}\left[\mathrm{d}s^2 \mid \mathbf{x} * t\right] = \mathrm{tr}\left[\mathbf{g}(\mathbf{x} * t, t)\mathbf{A}(\mathbf{x}_t, t)\right]\mathrm{d}t. \tag{3.42}$$

Under the diffusion-tensor convention,

$$\mathbb{E}\left[\mathrm{d}s^2 \mid \mathbf{x}_t\right] = 2\,\mathrm{tr}\left[\mathbf{g}\mathbf{D}\right]\mathrm{d}t. \tag{3.43}$$

This expression combines probabilistic mobility with structural meaning.

A useful metric-adjusted covariance operator is

$$\mathbf{A}_g = \mathbf{g}^{1/2}\mathbf{A}\mathbf{g}^{1/2}. \tag{3.44}$$

Its eigenvectors identify directions of mobility after weighting movements according to the structural metric.

One may also define generalized principal directions through

$$\mathbf{A}\mathbf{v} * i = \lambda * i\mathbf{g}^{-1}\mathbf{v}_i, \tag{3.45}$$

or equivalently,

$$\mathbf{g}\mathbf{A}\mathbf{v} * i = \lambda * i\mathbf{v}_i. \tag{3.46}$$

These generalized eigenvectors describe directions that are statistically mobile and structurally consequential.

State-Dependent Covariance

The covariance structure is generally not constant. It should be written as

$$\mathbf{A} = \mathbf{A}(\mathbf{x}, t). \tag{3.47}$$

This means that the shape of uncertainty changes across the manifold.

A borrower in a stable region may exhibit weak dependence among leverage, liquidity, and confidence. Near distress, those variables may become tightly coupled.

An operational system may show independent control failures under normal conditions. During a crisis, common staffing, infrastructure, and decision constraints may cause the controls to deteriorate together.

A legal position may be insulated from reputational risk until scrutiny changes the enforcement environment. Legal, regulatory, and reputational dimensions may then become aligned.

The local eigensystem is therefore state-dependent:

$$\mathbf{A}(\mathbf{x}, t)\mathbf{v} * i(\mathbf{x}, t) = \lambda * i(\mathbf{x}, t)\mathbf{v}_i(\mathbf{x}, t). \tag{3.48}$$

The total derivative of the covariance rate is

$$\frac{\mathrm{d}\mathbf{A}}{\mathrm{d}t} = \frac{\partial \mathbf{A}}{\partial t} + \sum_{k=1}^{n} \frac{\partial \mathbf{A}}{\partial x_k} \frac{\mathrm{d}x_k}{\mathrm{d}t}. \tag{3.49}$$

The first term captures explicit environmental change. The second captures change caused by movement through a heterogeneous state space.

Thus, covariance geometry can evolve because the world changes or because the system moves into a region with different stochastic structure.

Stress as Expansion, Rotation, and Rank Collapse

Stress is often represented as an increase in scalar volatility. The geometric interpretation is richer.

Under stress, the covariance ellipsoid may undergo several transformations.

Expansion

All principal variances may increase:

$$\lambda_i^{\text{stress}} > \lambda_i^{\text{normal}}. \tag{3.50}$$

Directional Stretching

One eigenvalue may increase disproportionately:

$$\frac{\lambda_1^{\text{stress}}}{\lambda_2^{\text{stress}}} \gg \frac{\lambda_1^{\text{normal}}}{\lambda_2^{\text{normal}}}. \tag{3.51}$$

Rotation

The principal directions may change:

$$\mathbf{v} * i^{\text{stress}} \neq \mathbf{v} * i^{\text{normal}}. \tag{3.52}$$

A direction that was previously benign may rotate toward an adverse boundary.

Rank Collapse

Several variables may begin to move through one common factor. A useful effective-dimension measure is

$$d_{\text{eff}} = \frac{[\text{tr}(\mathbf{A})]^2}{\text{tr}\left(\mathbf{A}^2\right)}. \tag{3.53}$$

When one eigenvalue dominates, d_{eff} approaches one.

This collapse means that apparently diverse risks are moving through one shared channel. Diversification weakens because exposures that appeared independent become aligned.

3.6 Drift, Diffusion, and Direction of Travel

Covariance describes random mobility, but the system also possesses drift.

Let

$$\mathbf{b} = \mathbf{b}(\mathbf{x}, t) \tag{3.54}$$

be the local drift vector.

Let $\mathbf{n}_A$ be a unit vector toward an adverse region. The adverse component of drift is

$$b_A = \mathbf{n}_A^\mathsf{T}\mathbf{b}. \tag{3.55}$$

The stochastic mobility toward the adverse region is

$$a_A = \mathbf{n} * A^\mathsf{T}\mathbf{A}\mathbf{n} * A. \tag{3.56}$$

The local drift-to-diffusion ratio may be written as

$$\chi_A = \frac{b_A}{\sqrt{a_A}}. \tag{3.57}$$

A large positive value indicates systematic movement toward danger. A value near zero indicates that stochastic mobility dominates. A large negative value indicates expected movement away from danger.

The most dangerous configuration occurs when both drift and covariance are aligned toward the adverse region.

Mobility Relative to a Barrier

Suppose a barrier is represented by the level set

$$\mathcal{B} = \{\mathbf{x} \in \mathcal{M} : h(\mathbf{x}) = 0\}. \tag{3.58}$$

The unit normal vector to the barrier is

$$\mathbf{n}_{\mathcal{B}} = \frac{\nabla h(\mathbf{x})}{|\nabla h(\mathbf{x})|}. \tag{3.59}$$

The covariance rate normal to the barrier is

$$a_{\mathcal{B}} = \mathbf{n} * \mathcal{B}^{\mathsf{T}} \mathbf{A} \mathbf{n} * \mathcal{B}. \tag{3.60}$$

This measures stochastic mobility through the barrier.

The projection onto the tangent space of the barrier is

$$\mathbf{P} * T = \mathbf{I} - \mathbf{n} * \mathcal{B} \mathbf{n}_{\mathcal{B}}^{\mathsf{T}}. \tag{3.61}$$

Tangential mobility is represented by

$$\mathbf{A} * T = \mathbf{P} * T \mathbf{A} \mathbf{P}_T. \tag{3.62}$$

A system may exhibit substantial variance while moving mainly along the barrier. In that case, total volatility is high but immediate penetration risk may be limited.

Conversely, modest total variance may be dangerous if it is concentrated in the normal direction.

3.7 A Local Geometric Fragility Index

The framework suggests a local fragility measure combining distance, drift, and mobility.

Let

$$d_{\mathcal{B}}(\mathbf{x}) \tag{3.63}$$

be the metric distance from the current state to the barrier.

Let

$$b_{\mathcal{B}} = \mathbf{n}_{\mathcal{B}}^{\top} \mathbf{b} \tag{3.64}$$

and

$$a_{\mathcal{B}} = \mathbf{n} * \mathcal{B}^{\top} \mathbf{A} \mathbf{n} * \mathcal{B}. \tag{3.65}$$

An illustrative fragility index is

$$\mathcal{F}(\mathbf{x}, t) = \frac{\alpha \sqrt{a_{\mathcal{B}}} + \beta \max (b_{\mathcal{B}}, 0)}{d_{\mathcal{B}}(\mathbf{x})}, \tag{3.66}$$

where α and β are calibration parameters.

The index increases when:

- the state is closer to the barrier;
- stochastic mobility is aligned with the barrier normal;
- deterministic drift points toward the barrier.

The formula is illustrative rather than universal. Its purpose is to show that fragility is relational. It depends on location, direction, and motion.

Covariance, Information Geometry, and Statistical Distance

Covariance also appears in information geometry, but again its role must be distinguished carefully.

For a parametric probability model

$$p(\mathbf{x} \mid \boldsymbol{\theta}), \tag{3.67}$$

the Fisher information metric is

$$\mathcal{I} * ij(\boldsymbol{\theta}) = \mathbb{E}\left[\frac{\partial \log p}{\partial \theta * i} \frac{\partial \log p}{\partial \theta_j}\right]. \tag{3.68}$$

This metric measures distinguishability among probability distributions.

The Fisher metric, the inverse covariance metric, and the structural risk metric may coincide in special settings, particularly for Gaussian families, but they are conceptually distinct.

The hierarchy is:

$$\boxed{\begin{aligned} \mathbf{A} &: \text{stochastic mobility,} \\ \mathbf{C}^{-1} &: \text{statistical distance relative to variability,} \\ \boldsymbol{\mathcal{I}} &: \text{information distance among distributions,} \\ \mathbf{g} &: \text{structural risk distance.} \end{aligned}} \tag{3.69}$$

Keeping these objects separate preserves the logical clarity of the framework.

3.8 Covariance Is Not Interference

The book will later introduce complex amplitudes, phase, superposition, and interference. Covariance does not provide these features.

Covariance can describe:

- positive or negative association;
- common response to shocks;
- shared exposure;
- state-dependent alignment;
- changing factor structure;
- nonlinear dependence when extended beyond second moments.

It remains a real-valued object derived from a probability distribution.

Suppose two classical channels contribute probabilities p_1 and p_2. A classical mixture may assign

$$p = w_1 p_1 + w_2 p_2, \tag{3.70}$$

where

$$w_1 + w_2 = 1. \tag{3.71}$$

A complex-amplitude representation instead combines amplitudes:

$$\Psi = \Psi_1 + \Psi_2. \tag{3.72}$$

The resulting probability is

$$|\Psi_1 + \Psi_2|^2 = |\Psi_1|^2 + |\Psi_2|^2 + 2\,\mathrm{Re}\left(\Psi_1^* \Psi_2\right). \tag{3.73}$$

The cross term depends on relative phase. It has no direct equivalent in an ordinary covariance matrix.

Classical nonlinear models can represent amplification, suppression, feedback, and interaction. Therefore, non-additivity alone does not justify a quantum representation.

The move to complex amplitudes will be justified only if phase, coherence, and interference produce explanatory or predictive value beyond classical dependence.

That conceptual jump belongs in Chapter 5.

From Classical Density to Amplitude Geometry

A classical probability density over the state manifold is

$$\rho(\mathbf{x}, t) \geq 0, \tag{3.74}$$

with

$$\int_{\mathcal{M}} \rho(\mathbf{x}, t), \mathrm{d}\mu(\mathbf{x}) = 1. \tag{3.75}$$

Its covariance is

$$\mathbf{C} * \rho = \int *\mathcal{M} \, (\mathbf{x} - \boldsymbol{\mu}) \, (\mathbf{x} - \boldsymbol{\mu})^{\mathsf{T}} \, \rho(\mathbf{x}, t), \mathrm{d}\mu(\mathbf{x}), \tag{3.76}$$

where

$$\boldsymbol{\mu} = \int_{\mathcal{M}} \mathbf{x} \rho(\mathbf{x}, t), \mathrm{d}\mu(\mathbf{x}). \tag{3.77}$$

A complex amplitude is

$$\Psi(\mathbf{x}, t) = R(\mathbf{x}, t) \exp\left[\mathrm{i}\theta(\mathbf{x}, t)\right]. \tag{3.78}$$

Its observable density is

$$\rho(\mathbf{x}, t) = |\Psi(\mathbf{x}, t)|^2 = R^2(\mathbf{x}, t). \tag{3.79}$$

Two systems can therefore share the same probability density and covariance while differing in phase structure.

This is the precise reason the complex representation is richer.

Covariance describes the geometry of observed or modelled stochastic dispersion. Phase describes relational information among amplitudes that is not recoverable from the covariance alone.

Estimating Covariance and Mobility

An operational implementation requires estimation.

Rolling Covariance

Observed state increments may be used to estimate the mean

$$\widehat{\boldsymbol{\mu}} * t = \frac{1}{N} \sum *k = 1^N \Delta \mathbf{x}_{t-k}, \tag{3.80}$$

and covariance

$$\widehat{\mathbf{C}} * t = \frac{1}{N-1} \sum *k = 1^N \left(\Delta \mathbf{x} * t - k - \widehat{\boldsymbol{\mu}} * t\right) \left(\Delta \mathbf{x} * t - k - \widehat{\boldsymbol{\mu}} * t\right)^{\mathsf{T}}. \tag{3.81}$$

Factor Models

A lower-dimensional factor representation is

$$\Delta \mathbf{x} * t = \mathbf{B}\mathbf{f} * t + \boldsymbol{\varepsilon}_t. \tag{3.82}$$

The covariance becomes

$$\mathbf{C} = \mathbf{B}\mathbf{C} * f \mathbf{B}^{\mathsf{T}} + \mathbf{C} * \varepsilon. \tag{3.83}$$

Regime-Dependent Covariance

The covariance may depend on a latent regime z_t:

$$\mathbf{C} * t = \mathbf{C}^{(z*t)}. \tag{3.84}$$

Local Covariance

When mobility varies continuously over the manifold, one estimates

$$\mathbf{C} = \mathbf{C}(\mathbf{x}, t). \tag{3.85}$$

Structural Mobility

When probability data are insufficient, experts may specify a structural mobility tensor

$$\mathbf{K}(\mathbf{x}, t) \tag{3.86}$$

using network structure, legal pathways, control dependencies, or simulated scenarios.

The distinction remains explicit: $\mathbf{C}$ is covariance; $\mathbf{K}$ is structural mobility.

3.9 Governance Implications

The geometric interpretation changes the questions risk managers should ask.

They should not ask only whether volatility has increased. They should ask:

- Which principal directions are expanding?
- Are the dominant eigenvectors rotating toward adverse regions?
- Is effective dimensionality collapsing?
- Are previously independent controls becoming aligned?
- Is mobility concentrated normal to a protective barrier?
- Has drift become aligned with diffusion?
- Is the metric assigning high institutional significance to a statistically small movement?
- Are covariance, structural mobility, and risk distance being confused?

A sound governance framework should therefore report separately:

1. the state of the system;
2. its drift;
3. its covariance or diffusion structure;
4. the structural mobility map;
5. the risk metric;
6. the position and orientation of nearby barriers.

The warning signal is not always movement of the state. It may be a change in the directions in which movement has become possible.

3.9.1 The Minimum Mobility Architecture

The concepts of this chapter can be summarized through the local mobility architecture

$$\mathfrak{M}(\mathbf{x}, t) = (\mathbf{b}, \mathbf{A}, \mathbf{D}, \mathbf{K}, \mathbf{g}, \mathcal{B}), \tag{3.87}$$

where:

- $\mathbf{b}$ is the drift;
- $\mathbf{A}$ is the instantaneous covariance rate;
- $\mathbf{D}$ is the diffusion tensor;

- $\mathbf{K}$ is an optional structural mobility tensor;
- $\mathbf{g}$ is the structural risk metric;
- $\mathcal{B}$ denotes nearby barriers and boundaries.

The architecture may be summarized as

$$\boxed{\begin{aligned}\text{Local Risk Mobility} &= \text{Drift} + \text{Stochastic Covariance}\\ &\quad + \text{Structural Mobility} + \text{Risk Metric} + \text{Barrier Alignment}.\end{aligned}} \tag{3.88}$$

3.10 Chapter Synthesis

This chapter has clarified the status of covariance within the geometric risk framework.

Covariance remains the classical covariance of a random vector or stochastic process. Its origin lies in probability theory. It is the second central moment of state increments.

In continuous time, the covariance of infinitesimal increments is proportional to $\mathrm{d}t$. The matrix $\boldsymbol{\Sigma}\boldsymbol{\Sigma}^{\mathsf{T}}$ is the instantaneous covariance rate, while one-half of that matrix is often called the diffusion tensor.

When placed on the tangent space of the state manifold, the covariance rate becomes a geometric tensor describing local stochastic mobility. It defines the orientation and scale of the uncertainty ellipsoid and transforms consistently under changes of coordinates.

Covariance is not the same as the metric. The metric defines the meaning of distance. Covariance defines the directions and intensity of random movement. Their interaction determines how much structurally meaningful movement the stochastic dynamics generate.

The inverse covariance can induce a Mahalanobis metric, but that statistical metric should not automatically be identified with the institutional risk metric. Statistical commonness and institutional significance are different concepts.

When data are insufficient, a structurally defined mobility tensor may be introduced. Such an object can be geometric without being probabilistic, but it should not be called covariance unless it is actually derived from a probability law.

Finally, covariance does not contain complex phase and cannot produce genuine quantum-style interference. It describes classical dependence and directional stochastic mobility. The richer transition to complex amplitudes, coherence, and interference will occur in Chapter 5.

The central proposition of Chapter 3 is therefore:

> Covariance is mapped from probability into geometry: it begins as a classical second moment and becomes, on the state manifold, a tensor describing the directions and intensity of stochastic movement.

Chapter 4 will now turn from movement to protection. It will examine barriers, boundaries, controls, and the geometric structures that impede transition into adverse regions.

Part II

Technical Foundations of Tunneling

Chapter 4

Barriers, Boundaries, and Protective Structures

4.1 From Thresholds to Barriers

Risk management frequently relies on thresholds. A borrower breaches a covenant, a liquidity ratio falls below a regulatory minimum, a system reaches capacity, or a legal exposure exceeds a materiality limit. Thresholds are useful because they create decision points. They divide one category from another and allow institutions to define triggers, limits, escalation procedures, and intervention rules.

Yet a threshold is not the same as a barrier.

A threshold is ordinarily represented as a boundary. It identifies the point at which classification changes. A barrier, by contrast, occupies a region of the state space and resists movement through that region. It has internal structure. It may be high or low, wide or narrow, continuous or fragmented, stable or deteriorating. It may resist movement strongly in one direction and weakly in another.

Suppose that the state of a risk-bearing system is

$$\mathbf{x}_t \in \mathcal{M}, \tag{4.1}$$

where $\mathcal{M}$ is the state manifold. Let $S \subset \mathcal{M}$ denote a stable region and $A \subset \mathcal{M}$ an adverse region.

A simple boundary separating the two may be represented as

$$\partial A = \left\{ \mathbf{x} \in \mathcal{M} : h(\mathbf{x}, t) = 0 \right\}, \tag{4.2}$$

where (h) is a classification function.

The safe side may satisfy

$$h(\mathbf{x}, t) < 0, \tag{4.3}$$

while the adverse side satisfies

$$h(\mathbf{x}, t) > 0. \tag{4.4}$$

This construction defines where the adverse region begins. It does not yet explain what impedes movement toward it.

A barrier is better represented as a subset

$$\mathcal{B}_t \subset \mathcal{M}, \tag{4.5}$$

occupying the transition region between stable and adverse states. It may be associated with a resistance field

$$V_{\mathcal{B}} : \mathcal{B} * t \times \mathbb{R} * + \longrightarrow \mathbb{R}_+, \tag{4.6}$$

which assigns a degree of transition difficulty to each point in the barrier.

The central distinction is therefore:

> A threshold identifies where danger begins. A barrier describes the structure that makes danger difficult to reach.

4.1.1 Protective Structures as Geometric Objects

Many familiar forms of protection can be interpreted as barriers.

In credit risk, the barrier may include capital, collateral, guarantees, liquidity reserves, covenant headroom, refinancing capacity, and shareholder support.

In operational risk, it may include segregation of duties, authorization controls, system redundancy, backup infrastructure, manual overrides, and recovery procedures.

In legal and regulatory risk, it may include contractual language, documentary evidence, precedent, legal opinions, jurisdictional protections, and compliance controls.

In institutional and systemic risk, barriers may consist of capital buffers, governance mechanisms, circuit breakers, resolution systems, information controls, and network redundancy.

The geometric interpretation changes how these protections are evaluated.

A conventional control inventory asks whether a safeguard exists. The geometric approach asks:

- where the safeguard operates;
- which directions of movement it impedes;
- how much deterioration it can absorb;
- how long the protection persists;
- whether several layers share one dependency;

- whether hidden passages remain open.

A protective structure may be represented by a barrier potential

$$V_{\mathcal{B}}(\mathbf{x}, t), \tag{4.7}$$

defined over the transition region. High values indicate strong resistance. Low values indicate weakness.

The barrier is therefore not one scalar number. It is a field over the state manifold.

4.2 Barrier Height

Barrier height represents the severity of deterioration, stress, or force required to move through the protective region.

Suppose the system occupies a stable state $\mathbf{x} * S$, and let γ be a path connecting $\mathbf{x} * S$ to the adverse region. The maximum barrier resistance along that path may be written as

$$H_{\mathcal{B}}[\gamma, t] = \max_{\mathbf{x} \in \gamma \cap \mathcal{B} * t} \left[V * \mathcal{B}(\mathbf{x}, t) - V_{\mathcal{B}}(\mathbf{x}_S, t) \right]. \tag{4.8}$$

This quantity measures the highest resistance encountered along the path.

In credit risk, a high barrier may represent substantial loss-absorbing capital, strong liquidity support, and valuable collateral. In operational risk, it may represent several genuinely independent control layers. In legal risk, it may represent clear contractual language reinforced by precedent, documentation, and consistent conduct.

Barrier height is path-dependent. A barrier may be high against one transition and low against another.

Capital, for example, may create a high barrier against ordinary credit losses but a much weaker barrier against a sudden liquidity run. A contract may resist one category of claim while remaining vulnerable to a different interpretation grounded in conduct, fiduciary duty, or public policy.

It is therefore more precise to write

$$H_{\mathcal{B}} = H_{\mathcal{B}}[\gamma, t]. \tag{4.9}$$

There is generally no single universal barrier height.

4.2.1 Barrier Width

Barrier width measures how much transition space must be traversed before the adverse region is reached.

Let γ be a path crossing the barrier. Its effective width is the metric length of the portion inside $\mathcal{B}_t$:

$$W_{\mathcal{B}}[\gamma, t] = \int_{\gamma \cap \mathcal{B}_t} \mathrm{d}s, \tag{4.10}$$

where

$$\mathrm{d}s^2 = \mathrm{d}\mathbf{x}^\mathsf{T} \mathbf{g}(\mathbf{x}, t) \mathrm{d}\mathbf{x}. \tag{4.11}$$

Barrier width has a natural institutional interpretation. A wide barrier may require prolonged deterioration, repeated failures, several legal developments, sustained funding pressure, or multiple stages of contagion. A narrow barrier may require only one short sequence of events.

The distinction between height and width is essential.

A high but narrow barrier may look strong while remaining vulnerable to a rapid transition through a specific channel. A lower but wider barrier may provide more practical resilience because the system must sustain deterioration over a longer path before reaching the adverse region.

A simple integrated resistance measure is

$$\mathcal{R} * \mathcal{B}[\gamma, t] = \int * \gamma \cap \mathcal{B} * tV * \mathcal{B}(\mathbf{x}, t), \mathrm{d}s. \tag{4.12}$$

This combines the intensity and extent of resistance.

The later theory of tunneling will replace this simple integral with a more precise under-barrier action. At this stage, the key point is that effective protection depends on the entire barrier profile, not merely on its highest point.

Width should therefore be estimated in units that preserve institutional meaning. In credit risk it may reflect months of liquidity, the number of refinancing stages, or the distance between a covenant warning and payment failure. In operational risk it may represent independent detection, containment, recovery, and verification stages. In legal risk it may correspond to the sequence of notice, evidence, review, appeal, and enforcement. These interpretations should not be mixed without an explicit metric, because a count of controls is not automatically equivalent to elapsed time or economic distance.

Validation should examine whether wider measured barriers actually provide longer warning periods, more intervention opportunities, or lower adverse transition rates after controlling for height. Shared dependencies must reduce effective width: four nominal stages that rely on the same data source, authority, vendor, or funding facility may behave like one stage under stress. Analysts should also distinguish width that delays failure from width that merely delays observation. A slow reporting chain can make a barrier look wide while leaving little practical response time. Stress tests should compress timing, remove intermediate controls, and impose simultaneous demand to determine whether the estimated width survives adverse conditions. A governed width measure should identify the stages involved, their independence, their activation time, and the intervention that can add meaningful distance before the adverse region is reached.

4.3 Barrier Shape, Curvature, and Direction

Barriers need not be flat or uniform.

They may curve around stable regions, form ridges between basins, contain saddle points, or narrow into channels. The shape of the barrier affects the paths available to the system.

Suppose the barrier surface is represented by

$$h(\mathbf{x}, t) = 0. \tag{4.13}$$

The unit normal vector is

$$\mathbf{n}_{\mathcal{B}} = \frac{\nabla h(\mathbf{x}, t)}{|\nabla h(\mathbf{x}, t)|}. \tag{4.14}$$

This vector identifies the direction perpendicular to the barrier.

The tangent space consists of directions $\mathbf{v}$ satisfying

$$\mathbf{n}_{\mathcal{B}}^{\mathsf{T}} \mathbf{v} = 0. \tag{4.15}$$

The distinction between normal and tangential directions matters because movement along a barrier differs from movement through it.

A system may experience substantial volatility while remaining tangential to the protective structure. Such movement may change the location of future weakness without producing immediate penetration.

Curvature may also concentrate paths. A curved barrier can guide trajectories toward a narrow saddle point or weak passage. In this sense, the geometry of protection can itself concentrate risk.

Anisotropic Barriers

A barrier is anisotropic when its resistance depends on direction.

Let

$$\mathbf{R}_{\mathcal{B}}(\mathbf{x}, t) \tag{4.16}$$

be a symmetric positive-definite resistance tensor. The effective resistance to an infinitesimal movement $\mathrm{d}\mathbf{x}$ may be written as

$$\mathrm{d}r^2 = \mathrm{d}\mathbf{x}^{\mathsf{T}} \mathbf{R}_{\mathcal{B}}(\mathbf{x}, t) \mathrm{d}\mathbf{x}. \tag{4.17}$$

High resistance in one direction and low resistance in another create directional asymmetry.

This is common in practical risk systems. Capital may absorb credit losses but provide limited protection against funding collapse. Internal controls may prevent unauthorized payments while offering little resistance to model error. Legal documentation may protect against direct contractual claims but remain weak against regulatory enforcement.

For a unit direction $\mathbf{v}$, directional barrier resistance is

$$r_{\mathcal{B}}(\mathbf{v}) = \mathbf{v}^{\mathsf{T}}\mathbf{R}_{\mathcal{B}}\mathbf{v}. \tag{4.18}$$

The weakest direction is

$$\mathbf{v} * \min = \arg, \min *|\mathbf{v}| = 1\mathbf{v}^{\mathsf{T}}\mathbf{R}_{\mathcal{B}}\mathbf{v}. \tag{4.19}$$

This direction corresponds to the eigenvector associated with the smallest eigenvalue of the resistance tensor.

The relevant risk question is not whether the barrier is strong in general, but whether the system's drift and mobility align with its weakest direction.

Barrier Continuity, Layers, and Hidden Passages

Protective structures are often layered.

A layered barrier may be written as

$$\mathcal{B} * t = \bigcup *k = 1^{K}\mathcal{B}_{k,t}, \tag{4.20}$$

where each $\mathcal{B}_{k,t}$ is one protective layer.

If the layers are independent and sequential, they create genuine depth. If they share dependencies, visible multiplicity may conceal fragility.

Three control layers may all rely on the same data source. Two legal protections may fail under the same judicial interpretation. Several liquidity facilities may depend on the same market remaining open.

Barrier continuity can be represented through a coverage function

$$c_{\mathcal{B}}(\mathbf{x}, t) \in [0, 1], \tag{4.21}$$

where values near (1) indicate strong continuous protection and values near (0) indicate a gap.

A gap region may be defined as

$$\mathcal{G} * t = \left\{\mathbf{x} \in \mathcal{B} * t : c_{\mathcal{B}}(\mathbf{x}, t) < c_{\min}\right\}. \tag{4.22}$$

Hidden passages arise when gaps create low-resistance paths through the barrier.

The minimum-resistance path is

$$\gamma^* = \arg, \min * \gamma \mathcal{R} * \mathcal{B}[\gamma, t]. \tag{4.23}$$

This path may pass through a small region that is nearly invisible when the barrier is assessed only in aggregate.

The implication is fundamental:

> A barrier can be strong on average and still fail through one narrow corridor.

4.4 Permeability and Integrity

Barrier permeability describes the ease with which effective transition occurs through the protective region.

Let

$$\Pi_{\mathcal{B}} : \mathcal{B} * t \times \mathbb{R} * + \longrightarrow \mathbb{R}_+ \tag{4.24}$$

be a permeability field.

Higher values indicate greater ease of passage.

A simple illustrative relation is

$$\Pi_{\mathcal{B}}(\mathbf{x}, t) = \exp\left[-\alpha V_{\mathcal{B}}(\mathbf{x}, t)\right], \qquad \alpha > 0. \tag{4.25}$$

For a path,

$$\Pi_{\mathcal{B}}[\gamma, t] = \exp\left[-\alpha \mathcal{R}_{\mathcal{B}}[\gamma, t]\right]. \tag{4.26}$$

This exponential expression anticipates the mathematical form of tunneling without yet asserting a quantum law.

Barrier integrity is broader than permeability. It represents whether the protective structure remains capable of performing its intended function.

Let

$$I_{\mathcal{B}}(t) \in [0, 1] \tag{4.27}$$

be a normalized integrity measure.

A barrier may remain unbreached while its integrity is deteriorating. It may become thinner, more fragmented, more correlated with other controls, or more aligned with the system's dominant mobility direction.

The purpose of barrier monitoring is therefore not only to detect breach, but to detect structural weakening before breach.

Dynamic Barriers

Barriers evolve through time.

Let

$$V_{\mathcal{B}} = V_{\mathcal{B}}(\mathbf{x}, t). \tag{4.28}$$

Along the system's path,

$$\frac{\mathrm{d}}{\mathrm{d}t} V_{\mathcal{B}}(\mathbf{x} * t, t) = \frac{\partial V * \mathcal{B}}{\partial t} + \nabla V_{\mathcal{B}}^{\mathsf{T}} \dot{\mathbf{x}}_t. \tag{4.29}$$

The first term captures explicit deformation of the barrier. The second captures the effect of movement through a heterogeneous resistance field.

Barrier deterioration may arise from capital depletion, collateral decline, control fatigue, staff turnover, software obsolescence, legal reinterpretation, weakening market liquidity, or growing network concentration.

Barrier strengthening may arise from recapitalization, diversification, legal clarification, redundancy, circuit breakers, or control redesign.

Risk can increase because

$$\mathbf{x} * t \longrightarrow \mathcal{B} * t, \tag{4.30}$$

or because

$$\mathcal{B} * t \longrightarrow \mathbf{x} * t. \tag{4.31}$$

The second mechanism is frequently overlooked.

If the barrier is defined implicitly by

$$h(\mathbf{x}, t) = 0, \tag{4.32}$$

then a point on the moving boundary satisfies

$$\nabla h^{\mathsf{T}} \dot{\mathbf{x}} + \frac{\partial h}{\partial t} = 0. \tag{4.33}$$

The normal velocity of the barrier is

$$v_{\mathcal{B}}^{\perp} = -\frac{\partial h / \partial t}{|\nabla h|}. \tag{4.34}$$

This quantity indicates whether the barrier is moving toward or away from the current state.

4.5 Alignment with Drift and Diffusion

The vulnerability of a barrier depends on how the system approaches it.

Let $\mathbf{n}_{\mathcal{B}}$ be the unit normal pointing toward the adverse region. The drift component normal to the barrier is

$$b_{\mathcal{B}} = \mathbf{n}_{\mathcal{B}}^{\top}\mathbf{b}. \tag{4.35}$$

The covariance rate normal to the barrier is

$$a_{\mathcal{B}} = \mathbf{n} * \mathcal{B}^{\top}\mathbf{A}\mathbf{n} * \mathcal{B}. \tag{4.36}$$

The tangential covariance is

$$\mathbf{A} * T = \mathbf{P} * T\mathbf{A}\mathbf{P}_{T}, \tag{4.37}$$

where

$$\mathbf{P} * T = \mathbf{I} - \mathbf{n} * \mathcal{B}\mathbf{n}_{\mathcal{B}}^{\top}. \tag{4.38}$$

A high value of $a_{\mathcal{B}}$ means that stochastic movement is directed through the barrier. High tangential mobility means that the system moves mainly along it.

A useful alignment index is

$$\mathcal{A} * \mathcal{B} = \frac{\mathbf{n} * \mathcal{B}^{\top}\mathbf{A}\mathbf{n}_{\mathcal{B}}}{\mathrm{tr}(\mathbf{A})}. \tag{4.39}$$

When $\mathcal{A}_{\mathcal{B}}$ is high, a large share of total stochastic mobility is directed toward penetration.

Barrier vulnerability is therefore relational. It depends on the interaction between barrier geometry and mobility geometry.

A Minimal Practical Example: Constructing a Credit-Risk Manifold

The preceding discussion is abstract. It is useful to see how a manifold and a barrier can be constructed in a minimal practical example.

Consider a corporate borrower described by two state variables:

$$\mathbf{x} * t = \begin{bmatrix} \ell * t \ q_t \end{bmatrix}, \tag{4.40}$$

where:

- ℓ_t is leverage, such as debt divided by EBITDA;
- q_t is liquidity, such as liquid resources divided by short-term obligations.

The ambient coordinate space is initially

$$(\ell, q) \in \mathbb{R}_+^2. \tag{4.41}$$

Not every pair (ℓ, q) is economically feasible. Very high leverage may restrict access to funding, while very low liquidity may trigger covenant action or refinancing stress. The admissible borrower states form a constrained subset:

$$\mathcal{M} * \text{credit} = \left\{ (\ell, q) \in \mathbb{R} * +^2 : F(\ell, q) \leq 0 \right\}, \tag{4.42}$$

where $F(\ell, q)$ summarizes accounting, financing, and institutional constraints.

In practice, the manifold can be approximated by a cloud of observed or simulated states:

$$\mathcal{X} = \{(\ell_{i,t}, q_{i,t})\}. \tag{4.43}$$

Each point represents one feasible borrower configuration. A smooth surface, local chart, or density-supported region can then be fitted through the cloud.

The manifold is therefore not initially discovered through sophisticated differential geometry. It begins with feasible states.

4.6 Defining Stable, Fragile, and Adverse Regions

The next step is to assign risk meaning to different parts of the manifold.

A simple stable region may be defined as

$$S = \{(\ell, q) : \ell < \ell_{\text{safe}}, \quad q > q_{\text{safe}}\}. \tag{4.44}$$

An adverse region may be defined as

$$A = \{(\ell, q) : \ell > \ell_{\text{critical}} \quad \text{or} \quad q < q_{\text{critical}}\}. \tag{4.45}$$

The fragile region lies between them.

For illustration, one might use

$$\ell_{\text{safe}} = 3, \tag{4.46}$$

$$\ell_{\text{critical}} = 6, \tag{4.47}$$

$$q_{\text{safe}} = 1.5, \tag{4.48}$$

$$q_{\text{critical}} = 0.8. \tag{4.49}$$

These values are illustrative. In practice, they would be calibrated from historical defaults, covenant structures, expert judgment, and market evidence.

Constructing the Risk Landscape

A scalar potential can be assigned to the manifold:

$$V_{\mathrm{R}}(\ell, q) = \alpha\ell^2 + \frac{\beta}{q^2} - \gamma q, \qquad \alpha, \beta, \gamma > 0. \tag{4.50}$$

The function rises when leverage increases and liquidity falls.

Low values of V_{R} correspond to stable borrower configurations. High values correspond to fragile or adverse configurations.

The surface

$$z = V_{\mathrm{R}}(\ell, q) \tag{4.51}$$

provides a visual risk landscape. Valleys represent stable basins. Ridges represent transition resistance. High regions correspond to fragile or adverse configurations.

The borrower is represented by a point moving across this landscape.

4.7 Defining the Law of Motion

The state must evolve through time. A minimal stochastic model is

$$\mathrm{d}\ell_t = b_\ell(\ell_t, q_t), \mathrm{d}t + \sigma_\ell, \mathrm{d}W_{\ell,t}, \tag{4.52}$$

$$\mathrm{d}q_t = b_q(\ell_t, q_t), \mathrm{d}t + \sigma_q, \mathrm{d}W_{q,t}. \tag{4.53}$$

A simple specification is

$$b_\ell(\ell, q) = a_1 - a_2 q, \tag{4.54}$$

$$b_q(\ell, q) = c_1 - c_2 \ell. \tag{4.55}$$

The interpretation is intuitive:

- low liquidity increases leverage pressure;
- high leverage weakens liquidity;
- random shocks move the borrower around the expected trajectory.

The manifold now contains not only locations, but possible movements.

Constructing the Credit Barrier

Suppose the borrower is protected by:

- available liquidity reserves;
- unused committed credit lines;
- collateral;
- covenant headroom;
- shareholder support.

A minimal barrier function may be defined as

$$B(\ell, q) = q + \delta C + \eta L - \kappa\ell, \tag{4.56}$$

where:

- (C) represents collateral support;
- (L) represents committed liquidity support;
- $\delta, \eta, \kappa > 0$ are calibration parameters.

The boundary is

$$B(\ell, q) = 0. \tag{4.57}$$

The safe side satisfies

$$B(\ell, q) > 0, \tag{4.58}$$

while the unsafe side satisfies

$$B(\ell, q) < 0. \tag{4.59}$$

The protective barrier should not be interpreted only as the line (B=0). The full barrier region may be defined as

$$\mathcal{B} = \{(\ell, q) : 0 \leq B(\ell, q) \leq \varepsilon\}, \tag{4.60}$$

where $\varepsilon > 0$ defines the effective width of the transition zone.

The borrower is not necessarily safe merely because it has not crossed the boundary. It may already be inside the barrier region, where protection is being consumed.

4.7.1 Interpreting Height, Width, and Weakest Passage

Barrier height may be associated with the amount of available protection:

$$H_{\mathcal{B}} = q + \delta C + \eta L. \tag{4.61}$$

Greater liquidity, collateral, and committed support increase the barrier.

Barrier width may be interpreted as the metric distance across the protective region:

$$W_{\mathcal{B}} = d_{\mathcal{M}}\left(\partial S, \partial A\right). \tag{4.62}$$

A borrower may therefore possess:

- a high but narrow barrier, representing substantial protection that can disappear quickly;
- a lower but wide barrier, representing modest protection that erodes only after prolonged deterioration;
- a fragmented barrier, representing multiple protections sharing one hidden dependency.

For each path γ connecting the current state to the adverse region, define integrated resistance as

$$\mathcal{R} * \mathcal{B}[\gamma] = \int *\gamma \cap \mathcal{B} V_{\mathcal{B}}(\ell, q), \mathrm{d}s. \tag{4.63}$$

The weakest passage is

$$\gamma^* = \arg, \min *\gamma \mathcal{R} * \mathcal{B}[\gamma]. \tag{4.64}$$

This path may correspond to a liquidity crisis rather than gradual insolvency.

A borrower with moderate leverage may still possess a short path to default if market funding disappears and liquidity falls rapidly. The geometric model reveals that the most accessible path to failure need not be the path of highest leverage.

4.8 The Minimal Construction Procedure

The practical construction can be summarized in seven steps:

1. Select two or three state variables.
2. Generate or observe feasible state combinations.
3. Construct the admissible state manifold.
4. Define stable, fragile, and adverse regions.
5. Assign a potential or stability function.

6. Construct the barrier from capital, liquidity, controls, contracts, or other protections.

7. Identify the minimum-resistance path from the current state to the adverse region.

The resulting model is

$$\mathfrak{R} * \text{credit} = (\mathcal{M}, \mathbf{x} * t, V_{\text{R}}, \mathcal{D}, \mathcal{B}, \gamma^*)\,, \tag{4.65}$$

where:

- $\mathcal{M}$ is the feasible borrower manifold;
- $\mathbf{x} * t$ is the current borrower state;
- $V * \text{R}$ is the risk landscape;
- $\mathcal{D}$ is the law of motion;
- $\mathcal{B}$ is the protective barrier;
- γ^* is the weakest transition pathway.

This example shows that constructing a manifold and barrier does not initially require sophisticated differential geometry. It can begin with a small number of variables, a cloud of feasible states, a fitted surface, a transition law, and a set of protective constraints.

The sophistication can be added later. The essential requirement is that the model identify where the system is, how it moves, where the adverse region lies, and what structure resists the transition.

Classical Barrier Crossing

Before introducing tunneling, it is necessary to distinguish ordinary barrier crossing.

Suppose a system has an effective stress capacity E_{R}, while the barrier has maximum resistance $V_{\mathcal{B}}^{\max}$.

A classical crossing condition may be represented as

$$E_{\text{R}} \geq V_{\mathcal{B}}^{\max}. \tag{4.66}$$

The system crosses because a sufficiently large shock, accumulated momentum, or sustained deterioration carries it over the barrier.

Examples include losses exceeding capital, demand exceeding system capacity, fraud overwhelming control limits, or liquidity demand exhausting available reserves.

This is an ordinary adverse transition. It may be rare, but it does not require tunneling.

The tunneling question arises when

$$E_{\mathrm{R}} < V_{\mathcal{B}}^{\max}, \tag{4.67}$$

yet effective transmission into the adverse region still occurs.

Chapter 5 will examine the mathematics of such transmission.

4.9 The Minimum Barrier Architecture

A complete barrier model requires:

1. a stable region (S);
2. an adverse region (A);
3. a barrier region $\mathcal{B} * t$;
4. a boundary function $h(\mathbf{x}, t)$;
5. a barrier potential $V * \mathcal{B}(\mathbf{x}, t)$;
6. a metric $\mathbf{g}(\mathbf{x}, t)$;
7. a resistance tensor $\mathbf{R} * \mathcal{B}(\mathbf{x}, t)$;
8. a continuity function $c * \mathcal{B}(\mathbf{x}, t)$;
9. a permeability field $\Pi_{\mathcal{B}}(\mathbf{x}, t)$;
10. a time-dependent integrity measure $I_{\mathcal{B}}(t)$;
11. the local drift and diffusion relative to the barrier.

These objects may be summarized as

$$\mathfrak{B} * t = (\mathcal{B} * t, h_t, V_{\mathcal{B},t}, \mathbf{R} * \mathcal{B}, t, c * \mathcal{B}, t, \Pi_{\mathcal{B},t}, I_{\mathcal{B},t}) . \tag{4.68}$$

The interaction with movement is

$$\mathfrak{I} * t = (b * \mathcal{B}, t, a_{\mathcal{B},t}, \mathcal{A} * \mathcal{B}, t, d * \mathcal{B}, t) , \tag{4.69}$$

where $d_{\mathcal{B},t}$ is the distance to the barrier.

The complete relationship is therefore

$$\boxed{\begin{aligned}\text{Barrier Risk} = {} & \text{Barrier Geometry} + \text{Barrier Integrity} \\ & + \text{Mobility Alignment} + \text{Time Deformation.}\end{aligned}} \tag{4.70}$$

4.9.1 Governance Implications

The geometric interpretation changes how protective structures should be governed.

Boards, risk committees, and control functions should not ask only whether controls are present. They should ask:

- Where is the barrier weakest?
- Which transition path encounters the least resistance?
- Is the barrier high but narrow?
- Are multiple layers genuinely independent?
- Is the barrier moving toward the current state?
- Is the adverse region expanding?
- Is drift aligned with the barrier normal?
- Is stochastic mobility concentrated through a weak section?
- Has permeability increased before any formal breach?
- Are legal, operational, and financial protections sharing one dependency?

A control that has not failed may nevertheless be deteriorating. A barrier that remains nominally intact may have become thinner, more fragmented, or more aligned with the system's dominant mobility direction.

The objective of governance is therefore not merely to record breaches. It is to monitor barrier geometry before breach.

4.10 Chapter Synthesis

This chapter has transformed protection from a binary condition into a geometric structure.

A threshold marks the point at which classification changes. A barrier occupies a region of transition resistance. It has height, width, curvature, continuity, anisotropy, permeability, and time-dependent integrity.

Capital, collateral, liquidity, contracts, controls, governance mechanisms, and technological safeguards can all be represented as barriers separating stable and adverse regions.

Barrier height measures the severity of deterioration required to cross. Barrier width measures the persistence or institutional distance that must be traversed. Integrated resistance combines the two.

Barriers may be anisotropic and fragmented. They can resist movement strongly in one direction while remaining weak in another. Multiple layers may appear independent while sharing one hidden dependency.

The chapter's practical credit example demonstrated how to construct a minimal manifold from leverage and liquidity, define stable and adverse regions, assign a potential landscape, specify a stochastic law of motion, build a barrier from collateral and liquidity support, and identify the weakest transition path.

Barrier integrity changes through time. Risk can rise because the system moves toward the barrier or because the barrier moves toward the system.

The effectiveness of a barrier also depends on the movement structure developed in Chapter 3. Drift and diffusion directed normal to the barrier are more relevant to penetration than volatility moving tangentially along it.

The central proposition of Chapter 4 is:

> Protection is not a binary state. It is a dynamic geometric structure whose effectiveness depends on height, width, orientation, continuity, permeability, and alignment with the system's motion.

Chapter 5 will now make the conceptual jump from classical geometry and stochastic mobility to complex amplitudes, phase, interference, and the mathematics of tunneling.

Chapter 5

The Mathematics and Intuition of Tunneling

5.1 The Conceptual Turning Point

The preceding chapters developed a classical geometric description of risk.

Chapter 2 introduced the state manifold: the set of feasible configurations that a risk-bearing system may occupy. Chapter 3 described movement over that manifold through drift, covariance, diffusion, and structural mobility. Chapter 4 introduced barriers as protective regions separating stable states from adverse states.

The present chapter makes the book's most important conceptual transition.

Until now, adverse movement has been interpreted classically. A borrower deteriorates because leverage rises, liquidity falls, cash flow weakens, collateral loses value, or market access disappears. The system moves through its state space until it reaches or crosses a protective barrier.

In a classical representation, barrier crossing occurs because the system accumulates enough stress, momentum, deterioration, or shock intensity to overcome the barrier.

The tunneling question is different:

> Can nonzero transition weight appear in an adverse region even when the system does not appear to possess sufficient classical capacity to cross the measured barrier?

This question must be approached carefully.

An unexpected default is not automatically evidence of tunneling. A sudden transition may instead arise because:

- a relevant state variable was omitted;
- the barrier was incorrectly measured;
- a hidden structural gap existed;
- the barrier deteriorated through time;
- a large jump occurred;

- a common dependency caused several controls to fail together;
- the probability model was misspecified;
- available information was incomplete or delayed.

A tunneling model should therefore not be introduced as an exotic substitute for ordinary risk analysis. It should be introduced only after classical explanations have been stated clearly and tested seriously.

The purpose of the chapter is not to assert that credit systems obey quantum physics. It is to investigate whether the mathematical architecture developed for quantum tunneling offers a useful representation of hidden transition pathways, interacting mechanisms, timing effects, and nonzero transmission through protective structures.

The chapter proceeds in five stages:

1. establish the classical benchmark;
2. derive the quantum-mechanical barrier problem;
3. explain complex amplitudes, phase, and interference;
4. map each mathematical object into the credit-risk example;
5. define a disciplined procedure for encoding real variables into an effective quantum language.

The objective is that, by the end of the chapter, the reader can move in both directions:

$$\boxed{\text{Quantum Mathematics} \longleftrightarrow \text{Credit-Risk Meaning.}} \tag{5.1}$$

The Classical Credit-Risk Benchmark

Before introducing amplitudes, it is necessary to state what a classical model already explains.

Recall the minimal borrower state introduced in Chapter 4:

$$\mathbf{x} * t = \left[\ell * t \ q_t\right], \tag{5.2}$$

where:

- ℓ_t is leverage;
- q_t is liquidity coverage.

The borrower occupies a feasible state manifold

$$\mathbf{x} * t \in \mathcal{M} * \text{credit}. \tag{5.3}$$

A classical stochastic evolution may be written as

$$\mathrm{d}\mathbf{x} * t = \mathbf{b}(\mathbf{x} * t, t), \mathrm{d}t + \boldsymbol{\Sigma}(\mathbf{x} * t, t), \mathrm{d}\mathbf{W} * t + \mathrm{d}\mathbf{J}_t, \tag{5.4}$$

where:

- $\mathbf{b}$ represents expected deterioration or recovery;
- $\boldsymbol{\Sigma}\mathrm{d}\mathbf{W} * t$ represents continuous random shocks;
- $\mathrm{d}\mathbf{J} * t$ represents discontinuous jumps.

A default or severe-distress region is

$$A \subset \mathcal{M}_{\text{credit}}. \tag{5.5}$$

The borrower may enter (A) through several classical mechanisms.

Gradual Deterioration

The drift may point persistently toward distress:

$$\mathbf{b}(\mathbf{x} * t, t) \cdot \mathbf{n} * A > 0, \tag{5.6}$$

where $\mathbf{n}_A$ denotes the direction toward the adverse region.

Leverage rises, liquidity declines, and the borrower gradually approaches the barrier.

Diffusive Escape

Even if expected drift is modest, random shocks may push the borrower across the barrier.

This is represented through the diffusion tensor

$$\mathbf{D} = \frac{1}{2}\boldsymbol{\Sigma}\boldsymbol{\Sigma}^{\mathsf{T}}. \tag{5.7}$$

The crossing is still classical. The state moves through the barrier because cumulative random motion becomes sufficiently large.

Jump-Driven Crossing

A discontinuous event may move the state directly:

$$\Delta\mathbf{x} * t = \mathbf{J} * t. \tag{5.8}$$

Examples include fraud, litigation, loss of a major customer, cancellation of a credit facility, or sudden regulatory action.

Again, this is not tunneling. The state crosses because a large jump occurred.

Barrier Erosion

The borrower may remain nearly stationary while the barrier deteriorates:

$$\mathcal{B} * t \longrightarrow \mathbf{x} * t. \tag{5.9}$$

Collateral loses value, committed funding becomes unreliable, covenant headroom contracts, or legal enforceability weakens.

The borrower appears to cross unexpectedly only because the barrier was moving.

Hidden Passage

The barrier may contain a gap:

$$\mathcal{G} \subset \mathcal{B}. \tag{5.10}$$

A common dependency, documentation defect, technological concentration, or liquidity mismatch may create a low-resistance route.

This is also classical. The measured barrier was incomplete.

These possibilities should be investigated before invoking an amplitude model.

When a Tunneling Representation Becomes Interesting

A tunneling representation becomes interesting when the following pattern appears.

First, the borrower remains on the nominally safe side of the measured barrier.

Second, the observed stress measure appears below the amount required for classical crossing.

Third, nonzero transition weight nevertheless develops in the adverse region.

Fourth, the transition appears to depend on the interaction, sequencing, and alignment of several pathways rather than on one visible shock.

This pattern can be represented schematically as

$$E_{\mathrm{R}} < V_{\mathcal{B}}^{\max}, \qquad P(A, t + \Delta t) > 0, \tag{5.11}$$

where:

- E_{R} is an effective measure of the system's classical transition capacity;
- $V_{\mathcal{B}}^{\max}$ is the measured barrier resistance;
- $P(A, t + \Delta t)$ is the probability of the adverse region at a later time.

The challenge is to represent how this nonzero adverse-state probability is formed.

Classical probability assigns probabilities directly. Quantum mechanics first assigns amplitudes and obtains probabilities from their squared magnitudes.

That additional mathematical layer makes interaction among pathways possible before the final probability is formed.

5.2 From Probability to Amplitude

A classical probability density over the state manifold is

$$\rho(\mathbf{x}, t) \geq 0, \tag{5.12}$$

with normalization

$$\int_{\mathcal{M}} \rho(\mathbf{x}, t), \mathrm{d}\mu(\mathbf{x}) = 1. \tag{5.13}$$

The probability that the system lies in a region $R \subset \mathcal{M}$ is

$$P(R, t) = \int_{R} \rho(\mathbf{x}, t), \mathrm{d}\mu(\mathbf{x}). \tag{5.14}$$

This is a complete classical probability description.

A quantum representation begins with a complex amplitude

$$\Psi(\mathbf{x}, t) \in \mathbb{C}. \tag{5.15}$$

The probability density is obtained through the Born rule:

$$\rho(\mathbf{x}, t) = |\Psi(\mathbf{x}, t)|^2 . \tag{5.16}$$

The amplitude may be written in polar form:

$$\Psi(\mathbf{x}, t) = R(\mathbf{x}, t) \exp\left[\mathrm{i}\theta(\mathbf{x}, t)\right], \tag{5.17}$$

where:

- $R(\mathbf{x}, t) \geq 0$ is the amplitude magnitude;
- $\theta(\mathbf{x}, t)$ is the phase.

The density is

$$\rho(\mathbf{x}, t) = R^2(\mathbf{x}, t). \tag{5.18}$$

The phase does not appear directly in the local density. It matters when amplitudes from multiple pathways combine.

This is the fundamental reason for introducing a complex object.

Why Probability Alone May Be Insufficient

Suppose two pathways lead from the current borrower state to distress.

The first is a refinancing pathway. The second is a confidence pathway.

In a simple classical additive model, the contributions might be written as

$$p_{\text{total}} = p_{\text{ref}} + p_{\text{conf}}, \tag{5.19}$$

possibly adjusted for overlap.

A more sophisticated classical model may use conditional probabilities, copulas, interaction terms, networks, or nonlinear hazards. These methods can represent a great deal of dependence and should not be dismissed.

An amplitude model instead assigns

$$\Psi_{\text{ref}} = R_{\text{ref}} \exp\left(\mathrm{i}\theta_{\text{ref}}\right), \tag{5.20}$$

$$\Psi_{\text{conf}} = R_{\text{conf}} \exp\left(\mathrm{i}\theta_{\text{conf}}\right). \tag{5.21}$$

The total amplitude is

$$\Psi_{\text{total}} = \Psi_{\text{ref}} + \Psi_{\text{conf}}. \tag{5.22}$$

The resulting probability is

$$\begin{aligned} \left|\Psi_{\text{total}}\right|^2 = {} & \left|\Psi_{\text{ref}}\right|^2 + \left|\Psi_{\text{conf}}\right|^2 \\ & + 2R_{\text{ref}}R_{\text{conf}} \cos\left(\theta_{\text{ref}} - \theta_{\text{conf}}\right). \end{aligned} \tag{5.23}$$

The last term is the interference term.

If

$$\theta_{\text{ref}} - \theta_{\text{conf}} \approx 0, \tag{5.24}$$

then

$$\cos\left(\theta_{\text{ref}} - \theta_{\text{conf}}\right) \approx 1, \tag{5.25}$$

and the pathways reinforce one another.

If

$$\theta_{\text{ref}} - \theta_{\text{conf}} \approx \pi, \tag{5.26}$$

then

$$\cos\left(\theta_{\text{ref}} - \theta_{\text{conf}}\right) \approx -1, \tag{5.27}$$

and the pathways partially suppress one another.

The phase therefore carries relational information.

What Phase Can Mean in Credit Risk

Phase must not be introduced as an unexplained or mystical variable.

In a credit application, phase may encode one or more of the following:

- timing differences among risk mechanisms;
- ordering of events;
- synchronization of stress channels;
- compatibility of institutional responses;
- persistence of one mechanism relative to another;
- delay between information and reaction;
- whether one pathway activates or suppresses another.

Consider two examples.

Constructive Interaction

A borrower must refinance within three months. At the same time, negative information begins to spread among lenders.

The refinancing pathway and confidence pathway become synchronized.

The confidence shock reduces market access precisely when refinancing demand becomes urgent. Each mechanism increases the effectiveness of the other.

The relative phase is near constructive alignment.

Destructive Interaction

A borrower experiences temporary market concern, but a committed facility becomes available before the refinancing deadline.

The liquidity-support pathway activates before the confidence pathway fully matures.

The stabilizing mechanism partially offsets the adverse pathway.

The relative phase is closer to destructive alignment.

Phase is therefore a compact representation of sequence and synchronization.

It does not replace a causal explanation. It summarizes the relational structure that the causal explanation must support.

5.3 The Quantum State and Hilbert Space

In ordinary quantum mechanics, the state is represented by a vector

$$|\psi(t)\rangle \in \mathcal{H}, \tag{5.28}$$

where $\mathcal{H}$ is a complex Hilbert space.

In the position representation,

$$\psi(x,t) = \langle x \mid \psi(t)\rangle. \tag{5.29}$$

For the risk model, we may define an effective state

$$|\Psi(t)\rangle \in \mathcal{H}_R, \tag{5.30}$$

where

$$\mathcal{H} * R = L^2\left(\mathcal{M} * \mathrm{credit}, \mathbb{C}\right). \tag{5.31}$$

This means that Ψ is a square-integrable complex-valued function over the credit manifold.

The normalization condition is

$$\langle \Psi \mid \Psi\rangle = \int_{\mathcal{M}_{\mathrm{credit}}} |\Psi(\mathbf{x},t)|^2 \,, \mathrm{d}\mu(\mathbf{x}) = 1. \tag{5.32}$$

This construction should be interpreted as a modelling architecture.

It does not imply that the borrower possesses a physical wave function. It means that the model represents state weight through a complex amplitude whose squared magnitude produces observable probabilities.

The Schrödinger Equation

In quantum mechanics, state evolution is governed by the Schrödinger equation:

$$\mathrm{i}\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = \widehat{H}|\psi(t)\rangle. \tag{5.33}$$

In one spatial dimension,

$$\mathrm{i}\hbar\frac{\partial\psi(x,t)}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\psi(x,t)}{\partial x^2} + V(x)\psi(x,t). \tag{5.34}$$

The Hamiltonian is

$$\widehat{H} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x). \tag{5.35}$$

It contains two terms.

The first is the kinetic operator:

$$\widehat{T} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}. \tag{5.36}$$

The second is the potential operator:

$$\widehat{V} = V(x). \tag{5.37}$$

The kinetic term governs propagation and curvature of the wave function. The potential term represents the landscape through which the state evolves.

The Stationary Schrödinger Equation

For a time-independent potential, assume a separable solution

$$\psi(x,t) = \phi(x)\exp\left(-\frac{\mathrm{i}Et}{\hbar}\right). \tag{5.38}$$

Substitution into Equation (5.34) gives

$$-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2\phi(x)}{\mathrm{d}x^2} + V(x)\phi(x) = E\phi(x). \tag{5.39}$$

Rearranging,

$$\frac{\mathrm{d}^2\phi(x)}{\mathrm{d}x^2} + \frac{2m}{\hbar^2}\left[E - V(x)\right]\phi(x) = 0. \tag{5.40}$$

The mathematical character of the solution depends on the sign of

$$E - V(x). \tag{5.41}$$

When

$$E > V(x), \tag{5.42}$$

the solutions are oscillatory.

When

$$E < V(x), \tag{5.43}$$

the solutions are exponential.

This second case creates the mathematical foundation of tunneling.

5.4 The Rectangular Barrier

Consider the potential

$$V(x) = \begin{cases} 0, & x < 0, \\ V_0, & 0 \le x \le a, \\ 0, & x > a. \end{cases} \tag{5.44}$$

Assume

$$0 < E < V_0. \tag{5.45}$$

Classically, a particle with energy (E) cannot cross a barrier of height $V_0 > E$.

Quantum mechanically, the wave function penetrates the barrier.

Region I: Before the Barrier

For (x<0),

$$V(x) = 0. \tag{5.46}$$

Define

$$k = \frac{\sqrt{2mE}}{\hbar}. \tag{5.47}$$

The solution is

$$\phi_I(x) = A\exp(\mathrm{i}kx) + B\exp(-\mathrm{i}kx). \tag{5.48}$$

The first term represents the incident wave. The second represents the reflected wave.

Region II: Inside the Barrier

For $0 \le x \le a$,

$$V(x) = V_0. \tag{5.49}$$

Define

$$\kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar}. \tag{5.50}$$

The solution is

$$\phi_{II}(x) = C\exp(\kappa x) + D\exp(-\kappa x). \tag{5.51}$$

The solution is not oscillatory. It is exponential.

Inside the classically forbidden barrier, the amplitude decays rather than disappearing immediately.

This is the critical mathematical fact.

Region III: Beyond the Barrier

For (x>a),

$$V(x) = 0. \tag{5.52}$$

Assuming no wave arrives from the right, the solution is

$$\phi_{III}(x) = F\exp(\mathrm{i}kx). \tag{5.53}$$

The coefficient (F) is the transmitted amplitude.

It is generally nonzero.

Boundary Conditions

The wave function and its first derivative must be continuous at the interfaces.

At (x=0),

$$\phi_I(0) = \phi_{II}(0), \tag{5.54}$$
$$\phi'_I(0) = \phi'_{II}(0). \tag{5.55}$$

At (x=a),

$$\phi_{II}(a) = \phi_{III}(a), \tag{5.56}$$
$$\phi'_{II}(a) = \phi'_{III}(a). \tag{5.57}$$

These conditions connect the incident, reflected, under-barrier, and transmitted amplitudes.

The transmitted amplitude is not inserted by assumption. It emerges from solving the differential equation across the complete barrier.

Probability Current and Transmission

The probability current in one dimension is

$$j = \frac{\hbar}{2m\mathrm{i}}\left[\psi^* \frac{\partial \psi}{\partial x} - \psi \frac{\partial \psi^*}{\partial x}\right]. \tag{5.58}$$

The transmission coefficient is

$$T = \frac{j_{\text{transmitted}}}{j_{\text{incident}}}. \tag{5.59}$$

For the rectangular barrier with $E < V_0$,

$$T = \left[1 + \frac{V_0^2 \sinh^2(\kappa a)}{4E(V_0 - E)}\right]^{-1}. \tag{5.60}$$

For a sufficiently high or wide barrier,

$$\kappa a \gg 1, \tag{5.61}$$

and the leading behaviour is approximately

$$T \propto \exp\left(-2\kappa a\right). \tag{5.62}$$

Substituting Equation (5.50),

$$T \propto \exp\left[-\frac{2a\sqrt{2m(V_0 - E)}}{\hbar}\right]. \tag{5.63}$$

The transmission probability declines exponentially with:

- barrier width (a);
- the square root of the barrier excess $V_0 - E$;
- the effective mass (m).

It increases with the scale parameter $\hbar$.

5.5 The Central Intuition of Tunneling

The central intuition is not that the particle temporarily acquires enough energy to climb over the barrier.

Rather, the quantum state is described by a spatially extended amplitude. The amplitude penetrates the classically forbidden region, decays inside it, and remains nonzero at the far boundary. If the barrier is finite, a nonzero transmitted amplitude can emerge on the other side.

The probability of transmission depends on the full barrier profile.

A barrier can be:

- high but narrow;
- low but wide;
- irregular;
- multidimensional;
- composed of several segments.

The relevant object is not merely the maximum height. It is the integrated difficulty of the under-barrier path.

The WKB Approximation

For a smooth potential (V(x)), suppose there are turning points x_1 and x_2 satisfying

$$V(x_1) = V(x_2) = E. \tag{5.64}$$

Between them,

$$V(x) > E. \tag{5.65}$$

The semiclassical under-barrier action is

$$S_{\mathcal{B}} = \int_{x_1}^{x_2} \sqrt{2m\,[V(x) - E]}, \mathrm{d}x. \tag{5.66}$$

The tunneling probability is approximately

$$T \approx \exp\left(-\frac{2S_{\mathcal{B}}}{\hbar}\right). \tag{5.67}$$

The action combines barrier height and width into one integrated quantity.

The dominant path is the one with the smallest under-barrier action.

The Classical Large-Deviation Counterpart

The exponential minimum-action structure is not unique to quantum mechanics. Consider a classical small-noise diffusion

$$d\mathbf{X}_t^\varepsilon = \mathbf{b}(\mathbf{X}_t^\varepsilon)\, dt + \sqrt{\varepsilon}\, \boldsymbol{\sigma}(\mathbf{X}_t^\varepsilon)\, d\mathbf{W}_t, \tag{5.68}$$

with covariance tensor $\mathbf{a} = \boldsymbol{\sigma}\boldsymbol{\sigma}^\mathsf{T}$. Freidlin–Wentzell theory assigns an action to an absolutely continuous path ϕ:

$$I_{0T}[\phi] = \frac{1}{2}\int_0^T \left(\dot{\phi} - \mathbf{b}(\phi)\right)^\mathsf{T} \mathbf{a}(\phi)^{-1} \left(\dot{\phi} - \mathbf{b}(\phi)\right)\, dt. \tag{5.69}$$

In the small-noise limit, rare-path probabilities scale as

$$\mathbb{P}\left(\mathbf{X}^\varepsilon \approx \phi\right) \asymp \exp\left[-\frac{I_{0T}[\phi]}{\varepsilon}\right]. \tag{5.70}$$

The quasipotential is obtained by minimizing this action over admissible paths and transition times. Kramers escape theory likewise produces an exponentially small classical escape rate governed, to leading order, by barrier height divided by noise intensity. Numerical minimum-action methods identify the most probable classical rare-event route [Freidlin and Wentzell, 2012, Kramers, 1940, E et al., 2004, Heymann and Vanden-Eijnden, 2008].

The WKB and Freidlin–Wentzell actions are not generally the same functional: WKB contains the square-root momentum integral associated with a Hamiltonian wave problem, whereas the diffusion action penalizes departure from classical drift in the inverse covariance metric. In gradient systems and suitable reductions, however, both lead to analogous conclusions: exponential barrier sensitivity, a dominant minimum-action path, and large changes in accessibility after modest barrier compression. Those conclusions therefore belong to the classical null as well as to tunneling theory.

This distinction changes the interpretation used in the remainder of the book. Minimum action, dominant paths, and exponential accessibility are treated first as classical large-deviation quantities. A quantum-like extension is justified only by additional, normalized phase-dependent structure that improves out-of-sample explanation or prediction beyond the fitted classical action model.

5.6 Multidimensional Tunneling

The credit state space is multidimensional.

Let

$$\mathbf{x} \in \mathcal{M}_{\text{credit}}. \tag{5.71}$$

Let γ be a path through the barrier. A generalized barrier action may be written as

$$S_{\mathcal{B}}[\gamma] = \int_{\gamma} \sqrt{2\left[V_{\mathrm{R}}(\mathbf{x},t) - E_{\mathrm{R}}\right]}, \mathrm{d}s_{\mathbf{M}}, \tag{5.72}$$

where

$$\mathrm{d}s_{\mathbf{M}}^2 = \mathrm{d}\mathbf{x}^{\mathsf{T}}\mathbf{M}(\mathbf{x},t)\mathrm{d}\mathbf{x}. \tag{5.73}$$

The matrix $\mathbf{M}$ is an effective resistance or inertia tensor.

The dominant path is

$$\gamma^* = \arg, \min *\gamma S * \mathcal{B}[\gamma]. \tag{5.74}$$

The baseline classical accessibility indicator is

$$T_{\mathrm{R}}^{(0)} \approx \exp\left[-\frac{2S_{\mathcal{B}}[\gamma^*]}{\eta}\right]. \tag{5.75}$$

The parameter η is initially interpreted as a fitted classical noise or dispersion scale. This indicator has the same exponential architecture as a Freidlin–Wentzell or Kramers approximation and is not evidence for a quantum-like mechanism.

This expression is not yet a calibrated credit model. It is the structural form that the empirical model must operationalize as its classical minimum-action benchmark. Only a residual phase or interference component that is defined independently, normalized, and validated against this benchmark can support the extended representation.

The Effective Risk Schrödinger Equation

We now define an effective evolution equation over the credit manifold:

$$\mathrm{i}\eta\frac{\partial}{\partial t}\Psi(\mathbf{x},t) = \widehat{H}_{\mathrm{R}}\Psi(\mathbf{x},t). \tag{5.76}$$

A simple effective Hamiltonian is

$$\widehat{H} * \mathrm{R} = -\frac{\eta^2}{2}\nabla^{\mathsf{T}}\mathbf{M}^{-1}(\mathbf{x},t)\nabla + V * \mathrm{R}(\mathbf{x},t). \tag{5.77}$$

Each term must be translated explicitly.

The Risk Amplitude

$$\Psi(\mathbf{x},t) \tag{5.78}$$

represents the complex weight associated with the borrower occupying or transitioning through state $\mathbf{x}$ at time (t).

Its squared magnitude is the observable state density:

$$\rho(\mathbf{x}, t) = |\Psi(\mathbf{x}, t)|^2 . \tag{5.79}$$

The Effective Scale Parameter

$$\eta \tag{5.80}$$

is not Planck's constant.

It controls the relative importance of amplitude dispersion, phase sensitivity, and under-barrier transmission in the model.

A small value of η produces strongly localized, nearly classical behaviour. A larger value permits greater amplitude penetration and stronger pathway effects.

It must be estimated or calibrated.

The Effective Mass or Resistance Tensor

$$\mathbf{M}(\mathbf{x}, t) \tag{5.81}$$

does not represent physical mass.

It represents resistance to movement across the risk state space.

Large resistance means that the borrower state changes slowly in a given direction. Small resistance means that the state can move more easily.

For the two-dimensional credit example,

$$\mathbf{M} = \begin{bmatrix} m_\ell & m_{\ell q} \\ m_{\ell q} & m_q \end{bmatrix} . \tag{5.82}$$

The diagonal terms represent resistance to leverage and liquidity movement. The off-diagonal term represents coupling between those directions.

The Risk Potential

$$V_{\mathrm{R}}(\mathbf{x}, t) \tag{5.83}$$

represents the risk landscape.

Low-potential regions correspond to stable configurations. High-potential regions correspond to transition resistance, fragility, or adverse states, depending on the model design.

The protective barrier is encoded as a region in which

$$V_{\mathrm{R}}(\mathbf{x}, t) > E_{\mathrm{R}}. \tag{5.84}$$

The Differential Operator

The term

$$-\frac{\eta^2}{2}\nabla^{\mathsf{T}}\mathbf{M}^{-1}\nabla \tag{5.85}$$

describes propagation, spreading, and curvature of the amplitude over the manifold.

It is the mathematical mechanism through which nonzero amplitude can extend into neighbouring states, including states inside a barrier.

A Translation Table

The mapping can be summarized as follows:

Table 5.1: Translation from quantum-mechanical objects to credit-risk interpretations.

Quantum object	**Credit-risk interpretation**
Position x	Borrower state, such as leverage and liquidity
State space	Feasible credit manifold $\mathcal{M}_{\text{credit}}$
Wave function ψ	Effective complex risk amplitude Ψ
Probability density $\lvert\psi\rvert^2$	Probability density over borrower states
Phase θ	Timing, sequencing, and relational alignment of pathways
Mass m	Resistance or inertia against state movement
Potential V	Credit-risk landscape and protective resistance
Energy E	Effective transition capacity or stress level
Barrier height	Strength of capital, liquidity, collateral, covenants, or support
Barrier width	Number, duration, or distance of reinforcing steps required for failure
Transmission T	Probability weight reaching the adverse region
Reflection	Stabilisation, recovery, or return toward safer states
Interference	Reinforcement or suppression among transition pathways

The translation is useful only if every row can be operationalized.

5.7 Encoding Real Credit Variables

The central practical problem is now addressed:

> How do we encode real borrower variables into the amplitude framework?

The process should be performed in a disciplined sequence.

Step 1: Select the State Variables

Begin with a real-valued credit state vector:

$$\mathbf{x} * t = \Big[\ell * t \ q_t \ c_t \ r_t \ s_t\Big], \tag{5.86}$$

where, for example:

- ℓ_t: leverage;
- q_t: liquidity coverage;
- c_t: collateral coverage;
- r_t: refinancing access;
- s_t: market or lender confidence.

The state vector should remain interpretable.

Adding variables without a clear structural role makes the manifold difficult to estimate and the amplitude difficult to govern.

Step 2: Normalize the Variables

Variables have different units and scales. Define normalized coordinates

$$z_{i,t} = \frac{x_{i,t} - \mu_i}{\sigma_i}, \tag{5.87}$$

or use institutionally meaningful transformations.

For liquidity, it may be preferable to define a stress coordinate

$$z_{q,t} = -\log\left(\frac{q_t}{q_{\text{safe}}}\right), \tag{5.88}$$

so that increasing values consistently represent increasing risk.

The coordinate system should make adverse movement directionally coherent.

Step 3: Construct the Manifold

Use historical observations, peer data, simulations, and accounting constraints to identify feasible states.

The empirical state cloud is

$$\mathcal{X} = \{\mathbf{z}_{i,t}\}. \tag{5.89}$$

The manifold may be approximated through:

- a constrained parametric surface;
- principal components;
- diffusion maps;
- autoencoder embeddings;

- local neighbourhood graphs;
- scenario-generated feasible regions.

The chosen method should preserve economic meaning.

Step 4: Define Stable and Adverse Regions

Define

$$S \subset \mathcal{M}_{\text{credit}} \tag{5.90}$$

and

$$A \subset \mathcal{M}_{\text{credit}}. \tag{5.91}$$

The adverse region may correspond to:

- default;
- distressed exchange;
- covenant breach;
- liquidity exhaustion;
- rating migration below a defined grade;
- loss exceeding a committee tolerance.

The definition must be operational and observable.

Step 5: Construct the Risk Potential

A simple potential may be built from a conventional credit score:

$$V_{\mathrm{R}}(\mathbf{z}, t) = f\left(\mathrm{PD}(\mathbf{z}, t)\right), \tag{5.92}$$

where (f) is a monotonic transformation.

For example,

$$V_{\mathrm{R}}(\mathbf{z}, t) = -\log\left[1 - \mathrm{PD}(\mathbf{z}, t)\right]. \tag{5.93}$$

Alternatively, the potential may combine structural components:

$$V_{\mathrm{R}} = w_\ell V_\ell + w_q V_q + w_c V_c + w_r V_r + w_s V_s + V_{\text{interaction}}. \tag{5.94}$$

The potential should be calibrated so that known stable states lie in low-potential regions and known adverse states lie beyond the relevant barrier structure.

Step 6: Encode the Protective Barrier

Suppose protection is provided by collateral, liquidity facilities, covenant headroom, and shareholder support.

Define a protection score

$$P_{\mathcal{B}} = \delta C + \eta_L L + \eta_H H + \eta_S S, \tag{5.95}$$

where:

- (C) is collateral support;
- (L) is committed liquidity;
- (H) is covenant headroom;
- (S) is shareholder or sponsor support.

The barrier potential may be defined as

$$V_{\mathcal{B}}(\mathbf{z}, t) = \varphi \left[P_{\mathcal{B}}(\mathbf{z}, t) \right], \tag{5.96}$$

where φ maps protection into transition resistance.

The barrier should have finite width. It is not merely a boundary.

Step 7: Define the Initial Probability Density

A classical probability model provides an initial density

$$\rho_0(\mathbf{z}) = \rho(\mathbf{z}, t_0). \tag{5.97}$$

This may come from:

- a Bayesian posterior;
- a state-space filter;
- a scenario distribution;
- a transition model;
- a market-implied distribution;
- an ensemble of borrower forecasts.

The initial amplitude magnitude is

$$R_0(\mathbf{z}) = \sqrt{\rho_0(\mathbf{z})}. \tag{5.98}$$

This is the simplest bridge from classical probability to amplitude.

Step 8: Assign the Initial Phase

The initial phase is

$$\theta_0(\mathbf{z}). \tag{5.99}$$

This is the most novel and delicate step.

The phase should be tied to observable relational structure. One possible form is

$$\theta_0(\mathbf{z}) = \sum_{k=1}^{K} \omega_k \tau_k(\mathbf{z}) + \sum_{j<k} \omega_{jk} \mathcal{S}_{jk}(\mathbf{z}), \tag{5.100}$$

where:

- τ_k represents timing or maturity of pathway (k);
- $\mathcal{S} * jk$ represents synchronization between pathways (j) and (k);
- $\omega * k$ and ω_{jk} are estimated parameters.

For example,

$$\tau_{\text{ref}} = \frac{\text{time to refinancing need}}{\text{reference horizon}}, \tag{5.101}$$

while

$$\mathcal{S}_{\text{ref,conf}} = \text{measured overlap between refinancing pressure and confidence decline.} \tag{5.102}$$

Phase is therefore constructed from timing and interaction data rather than chosen arbitrarily.

Step 9: Form the Initial Complex Amplitude

The initial risk amplitude is

$$\Psi(\mathbf{z}, t_0) = \sqrt{\rho_0(\mathbf{z})} \exp\left[\mathrm{i}\theta_0(\mathbf{z})\right]. \tag{5.103}$$

This equation is the practical encoding step.

It combines:

- classical state probability through ρ_0;
- pathway timing and relational structure through θ_0.

Step 10: Estimate the Effective Resistance Tensor

The resistance tensor may be related to observed mobility.

If the local covariance rate is $\mathbf{A}$, one possible parameterization is

$$\mathbf{M}^{-1} = \lambda \mathbf{A}, \tag{5.104}$$

where $\lambda > 0$ is a scaling parameter.

This means that directions with high observed mobility have lower effective resistance.

However, the equality should not be automatic. Structural resistance may differ from historical covariance. A hybrid form is

$$\mathbf{M}^{-1} = \lambda_1 \mathbf{A} + \lambda_2 \mathbf{K}, \tag{5.105}$$

where $\mathbf{K}$ is a structural mobility tensor.

Step 11: Calibrate the Scale Parameter

The parameter η controls the degree of amplitude dispersion and barrier penetration.

It may be estimated by minimizing a loss function such as

$$\widehat{\eta} = \arg, \min *\eta \sum *t \left[P_{\text{obs}}(A,t) - P_\Psi(A,t;\eta)\right]^2, \tag{5.106}$$

where

$$P_\Psi(A,t;\eta) = \int_A |\Psi(\mathbf{z},t;\eta)|^2, \mathrm{d}\mu(\mathbf{z}). \tag{5.107}$$

The calibration may also use likelihood, cross-entropy, Brier score, or transition-frequency matching.

Step 12: Evolve the State

The amplitude is propagated through

$$\mathrm{i}\eta \frac{\partial \Psi}{\partial t} = \widehat{H}_{\mathrm{R}} \Psi. \tag{5.108}$$

At each horizon, the adverse-region probability is

$$P_{\text{default}}(t) = \int_A |\Psi(\mathbf{z},t)|^2, \mathrm{d}\mu(\mathbf{z}). \tag{5.109}$$

The contribution attributable to pathway interference can be reported separately.

A Two-Path Credit Example

Consider two pathways:

1. refinancing stress;
2. lender-confidence deterioration.

Assign magnitudes

$$R_{\text{ref}} = \sqrt{p_{\text{ref}}}, \qquad R_{\text{conf}} = \sqrt{p_{\text{conf}}}. \tag{5.110}$$

Suppose

$$p_{\text{ref}} = 0.04, \qquad p_{\text{conf}} = 0.03. \tag{5.111}$$

Then

$$R_{\text{ref}} = 0.20, \qquad R_{\text{conf}} \approx 0.173. \tag{5.112}$$

If the relative phase is

$$\Delta\theta = 0, \tag{5.113}$$

the combined probability is

$$\begin{aligned} p_{\text{combined}} &= 0.04 + 0.03 + 2(0.20)(0.173) \\ &\approx 0.139. \end{aligned} \tag{5.114}$$

If

$$\Delta\theta = \pi, \tag{5.115}$$

then

$$\begin{aligned} p_{\text{combined}} &= 0.04 + 0.03 - 2(0.20)(0.173) \\ &\approx 0.001. \end{aligned} \tag{5.116}$$

These values do not imply that the true default probability must behave this way. They illustrate how the same pathway magnitudes can produce different aggregate probabilities when relational phase differs.

The credit interpretation is that timing and synchronization can strongly amplify or suppress the combined transition.

Encoding Pathways as Basis States

An alternative representation uses discrete pathway states.

Let

$$\{|k\rangle\}_{k=1}^{K} \tag{5.117}$$

be a basis corresponding to pathways such as:

- operating deterioration;
- refinancing stress;
- collateral decline;
- confidence withdrawal;
- covenant acceleration;
- sponsor support;
- restructuring.

The borrower state may be represented as

$$|\Psi\rangle = \sum_{k=1}^{K} \alpha_k |k\rangle, \tag{5.118}$$

with

$$\alpha_k = R_k \exp\left(\mathrm{i}\theta_k\right). \tag{5.119}$$

The coefficients satisfy

$$\sum_{k=1}^{K} |\alpha_k|^2 = 1. \tag{5.120}$$

A transition operator

$$\widehat{U}(t) = \exp\left(-\frac{\mathrm{i}\widehat{H}_R t}{\eta}\right) \tag{5.121}$$

evolves the pathway state.

This discrete formulation may be easier to explain to a credit committee than a continuous partial differential equation.

5.8 Observables for the Credit Committee

In quantum mechanics, measurable quantities are represented by operators.

In the credit model, define an observable

$$\widehat{O} \tag{5.122}$$

for a committee-relevant quantity.

Examples include:

- probability of default;
- probability of covenant breach;
- probability of liquidity exhaustion;
- expected recovery;
- probability of rating migration;
- expected time in the fragile region.

The expected value is

$$\mathbb{E}_\Psi[O] = \langle \Psi \mid \widehat{O} \mid \Psi \rangle. \tag{5.123}$$

For the adverse-region projector

$$\widehat{\Pi}_A, \tag{5.124}$$

the adverse-state probability is

$$P(A, t) = \langle \Psi(t) \mid \widehat{\Pi}_A \mid \Psi(t) \rangle. \tag{5.125}$$

This gives the credit committee an ordinary probability, even though the internal model uses complex amplitudes.

How to Explain the Model to a Credit Committee

The model should not be presented by beginning with quantum terminology.

A practical committee explanation should proceed in the following order.

First: State the Conventional Risk View

> The borrower has a current probability distribution over feasible credit states based on leverage, liquidity, collateral, refinancing access, and lender confidence.

Second: State the Barrier View

> The borrower is protected by collateral, liquidity facilities, covenant headroom, and sponsor support. These protections form a barrier separating the current state from distress.

Third: State the Pathway Problem

> Several routes to distress exist. Their effect depends not only on their individual probability, but on whether they occur in reinforcing or offsetting sequences.

Fourth: Introduce the Amplitude

> We represent each pathway with a magnitude and a timing-alignment component. The magnitude is related to pathway probability. The alignment component represents whether pathways reinforce or suppress each other.

Fifth: Report Ordinary Outputs

The committee should receive:

- conventional probability of default;
- amplitude-model probability of default;
- interference contribution;
- dominant pathway;
- barrier action;
- sensitivity to phase assumptions;
- comparison with classical alternatives.

The committee does not need to accept a metaphysical claim. It needs to evaluate whether the additional structure improves explanation, prediction, and governance.

Model Validation

The amplitude model should compete against conventional alternatives.

These include:

- logistic default models;
- survival and hazard models;

- transition matrices;
- hidden Markov models;
- regime-switching models;
- jump-diffusion models;
- copula models;
- Bayesian networks;
- structural credit models;
- machine-learning classifiers.

The amplitude framework should be accepted only if it provides measurable incremental value.

Validation criteria may include:

- calibration;
- discrimination;
- Brier score;
- log loss;
- out-of-sample likelihood;
- stability across regimes;
- scenario consistency;
- explanatory coherence;
- sensitivity to parameter perturbation.

The null hypothesis should be that classical modelling is sufficient.

The amplitude model must earn its complexity.

5.9 Guardrails Against Misinterpretation

Several safeguards are essential.

Do Not Call Every Surprise Tunneling

Unexpected failure may be caused by missing data or poor barrier measurement.

Do Not Treat Phase as an Arbitrary Fitting Parameter

Phase should be tied to observable timing, ordering, synchronization, or intervention structure.

Do Not Confuse Analogy with Physical Identity

The borrower is not claimed to be a quantum particle.

Do Not Hide Classical Mechanisms

Jump risk, contagion, nonlinear dependence, and barrier erosion should remain explicit competing explanations.

Do Not Report Only the Final Probability

The committee should see the pathway decomposition, barrier profile, interference term, and parameter uncertainty.

A Minimal End-to-End Encoding Example

The entire procedure can be summarized through a minimal borrower model.

Real Variables

$$\mathbf{x} * t = (\ell * t, q_t, r_t, s_t)\,. \tag{5.126}$$

Normalized Coordinates

$$\mathbf{z} * t = \mathcal{N}\,(\mathbf{x} * t)\,. \tag{5.127}$$

Classical Density

$$\rho_0(\mathbf{z}) = p\,(\mathbf{z} \mid \mathcal{I}_t)\,, \tag{5.128}$$

where $\mathcal{I}_t$ is the available information set.

Phase Function

$$\theta_0(\mathbf{z}) = \omega_1 \tau_{\text{ref}} + \omega_2 \tau_{\text{conf}} + \omega_3 \mathcal{S}_{\text{ref,conf}}. \tag{5.129}$$

Complex Amplitude

$$\Psi_0(\mathbf{z}) = \sqrt{\rho_0(\mathbf{z})}\,\exp\left[\mathrm{i}\theta_0(\mathbf{z})\right]. \tag{5.130}$$

Barrier Potential

$$V_{\mathcal{B}} = \varphi\,(C, L, H, S)\,. \tag{5.131}$$

Effective Hamiltonian

$$\widehat{H} * R = -\frac{\eta^2}{2}\nabla^{\mathsf{T}}\mathbf{M}^{-1}\nabla + V * R. \tag{5.132}$$

Evolution

$$\mathrm{i}\eta\frac{\partial \Psi}{\partial t} = \widehat{H}_R \Psi. \tag{5.133}$$

Committee Output

$$P_{\text{default}}(t) = \int_A |\Psi(\mathbf{z},t)|^2 \,, \mathrm{d}\mu(\mathbf{z}). \tag{5.134}$$

The full encoding chain is therefore

$$\boxed{\begin{aligned} \text{Real borrower data} &\longrightarrow \text{normalized state coordinates} \\ &\longrightarrow \text{classical probability density} \\ &\longrightarrow \text{phase from timing and pathway alignment} \\ &\longrightarrow \text{complex risk amplitude} \\ &\longrightarrow \text{evolution through the risk Hamiltonian} \\ &\longrightarrow \text{ordinary adverse-state probabilities.} \end{aligned}} \tag{5.135}$$

5.9.1 What the Reader Should Now Understand

At this point, the reader should have a concrete intuition for the role of every object.

The manifold is the space of feasible borrower configurations.

The probability density describes how current uncertainty is distributed over those configurations.

The amplitude magnitude is the square root of that density.

The phase represents timing, sequencing, and relational alignment among transition mechanisms.

The Hamiltonian specifies how the amplitude evolves.

The effective mass tensor represents resistance to movement.

The potential represents the risk landscape and protective barrier.

The tunneling action measures the integrated difficulty of a latent path through the barrier.

The transmission coefficient represents the probability weight reaching the adverse region.

The observable result remains an ordinary probability suitable for a credit decision.

These objects form a chain of modelling commitments rather than a vocabulary that can be selected independently. If the manifold is poorly specified, distance and barrier placement lose meaning. If the classical density is uncalibrated, the amplitude magnitude has no reliable empirical anchor. If phase is assigned without observable timing evidence, interference becomes an unrestricted interaction term. If the Hamiltonian, effective mass, or potential is chosen only for mathematical convenience, the resulting action and transmission may be numerically precise but institutionally empty. Each object must therefore have a stated role, an estimation method, a unit or normalization, and a validation test.

The reader should also understand the hierarchy of evidence. Classical explanations remain the benchmark: drift, diffusion, jumps, regime changes, omitted variables, and visible barrier gaps must be tested first. The quantum-like extension is justified only when the additional structure produces stable incremental value, such as earlier pathway detection, improved calibration, or a clearer intervention target. Large interference or transmission effects require especially strong sensitivity analysis because they can arise from flexible phase choices or poorly scaled action. Out-of-sample performance, ablation tests, and comparison with nonlinear classical models are indispensable.

Finally, the model's output must return to ordinary governance. Committees need to know which evidence changed, which pathway dominates, how uncertainty affects the estimate, and which action changes height, width, permeability, or timing. A probability without this decomposition recreates the compression problem that motivated the geometric framework. Understanding the mathematics therefore means understanding both its analytical possibilities and its limits: the formalism is useful only when every transformation remains traceable from observed state evidence to a decision that can be tested after implementation.

5.10 Chapter Synthesis

This chapter has introduced the mathematics and intuition of tunneling and translated each major quantum object into a credit-risk modelling construct.

The classical benchmark was established first. Gradual deterioration, diffusion, jumps, barrier erosion, and hidden structural gaps remain ordinary explanations for adverse transitions. A tunneling interpretation becomes relevant only when nonzero transition weight develops across a measured barrier without an apparently sufficient classical crossing mechanism.

The key mathematical transition is from probability density to complex amplitude:

$$\Psi = \sqrt{\rho}\,\exp\left(\mathrm{i}\theta\right). \tag{5.136}$$

The magnitude carries state probability. The phase carries relational information about timing, sequencing, and alignment among pathways.

Quantum tunneling was derived through the rectangular barrier. Inside the classically forbidden region, the amplitude decays exponentially but does not vanish. Continuity conditions produce a nonzero transmitted amplitude beyond the barrier.

For a general barrier, transmission is governed approximately by the under-barrier action:

$$T \approx \exp\left(-\frac{2S_{\mathcal{B}}}{\eta}\right). \tag{5.137}$$

The dominant pathway is the path minimizing that action.

The chapter then showed how to encode real credit variables. The process begins with leverage, liquidity, collateral, refinancing access, and confidence. These variables are normalized and placed on a feasible manifold. A classical density is estimated. Its square root becomes the amplitude magnitude. Observable timing and synchronization variables define the phase. A risk potential, effective resistance tensor, and scale parameter complete the effective Hamiltonian.

The model ultimately produces an ordinary probability of default:

$$P_{\text{default}}(t) = \int_A \left|\Psi(\mathbf{x},t)\right|^2 , \mathrm{d}\mu(\mathbf{x}). \tag{5.138}$$

The amplitude architecture is therefore not a replacement for probability. It is a deeper layer from which probability is constructed.

Its use is justified only when phase and pathway interaction provide explanatory or predictive value beyond classical alternatives.

The central proposition of Chapter 5 is:

> Real credit variables can be encoded in a quantum-like language by mapping borrower states to a manifold, classical probabilities to amplitude magnitudes, timing and pathway alignment to phase, protective structures to a potential barrier, and adverse-state probability to the squared magnitude of the evolved amplitude.

Chapter 6 will translate this formalism into a general theory of risk tunneling and define more precisely when an observed transition should, and should not, be classified as a tunneling event.

Chapter 6

From Physical Tunneling to Risk Tunneling

6.1 The Purpose of the Translation

Chapter 5 developed the mathematical structure of tunneling. It introduced complex amplitudes, phase, interference, the Schrödinger equation, barrier penetration, transmission coefficients, and minimum-action pathways. It also showed how observable credit variables could be encoded into a quantum-like language.

The present chapter asks a different question:

> When is it intellectually and operationally legitimate to describe a real adverse transition as risk tunneling?

This question is more difficult than the mathematical derivation itself. It is relatively easy to take a quantum equation, replace physical symbols with financial variables, and produce an attractive analogy. It is much harder to define when the resulting model contributes explanatory or predictive value.

Risk tunneling should not become a metaphor for every surprising failure. A borrower that defaults unexpectedly may have experienced an omitted shock. A control system that collapses may contain an unidentified dependency. A legal protection may have been incorrectly interpreted. A liquidity barrier may have deteriorated before the transition became visible.

All these possibilities must be investigated before a tunneling interpretation is considered.

The central purpose of Chapter 6 is therefore to define risk tunneling as a disciplined classification of transition structure.

A preliminary definition is:

> Risk tunneling is a modelled transition in which material probability weight reaches an adverse region through a nominally protective barrier even though the observed classical state and measured transition capacity appear insufficient to explain ordinary over-the-barrier crossing.

The definition contains several qualifications.

First, risk tunneling is a *modelled* transition. It is not presented as a directly observable physical phenomenon.

Second, the barrier must be more than a threshold. It must represent a measurable protective structure with height, width, continuity, and directional resistance.

Third, the observed classical capacity must appear insufficient relative to the barrier.

Fourth, material probability weight must nevertheless be assigned to the adverse region.

Finally, the classification should survive comparison with credible classical alternatives.

6.1.1 The Classical Null Hypothesis

A disciplined theory of risk tunneling should begin with a classical null hypothesis:

$$H_0 : \text{The adverse transition is adequately explained by classical mechanisms.} \tag{6.1}$$

The alternative hypothesis is

$$H_1 : \text{An amplitude-based sub-barrier representation adds material explanatory value.} \tag{6.2}$$

The burden of proof rests with the tunneling model.

This is essential because classical models are already capable of representing a wide variety of complex phenomena. Nonlinearity, path dependence, feedback, common factors, hidden regimes, jumps, contagion, and changing barriers can all be modelled without complex amplitudes.

An event should not be called tunneling merely because it is rare, fast, nonlinear, or difficult to predict.

The analyst should first examine the following classical explanations.

Gradual Drift

The system may have been deteriorating gradually, but the selected indicators did not capture the relevant direction of movement.

Formally,

$$\mathbf{b}(\mathbf{x} * t, t) \cdot \mathbf{n} * A > 0 \tag{6.3}$$

may have persisted for some time even if headline metrics appeared stable.

Diffusive Crossing

A sequence of moderate random shocks may have accumulated sufficiently to push the state through the barrier.

The crossing probability may be small but nonzero under the classical stochastic law

$$\mathrm{d}\mathbf{x} * t = \mathbf{b}, \mathrm{d}t + \mathbf{\Sigma}, \mathrm{d}\mathbf{W} * t. \tag{6.4}$$

A low-probability diffusion event is not tunneling.

Jump Risk

A discontinuous shock may have moved the system directly into the adverse region:

$$\Delta\mathbf{x} * t = \mathbf{J} * t. \tag{6.5}$$

Fraud, litigation, cyberattack, regulatory intervention, loss of a major customer, and withdrawal of funding can all generate jumps.

Omitted Variables

The apparent barrier may exist only in the reduced model.

Suppose the observed state is

$$\mathbf{x} * t^{\mathrm{obs}} = (\ell * t, q_t), \tag{6.6}$$

but the true relevant state is

$$\mathbf{x} * t^{\mathrm{true}} = (\ell * t, q_t, r_t, s_t, c_t), \tag{6.7}$$

where refinancing access, confidence, and collateral quality are omitted.

A transition that appears impossible in the two-dimensional projection may be entirely classical in the larger state space.

Barrier Erosion

The barrier may have become lower, narrower, or discontinuous.

If

$$V_{\mathcal{B}} = V_{\mathcal{B}}(\mathbf{x}, t), \tag{6.8}$$

then the apparent sub-barrier condition may disappear when the time dependence is modelled correctly.

Hidden Structural Gaps

A barrier may be strong in aggregate but contain a narrow passage

$$\mathcal{G} \subset \mathcal{B} \tag{6.9}$$

with low classical resistance.

A transition through such a passage is not tunneling. It is ordinary passage through an incompletely mapped barrier.

Regime Change

The governing dynamics may have changed:

$$\mathcal{D} * 0 \longrightarrow \mathcal{D} * 1. \tag{6.10}$$

Historical covariance, drift, and control effectiveness may no longer describe the new regime.

Model Misspecification

A poor functional form, incorrect calibration, or unstable estimation procedure may produce the illusion of an inexplicable transition.

Freidlin–Wentzell and Kramers Escape

The strongest classical null is not merely ordinary drift or diffusion near the centre of a distribution. Small-noise diffusion theory already describes rare transitions through a pathwise action. Consider

$$\mathrm{d}\mathbf{X}_t^{\varepsilon} = \mathbf{b}(\mathbf{X}_t^{\varepsilon})\,\mathrm{d}t + \sqrt{\varepsilon}\,\boldsymbol{\sigma}(\mathbf{X}_t^{\varepsilon})\,\mathrm{d}\mathbf{W}_t. \tag{6.11}$$

Freidlin–Wentzell theory assigns exponentially small weight to rare paths through an action functional, while the associated quasipotential identifies the least-cost route between stable and adverse regions. Kramers escape theory likewise produces exponentially sensitive barrier-crossing rates. These classical results generate minimum-action paths, dominant routes, and strong sensitivity to barrier height and effective width without requiring complex amplitudes [Freidlin and Wentzell, 2012, Kramers, 1940, E et al., 2004, Heymann and Vanden-Eijnden, 2008].

Consequently, an expression of the form $\exp(-S/\eta)$, a dominant path, or a large response to modest barrier compression is not evidence of risk tunneling. It is part of the mandatory classical rare-event benchmark. The diagnostic hierarchy must therefore include a fitted large-deviation or quasipotential model. What may remain for an extended model is narrower: a prespecified, normalized phase structure that adds reproducible information beyond this strong classical action null.

6.2 A Formal Definition of Risk Tunneling

Let the system occupy a state

$$\mathbf{x}_t \in \mathcal{M}, \tag{6.12}$$

where $\mathcal{M}$ is the relevant state manifold.

Let $S \subset \mathcal{M}$ be the stable region, $A \subset \mathcal{M}$ the adverse region, and $\mathcal{B} \subset \mathcal{M}$ the protective barrier separating them.

Let

$$E_{\mathrm{R}}(\mathbf{x}_t, t) \tag{6.13}$$

denote an effective classical transition capacity. This may summarize current stress, momentum, shock intensity, or available deterioration relative to the barrier.

Let

$$V_{\mathrm{R}}(\mathbf{x}, t) \tag{6.14}$$

denote the risk potential.

A path γ is under the barrier when

$$V_{\mathrm{R}}(\mathbf{x}, t) > E_{\mathrm{R}}(\mathbf{x}_t, t) \qquad \text{for } \mathbf{x} \in \gamma \cap \mathcal{B}. \tag{6.15}$$

The WKB-shaped risk barrier proxy along γ is

$$S_{\mathrm{R}}[\gamma] = \int_{\gamma \cap \mathcal{B}} \sqrt{2\left[V_{\mathrm{R}}(\mathbf{x}, t) - E_{\mathrm{R}}\right], \mathrm{d}s_{\mathbf{M}}}, \tag{6.16}$$

where

$$\mathrm{d}s_{\mathbf{M}}^2 = \mathrm{d}\mathbf{x}^{\mathsf{T}} \mathbf{M}(\mathbf{x}, t) \mathrm{d}\mathbf{x}. \tag{6.17}$$

The dominant sub-barrier path is

$$\gamma^* = \arg, \min *\gamma S * \mathrm{R}[\gamma]. \tag{6.18}$$

The corresponding baseline accessibility indicator is

$$T_{\mathrm{R}}^{(0)} \approx \exp\left[-\frac{2S_{\mathrm{R}}[\gamma^*]}{\eta}\right]. \tag{6.19}$$

This exponential indicator must first be interpreted and tested as a classical rare-event object. Its substantive conclusions about minimum action, dominant paths, and barrier sensitivity can be reproduced by Freidlin–Wentzell and Kramers models. The notation is retained as a compact structural proxy, but it does not by itself establish a tunneling extension.

A candidate phase-extended risk-tunneling model therefore satisfies four conditions.

Condition One: A Measurable Barrier Exists

There must be a protective structure with identifiable height, width, integrity, and directionality.

Without a barrier, there can be no tunneling classification.

Condition Two: A Strong Classical Action Null Is Estimated

Along the relevant path,

$$E_{\mathrm{R}} < V_{\mathrm{R}}(\mathbf{x}, t). \tag{6.20}$$

The observed state should not possess sufficient measured capacity for ordinary crossing. More importantly, a classical small-noise action, quasipotential, or Kramers-type escape model must be estimated where the data permit. The extended model cannot claim the exponential action law, the minimum-action path, or barrier compression as distinctive results because the classical null already contains them.

Condition Three: Phase Is Anchored and State Weight Is Normalized

If phase is used, its time origin τ_0 and period or decision horizon H must be fixed before evaluation. A timing proxy may then be mapped by

$$\theta_{k,t} = 2\pi \left[\frac{\tau_{k,t} - \tau_0}{H} \bmod 1 \right]. \tag{6.21}$$

Under this mapping, a delay of $H/2$ corresponds to a phase difference of π. Without a defensible H, the cosine of a phase difference has no fixed empirical meaning and ordinary timing covariates should be used instead.

Any amplitude interpreted through the Born rule must also be normalized over the complete state space:

$$\int_{\mathcal{M}} |\Psi(\mathbf{x}, t)|^2 \, \mathrm{d}\mu(\mathbf{x}) = 1. \tag{6.22}$$

Square roots of separately estimated marginal pathway probabilities may be combined into a raw structural alignment score, but that score is not itself an event probability. A probability must come from a normalized full-state model or a separately calibrated prediction layer.

Condition Four: The Phase Extension Adds Value

The anchored and normalized phase extension must outperform or materially enrich the strong classical action benchmark.

This may be expressed as

$$\Delta\mathcal{L} = \mathcal{L} * \text{classical action} - \mathcal{L} * \text{phase extended} > \tau_{\mathcal{L}}, \tag{6.23}$$

where $\mathcal{L}$ is an appropriate validation loss.

The four conditions can be summarized as

$$\boxed{\begin{aligned}\text{Phase-Extended Candidate} \iff ;&\text{Barrier exists}\\ &\wedge \text{strong classical action null is fitted}\\ &\wedge \text{phase is anchored and state weight normalized}\\ &\wedge \text{phase extension adds out-of-sample value.}\end{aligned}} \tag{6.24}$$

6.3 Over-the-Barrier and Through-the-Barrier Transitions

A central objective is to distinguish two geometrically different mechanisms.

Over-the-Barrier Transition

An ordinary classical crossing occurs when

$$E_{\mathrm{R}} \geq V_{\mathcal{B}}^{\max}. \tag{6.25}$$

The system has sufficient transition capacity to overcome the measured barrier.

Examples include losses exceeding capital, liquidity demand exhausting reserves, or operational load exceeding capacity.

Through-the-Barrier Transition

A tunneling representation applies when

$$E_{\mathrm{R}} < V_{\mathcal{B}}^{\max}, \tag{6.26}$$

yet

$$T_{\mathrm{R}} > 0. \tag{6.27}$$

The distinction is not merely linguistic.

The over-the-barrier mechanism depends mainly on whether stress exceeds protection. The through-the-barrier mechanism depends on the integrated barrier profile, the effective resistance tensor, the amplitude scale, and the interaction among pathways.

A high barrier may still permit material tunneling if it is sufficiently narrow. A moderate barrier may strongly suppress tunneling if it is wide and coherent.

This introduces a different interpretation of resilience:

> Protection depends not only on the amount of resistance, but on how resistance is distributed along every plausible transition path.

6.3.1 A Taxonomy of Risk-Tunneling Architectures

Risk tunneling need not have one universal form. Several architectures are possible.

Single-Path Barrier Tunneling

The simplest case involves one dominant path through a protective barrier.

The transmission is approximately

$$T_{\mathrm{R}} \approx \exp\left[-\frac{2S_{\mathrm{R}}[\gamma^*]}{\eta}\right]. \tag{6.28}$$

In credit risk, this may represent a narrow refinancing channel through otherwise strong protections.

In cyber risk, it may represent one credential pathway penetrating several nominal control layers.

Multi-Path Interference Tunneling

Several pathways may contribute amplitudes:

$$\Psi_A = \sum_{k=1}^{K} \Psi_k. \tag{6.29}$$

The adverse probability is

$$P(A) = \left|\sum_{k=1}^{K} \Psi_k\right|^2. \tag{6.30}$$

Expanding,

$$P(A) = \sum_{k=1}^{K} |\Psi_k|^2 + \sum_{j\neq k} \Psi_j^* \Psi_k. \tag{6.31}$$

The cross terms represent constructive or destructive interference.

In credit risk, refinancing pressure, market confidence, collateral decline, and covenant acceleration may reinforce one another.

In operational risk, model failure, data error, staffing pressure, and governance delay may become synchronized.

Resonant Tunneling

A layered barrier may contain an intermediate region in which amplitude becomes temporarily concentrated.

Suppose two barriers surround an intermediate state region (I). Under certain phase conditions, transmission through the combined structure may be substantially larger than transmission through either barrier considered independently.

In risk terms, an intermediate restructuring, waiver, temporary liquidity facility, or legal standstill may not always reduce risk. It may create a temporary configuration that aligns subsequent pathways and facilitates later transition.

This should not be assumed, but the architecture is conceptually important.

Moving-Barrier Tunneling

The barrier itself may change while amplitude propagates.

The tunneling coefficient becomes time-dependent:

$$T_{\mathrm{R}}(t) \approx \exp\left[-\frac{2S_{\mathrm{R}}[\gamma^*, t]}{\eta}\right]. \tag{6.32}$$

A shrinking or eroding barrier reduces the action and increases transmission.

This architecture may combine ordinary barrier deterioration with amplitude effects. The two mechanisms should be decomposed rather than conflated.

Network Tunneling

In a networked system, amplitude may propagate across nodes and links.

Let the system have graph

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}), \tag{6.33}$$

with weighted adjacency matrix $\mathbf{W}$.

A discrete effective Hamiltonian may be written as

$$\widehat{H} * \mathrm{R} = -\eta^2 \mathbf{L} * \mathcal{G} + \mathbf{V}, \tag{6.34}$$

where $\mathbf{L}_{\mathcal{G}}$ is the graph Laplacian and $\mathbf{V}$ is the diagonal risk-potential matrix.

This permits modelling of amplitude propagation through institutions, suppliers, legal entities, technological systems, or control networks.

Control-Layer Tunneling

A system may contain several control layers:

$$\mathcal{B} = \bigcup_{k=1}^{K} \mathcal{B}_k. \tag{6.35}$$

If the layers share dependencies, the effective action may be much smaller than a simple additive model suggests.

Tunneling language is relevant only when the amplitude formulation captures interaction among those layers more effectively than an ordinary dependency model.

6.4 Credit-Risk Tunneling in the Minimal Example

Return to the borrower described by leverage and liquidity:

$$\mathbf{x} * t = (\ell * t, q_t) . \tag{6.36}$$

Suppose the borrower also has collateral (C), committed liquidity (L), covenant headroom (H), and sponsor support (S).

The barrier function may be

$$B(\ell, q) = q + \delta C + \eta_L L + \eta_H H + \eta_S S - \kappa \ell. \tag{6.37}$$

Assume the current borrower remains on the safe side:

$$B(\ell_t, q_t) > 0. \tag{6.38}$$

A classical analysis may conclude that available protection exceeds current stress.

However, consider three pathways:

1. refinancing pressure;
2. lender-confidence deterioration;
3. collateral revaluation.

Assign amplitudes

$$\Psi_k = R_k \exp\left(\mathrm{i}\theta_k\right), \qquad k \in \mathrm{ref}, \mathrm{conf}, \mathrm{coll}. \tag{6.39}$$

The total adverse-path amplitude is

$$\Psi_A = \Psi_{\mathrm{ref}} + \Psi_{\mathrm{conf}} + \Psi_{\mathrm{coll}}. \tag{6.40}$$

Suppose refinancing pressure and confidence deterioration are closely synchronized, while collateral revaluation follows with a short delay. Their prespecified phases may become sufficiently aligned to raise the raw structural amplification score.

That score may exceed the sum of the marginal pathway probabilities, precisely because it is not itself a probability. It must be normalized as a bounded alignment measure and passed through a separately calibrated prediction model before any claim about adverse-event probability is made.

The interpretation is not that the borrower violated the laws of classical finance. It is that the amplitude model may capture timing-dependent pathway interaction that the benchmark model does not represent adequately.

A credit-tunneling diagnosis would still require evidence that:

- no large omitted shock explains the transition;
- the protection values were measured correctly;
- the barrier did not simply collapse;
- the pathway phases are linked to observable timing data;
- the amplitude model improves out-of-sample performance.

6.5 Operational and Cyber Risk

The same framework can be translated into operational and cyber systems.

The state vector may contain

$$\mathbf{x} * t = (u * t, c_t, d_t, r_t, h_t)\,, \tag{6.41}$$

where:

- u_t is system utilization;
- c_t is control effectiveness;
- d_t is dependency concentration;
- r_t is recovery capacity;
- h_t is human or governance reliability.

The barrier may consist of authentication, segregation, redundancy, monitoring, incident response, and manual controls.

A cyber event may appear to penetrate strong controls without any single control being visibly overwhelmed.

Before calling this tunneling, analysts should examine credential theft, hidden access, software vulnerability, configuration error, and common infrastructure.

An amplitude representation becomes interesting when multiple weak pathways interact in timing-dependent ways. For example, stolen credentials, delayed detection, cloud concentration, and response latency may reinforce one another and produce material transition weight across the control barrier.

Legal, Regulatory, and Conduct Risk

Legal and regulatory barriers are especially dynamic.

A state vector may include:

$$\mathbf{x} * t = (a * t, e_t, p_t, c_t, r_t)\,, \tag{6.42}$$

where:

- a_t is contractual ambiguity;
- e_t is evidentiary strength;
- p_t is precedent alignment;
- c_t is conduct consistency;
- r_t is regulatory posture.

The protective barrier may consist of contractual language, legal opinions, documentation, precedent, jurisdiction, and compliance controls.

An adverse outcome may appear to cross a strong legal barrier because several interpretative and conduct pathways align.

Yet legal tunneling should be invoked cautiously. Legal barriers often move through changes in doctrine, enforcement, public policy, or factual interpretation. A moving-boundary model may explain the event better than amplitude transmission.

The tunneling framework contributes only when pathway interaction and sequencing add material explanatory value beyond ordinary legal uncertainty.

6.6 Institutional and Systemic Risk

At the institutional level, the state space includes multiple entities and network relationships.

Let

$$\mathbf{x} * t = (\mathbf{x} * 1, t, \mathbf{x} * 2, t, \ldots, \mathbf{x} * N, t)\,. \tag{6.43}$$

Barriers may include capital buffers, clearing arrangements, liquidity facilities, circuit breakers, governance, resolution rules, and information controls.

A systemic transition may occur even when no individual institution appears to have crossed its own barrier.

This can arise classically through network contagion, common shocks, or fire-sale feedback. These explanations remain the baseline.

A systemic tunneling model becomes relevant when the amplitude architecture captures coherent propagation across several latent pathways and institutions, particularly when timing and network structure produce transmission not well represented by additive probabilities.

6.7 Detecting Candidate Tunneling Events

A practical diagnostic process should proceed in stages.

Stage One: Define the Event

Specify the adverse event precisely.

Examples include default, liquidity exhaustion, operational shutdown, regulatory sanction, or systemic impairment.

Stage Two: Reconstruct the State Path

Estimate the sequence

$$\{\mathbf{x} * t\} * t = {t_0}^{t_1}. \tag{6.44}$$

The reconstruction should include revisions to information that became available only after the event.

Stage Three: Reconstruct the Barrier

Estimate

$$\mathcal{B} * t, \qquad V * \mathcal{B}(\mathbf{x}, t), \qquad I_{\mathcal{B}}(t). \tag{6.45}$$

This step is essential. A tunneling diagnosis based on an incorrectly reconstructed barrier is meaningless.

Stage Four: Test Classical Models

Fit credible models of drift, diffusion, jumps, regimes, hidden states, networks, and barrier deterioration.

The objective is to determine whether the event remains anomalous after classical mechanisms are incorporated.

Stage Five: Construct Pathway Amplitudes

Define pathway magnitudes and phases from observable data:

$$\Psi_k = R_k \exp\left(\mathrm{i}\theta_k\right). \tag{6.46}$$

The phases should reflect timing, sequencing, and synchronization.

Stage Six: Estimate the Barrier Action

Calculate or approximate

$$S_{\mathrm{R}}[\gamma^*]. \tag{6.47}$$

This identifies the dominant sub-barrier path.

Stage Seven: Compare Models

Evaluate calibration, discrimination, likelihood, stability, and explanatory coherence.

The result should be a model comparison, not a narrative declaration.

6.7.1 Falsifiable Implications

A useful theory must generate implications that can be contradicted by evidence.

A risk-tunneling model should predict at least some of the following.

Falsifiability begins with precommitting the unit of analysis, adverse event, horizon, barrier definition, and classical benchmark. A model cannot be contradicted if the adverse boundary or pathway is redefined after every observation. The empirical protocol should specify which quantities are estimated in the training period, which thresholds are fixed before testing, and which outcomes count as failures, recoveries, or censored observations. Predictions should include direction and, where possible, magnitude: lower action should precede higher transition frequency; wider reliable barriers should reduce transmission; and destructive timing changes should reduce aggregate risk even when standalone pathway probabilities remain stable.

The implications must also survive alternative explanations. Exponential sensitivity may be reproduced by logistic links, threshold models, or nonlinear hazards. Phase-dependent amplification may be captured by interactions, common shocks, or latent regimes. Dominant-path concentration may reflect data sparsity or model regularization rather than a real structural route. Each claimed signature should therefore be compared with strong classical alternatives using temporal holdouts, entity holdouts, placebo outcomes, ablation, and sensitivity to pathway definitions. Evidence supporting the tunneling extension should remain stable across plausible specifications and should weaken when phase, barrier, or path information is deliberately destroyed.

Intervention evidence provides a particularly demanding test. If the model identifies a barrier segment or timing relationship as causal, an intervention targeted at that mechanism should alter subsequent action, transmission, or event frequency in the predicted direction. Natural experiments, staged rollouts, controlled exercises, or carefully matched comparisons may supply evidence. Failure of the predicted response should reduce confidence in the model rather than be explained away retrospectively. A falsifiable tunneling model is therefore one that accepts clear conditions for rejection and records those conditions before the observed outcome is known.

Exponential Sensitivity to Barrier Width

Small changes in effective barrier width should produce nonlinear changes in transmission:

$$\frac{\partial \log T_{\mathrm{R}}}{\partial W_{\mathcal{B}}} < 0. \tag{6.48}$$

Sensitivity to Barrier Height

Transmission should decline as the barrier excess rises:

$$\frac{\partial T_{\mathrm{R}}}{\partial \left(V_{\mathrm{R}} - E_{\mathrm{R}}\right)} < 0. \tag{6.49}$$

Phase-Dependent Amplification

Changing pathway synchronization should alter the predicted adverse probability even when marginal pathway magnitudes remain unchanged.

Dominant-Path Concentration

The model should identify one or a small number of paths contributing most of the transmission.

Intervention Effects

An intervention that changes timing or phase without materially changing marginal probabilities may nevertheless change aggregate risk.

For example, advancing the availability of a liquidity facility may disrupt constructive alignment among adverse channels.

These implications distinguish the amplitude model from a purely rhetorical use of tunneling.

6.8 Governance and Model-Risk Controls

Risk tunneling is a complex modelling construct and therefore requires strong governance.

A committee should receive the following information:

1. the definition of the adverse event;
2. the current state and state uncertainty;
3. the measured barrier profile;
4. the classical transition benchmark;
5. the pathways included in the amplitude model;
6. the interpretation and estimation of phase;
7. the dominant barrier action;
8. the tunneling transmission estimate;
9. the raw phase cross-term and bounded alignment diagnostics;

10. sensitivity to alternative specifications;
11. out-of-sample performance;
12. rejected classical explanations.

The model should never report only one tunneling probability.

A suitable decomposition is

$$P_{\text{total}} = P_{\text{classical}} + \Delta P_{\text{pathway}} + \Delta P_{\text{interference}} + \varepsilon_{\text{model}}, \tag{6.50}$$

where the terms distinguish benchmark risk, additional pathway structure, interference effects, and residual model uncertainty.

The committee must also be told when the tunneling interpretation is weak.

A classification confidence score may be defined as

$$\mathcal{C} * \mathrm{T} = f\left(\Delta\mathcal{L}, \mathcal{S} * \text{phase}, \mathcal{R} * \text{barrier}, \mathcal{U} * \text{model}\right), \tag{6.51}$$

where:

- $\Delta\mathcal{L}$ is incremental model performance;
- $\mathcal{S} * \text{phase}$ is stability of phase estimates;
- $\mathcal{R} * \text{barrier}$ is reliability of barrier reconstruction;
- $\mathcal{U}_{\text{model}}$ is model uncertainty.

The classification should remain provisional when these quantities are weak.

6.9 What Risk Tunneling Is Not

The following distinctions should remain explicit.

Risk tunneling is not synonymous with tail risk.

It is not simply a jump.

It is not any event that crosses a threshold.

It is not a substitute for omitted-variable analysis.

It is not a label for model failure.

It is not evidence that organizations are physical quantum systems.

It is not justified merely because a complex-amplitude model can fit historical data.

Risk tunneling is a specific structural claim:

A sub-barrier amplitude representation, including pathway phase and transmission, explains material aspects of the adverse transition that remain inadequately captured by credible classical alternatives.

6.9.1 The Minimum Risk-Tunneling Object

The theory developed in this chapter can be summarized through the object

$$\mathfrak{T} * \mathrm{R} = \left(\mathcal{M}, S, A, \mathcal{B}, V * \mathrm{R}, E_{\mathrm{R}}, \mathbf{M}, \eta, \Psi, \gamma^{*}, T_{\mathrm{R}}\right), \tag{6.52}$$

where:

- $\mathcal{M}$ is the state manifold;
- (S) is the stable region;
- (A) is the adverse region;
- $\mathcal{B}$ is the protective barrier;
- V_{R} is the risk potential;
- E_{R} is effective transition capacity;
- $\mathbf{M}$ is the resistance tensor;
- η is the amplitude scale;
- Ψ is the risk amplitude;
- γ^{*} is the dominant under-barrier path;
- T_{R} is the transmission coefficient.

A complete diagnosis additionally requires the classical benchmark set

$$\mathfrak{C} = (\text{drift}, \text{diffusion}, \text{jumps}, \text{hidden states}, \text{barrier deformation}, \text{network effects}). \tag{6.53}$$

The classification depends on the comparison

$$\mathfrak{T}_{\mathrm{R}} \quad \text{versus} \quad \mathfrak{C}. \tag{6.54}$$

6.10 Chapter Synthesis

This chapter has translated the mathematics of physical tunneling into a disciplined theory of risk tunneling.

Risk tunneling is not defined by surprise alone. It describes a candidate sub-barrier transition in which material probability weight reaches an adverse region even though measured classical transition capacity appears insufficient to overcome a nominally protective barrier.

The classification begins with a classical null hypothesis. Drift, diffusion, jumps, omitted variables, barrier erosion, hidden gaps, regime change, network contagion, and model misspecification must all be examined first.

A candidate tunneling event requires four elements:

1. a measurable protective barrier;
2. a classically sub-barrier transition path;
3. material adverse-state probability;
4. demonstrable value added by the amplitude model.

The chapter distinguished over-the-barrier transitions from through-the-barrier transmission. The former occurs when measured stress exceeds protection. The latter depends on the integrated barrier action and the propagation of amplitude through a finite protective structure.

Several architectures were introduced: single-path tunneling, multi-path interference, resonant tunneling, moving-barrier tunneling, network tunneling, and control-layer tunneling.

The credit example showed how refinancing pressure, confidence deterioration, and collateral revaluation may be encoded as amplitudes whose phases represent timing and synchronization. The framework was then extended to operational, cyber, legal, regulatory, institutional, and systemic risk.

Most importantly, the theory was made falsifiable. It should predict sensitivity to barrier height and width, phase-dependent amplification, dominant pathways, and intervention effects. It must also outperform or materially enrich classical alternatives.

The central proposition of Chapter 6 is:

> Risk tunneling is a disciplined and testable classification of sub-barrier transition structure, justified only when an amplitude-based model explains the geometry, timing, and interaction of adverse pathways more effectively than credible classical models.

Chapter 7 will apply this framework directly to credit risk, where borrowers, lenders, collateral, liquidity, covenants, confidence, and refinancing channels provide a concrete environment for studying tunneling mechanisms in practice.

Part III

Applications to Risk Management

Chapter 7

Credit-Risk Tunneling Across Lending Classes

7.1 From a General Theory to Specific Credit Models

The previous chapters developed a general theory of risk tunneling. They introduced state manifolds, stochastic mobility, protective barriers, complex amplitudes, phase, interference, minimum-action pathways, and the distinction between over-the-barrier and through-the-barrier transitions.

The present chapter applies that architecture to credit risk.

The central challenge is that credit risk is not one homogeneous phenomenon. A corporate loan, a consumer loan, and a sovereign exposure may all generate default losses, but they do so through different state variables, institutional mechanisms, protective structures, and transition pathways.

A corporate borrower may fail because operating cash flow weakens, refinancing becomes unavailable, collateral loses value, and lenders withdraw confidence. A consumer borrower may default because income falls, debt-service burden rises, savings are exhausted, and payment behaviour deteriorates. A sovereign may experience distress because fiscal space contracts, foreign-exchange reserves fall, political capacity weakens, and market access disappears.

The general tunneling architecture can therefore be preserved only at a sufficiently abstract level:

$$\mathfrak{T}_{\mathrm{credit}} = (\mathcal{M}, \mathbf{x}_t, S, F, A, \mathcal{B}, V_{\mathrm{R}}, \mathbf{M}, \Psi, \gamma^*, T_{\mathrm{R}}), \tag{7.1}$$

where:

- $\mathcal{M}$ is the relevant credit-state manifold;
- $\mathbf{x}_t$ is the current borrower state;
- S, F, and A are stable, fragile, and adverse regions;
- $\mathcal{B}$ is the protective barrier;

- V_{R} is the risk potential;
- $\mathbf{M}$ is the effective resistance tensor;
- Ψ is the complex risk amplitude;
- γ^* is the dominant transition path;
- T_{R} is the corresponding transmission measure.

The meaning of each object must then be specialized to the lending class.

The principal proposition of this chapter is:

> Credit-risk tunneling is not one model applied mechanically to every borrower. It is a common modelling grammar whose vocabulary must be reconstructed for each class of credit exposure.

7.2 A Common Credit-Tunneling Architecture

Before differentiating among lending classes, it is useful to state the common architecture.

Every credit-tunneling model requires six practical constructions.

7.2.1 A Borrower State

The borrower must be represented by a vector of economically meaningful variables:

$$\mathbf{x}_t = (x_{1,t}, x_{2,t}, \ldots, x_{n,t}) . \tag{7.2}$$

The variables should describe capacity, obligations, liquidity, support, behaviour, and access to future resources.

7.2.2 A Feasible Manifold

Not every combination of variables is economically possible. The feasible states define a manifold or constrained state region

$$\mathcal{M} = \{\mathbf{x} : F(\mathbf{x}) \leq 0\} . \tag{7.3}$$

7.2.3 Risk Regions

The manifold is partitioned into

$$\mathcal{M} = S \cup F \cup A, \tag{7.4}$$

where S is stable, F is fragile, and A is adverse.

The adverse region must correspond to an operationally defined event, such as default, distressed restructuring, delinquency, debt reprofiling, or loss above tolerance.

7.2.4 A Protective Barrier

The barrier consists of the mechanisms that prevent movement into the adverse region.

It is represented by a barrier potential

$$V_{\mathcal{B}}(\mathbf{x}, t) \tag{7.5}$$

or a protection function

$$B(\mathbf{x}, t). \tag{7.6}$$

7.2.5 Transition Pathways

The model identifies distinct pathways

$$\Gamma = \{\gamma_1, \gamma_2, \dots, \gamma_K\} \tag{7.7}$$

connecting the current state to the adverse region.

Each pathway receives an amplitude

$$\Psi_k = R_k \exp\left(\mathrm{i}\theta_k\right). \tag{7.8}$$

The magnitude R_k reflects pathway weight. The phase θ_k represents timing, sequencing, synchronization, or alignment.

7.2.6 Decision Outputs

The model ultimately produces classical decision quantities:

- probability of adverse transition;
- dominant pathway;
- barrier weakness;
- interference contribution;
- sensitivity to intervention;
- expected loss or required protection.

The internal model may use amplitudes, but the committee output remains expressed in probabilities, losses, limits, and actions.

7.3 Corporate Lending

Corporate lending provides the most direct application of the framework because the borrower possesses an identifiable balance sheet, income statement, financing structure, covenant package, and set of protective arrangements.

7.3.1 The Corporate State Vector

A useful corporate state vector is

$$\mathbf{x}_t^{\text{corp}} = \begin{bmatrix} \ell_t \\ q_t \\ m_t \\ c_t \\ r_t \\ h_t \\ s_t \end{bmatrix}, \tag{7.9}$$

where:

- ℓ_t is leverage;
- q_t is liquidity coverage;
- m_t is operating margin or cash-flow generation;
- c_t is collateral coverage;
- r_t is refinancing access;
- h_t is covenant headroom;
- s_t is sponsor, shareholder, or lender support.

The state manifold is

$$\mathcal{M}_{\text{corp}} = \left\{ \mathbf{x}^{\text{corp}} : F_{\text{corp}} \left(\mathbf{x}^{\text{corp}} \right) \leq 0 \right\}, \tag{7.10}$$

where the constraint function captures accounting identities, debt contracts, operating feasibility, and financing conditions.

7.3.2 Corporate Risk Regions

The stable region may be characterized by moderate leverage, sufficient liquidity, positive cash generation, covenant headroom, and reliable market access.

The fragile region includes borrowers that remain current but depend increasingly on refinancing, collateral support, or lender tolerance.

The adverse region may be defined as

$$A_{\text{corp}} = A_{\text{default}} \cup A_{\text{distressed exchange}} \cup A_{\text{liquidity exhaustion}} \cup A_{\text{covenant acceleration}}. \tag{7.11}$$

This is broader than legal bankruptcy. A credit committee may care about distressed restructuring or substantial impairment before formal default.

7.3.3 The Corporate Barrier

The corporate protection function may be written as

$$\begin{aligned} B_{\text{corp}} = {} & w_q q + w_c c + w_h h + w_r r + w_s s \\ & - w_\ell \ell - w_v v, \end{aligned} \tag{7.12}$$

where v represents cash-flow volatility or business instability.

The barrier contains several distinct layers:

- internally generated cash flow;
- cash and liquid investments;
- committed credit lines;
- collateral;
- covenant flexibility;
- sponsor support;
- access to debt or equity markets.

These layers are not equivalent.

A committed facility may appear to provide liquidity but contain material adverse change clauses. Collateral may have substantial book value but weak liquidation value. Sponsor support may be politically or reputationally credible but not legally binding.

The barrier should therefore include an enforceability or reliability adjustment:

$$\widetilde{P}_j = \omega_j^{\text{rel}} P_j, \qquad 0 \le \omega_j^{\text{rel}} \le 1. \tag{7.13}$$

Nominal protection and effective protection should not be confused.

7.3.4 Corporate Transition Pathways

A practical corporate model may define the following pathways:

1. operating deterioration;
2. refinancing failure;

3. liquidity exhaustion;
4. collateral impairment;
5. covenant acceleration;
6. confidence withdrawal;
7. sponsor disengagement;
8. legal or regulatory shock.

The refinancing amplitude may be written as

$$\Psi_{\text{ref}} = R_{\text{ref}} \exp\left(\mathrm{i}\theta_{\text{ref}}\right), \tag{7.14}$$

while the confidence amplitude is

$$\Psi_{\text{conf}} = R_{\text{conf}} \exp\left(\mathrm{i}\theta_{\text{conf}}\right). \tag{7.15}$$

Their relative phase may depend on the difference between the refinancing horizon and the speed of lender-confidence deterioration:

$$\Delta\theta_{\text{ref,conf}} = \omega_\tau \left(\tau_{\text{ref}} - \tau_{\text{conf}}\right). \tag{7.16}$$

When confidence declines at the same moment a large maturity approaches, the pathways are constructively aligned.

When committed support becomes available before the maturity, the adverse sequence may be disrupted.

7.3.5 The Corporate Dominant Path

The dominant corporate pathway is

$$\gamma^*_{\text{corp}} = \arg\min_{\gamma} S_{\text{corp}}[\gamma]. \tag{7.17}$$

It may not correspond to the largest accounting weakness.

A borrower with moderate leverage but concentrated maturities may have a lower-action path through liquidity and confidence than through gradual insolvency.

This is a central practical implication:

> The visible weakness and the most accessible failure path need not be the same.

7.3.6 Corporate Credit-Committee Use

The corporate credit committee should receive:

- current position on the corporate manifold;
- probability of movement into the fragile and adverse regions;
- effective barrier composition;
- reliability-adjusted protection;
- dominant transition pathways;
- tunneling and interference contributions;
- effects of proposed covenants, collateral, or liquidity conditions.

Possible actions include:

- shortening or extending maturity;
- requiring additional liquidity;
- modifying covenant design;
- increasing collateral;
- limiting distributions;
- requiring sponsor commitment;
- adjusting price or exposure.

The purpose is not simply to predict default. It is to redesign the barrier.

7.4 Consumer Lending

Consumer lending requires a different model.

The borrower is an individual or household rather than an operating firm. Data are often more granular and plentiful, but the state is less fully observed. Income, employment stability, payment behaviour, savings, household obligations, and access to informal support may matter.

7.4.1 The Consumer State Vector

A minimal consumer state vector is

$$\mathbf{x}_t^{\text{cons}} = \begin{bmatrix} y_t \\ d_t \\ a_t \\ e_t \\ p_t \\ u_t \end{bmatrix}, \tag{7.18}$$

where:

- y_t is disposable income;
- d_t is debt-service burden;
- a_t is liquid savings or available assets;
- e_t is employment stability;
- p_t is payment behaviour;
- u_t is unused credit or emergency borrowing capacity.

The consumer manifold is constrained by household budget identities:

$$y_t = c_t + s_t + \mathrm{DS}_t, \tag{7.19}$$

where c_t is consumption, s_t is saving, and DS_t is debt service.

A borrower cannot sustain arbitrary combinations of consumption, saving, and debt service. These constraints shape

$$\mathcal{M}_{\text{cons}}. \tag{7.20}$$

7.4.2 Consumer Risk Regions

The stable region includes sufficient income, low debt-service burden, positive savings, and current payment behaviour.

The fragile region includes households that remain current only by reducing consumption, drawing savings, or using revolving credit.

The adverse region may include

$$A_{\text{cons}} = A_{\text{delinquency}} \cup A_{\text{chargeoff}} \cup A_{\text{restructuring}}. \tag{7.21}$$

The transition into delinquency may occur before household insolvency in any formal sense.

7.4.3 The Consumer Barrier

The protective barrier may include:

- disposable income surplus;
- liquid savings;
- employment continuity;
- insurance;
- family support;

- unused credit;
- payment flexibility or restructuring options.

A simple protection function is

$$\begin{aligned} B_{\text{cons}} = {} & \alpha_y (y - \text{DS}) + \alpha_a a + \alpha_e e + \alpha_u u \\ & - \alpha_v v_y - \alpha_d d, \end{aligned} \tag{7.22}$$

where v_y is income volatility.

The barrier is highly time-sensitive.

A household may appear protected by unused credit, but using that credit may increase future debt service. One barrier component can therefore weaken another.

This creates endogenous barrier erosion.

7.4.4 Consumer Transition Pathways

Relevant pathways include:

1. job loss or income reduction;
2. medical or household expense shock;
3. interest-rate reset;
4. savings exhaustion;
5. revolving-credit saturation;
6. behavioural payment deterioration;
7. household separation or dependency shock.

The income-loss amplitude may be

$$\Psi_{\text{inc}} = R_{\text{inc}} \exp (\mathrm{i}\theta_{\text{inc}}), \tag{7.23}$$

while the savings-exhaustion amplitude is

$$\Psi_{\text{sav}} = R_{\text{sav}} \exp (\mathrm{i}\theta_{\text{sav}}). \tag{7.24}$$

The relative phase may reflect whether savings are exhausted before income recovers.

A temporary income shock may not cause delinquency when savings persist long enough. The same shock may become highly damaging when savings depletion and payment dates are synchronized.

7.4.5 Consumer Tunneling and Population Heterogeneity

Consumer portfolios require attention to heterogeneity.

The same observed credit score may correspond to borrowers with different hidden states.

Let z denote an unobserved household type:

$$z \in \{\text{stable income}, \text{volatile income}, \text{high support}, \text{low support}\}. \tag{7.25}$$

A hierarchical model may define

$$\Psi_{\text{cons}} = \sum_z \alpha_z \Psi(\mathbf{x} \mid z). \tag{7.26}$$

This allows the model to distinguish population segments whose pathways have different timing and resistance.

However, fairness and explainability constraints are especially important. Variables should not encode prohibited attributes directly or indirectly. The model must be tested for discriminatory outcomes and unstable proxy effects.

7.4.6 Consumer Lending Decisions

The consumer-lending output may influence:

- approval;
- line assignment;
- pricing;
- term;
- payment schedule;
- early-warning intervention;
- restructuring offers.

A barrier-oriented intervention may be more effective than simply reducing exposure.

For example, payment-date flexibility may disrupt alignment between income timing and debt-service obligations. Small emergency liquidity may widen the barrier and prevent savings exhaustion from becoming synchronized with delinquency.

The model therefore supports intervention design, not only classification.

7.5 Sovereign Lending

Sovereign lending is the most complex of the three classes because the borrower can tax, issue currency under some conditions, regulate domestic institutions, impose capital controls, and restructure debt through political processes.

The state space is partly economic, partly financial, and partly institutional.

7.5.1 The Sovereign State Vector

A minimal sovereign state vector may be

$$\mathbf{x}_t^{\mathrm{sov}} = \begin{bmatrix} f_t \\ g_t \\ r_t \\ b_t \\ x_t \\ p_t \\ m_t \end{bmatrix}, \tag{7.27}$$

where:

- f_t is fiscal balance or fiscal space;
- g_t is economic growth;
- r_t is foreign-exchange reserve adequacy;
- b_t is public debt burden;
- x_t is external financing exposure;
- p_t is political and institutional capacity;
- m_t is market access.

The sovereign manifold is constrained by debt dynamics:

$$\Delta b_t \approx (i_t - g_t)\, b_{t-1} - \mathrm{PB}_t + \mathrm{SF}_t, \tag{7.28}$$

where i_t is the effective interest rate, PB_t is the primary balance, and SF_t is a stock-flow adjustment.

Reserve dynamics, exchange rates, banking-system exposure, and political constraints add further dimensions.

7.5.2 Sovereign Risk Regions

The stable region includes sustainable debt dynamics, adequate reserves, credible institutions, manageable refinancing needs, and market access.

The fragile region includes rising rollover needs, declining reserves, political constraints on adjustment, and increasing reliance on external or official support.

The adverse region may include:

$$A_{\text{sov}} = A_{\text{restructuring}} \cup A_{\text{arrears}} \cup A_{\text{currency crisis}} \cup A_{\text{official rescue}}. \tag{7.29}$$

A sovereign may avoid legal default but still experience severe economic distress.

7.5.3 The Sovereign Barrier

Protective structures include:

- fiscal revenue capacity;
- foreign-exchange reserves;
- domestic investor base;
- central-bank capacity;
- multilateral support;
- market access;
- political capacity to adjust;
- debt maturity structure;
- currency composition.

A sovereign protection function may be written as

$$\begin{aligned} B_{\text{sov}} = \beta_r r + \beta_f f + \beta_m m + \beta_p p + \beta_o o \\ - \beta_b b - \beta_x x - \beta_v v, \end{aligned} \tag{7.30}$$

where o represents official-sector support and v represents macroeconomic volatility.

Several protections are conditional.

Official support may require policy adjustment. Reserve use can weaken future protection. Monetary financing may reduce nominal default risk while increasing currency or inflation risk.

The barrier therefore changes form as the sovereign chooses among policy responses.

7.5.4 Sovereign Transition Pathways

Important pathways include:

1. fiscal deterioration;
2. rollover failure;
3. reserve depletion;

4. exchange-rate collapse;
5. banking-sovereign feedback;
6. political inability to adjust;
7. official-sector support failure;
8. investor-confidence withdrawal.

The rollover amplitude may be

$$\Psi_{\mathrm{roll}} = R_{\mathrm{roll}} \exp\left(\mathrm{i}\theta_{\mathrm{roll}}\right), \tag{7.31}$$

and the political-capacity amplitude may be

$$\Psi_{\mathrm{pol}} = R_{\mathrm{pol}} \exp\left(\mathrm{i}\theta_{\mathrm{pol}}\right). \tag{7.32}$$

Constructive alignment may occur when large maturities coincide with political transition, reserve decline, or external shocks.

Destructive alignment may arise when credible official support arrives before market access is lost.

7.5.5 Multiple Adverse Regions

Sovereign risk differs because the state may transition into different forms of distress.

Define projectors

$$\widehat{\Pi}_D, \qquad \widehat{\Pi}_X, \qquad \widehat{\Pi}_I, \tag{7.33}$$

for default, currency crisis, and inflationary adjustment.

The probabilities are

$$P_D(t) = \langle \Psi(t) | \widehat{\Pi}_D | \Psi(t) \rangle, \tag{7.34}$$
$$P_X(t) = \langle \Psi(t) | \widehat{\Pi}_X | \Psi(t) \rangle, \tag{7.35}$$
$$P_I(t) = \langle \Psi(t) | \widehat{\Pi}_I | \Psi(t) \rangle. \tag{7.36}$$

An intervention may reduce one adverse probability while increasing another.

This is an important reason not to reduce sovereign analysis to one probability of default.

7.5.6 Sovereign Credit-Committee Use

The sovereign committee should evaluate:

- fiscal and external state;
- reserve and market-access barriers;
- political capacity;
- dominant distress pathways;
- interaction among default, currency, and banking risks;
- role of official support;
- scenario-dependent barrier deformation.

Decisions may include:

- country limits;
- tenor;
- currency composition;
- collateral or guarantee structure;
- political-risk insurance;
- concentration limits;
- pricing and capital allocation.

7.5.7 Comparing the Three Lending Classes

The three models share one architecture but differ materially in content.

Table 7.1: Comparison of the principal tunneling-model components across corporate, consumer, and sovereign lending.

Element	**Corporate**	**Consumer**	**Sovereign**
Core capacity	Cash flow and financing	Disposable income and savings	Fiscal and external capacity
Barrier	Liquidity, collateral, covenants, and support	Income surplus, savings, and credit flexibility	Reserves, fiscal space, market access, and official support
Typical pathway	Refinancing pressure and confidence withdrawal	Income loss and savings exhaustion	Rollover pressure, reserve depletion, and political constraints
Relevant phase	Maturity and lender-reaction timing	Income, payment, and savings timing	Debt-maturity, political, and official-support timing
Adverse event	Default or distressed exchange	Delinquency or charge-off	Restructuring, arrears, or macroeconomic adjustment
Decision	Exposure, covenants, collateral, and tenor	Approval, limit, term, and intervention	Country limit, tenor, currency, and concentration

The common mathematical form should not obscure these differences.

7.6 Class-Specific Calibration

The amplitude model must be calibrated differently in each class.

7.6.1 Corporate Calibration

Corporate calibration may use:

- financial statements;
- market spreads;
- ratings;
- covenant events;
- refinancing outcomes;
- collateral values;
- management and lender actions.

Path phases may be inferred from maturity schedules, spread changes, lender withdrawals, and timing of support.

7.6.2 Consumer Calibration

Consumer calibration may use:

- payment history;
- income deposits;
- utilization;
- delinquency transitions;
- savings balances;
- employment indicators;
- account-level behavioural data.

The model must include fairness, privacy, and explainability constraints.

7.6.3 Sovereign Calibration

Sovereign calibration may use:

- fiscal data;

- reserves;
- debt maturity profiles;
- market spreads;
- political events;
- capital flows;
- official-sector programmes;
- historical restructurings.

Data are less plentiful and regimes differ substantially. Expert scenarios and structural calibration therefore play a larger role.

7.7 A Unified Class-Conditional Model

The three models may be represented through a class indicator

$$c \in \{\mathrm{corp}, \mathrm{cons}, \mathrm{sov}\}. \tag{7.37}$$

The amplitude becomes

$$\Psi(\mathbf{x}, t \mid c), \tag{7.38}$$

and the Hamiltonian is

$$\widehat{H}_R^{(c)} = -\frac{\eta_c^2}{2}\nabla^{\mathsf{T}}\left[\mathbf{M}^{(c)}\right]^{-1}\nabla + V_R^{(c)}. \tag{7.39}$$

Each class has its own:

- manifold;
- state variables;
- resistance tensor;
- risk potential;
- barrier architecture;
- phase model;
- adverse-state definition;
- calibration scale.

The general theory is unified, but the empirical models remain class-conditional.

7.8 A Practical Credit-Tunneling Workflow

A lender can implement the framework through the following sequence.

7.8.1 Step One: Identify the Lending Class

Determine whether the exposure belongs to corporate, consumer, sovereign, or another sufficiently homogeneous class.

7.8.2 Step Two: Define the Adverse Event

The event may be default, delinquency, restructuring, liquidity exhaustion, or a committee-specific impairment state.

7.8.3 Step Three: Select Class-Specific Variables

The variables must represent the actual capacity, obligations, support, and behaviour relevant to the lending class.

7.8.4 Step Four: Build the Feasible Manifold

Use historical observations, structural constraints, simulations, and expert scenarios.

7.8.5 Step Five: Construct the Protective Barrier

Map real protections into barrier height, width, continuity, reliability, and directionality.

7.8.6 Step Six: Define Transition Pathways

Identify the economically distinct routes into the adverse region.

7.8.7 Step Seven: Encode Magnitudes and Phases

Set

$$R_k = \sqrt{p_k}, \tag{7.40}$$

as an initial magnitude mapping, and construct phase from timing and synchronization variables.

The phase construction must fix an origin τ_0 and a credit decision horizon H before evaluation:

$$\theta_k = 2\pi \left[\frac{\tau_k^{\text{eff}} - \tau_0}{H} \bmod 1 \right]. \tag{7.41}$$

Thus a timing difference of $H/2$ corresponds to π. If no defensible horizon exists, timing variables should remain ordinary covariates and no cosine-based interference claim should be made.

7.8.8 Step Eight: Estimate the Dominant Path

Calculate

$$\gamma^* = \arg\min_{\gamma} S_R[\gamma]. \tag{7.42}$$

7.8.9 Step Nine: Compare with Classical Models

The amplitude model should be compared with class-appropriate benchmarks.

For corporate lending, these include structural and hazard models. For consumer lending, scorecards and survival models. For sovereign lending, debt-sustainability and market-implied models.

7.8.10 Step Ten: Translate into a Credit Decision

The final output must connect to exposure, pricing, maturity, collateral, covenants, limits, monitoring, or intervention.

7.9 Governance Across Credit Classes

The governance framework must preserve comparability without forcing false uniformity.

A common reporting template may include:

1. current state;
2. classical probability of adverse transition;
3. calibrated probability from the phase-extended model;
4. bounded structural phase-alignment score;
5. barrier profile;
6. dominant path;
7. raw phase cross-term diagnostic;
8. sensitivity;
9. model uncertainty;
10. proposed mitigation.

The interpretation of each field will differ across classes.

A corporate committee may focus on refinancing and covenant design. A consumer committee may focus on affordability and early intervention. A sovereign committee may focus on rollover, reserves, political capacity, and concentration.

The model should not be used to create a false precision across fundamentally different forms of credit.

7.10 Chapter Synthesis

This chapter has adapted the general theory of risk tunneling to three specific classes of credit exposure: corporate, consumer, and sovereign lending.

The common architecture remains stable. Each model contains a state manifold, stable and adverse regions, a protective barrier, transition pathways, complex amplitudes, a dominant action path, and observable credit outputs.

However, the content of those objects differs substantially.

Corporate credit is organized around operating cash flow, leverage, liquidity, refinancing access, collateral, covenants, and sponsor support. Its dominant tunneling pathways often involve the interaction of refinancing pressure and lender confidence.

Consumer credit is organized around disposable income, debt-service burden, savings, employment stability, unused credit, and payment behaviour. Its most important pathways often involve the sequencing of income shocks, savings exhaustion, and payment obligations.

Sovereign credit is organized around fiscal space, reserves, debt burden, market access, political capacity, external exposure, and official support. It contains several possible adverse regions, including default, restructuring, currency crisis, and inflationary adjustment.

The chapter showed that the protective barrier must also be class-specific. Corporate barriers depend on enforceable funding and collateral. Consumer barriers depend on income surplus, savings, and payment flexibility. Sovereign barriers depend on reserves, fiscal capacity, maturity structure, market access, and official support.

The amplitude phase receives different interpretations in each class. In corporate lending, it may reflect the timing of maturities and confidence withdrawal. In consumer lending, it may reflect the alignment of income, expenses, savings depletion, and payment dates. In sovereign lending, it may reflect the synchronization of rollover needs, reserve decline, political events, and official support.

The central practical lesson is that the general tunneling theory should be implemented through class-conditional models:

$$\boxed{\begin{gathered}\text{Common Mathematical Grammar} + \text{Class-Specific Credit Meaning}\\ = \text{Operational Credit-Tunneling Model.}\end{gathered}} \tag{7.43}$$

The central proposition of Chapter 7 is:

> Credit-risk tunneling becomes operational only when the manifold, barrier, pathways, phase structure, and adverse events are reconstructed for a sufficiently specific class of lending rather than imposed through one universal credit model.

Chapter 8 will extend the same class-specific approach to operational and cyber risk, where controls, dependencies, attack pathways, detection delays, and recovery structures create a different family of tunneling environments.

Chapter 8

Operational and Cyber Risk Tunneling

8.1 From Credit Systems to Control Systems

The preceding chapter applied the tunneling framework to corporate, consumer, and sovereign credit. In each case, the system consisted of a borrower, a set of financial obligations, a protective structure, and several pathways leading toward an adverse credit event.

Operational and cyber risk require a different interpretation.

The protected object is no longer primarily a borrower. It may be:

- a business process;
- a payment platform;
- a trading system;
- a data environment;
- a cloud architecture;
- a model-production pipeline;
- an identity and access-management system;
- a critical third-party service;
- an institution's capacity to continue operating.

The adverse event may be unauthorized access, fraud, data corruption, service interruption, model failure, control breakdown, privacy loss, operational paralysis, or severe recovery delay.

The general tunneling architecture remains useful:

$$\mathfrak{T} * \mathrm{op} = (\mathcal{M} * \mathrm{op}, \mathbf{x} * t, S, F, A, \mathcal{B}, V * \mathrm{R}, \mathbf{M}, \Psi, \gamma^*, T_{\mathrm{R}}), \tag{8.1}$$

where:

- $\mathcal{M}_{\mathrm{op}}$ is the operational or cyber state manifold;
- $\mathbf{x}_t$ is the current system state;
- S, F, and A are stable, fragile, and adverse regions;
- $\mathcal{B}$ is the control barrier;
- V_{R} is the operational-risk potential;
- $\mathbf{M}$ is the effective resistance tensor;
- Ψ is the complex risk amplitude;
- γ^* is the dominant failure or attack pathway;
- T_{R} is the probability weight transmitted into the adverse region.

The economic meaning, however, changes substantially.

In operational and cyber risk, the barrier is formed by controls, redundancies, permissions, monitoring, response mechanisms, and recovery capacity. The transition pathways often involve a sequence of individually modest weaknesses. The critical question is therefore not merely whether a control exists, but whether several weaknesses can align into a coherent route through the control architecture.

The central proposition of this chapter is:

> Operational and cyber resilience depend not only on the number or nominal strength of controls, but on the geometry, independence, timing, and interaction of the pathways that connect ordinary operation to material failure.

8.1.1 The Operational State Manifold

An operational system can be represented through a state vector

$$\mathbf{x} * t^{\mathrm{op}} = \begin{bmatrix} u * t \\ c_t \\ v_t \\ d_t \\ h_t \\ m_t \\ r_t \end{bmatrix}, \tag{8.2}$$

where:

- u_t is system utilization or workload;
- c_t is control effectiveness;

- v_t is technical vulnerability;
- d_t is dependency concentration;
- h_t is human reliability;
- m_t is monitoring and detection capacity;
- r_t is recovery readiness.

The variables may be expanded or specialized for a particular system.

For a payment platform, relevant dimensions might include transaction load, fraud-screening effectiveness, settlement liquidity, identity assurance, vendor availability, and recovery time.

For a cloud environment, the state might include access-control quality, configuration integrity, dependency concentration, patch status, logging coverage, data redundancy, and incident-response readiness.

For a model-risk environment, the state might include data quality, model stability, validation coverage, deployment controls, monitoring effectiveness, human override capacity, and change-management discipline.

Not every combination of variables is feasible.

For example, extremely high utilization may reduce monitoring quality and human reliability. High dependency concentration may increase recovery time. Strong access restrictions may reduce operational flexibility. These relationships define the feasible manifold

$$\mathcal{M} * \mathrm{op} = \left\{ \mathbf{x}^{\mathrm{op}} : F * \mathrm{op} \left(\mathbf{x}^{\mathrm{op}} \right) \leq 0 \right\}. \tag{8.3}$$

The manifold may be constructed from:

- operational telemetry;
- incident histories;
- control-test results;
- vulnerability data;
- service-level indicators;
- recovery exercises;
- red-team simulations;
- expert-designed scenarios.

As in the earlier credit examples, the manifold does not initially require elaborate differential geometry. It begins with a set of feasible system states and their structural relationships.

8.1.2 Stable, Fragile, and Adverse Regions

The operational manifold should be divided into regions that correspond to meaningful system conditions.

The stable region S_{op} contains states in which the system performs within tolerance, controls function as designed, monitoring is effective, and recovery capacity remains available.

The fragile region F_{op} contains states in which the system continues to operate but depends increasingly on degraded controls, manual intervention, concentrated infrastructure, deferred maintenance, or delayed response.

The adverse region A_{op} may be defined as

$$\begin{aligned} A_{\text{op}} = A_{\text{fraud}} \cup A_{\text{intrusion}} \cup A_{\text{data loss}} \cup A_{\text{service failure}} \\ \cup A_{\text{model failure}} \cup A_{\text{recovery failure}}. \end{aligned} \tag{8.4}$$

The adverse region should be operationally observable.

A cyber intrusion that is detected and isolated before material impact may belong to the fragile region rather than the adverse region. A temporary service interruption may be tolerable, while a prolonged outage may cross the materiality threshold.

The event definition should therefore include severity, duration, propagation, and recovery criteria.

For example,

$$A_{\text{outage}} = \{\mathbf{x} : \text{duration} > \tau_{\text{outage}}, \quad \text{impact} > I_{\min}\}. \tag{8.5}$$

The distinction between stable, fragile, and adverse states is important because operational systems often remain functional while already inside the barrier region.

8.2 The Control Barrier

The protective barrier consists of the mechanisms that impede movement from ordinary operation into failure.

A general operational protection function may be written as

$$\begin{aligned} B_{\text{op}} = w_a a + w_g g + w_m m + w_r r + w_b b \\ - w_v v - w_d d - w_u u - w_h (1-h), \end{aligned} \tag{8.6}$$

where:

- (a) is access-control strength;

- (g) is governance and segregation effectiveness;
- (m) is monitoring and detection;
- (r) is response and recovery capability;
- (b) is redundancy or backup capacity;
- (v) is vulnerability;
- (d) is dependency concentration;
- (u) is workload or utilization;
- (h) is human reliability.

The safe side satisfies

$$B_{\mathrm{op}}(\mathbf{x}, t) > 0, \tag{8.7}$$

while the adverse side satisfies

$$B_{\mathrm{op}}(\mathbf{x}, t) < 0. \tag{8.8}$$

The full barrier is not merely the boundary $B_{\mathrm{op}} = 0$. It is a finite protective region

$$\mathcal{B} * \mathrm{op} = \{\mathbf{x} : 0 \le B * \mathrm{op}(\mathbf{x}, t) \le \varepsilon\}. \tag{8.9}$$

A system may therefore be technically operational while already consuming its control buffer.

8.2.1 Layered Controls and the Illusion of Defence in Depth

Operational and cyber systems are commonly protected through multiple control layers.

A simplified control architecture may contain:

1. identity verification;
2. authentication;
3. authorization;
4. transaction or activity monitoring;
5. anomaly detection;
6. incident response;
7. backup and recovery.

The nominal barrier is

$$\mathcal{B} * \mathrm{op} = \bigcup * k = 1^K \mathcal{B}_k. \tag{8.10}$$

If the layers are genuinely independent, the attacker or failure mechanism must traverse several separate regions of resistance.

In practice, the layers often share dependencies.

Authentication and authorization may rely on the same identity provider. Monitoring and response may depend on the same logging pipeline. Production and backup may share the same cloud region. Several manual controls may depend on the same small group of specialists.

A control-dependency matrix may be written as

$$\mathbf{G} * \mathrm{op} = [g * ij], \tag{8.11}$$

where g_{ij} measures shared dependency between controls (i) and (j).

The effective number of independent control layers may be approximated by

$$K_{\mathrm{eff}} = \frac{[\mathrm{tr}\,(\mathbf{G} * \mathrm{op})]^2}{\mathrm{tr}\,(\mathbf{G} * \mathrm{op}^2)}. \tag{8.12}$$

When one common dependency dominates, K_{eff} may be much smaller than the nominal count of controls.

This is a central operational insight:

> Several visible controls may form only one effective barrier when they depend on the same infrastructure, information, judgment, or response mechanism.

8.3 Barrier Height, Width, and Continuity

Control strength has several dimensions.

8.3.1 Barrier Height

Barrier height represents how much pressure, sophistication, workload, or control deterioration is required to defeat the protective structure.

Strong authentication, strict privilege management, high-quality monitoring, and robust recovery increase barrier height.

8.3.2 Barrier Width

Barrier width represents how many stages must be traversed and how long the harmful sequence must persist.

An attacker who must compromise credentials, escalate privileges, evade detection, access protected data, and maintain persistence faces a wider barrier than an attacker who can exploit one direct path.

A high but narrow barrier may be defeated quickly through a single severe vulnerability. A lower but wider barrier may be more resilient because several independent conditions must align.

8.3.3 Barrier Continuity

Barrier continuity describes whether protection covers the full operational surface.

A coverage function may be written as

$$c_{\mathcal{B}}(\mathbf{x}, t) \in [0, 1]. \tag{8.13}$$

A control gap exists where

$$c_{\mathcal{B}}(\mathbf{x}, t) < c_{\min}. \tag{8.14}$$

An unmonitored service account, an unpatched interface, an undocumented manual process, or an unsupported legacy system may create such a gap.

A transition through a visible gap is classical. It should not be called tunneling.

8.3.4 Classical Operational Failure Mechanisms

Before introducing tunneling, the analysis must test conventional explanations.

Extreme Workload

The system may fail because demand exceeds capacity:

$$u_t > u_{\max}. \tag{8.15}$$

This is ordinary over-the-barrier failure.

Technical Vulnerability

An unpatched vulnerability may provide a direct path into the system.

This is a hidden or known barrier gap.

Human Error

A mistaken configuration, approval, or deployment may generate a jump in the operational state.

Control Erosion

Monitoring may degrade, staff may become overloaded, or backup systems may become obsolete.

The barrier moves toward the system.

Common-Cause Failure

Several control layers may fail together because they share one dependency.

This is a structural correlation problem that may be modelled classically.

Delayed Detection

An incident may become severe because it remains undetected long enough to propagate.

A delay model may explain the outcome without requiring amplitudes.

The tunneling model should be introduced only if interaction among multiple pathways produces structure that credible classical models do not represent adequately.

8.3.5 Operational and Cyber Transition Pathways

A practical model may define the following pathways:

1. credential compromise;
2. privilege escalation;
3. software vulnerability exploitation;
4. configuration failure;
5. insider misuse;
6. monitoring failure;
7. delayed response;
8. backup or recovery failure;
9. third-party service failure;
10. model or data-pipeline corruption.

Each pathway receives an amplitude

$$\Psi_k = R_k \exp\left(\mathrm{i}\theta_k\right). \tag{8.16}$$

The magnitude R_k reflects the pathway's probability weight.

The phase θ_k represents timing, ordering, persistence, and compatibility with other pathways.

For example, define:

$$\Psi_{\text{cred}} = R_{\text{cred}} \exp\left(\mathrm{i}\theta_{\text{cred}}\right), \tag{8.17}$$

$$\Psi_{\text{priv}} = R_{\text{priv}} \exp\left(\mathrm{i}\theta_{\text{priv}}\right), \tag{8.18}$$

$$\Psi_{\text{detect}} = R_{\text{detect}} \exp\left(\mathrm{i}\theta_{\text{detect}}\right). \tag{8.19}$$

The total adverse amplitude is

$$\Psi_A = \Psi_{\text{cred}} + \Psi_{\text{priv}} + \Psi_{\text{detect}} + \cdots. \tag{8.20}$$

The squared magnitude of this sum is a raw structural alignment score. It is not an event probability unless the amplitude is normalized over the complete state space. Operational probabilities should therefore be generated by a separately calibrated prediction model that may use the bounded phase-alignment score as an input.

8.4 Constructive Interference in a Cyber Sequence

Consider four individually moderate weaknesses:

1. a user's credentials are compromised;
2. access privileges are broader than necessary;
3. anomaly detection is temporarily degraded;
4. incident response is delayed.

Each weakness alone may be insufficient to generate material loss.

Together, however, they may form a coherent pathway.

Let the amplitudes be

$$\Psi_1, \Psi_2, \Psi_3, \Psi_4. \tag{8.21}$$

The adverse probability is

$$P(A) = \left| \sum_{k=1}^{4} \Psi_k \right|^2. \tag{8.22}$$

Expanding,

$$P(A) = \sum_{k=1}^{4} |\Psi_k|^2 + \sum_{j \neq k} \Psi_j^* \Psi_k. \tag{8.23}$$

When the phases align, the cross terms are positive.

Operationally, phase alignment means that the events occur in the correct order and sufficiently close in time:

$$\text{credential compromise} \longrightarrow \text{privilege use} \longrightarrow \text{monitoring evasion} \longrightarrow \text{delayed response.} \tag{8.24}$$

The final event may appear disproportionate to the individual weakness scores because the sequence, rather than the components alone, creates the effective transition.

8.4.1 Destructive Interference and Control Intervention

Interventions may interrupt an adverse sequence without eliminating the individual vulnerabilities.

Suppose rapid detection activates before privilege escalation is completed.

The response pathway may be represented as

$$\Psi_{\text{response}} = R_{\text{response}} \exp\left(\mathrm{i}\theta_{\text{response}}\right). \tag{8.25}$$

The total amplitude becomes

$$\Psi_{\text{total}} = \Psi_{\text{attack}} + \Psi_{\text{response}}. \tag{8.26}$$

If the response pathway is out of phase with the harmful sequence, the raw structural alignment score may fall. A reduction in calibrated adverse probability must then be demonstrated by the prediction layer rather than inferred mechanically from the amplitude sum.

In practical terms, control design can change phase by changing timing.

Examples include:

- reducing detection latency;
- revoking credentials automatically;
- isolating affected systems;
- requiring step-up authentication;
- slowing high-risk transactions;
- triggering human review;
- restoring from clean backups before corruption propagates.

This is an important advantage of the amplitude language.

It directs attention not only to control strength, but to whether the intervention arrives at the right point in the sequence.

8.5 The Operational Risk Hamiltonian

An effective evolution equation may be written as

$$\mathrm{i}\eta_{\mathrm{op}}\frac{\partial \Psi_{\mathrm{op}}}{\partial t} = \widehat{H} * \mathrm{op}\Psi * \mathrm{op}. \tag{8.27}$$

The effective Hamiltonian is

$$\widehat{H} * \mathrm{op} = -\frac{\eta * \mathrm{op}^2}{2}\nabla^{\mathsf{T}}\mathbf{M} * \mathrm{op}^{-1}\nabla + V * \mathrm{op}(\mathbf{x}, t). \tag{8.28}$$

The terms have operational interpretations.

8.5.1 Amplitude

$$\Psi_{\mathrm{op}}(\mathbf{x}, t) \tag{8.29}$$

is the complex weight associated with the system occupying or transitioning through operational state $\mathbf{x}$.

8.5.2 Risk Potential

$$V_{\mathrm{op}}(\mathbf{x}, t) \tag{8.30}$$

represents the landscape of operational fragility and control resistance.

8.5.3 Resistance Tensor

$$\mathbf{M}_{\mathrm{op}} \tag{8.31}$$

represents resistance to movement across operational dimensions.

A high value along the access-control direction indicates that access conditions are difficult to change or defeat. A low value along the monitoring direction indicates that detection quality can deteriorate rapidly.

8.5.4 Scale Parameter

$$\eta_{\mathrm{op}} \tag{8.32}$$

controls the importance of amplitude dispersion and sub-barrier transmission.

It must be calibrated rather than interpreted physically.

8.5.5 Barrier Action and the Dominant Operational Path

Let γ be a path from the current operating state to the adverse region.

The operational barrier action is

$$S_{\mathrm{op}}[\gamma] = \int_{\gamma\cap\mathcal{B}*\mathrm{op}} \sqrt{2\left[V*\mathrm{op}(\mathbf{x},t) - E_{\mathrm{op}}\right]}, \mathrm{d}s_{\mathbf{M}_{\mathrm{op}}}, \tag{8.33}$$

where E_{op} is the effective transition capacity of the harmful mechanism.

The dominant path is

$$\gamma^*_{\mathrm{op}} = \arg, \min *\gamma S * \mathrm{op}[\gamma]. \tag{8.34}$$

The transmission is

$$T_{\mathrm{op}} \approx \exp\left[-\frac{2S_{\mathrm{op}}\left[\gamma^*_{\mathrm{op}}\right]}{\eta_{\mathrm{op}}}\right]. \tag{8.35}$$

The dominant path may not involve the weakest individual control.

A moderate weakness may become the dominant path because it is narrow, well connected to other weaknesses, and aligned with delayed response.

This distinction is important:

> The weakest control and the lowest-action route to failure are not necessarily the same object.

8.5.6 A Minimal Cyber Example

Consider a cloud-based payment system with four state variables:

$$\mathbf{x} * t = \left[a * t \ p_t \ m_t \ r_t\right], \tag{8.36}$$

where:

- a_t is identity-assurance quality;
- p_t is privilege-control quality;
- m_t is monitoring effectiveness;
- r_t is response readiness.

The adverse region is unauthorized payment execution or material data compromise.

A minimal protection function is

$$B_{\mathrm{cyber}} = \alpha_a a + \alpha_p p + \alpha_m m + \alpha_r r - \alpha_v v, \tag{8.37}$$

where (v) is attacker capability or vulnerability severity.

Suppose all four controls remain individually above their minimum thresholds.

A threshold model concludes that the system is protected.

However, the sequence

$$\text{credential compromise} \to \text{privilege escalation} \to \text{monitoring delay} \to \text{slow response} \tag{8.38}$$

may produce a low-action route through the combined barrier.

The pathway amplitudes are

$$\Psi_a = R_a e^{\mathrm{i}\theta_a}, \tag{8.39}$$
$$\Psi_p = R_p e^{\mathrm{i}\theta_p}, \tag{8.40}$$
$$\Psi_m = R_m e^{\mathrm{i}\theta_m}, \tag{8.41}$$
$$\Psi_r = R_r e^{\mathrm{i}\theta_r}. \tag{8.42}$$

The raw combined structural score is

$$Q_A^{\mathrm{amp}} = \left|\Psi_a + \Psi_p + \Psi_m + \Psi_r\right|^2. \tag{8.43}$$

For reporting, this score should be divided by $(R_a + R_p + R_m + R_r)^2$ to obtain a bounded alignment measure and then passed through an out-of-sample calibrated probability model. The raw value must not be compared directly with a union probability.

First construct an effective timing coordinate from observed delay and synchronization variables:

$$\tau_k^{\mathrm{eff}} = \omega_k \tau_k + \sum_{j \neq k} \omega_{jk} \mathcal{S}_{jk}, \tag{8.44}$$

where τ_k measures delay and $\mathcal{S}_{jk}$ measures synchronization.

Fix an incident origin τ_0 and response horizon H before evaluation, and map the timing coordinate to

$$\theta_k = 2\pi \left[\frac{\tau_k^{\mathrm{eff}} - \tau_0}{H} \bmod 1 \right]. \tag{8.45}$$

A delay of $H/2$ now has the fixed meaning π. Without a defensible response horizon, the timing variables should enter a classical model directly.

The model therefore converts operational telemetry into pathway magnitude and timing alignment.

8.6 Operational Failure Beyond Cyberattack

The same architecture applies to non-malicious operational failure.

Consider a model-production process.

The state variables may include:

- data quality;
- model stability;
- validation coverage;
- deployment-control strength;
- monitoring quality;
- human-review capacity;
- rollback readiness.

The adverse event may be material model error entering production and affecting decisions.

Relevant pathways include:

1. data drift;
2. coding defect;
3. validation failure;
4. change-management bypass;
5. monitoring delay;
6. human overreliance;
7. unsuccessful rollback.

Each pathway may be individually low probability. Yet a synchronized sequence can create a coherent route to loss.

The same logic applies to:

- payment-processing failure;
- trade-booking errors;
- vendor outages;
- accounting-control breakdowns;
- fraud events;
- supply-chain interruptions;
- business-continuity failures.

8.6.1 Detection and Recovery as Dynamic Barriers

Detection and recovery deserve special treatment because they operate after the initial harmful transition begins.

Let

$$\tau_{\text{detect}} \tag{8.46}$$

be detection time, and

$$\tau_{\text{recover}} \tag{8.47}$$

be recovery time.

The effective barrier may depend on both:

$$V_{\mathcal{B}} = V_{\mathcal{B}}\left(\mathbf{x}, \tau_{\text{detect}}, \tau_{\text{recover}}, t\right). \tag{8.48}$$

Longer detection time may narrow the barrier because the harmful process has more time to propagate.

Longer recovery time may increase the size of the adverse region because losses accumulate after the initial event.

A simple action adjustment is

$$S_{\text{op}}^{\text{adj}} = S_{\text{op}} - \beta_d \tau_{\text{detect}} - \beta_r \tau_{\text{recover}}, \tag{8.49}$$

with positive parameters β_d and β_r.

This is illustrative, but the intuition is important.

Control effectiveness is partly temporal. A control that responds too late may exist nominally while contributing little effective barrier resistance.

8.7 Network and Third-Party Tunneling

Modern operational systems are networked.

Let

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}) \tag{8.50}$$

represent applications, databases, users, vendors, infrastructure components, or business processes.

The graph Laplacian is

$$\mathbf{L} * \mathcal{G} = \mathbf{D} * \mathcal{G} - \mathbf{W}_{\mathcal{G}}, \tag{8.51}$$

where $\mathbf{W} * \mathcal{G}$ is the weighted adjacency matrix and $\mathbf{D} * \mathcal{G}$ is the degree matrix.

A network Hamiltonian may be written as

$$\widehat{H} * \mathrm{net} = -\eta * \mathrm{net}^2 \mathbf{L} * \mathcal{G} + \mathbf{V} * \mathrm{net}. \tag{8.52}$$

Amplitude can then propagate through connected nodes.

A third-party provider may create a low-action path across several institutions even when each institution has strong internal controls.

The external dependency changes the geometry of the combined barrier.

This is particularly relevant for:

- shared cloud providers;
- payment processors;
- identity platforms;
- market-data vendors;
- software libraries;
- outsourced operations;
- telecommunications infrastructure.

8.7.1 Distinguishing Tunneling from an Attack Graph

Attack graphs describe feasible sequences through which an attacker can move from one system state to another.

They are classical and highly useful.

A tunneling model should not replace an attack graph when the primary problem is identifying a direct technical path.

The amplitude framework adds a different layer.

It asks:

- how much weight is associated with each path;
- how timing changes the raw alignment score and the separately calibrated probability;
- whether pathways reinforce or suppress one another;
- how finite control barriers permit nonzero transmission;

- how interventions change phase and action.

The attack graph describes topology.

The amplitude model describes weighted, timed, and interacting propagation over that topology.

A hybrid formulation may assign an amplitude to each graph path:

$$\Psi_A = \sum_{\gamma \in \Gamma_A} R_\gamma \exp\left(\mathrm{i}\theta_\gamma\right), \tag{8.53}$$

where Γ_A is the set of paths to the adverse node.

The adverse probability is then

$$P(A) = \left|\Psi_A\right|^2. \tag{8.54}$$

8.7.2 Comparison with Classical Operational Models

The tunneling model should be compared with established approaches.

These include:

- fault trees;
- event trees;
- reliability models;
- attack graphs;
- Bayesian networks;
- Markov chains;
- hidden-state models;
- point-process models;
- jump processes;
- scenario analysis;
- Monte Carlo simulation.

A complex-amplitude model is justified only when it contributes value beyond these alternatives.

Possible evidence includes:

- improved incident-probability calibration;
- better prediction of severe multi-stage events;
- more stable representation of timing interactions;

- superior explanation of intervention effects;
- clearer identification of dominant cross-control pathways.

The null hypothesis remains that classical models are sufficient.

8.8 Estimating Operational Amplitudes

Pathway magnitudes may be derived from:

- incident frequencies;
- vulnerability-exploitation rates;
- red-team outcomes;
- control-test failures;
- false-negative rates;
- vendor-outage histories;
- expert scenarios;
- simulation results.

A simple initial magnitude is

$$R_k = \sqrt{p_k}. \tag{8.55}$$

Phase may be estimated from:

- event ordering;
- time between pathway stages;
- overlap of vulnerability windows;
- detection delay;
- response delay;
- persistence duration;
- synchronization across systems.

One possible phase model is

$$\theta_k = \alpha_k \tau_k + \sum_{j \neq k} \beta_{jk} \mathcal{S} * jk + \gamma * k \pi_k, \tag{8.56}$$

where:

- τ_k is timing;
- $\mathcal{S} * jk$ is synchronization;
- $\pi * k$ is persistence.

The parameters should be estimated and validated rather than chosen narratively.

8.8.1 Governance and Operational Decision-Making

The final output should remain understandable to operational-risk committees, cyber committees, technology executives, and boards.

A suitable report should contain:

1. the current operational state;
2. the definition of the adverse event;
3. the control-barrier architecture;
4. control dependencies;
5. the classical failure benchmark;
6. the dominant attack or failure pathways;
7. pathway magnitudes and phases;
8. tunneling transmission;
9. raw phase cross-term and bounded alignment diagnostics;
10. sensitivity to detection and recovery timing;
11. recommended interventions.

The committee output may be decomposed as

$$P_{\text{loss}} = P_{\text{classical}} + \Delta P_{\text{path}} + \Delta P_{\text{interference}} + \varepsilon_{\text{model}}. \tag{8.57}$$

This decomposition prevents the final probability from becoming a black box.

8.9 Designing Better Operational Barriers

The tunneling framework is most valuable when it changes control design.

Four broad interventions are possible.

8.9.1 Raise Barrier Height

Strengthen authentication, authorization, monitoring, segregation, or recovery capacity.

8.9.2 Increase Barrier Width

Add independent stages that a harmful process must traverse.

For example, require approval, secondary validation, delayed execution, and post-transaction review.

8.9.3 Reduce Barrier Permeability

Remove common dependencies, close hidden gaps, and reduce concentration.

8.9.4 Change Phase

Interrupt the timing and sequence of harmful pathways.

Rapid detection and automated containment can reduce risk even when the underlying vulnerabilities remain partially present.

This leads to a broader control-design principle:

$$\boxed{\begin{aligned}\text{Operational Resilience} = {} & \text{Control Strength} + \text{Control Independence} \\ & + \text{Sequence Interruption} + \text{Recovery Timing}.\end{aligned}} \tag{8.58}$$

8.9.5 What Operational Tunneling Is Not

Operational tunneling is not every cyberattack.

It is not an unpatched vulnerability.

It is not simply an insider event.

It is not an extreme workload failure.

It is not a substitute for attack-path analysis.

It is not evidence that software systems are physical quantum objects.

It is not justified merely because several controls fail.

The classification is appropriate only when:

- a measurable control barrier exists;
- the harmful mechanism appears individually sub-barrier;
- material transmission nevertheless occurs;
- pathway timing and interaction matter;
- the amplitude model adds value beyond classical alternatives.

8.9.6 A Practical Implementation Workflow

A practical operational or cyber implementation may follow ten steps.

1. Define the protected service and adverse event.
2. Select the operational state variables.
3. Construct the feasible state manifold.
4. Define stable, fragile, and adverse regions.
5. Map controls into barrier height, width, continuity, and dependency.
6. Identify classical attack and failure pathways.
7. Estimate pathway magnitudes and timing-based phases.
8. Calculate dominant action paths and transmission.
9. Compare with classical reliability and cyber models.
10. translate the result into control redesign and monitoring.

The class-specific model may be summarized as

$$\boxed{\mathfrak{T}_{\mathrm{op}} = \text{State Geometry} + \text{Control Barrier} + \text{Pathway Amplitudes} + \text{Timing} + \text{Recovery}.} \tag{8.59}$$

8.10 Chapter Synthesis

This chapter has adapted the general tunneling framework to operational and cyber risk.

The protected object may be a process, platform, data environment, model pipeline, vendor relationship, or critical service. The system is represented on an operational manifold whose coordinates describe workload, controls, vulnerabilities, dependencies, human reliability, detection, and recovery.

The stable region contains normal and resilient operating states. The fragile region contains states that remain functional but rely increasingly on degraded controls, manual workarounds, concentrated infrastructure, or delayed response. The adverse region contains material fraud, intrusion, data loss, service failure, model breakdown, or recovery failure.

Protective barriers are built from access controls, segregation, monitoring, redundancy, response, and recovery. These barriers are layered but often not independent. Shared infrastructure, data, personnel, and vendors can reduce many nominal controls to one effective layer.

The chapter distinguished classical failure mechanisms from tunneling. Extreme workload, direct vulnerabilities, human errors, barrier erosion, and hidden gaps remain ordinary explanations. Operational tunneling becomes relevant only when several individually sub-barrier pathways interact through timing and sequencing to produce material transition through a combined control structure.

Each pathway is encoded through a complex amplitude

$$\Psi_k = R_k e^{\mathrm{i}\theta_k}, \tag{8.60}$$

where magnitude represents pathway weight and phase represents timing, ordering, synchronization, and persistence.

Constructive interference occurs when weaknesses align into a reinforcing sequence. Destructive interference occurs when detection, isolation, response, or recovery interrupts that sequence.

The dominant operational path minimizes the under-barrier action:

$$\gamma^*_{\mathrm{op}} = \arg, \min *\gamma S * \mathrm{op}[\gamma]. \tag{8.61}$$

The associated transmission is

$$T_{\mathrm{op}} \approx \exp\left[-\frac{2S_{\mathrm{op}}\left[\gamma^*_{\mathrm{op}}\right]}{\eta_{\mathrm{op}}}\right]. \tag{8.62}$$

The practical value of the framework lies in control design. Institutions can raise barriers, widen them, reduce shared dependencies, close gaps, and change phase by detecting and interrupting harmful sequences earlier.

The central proposition of Chapter 8 is:

> Operational and cyber resilience depend not only on whether controls exist, but on whether their geometry, independence, timing, and interaction prevent individually modest weaknesses from forming a coherent pathway to material failure.

Chapter 9 will apply the same architecture to legal, regulatory, and conduct risk, where barriers are formed by contracts, precedent, evidence, institutional behaviour, governance, and enforcement structure.

Chapter 9

Legal, Regulatory, and Conduct-Risk Tunneling

9.1 The Distinctive Nature of Legal Risk

Legal, regulatory, and conduct risk differ fundamentally from credit and operational risk. A borrower may be described through leverage, liquidity, collateral, and cash flow. An operational system may be described through capacity, control effectiveness, dependencies, and recovery resources. Legal exposure, by contrast, emerges from the interaction of rules, facts, documents, behaviour, evidence, interpretation, jurisdiction, and enforcement.

The relevant state is not determined solely by what an institution has formally agreed to do. It also depends on what it actually did, how that conduct was documented, how third parties understood it, which legal principles apply, and how courts or regulators may interpret the resulting record.

A legal-risk state may therefore be represented as

$$\mathbf{x} * t^{\text{legal}} = \left[a * t \; d_t \; c_t \; e_t \; p_t \; r_t \; j_t \; s_t\right], \tag{9.1}$$

where:

- a_t is contractual or regulatory ambiguity;
- d_t is documentary quality;
- c_t is consistency of institutional conduct;
- e_t is evidentiary strength;
- p_t is alignment with precedent or authoritative interpretation;
- r_t is regulatory attention or enforcement posture;
- j_t is jurisdictional exposure;
- s_t is social, political, or reputational pressure.

The state belongs to a legal-risk manifold

$$\mathbf{x} * t^{\text{legal}} \in \mathcal{M} * \text{legal}. \tag{9.2}$$

This manifold is partly empirical and partly normative. It contains feasible combinations of documents, facts, institutional behaviour, legal doctrines, and enforcement environments.

The central proposition of this chapter is:

> Legal and conduct risk emerge from the geometry through which facts, rules, evidence, behaviour, interpretation, and enforcement interact—not merely from the binary violation of a clearly defined rule.

9.1.1 Legal States Are Interpretive States

A financial state can often be measured directly. Cash is held or it is not. Debt has a contractual amount. A payment has or has not been made.

Legal states are less direct.

A contract may contain language that appears clear until a dispute places the language in a different factual context. A compliance procedure may exist formally while failing to govern the economic substance of the relevant conduct. A legal opinion may be technically sound but depend on assumptions that later prove false.

The legal state is therefore interpretive.

Let the observable record be

$$\mathcal{F}_t = \{\text{documents}, \text{communications}, \text{transactions}, \text{decisions}, \text{conduct}\}. \tag{9.3}$$

Let the applicable legal and regulatory framework be

$$\mathcal{L}_t = \{\text{statutes}, \text{regulations}, \text{contracts}, \text{precedents}, \text{guidance}\}. \tag{9.4}$$

The legal-risk state is produced by an interpretive mapping

$$\mathbf{x} * t^{\text{legal}} = \mathcal{I}\left(\mathcal{F} * t, \mathcal{L} * t, \mathcal{E} * t\right), \tag{9.5}$$

where $\mathcal{E}_t$ denotes the enforcement and institutional environment.

The mapping $\mathcal{I}$ is not mechanically fixed. Different courts, regulators, counterparties, or internal reviewers may assign different meaning to the same factual record.

Legal risk therefore includes uncertainty over the map itself.

9.1.2 Stable, Fragile, and Adverse Legal Regions

The legal manifold can be partitioned into stable, fragile, and adverse regions:

$$\mathcal{M} * \text{legal} = S * \text{legal} \cup F_{\text{legal}} \cup A_{\text{legal}}. \tag{9.6}$$

A stable legal region may be characterized by:

- clear contractual rights;
- consistent conduct;
- complete documentation;
- strong evidentiary support;
- favourable precedent;
- low regulatory concern;
- reliable governance and escalation.

The fragile region includes states in which the institution remains formally compliant or contractually protected, but weaknesses have emerged. These may include ambiguity, inconsistent communications, incomplete records, exceptions to policy, or rising regulatory interest.

The adverse region may contain several different outcomes:

$$A_{\text{legal}} = A_{\text{liability}} \cup A_{\text{sanction}} \cup A_{\text{remediation}} \cup A_{\text{conduct}} \cup A_{\text{reputation}}. \tag{9.7}$$

These regions need not coincide.

An institution may avoid contractual liability while facing regulatory sanction. It may prevail legally while suffering reputational damage. It may avoid a fine but be required to compensate customers or redesign a product.

The model should therefore preserve multiple adverse regions rather than collapsing all legal exposure into one scalar outcome.

9.2 Legal Protections as Barriers

Legal protections form barriers separating acceptable or defensible conduct from liability, sanction, or remediation.

These protections include:

- contractual language;
- representations and warranties;
- disclosure;
- legal opinions;
- approvals;

- policies and procedures;
- monitoring and escalation;
- evidentiary records;
- jurisdictional structure;
- precedent;
- limitation clauses;
- indemnities and insurance.

A legal barrier may be represented by

$$\mathcal{B} * \text{legal} \subset \mathcal{M} * \text{legal}. \tag{9.8}$$

Its resistance field is

$$V_{\mathcal{B}}^{\text{legal}}(\mathbf{x}, t)\,. \tag{9.9}$$

High resistance corresponds to a strong and coherent legal defence. Low resistance corresponds to ambiguity, weak evidence, contradictory conduct, or an unfavourable enforcement environment.

A simple protection function may be written as

$$\begin{aligned} B_{\text{legal}} = {} & w_d d + w_e e + w_p p + w_g g + w_j j^+ \\ & - w_a a - w_{c^-} c^- - w_r r - w_s s, \end{aligned} \tag{9.10}$$

where:

- (g) represents governance and control quality;
- j^+ represents favourable jurisdictional protection;
- c^- represents inconsistent or adverse conduct;
- the remaining variables retain their earlier meanings.

The protected side may satisfy

$$B_{\text{legal}} > 0, \tag{9.11}$$

while the exposed side satisfies

$$B_{\text{legal}} < 0. \tag{9.12}$$

The barrier region itself is not merely the boundary $B_{\text{legal}} = 0$. It is the zone in which the defence remains plausible but increasingly contestable.

9.2.1 Formal Protection and Effective Protection

A recurring legal-risk problem is the distinction between formal and effective protection.

A contract may contain a limitation of liability. A product may have received legal approval. A compliance procedure may have been documented.

Yet formal protection may fail when:

- the factual assumptions supporting it are false;
- conduct contradicts the documented position;
- disclosures are incomplete;
- records are missing;
- the relevant clause is unenforceable;
- the institution cannot prove that controls operated;
- a regulator focuses on substance rather than form.

Let P_j denote nominal protection (j). Its effective value may be written as

$$\widetilde{P} * j = \omega * j^{\text{enf}} \omega_j^{\text{evid}} \omega_j^{\text{cons}} P_j, \tag{9.13}$$

where:

- ω_j^{enf} measures enforceability;
- ω_j^{evid} measures evidentiary support;
- ω_j^{cons} measures consistency with actual conduct.

Each adjustment lies between zero and one.

The equation shows why legal barriers are conditional. Protection exists only to the extent that the institution can invoke, prove, and sustain it.

9.3 Barrier Height, Width, and Continuity

Legal barrier height represents the strength of the defence that an adverse claimant or regulator must overcome.

It may depend on:

- clarity of drafting;

- strength of legal authority;
- quality of disclosure;
- consistency of the factual record;
- credibility of witnesses;
- reliability of approvals.

Legal barrier width represents the number and depth of steps required to establish liability or sanction.

A wide barrier may require proof of several elements:

$$\mathcal{E} * 1 \wedge \mathcal{E} * 2 \wedge \cdots \wedge \mathcal{E}_K. \tag{9.14}$$

For example, a claimant may need to prove duty, breach, causation, and loss. A regulator may need to establish jurisdiction, prohibited conduct, materiality, and attribution.

A narrow barrier may require only one decisive fact or one strict-liability element.

Barrier continuity concerns whether the protection applies across the full factual pathway.

A contract may be strong for routine transactions but silent on digital communications. A compliance process may cover product approval but not post-sale conduct. A policy may exist but fail to address exceptions.

The barrier can therefore be strong in one region and discontinuous in another.

9.3.1 The Directionality of Legal Barriers

Legal protections are anisotropic.

A disclosure may protect against a misrepresentation claim while offering little defence against unfair-conduct allegations. An indemnity may transfer contractual loss but not regulatory responsibility. A jurisdiction clause may control private litigation while having no effect on a regulator.

Let

$$\mathbf{R}_{\text{legal}}(\mathbf{x}, t) \tag{9.15}$$

be a legal resistance tensor.

For a transition direction $\mathbf{v}$,

$$r_{\text{legal}}(\mathbf{v}) = \mathbf{v}^{\mathsf{T}} \mathbf{R}_{\text{legal}} \mathbf{v}. \tag{9.16}$$

The weakest direction is

$$\mathbf{v} * \min = \arg, \min * |\mathbf{v}| = 1 r_{\text{legal}}(\mathbf{v}). \tag{9.17}$$

This direction may correspond to conduct risk rather than contractual risk, or to regulatory scrutiny rather than civil litigation.

The practical lesson is:

> A legal structure should not be evaluated only against the claim it was designed to prevent. It should be tested against the alternative legal and regulatory pathways through which the same conduct may be challenged.

9.3.2 The Main Legal and Conduct Pathways

The transition from a stable legal state to an adverse state may occur through several pathways.

A useful pathway set is

$$\Gamma_{\text{legal}} = \{\gamma_{\text{contract}}, \gamma_{\text{evidence}}, \gamma_{\text{conduct}}, \gamma_{\text{regulatory}}, \gamma_{\text{reputation}}, \gamma_{\text{governance}}\}. \tag{9.18}$$

Contractual Ambiguity

A disputed term may support more than one plausible interpretation.

The amplitude associated with contractual ambiguity may be

$$\Psi_{\text{contract}} = R_{\text{contract}} \exp\left(\mathrm{i}\theta_{\text{contract}}\right). \tag{9.19}$$

Its magnitude reflects the weight of the adverse interpretation. Its phase reflects how that interpretation aligns with other facts and doctrines.

Evidentiary Weakness

The institution may have a strong legal position but insufficient records to prove it.

Missing approvals, inconsistent files, undocumented exceptions, and unreliable data can create an evidentiary pathway.

Adverse Conduct

The institution's actual conduct may weaken its formal defence.

Repeated exceptions, aggressive communications, inconsistent treatment, or knowledge of customer harm may become legally significant.

Regulatory Escalation

A matter may move from routine supervision to investigation, enforcement, or public sanction.

Regulatory escalation has its own timing, thresholds, and institutional dynamics.

Reputational Amplification

Public attention may alter the practical legal environment. Complaints, media reports, political concern, or social pressure may affect enforcement intensity and settlement incentives.

Governance Failure

Failure to escalate, investigate, remediate, or disclose may create a separate pathway even when the original conduct was defensible.

9.4 Amplitude and Phase in Legal Risk

Each legal pathway may be assigned an amplitude

$$\Psi_k = R_k \exp\left(\mathrm{i}\theta_k\right). \tag{9.20}$$

The magnitude R_k may reflect:

- probability of an adverse interpretation;
- evidentiary weight;
- likelihood of enforcement;
- estimated pathway contribution;
- scenario frequency.

The phase θ_k may reflect:

- timing of the relevant event;
- sequencing of documents and conduct;
- synchronization of complaints and regulatory attention;
- compatibility among legal theories;
- alignment of evidence with an adverse narrative.

The total adverse amplitude is

$$\Psi_A = \sum_{k=1}^{K} \Psi_k. \tag{9.21}$$

The adverse probability is

$$P(A) = \left|\Psi_A\right|^2. \tag{9.22}$$

Expanding,

$$P(A) = \sum_{k=1}^{K} R_k^2 + \sum_{j \neq k} R_j R_k \cos(\theta_j - \theta_k)\,. \tag{9.23}$$

The cross terms represent interference among legal pathways.

9.4.1 Constructive Interference and the Adverse Narrative

Constructive interference occurs when separate facts, documents, conduct, and legal doctrines align into one coherent adverse narrative.

Consider the following facts:

1. the contract contains an ambiguous term;
2. internal communications show awareness of the ambiguity;
3. customer disclosures do not address it;
4. complaints reveal recurring harm;
5. management fails to remediate promptly;
6. the regulator has announced heightened concern in the area.

Each fact may be manageable in isolation.

Together, they may produce a coherent narrative:

> The institution understood the risk, failed to disclose it, continued the conduct, and did not respond adequately when harm became visible.

This is constructive interference in legal-risk language.

The pathways have become phase-aligned because their timing, meaning, and evidentiary role reinforce one another.

The combined exposure may be substantially greater than the sum of the isolated issues.

9.5 Destructive Interference and Legal Stabilization

Destructive interference occurs when evidence or intervention disrupts the adverse narrative.

Examples include:

- clear contemporaneous disclosure;
- consistent customer treatment;
- documented legal review;
- prompt remediation;
- independent governance challenge;

- credible reliance on authoritative guidance;
- compensation before regulatory escalation.

Suppose a conduct pathway and regulatory pathway are developing. Prompt remediation may alter their relative phase by changing the sequence of facts.

The regulator no longer observes an institution ignoring known harm. It observes an institution identifying, escalating, and correcting the issue.

The intervention may therefore reduce aggregate risk even if it does not erase the original event.

This illustrates an important feature of the amplitude framework:

> An intervention can reduce risk not only by lowering the magnitude of a pathway, but by disrupting the alignment among several pathways.

9.5.1 Legal Tunneling and the Protective Barrier

A candidate legal-tunneling event arises when a matter reaches an adverse region despite a nominally strong legal barrier.

Let the effective legal capacity of the adverse claim or enforcement pathway be

$$E_{\text{legal}}. \tag{9.24}$$

Let the legal barrier potential be

$$V_{\text{legal}}\left(\mathbf{x},t\right). \tag{9.25}$$

A sub-barrier condition exists when

$$E_{\text{legal}} < V_{\text{legal}}\left(\mathbf{x},t\right) \tag{9.26}$$

along the nominal defence path.

Yet a nonzero adverse probability may emerge through a different pathway:

$$T_{\text{legal}} \approx \exp\left[-\frac{2S_{\text{legal}}[\gamma^*]}{\eta_{\text{legal}}}\right] > 0. \tag{9.27}$$

The dominant pathway is

$$\gamma^* = \arg, \min *\gamma S * \text{legal}[\gamma]. \tag{9.28}$$

The key risk may therefore lie outside the strongest formal defence.

An institution may have a strong contractual position but a weak conduct pathway. It may have a strong legal opinion but poor evidence. It may be protected against private claims but exposed to regulatory interpretation.

9.5.2 Moving Legal Barriers

Many apparent tunneling events are better understood as moving-barrier events.

The legal barrier depends on time:

$$V_{\text{legal}} = V_{\text{legal}}\left(\mathbf{x}, t\right). \tag{9.29}$$

It may move because of:

- new legislation;
- regulatory guidance;
- judicial precedent;
- political expectations;
- changing enforcement priorities;
- shifts in accepted market practice;
- evolving social standards;
- new evidence.

The total change in legal resistance is

$$\frac{\mathrm{d}}{\mathrm{d}t} V_{\text{legal}}\left(\mathbf{x} * t, t\right) = \frac{\partial V * \text{legal}}{\partial t} + \nabla V_{\text{legal}}^{\mathsf{T}} \dot{\mathbf{x}}_t. \tag{9.30}$$

The first term represents change in the legal environment. The second represents change caused by institutional conduct.

A business practice may remain unchanged while the regulatory boundary moves toward it.

A model that ignores this movement may incorrectly classify the resulting exposure as tunneling.

The diagnostic order should therefore be:

$$\boxed{\text{Barrier movement; } \rightarrow \text{; hidden gap; } \rightarrow \text{; pathway interaction; } \rightarrow \text{; tunneling candidate.}} \tag{9.31}$$

9.6 A Practical Example: Product-Suitability Risk

Consider a financial institution offering a complex investment product to retail clients.

The institution possesses:

- approved product documentation;

- customer disclosures;
- suitability procedures;
- sales scripts;
- supervisory controls;
- legal opinions.

These form the nominal barrier.

The legal-risk state may be represented by

$$\mathbf{x} * t = (a * t, d_t, c_t, e_t, r_t, s_t) \,, \tag{9.32}$$

where ambiguity, documentation, conduct, evidence, regulatory attention, and reputational pressure are the principal variables.

Suppose the following pathways develop:

1. disclosure language proves difficult for customers to understand;
2. sales incentives encourage aggressive distribution;
3. suitability exceptions become frequent;
4. complaints increase;
5. internal escalation is delayed;
6. the regulator announces concern about similar products.

Assign amplitudes

$$\Psi_{\text{disc}}, \quad \Psi_{\text{sales}}, \quad \Psi_{\text{suit}}, \quad \Psi_{\text{compl}}, \quad \Psi_{\text{reg}}. \tag{9.33}$$

At first, the pathways may be weakly related.

As complaints accumulate and internal documents reveal awareness of sales-practice problems, the phases become aligned. The regulator can combine separate facts into a coherent conduct narrative.

The dominant path may not challenge the legality of the product itself. It may instead move through the conduct and governance barrier:

$$\gamma^* = \gamma_{\text{sales}} + \gamma_{\text{complaints}} + \gamma_{\text{governance}}. \tag{9.34}$$

The institution's strongest legal opinion may therefore provide limited protection.

The lowest-action route to sanction lies elsewhere.

9.6.1 A Practical Example: Contractual Protection and Adverse Conduct

Consider a lender with a strong contractual right to accelerate a loan after covenant breach.

The formal legal barrier appears strong:

- clear covenant;
- express acceleration right;
- governing-law clause;
- legal opinion;
- complete execution record.

However, the lender has repeatedly waived similar breaches, made inconsistent representations to the borrower, and delayed enforcement while encouraging continued investment.

The relevant pathways include:

$$\Gamma = \left\{ \gamma_{\text{contract}}, \gamma_{\text{waiver}}, \gamma_{\text{reliance}}, \gamma_{\text{conduct}} \right\}. \tag{9.35}$$

The contractual pathway favours the lender. The waiver and reliance pathways favour the borrower.

The resulting structural-alignment assessment depends on how these amplitudes interact. Any reported legal probability must be produced by a normalized full-state amplitude or, more practically, by a separately calibrated model using the alignment measures as predictors.

A formally strong right may be weakened by a coherent pattern of contrary conduct.

The example demonstrates why legal barrier strength cannot be inferred from the contract alone.

9.7 Multiple Adverse Legal Outcomes

The legal-risk amplitude may project into several adverse regions.

Define projectors:

$$\widehat{\Pi} * L, \qquad \widehat{\Pi} * R, \qquad \widehat{\Pi} * C, \qquad \widehat{\Pi} * S, \tag{9.36}$$

for civil liability, regulatory sanction, conduct remediation, and reputational harm.

The corresponding probabilities are

$$P_L(t) = \langle \Psi(t) | \widehat{\Pi} * L | \Psi(t) \rangle, \tag{9.37}$$
$$P * R(t) = \langle \Psi(t) | \widehat{\Pi} * R | \Psi(t) \rangle, \tag{9.38}$$
$$P * C(t) = \langle \Psi(t) | \widehat{\Pi} * C | \Psi(t) \rangle, \tag{9.39}$$
$$P * S(t) = \langle \Psi(t) | \widehat{\Pi}_S | \Psi(t) \rangle. \tag{9.40}$$

An intervention may reduce one probability while increasing another.

For example, public remediation may reduce regulatory exposure but increase short-term reputational visibility. Aggressive litigation may reduce settlement cost but increase conduct and reputational risk.

The model should make these trade-offs explicit.

9.7.1 Estimating Legal Pathway Magnitudes

Legal data are sparse, heterogeneous, and often non-comparable.

Pathway magnitudes may be estimated using:

- historical claims;
- regulatory actions;
- internal investigations;
- complaint data;
- legal opinions;
- precedent analysis;
- scenario assessment;
- expert elicitation;
- document and communication analysis.

A pathway magnitude may initially be written as

$$R_k = \sqrt{p_k}, \tag{9.41}$$

where p_k is a calibrated pathway probability or normalized evidentiary weight.

The quantity should not be presented as mechanically objective. It may contain substantial model and expert uncertainty.

A confidence interval or distribution should therefore accompany it.

9.7.2 Estimating Legal Phase

Phase is especially important in legal risk because sequence and narrative coherence matter.

An effective legal timing coordinate may be constructed as

$$\tau_k^{\text{eff}} = \omega_1 \tau_k + \omega_2 \nu_k + \omega_3 \chi_k + \omega_4 \zeta_k, \tag{9.42}$$

where:

- τ_k is timing;
- ν_k is narrative compatibility;
- χ_k is evidentiary consistency;
- ζ_k is synchronization with enforcement or public attention.

The non-temporal features must be trained or elicited as equivalent timing adjustments on a declared scale. With a prespecified origin τ_0 and legal decision horizon H, define

$$\theta_k = 2\pi \left[\frac{\tau_k^{\text{eff}} - \tau_0}{H} \bmod 1 \right]. \tag{9.43}$$

A separation of $H/2$ therefore corresponds to π. If that translation cannot be defended before evaluation, narrative compatibility and evidentiary consistency should remain classical predictors rather than phase components.

Relative phase is

$$\Delta\theta_{jk} = \theta_j - \theta_k. \tag{9.44}$$

The phase should be tied to observable events:

- date of approval;
- date of communication;
- date of customer harm;
- date of complaint;
- date of escalation;
- date of remediation;
- date of regulatory inquiry.

This converts the phase from an abstract mathematical variable into a disciplined representation of sequence.

9.8 Dominant Legal Path and Barrier Action

For each legal pathway γ, define a barrier action

$$S_{\text{legal}}[\gamma] = \int_{\gamma\cap\mathcal{B}*\text{legal}} \sqrt{2\,[V*\text{legal}(\mathbf{x},t) - E_{\text{legal}}]},\mathrm{d}s_{\mathbf{M}}. \tag{9.45}$$

The dominant pathway is

$$\gamma^*_{\text{legal}} = \arg,\min*\gamma S*\text{legal}[\gamma]. \tag{9.46}$$

This path identifies the route through which the institution is most vulnerable.

It may pass through:

- one ambiguous clause;
- one unrecorded approval;
- one inconsistent communication;
- one governance delay;
- one jurisdictional weakness;
- one cluster of customer complaints.

The dominant pathway is not necessarily the most dramatic issue. It is the one that encounters the least integrated resistance.

9.8.1 Classical Alternatives

A legal-tunneling model should be compared with classical approaches.

These include:

- legal issue trees;
- scenario analysis;
- Bayesian networks;
- event trees;
- litigation outcome models;
- expert scoring;
- regulatory risk matrices;
- causal graphs;
- text-based evidence classification;

- survival and escalation models.

The amplitude framework is justified only if it adds value by representing timing, pathway alignment, or interaction more coherently.

The null hypothesis remains:

$$H_0 : \text{Classical legal-risk methods are sufficient.} \tag{9.47}$$

The alternative is:

$$H_1 : \text{Amplitude and phase add material explanatory value.} \tag{9.48}$$

9.9 Governance and Professional Judgment

Legal-risk modelling must remain subordinate to professional legal judgment.

The model should not be used to:

- provide definitive legal conclusions;
- replace advice from qualified counsel;
- convert uncertain interpretation into false precision;
- conceal judgement behind mathematics;
- automate decisions requiring legal privilege or fiduciary responsibility.

Its purpose is to organize information, compare pathways, identify weak protections, and test scenarios.

A committee should receive:

1. the current legal-risk state;
2. the relevant factual record;
3. the applicable legal framework;
4. the barrier profile;
5. formal and effective protection;
6. the dominant pathways;
7. the role of phase and interference;
8. moving-boundary scenarios;
9. multiple adverse outcomes;

10. model uncertainty;

11. counsel's qualitative interpretation.

The final decision must combine model structure and legal reasoning.

9.9.1 Legal-Risk Interventions

The framework can support intervention design.

A legal-risk intervention may:

- increase barrier height;
- widen the barrier;
- close a gap;
- alter pathway phase;
- reduce evidentiary ambiguity;
- disrupt constructive interference;
- move the state toward a stable region.

Examples include:

- clarifying disclosure;
- correcting documentation;
- suspending problematic conduct;
- compensating affected customers;
- escalating to independent governance;
- preserving evidence;
- obtaining authoritative guidance;
- redesigning incentives;
- separating decision functions.

The intervention should be evaluated against all adverse regions.

A measure that improves the civil-litigation barrier may worsen regulatory or reputational exposure if implemented poorly.

9.9.2 A Minimum Legal-Tunneling Architecture

The chapter's framework may be summarized through

$$\mathfrak{T} * \text{legal} = \left(\mathcal{M} * \text{legal}, \mathcal{F} * t, \mathcal{L} * t, \mathcal{E} * t, \mathcal{B} * \text{legal}, V_{\text{legal}}, \mathbf{R} * \text{legal}, \Psi, \Gamma * \text{legal}, \gamma^*, \mathbf{P}_A\right), \quad (9.49)$$

where:

- $\mathcal{M} * \text{legal}$ is the legal-risk manifold;
- $\mathcal{F} * t$ is the factual record;
- $\mathcal{L} * t$ is the legal framework;
- $\mathcal{E} * t$ is the enforcement environment;
- $\mathcal{B} * \text{legal}$ is the protective barrier;
- $V * \text{legal}$ is the legal-risk potential;
- $\mathbf{R} * \text{legal}$ is the resistance tensor;
- Ψ is the complex legal-risk amplitude;
- $\Gamma * \text{legal}$ is the pathway set;
- γ^* is the minimum-action pathway;
- $\mathbf{P}_A$ is the vector of adverse-outcome probabilities.

The model may be summarized as

$$\boxed{\begin{aligned} \text{Legal Risk} = {} & \text{Facts} + \text{Rules} + \text{Evidence} + \text{Conduct} \\ & + \text{Interpretation} + \text{Enforcement} + \text{Pathway Interaction}. \end{aligned}} \quad (9.50)$$

9.10 Chapter Synthesis

This chapter has applied the tunneling framework to legal, regulatory, and conduct risk.

Legal-risk states are interpretive states. They are constructed from factual records, legal rules, documents, conduct, evidence, precedent, jurisdiction, and enforcement posture.

The legal manifold contains stable, fragile, and multiple adverse regions. These adverse regions include civil liability, regulatory sanction, conduct remediation, and reputational harm.

Legal protections form conditional barriers. Contracts, legal opinions, approvals, disclosures, policies, and compliance procedures provide protection only to the extent that they are enforceable, evidenced, and consistent with actual conduct.

The chapter distinguished nominal protection from effective protection. It also showed that legal barriers are directional. A contractual defence may remain strong while a conduct or regulatory pathway is weak.

The principal legal pathways include contractual ambiguity, evidentiary weakness, adverse conduct, regulatory escalation, reputational amplification, and governance failure.

Each pathway may be encoded through a complex amplitude. The magnitude represents pathway weight. The phase represents timing, sequencing, narrative compatibility, and synchronization with evidence or enforcement.

Constructive interference occurs when separate facts and pathways combine into a coherent adverse narrative. Destructive interference occurs when disclosure, evidence, remediation, governance, or authoritative guidance disrupt that narrative.

The chapter emphasized that many apparent legal-tunneling events are moving-barrier events. Regulations, precedents, enforcement priorities, political expectations, and social standards can change the location and strength of the legal barrier.

The product-suitability and contractual-conduct examples demonstrated how the dominant legal pathway may differ from the institution's most visible formal risk.

A strong contract does not necessarily create a strong conduct barrier. A legal opinion does not compensate for inconsistent facts. A compliance procedure does not protect the institution if it cannot demonstrate that the procedure operated effectively.

The dominant pathway is the path minimizing legal barrier action. It identifies the combination of facts, interpretations, evidence, and conduct that encounters the least resistance on the route to an adverse outcome.

The model must remain governed by professional judgment. It does not replace legal counsel or convert uncertainty into objective truth. Its value lies in structuring complex evidence, comparing pathways, identifying weak barriers, and evaluating interventions.

The central proposition of Chapter 9 is:

> Legal, regulatory, and conduct risk arise not only from violating a rule, but from the evolving geometry through which facts, conduct, evidence, interpretation, and enforcement combine to make nominal legal protections permeable.

Chapter 10 will extend the framework to institutional and systemic risk, where barriers and pathways are distributed across organizations, markets, infrastructures, and networks rather than contained within one institution or transaction.

Chapter 10

Institutional and Systemic Risk Tunneling

10.1 From Individual Failure to Systemic Transition

The preceding chapters examined tunneling in credit, operational, cyber, legal, regulatory, and conduct environments. In each case, the analysis began with an identifiable system: a borrower, a technological platform, a control architecture, or a legally defensible institutional position.

Systemic risk requires a wider unit of analysis.

The relevant object is no longer one institution considered in isolation. It is a network of institutions, markets, infrastructures, legal arrangements, and behavioural responses whose interactions create collective states that cannot be inferred by examining each component separately.

A bank may appear adequately capitalized. A clearing house may appear operationally resilient. A sovereign may appear capable of refinancing. A payment system may remain functional. Yet the combined system may still contain a low-resistance pathway toward collective distress.

This occurs because systemic risk is relational.

It depends on:

- who is exposed to whom;
- which assets are held in common;
- which funding sources are shared;
- which infrastructures are concentrated;
- which institutions respond similarly to stress;
- which barriers become crowded or unreliable during crisis.

The general tunneling object becomes

$$\mathfrak{T} * \mathrm{sys} = (\mathcal{M} * \mathrm{sys}, \mathbf{X} * t, \mathcal{G} * t, S, F, A, \mathcal{B} * \mathrm{sys}, V * \mathrm{sys}, \mathbf{M} * \mathrm{sys}, \Psi * \mathrm{sys}, \gamma^*, T_{\mathrm{sys}}) \,, \tag{10.1}$$

where:

- $\mathcal{M} * \mathrm{sys}$ is the systemic state manifold;
- $\mathbf{X} * t$ is the collective state;
- $\mathcal{G} * t$ is the network of relationships;
- (S), (F), and (A) are stable, fragile, and adverse systemic regions;
- $\mathcal{B} * \mathrm{sys}$ is the collective protective barrier;
- V_{sys} is the systemic risk potential;
- $\mathbf{M} * \mathrm{sys}$ is the systemic resistance tensor;
- $\Psi * \mathrm{sys}$ is the systemic amplitude;
- γ^* is the dominant systemic pathway;
- T_{sys} is the associated transmission measure.

The central proposition of this chapter is:

> Systemic resilience cannot be inferred from the resilience of individual institutions alone. It depends on whether network structure, synchronized behaviour, and shared dependencies create coherent pathways through collective protective barriers.

10.1.1 The Systemic State

Consider (N) institutions. Institution (i) has state vector

$$\mathbf{x} * i, t \in \mathcal{M} * i. \tag{10.2}$$

The aggregate state is

$$\mathbf{X} * t = (\mathbf{x} * 1, t, \mathbf{x} * 2, t, \ldots, \mathbf{x} * N, t) \,. \tag{10.3}$$

A simple product-space representation is

$$\mathcal{M} * 0 = \mathcal{M} * 1 \times \mathcal{M} * 2 \times \cdots \times \mathcal{M} * N. \tag{10.4}$$

This product space is not yet sufficient because institutions are connected.

The relevant systemic state must also include the relationship network

$$\mathcal{G} * t = (\mathcal{V}, \mathcal{E} * t, \mathbf{W}_t)\,, \tag{10.5}$$

where:

- $\mathcal{V}$ is the set of institutions or infrastructures;
- $\mathcal{E} * t$ is the set of links;
- $\mathbf{W} * t$ contains the strength and direction of relationships.

The links may represent:

- bilateral credit exposure;
- secured funding;
- derivatives;
- collateral dependence;
- payment obligations;
- common asset holdings;
- shared service providers;
- legal guarantees;
- confidence relationships.

The systemic manifold is therefore better written as

$$\mathcal{M} * \mathrm{sys} = \{(\mathbf{X}, \mathcal{G}) : F * \mathrm{sys}\,(\mathbf{X}, \mathcal{G}) \leq 0\}\,. \tag{10.6}$$

The feasible systemic states are constrained by accounting identities, balance-sheet consistency, market clearing, collateral rules, payment obligations, regulatory requirements, and behavioural responses.

10.2 Institutional and Systemic State Variables

A minimal state for institution (i) may include

$$\mathbf{x}_{i,t} = \begin{bmatrix} k_{i,t} \\ l_{i,t} \\ f_{i,t} \\ a_{i,t} \\ c_{i,t} \\ o_{i,t} \end{bmatrix}, \tag{10.7}$$

where:

- $k_{i,t}$ is capital adequacy;
- $l_{i,t}$ is liquidity;
- $f_{i,t}$ is funding stability;
- $a_{i,t}$ is asset quality;
- $c_{i,t}$ is confidence or market access;
- $o_{i,t}$ is operational continuity.

System-level variables should also be included:

$$\mathbf{z}_t^{\text{sys}} = \begin{bmatrix} \lambda_t \\ \chi_t \\ \mu_t \\ \rho_t \\ \omega_t \end{bmatrix}, \tag{10.8}$$

where:

- λ_t is network leverage or interconnectedness;
- χ_t is funding concentration;
- μ_t is market liquidity;
- ρ_t is behavioural synchronization;
- ω_t is infrastructure concentration.

The complete systemic state is

$$\mathbf{Y} * t = (\mathbf{X} * t, \mathbf{z} * t^{\text{sys}}, \mathcal{G} * t). \tag{10.9}$$

This representation makes explicit that systemic fragility is not merely the sum of institutional fragilities.

10.2.1 Stable, Fragile, and Adverse Systemic Regions

The stable region S_{sys} contains states in which institutions can absorb losses, funding markets remain open, collateral circulates, infrastructures operate, and confidence is sufficiently differentiated.

The fragile region F_{sys} contains states in which the system continues to function but depends increasingly on:

- concentrated funding;

- common collateral;
- central-bank liquidity;
- a small number of infrastructures;
- correlated asset valuations;
- synchronized expectations.

The adverse systemic region may be

$$\begin{aligned} A_{\text{sys}} = A_{\text{multiple default}} \cup A_{\text{market freeze}} \cup A_{\text{payment disruption}} \\ \cup A_{\text{fire sale}} \cup A_{\text{infrastructure failure}} \cup A_{\text{confidence collapse}}. \end{aligned} \tag{10.10}$$

A systemic event need not require the legal insolvency of many institutions. A market can become dysfunctional while most balance sheets remain formally solvent.

The adverse region should therefore be defined through collective impairment, not merely through individual default counts.

10.2.2 Collective Protective Barriers

Systemic barriers are the structures that prevent local stress from becoming collective failure.

These may include:

- capital and liquidity requirements;
- deposit insurance;
- lender-of-last-resort facilities;
- central clearing;
- margining and collateral systems;
- resolution regimes;
- circuit breakers;
- payment-system safeguards;
- diversification requirements;
- macroprudential limits;
- emergency fiscal capacity.

A systemic protection function may be written as

$$
\begin{aligned}
B_{\text{sys}} = {} & \alpha_K K + \alpha_L L + \alpha_R R + \alpha_C C + \alpha_I I \\
& - \alpha_X X - \alpha_H H - \alpha_S S,
\end{aligned} \tag{10.11}
$$

where:

- (K) is aggregate usable capital;
- (L) is available system liquidity;
- (R) is resolution capacity;
- (C) is clearing and collateral resilience;
- (I) is infrastructure redundancy;
- (X) is interconnected exposure;
- (H) is asset-holding concentration;
- (S) is behavioural synchronization.

This barrier is collective because its components may be shared.

A central liquidity facility may protect one institution effectively but become insufficient when many institutions draw simultaneously.

A clearing mechanism may reduce bilateral counterparty risk while concentrating operational and liquidity risk in one central node.

A resolution regime may be credible for one failure but politically or operationally difficult to apply to several institutions at once.

Thus, collective barriers are often capacity-constrained.

10.3 The Fallacy of Individual Resilience

Suppose every institution satisfies

$$B_i\left(\mathbf{x}_{i,t}\right) > 0. \tag{10.12}$$

It does not follow that

$$B_{\text{sys}}\left(\mathbf{Y}_t\right) > 0. \tag{10.13}$$

Individual safety may coexist with systemic fragility.

This can occur when:

- institutions hold the same assets;

- they depend on the same short-term funding;
- they use the same risk models;
- they respond to regulation in similar ways;
- they rely on one infrastructure provider;
- they hedge through the same counterparties;
- they seek liquidity at the same time.

A system may therefore possess high institutional barrier height but low systemic barrier width.

Each institution can absorb a moderate isolated shock. Yet only a short sequence separates the system from collective failure once reactions become synchronized.

10.3.1 Systemic Transition Pathways

The principal systemic pathways include:

1. direct counterparty loss;
2. wholesale funding withdrawal;
3. collateral and margin spirals;
4. common-asset fire sales;
5. deposit or investor runs;
6. payment-system disruption;
7. operational infrastructure failure;
8. sovereign-financial feedback;
9. regulatory amplification;
10. confidence contagion.

Each pathway may be assigned an amplitude

$$\Psi_k^{\mathrm{sys}} = R_k^{\mathrm{sys}} \exp\left(\mathrm{i}\theta_k^{\mathrm{sys}}\right). \tag{10.14}$$

The magnitude represents the individual weight of the pathway.

The phase represents:

- timing across institutions;
- ordering of market reactions;
- synchronization of balance-sheet adjustment;

- persistence of funding pressure;
- alignment between market and policy responses.

These features must first be mapped to an effective timing coordinate $\tau_{k,t}^{\text{eff}}$. Given a prespecified crisis origin τ_0 and policy horizon H, phase is defined by

$$\theta_{k,t}^{\text{sys}} = 2\pi \left[\frac{\tau_{k,t}^{\text{eff}} - \tau_0}{H} \bmod 1 \right]. \tag{10.15}$$

The period cannot be selected after observing the event. If no defensible H exists, synchronization should be modelled through ordinary time-series or network interaction terms.

The total systemic amplitude is

$$\Psi_A^{\text{sys}} = \sum_{k=1}^{K} \Psi_k^{\text{sys}}. \tag{10.16}$$

The systemic adverse probability is

$$P_{\text{sys}}(A, t) = |\Psi_A^{\text{sys}}|^2. \tag{10.17}$$

10.4 Synchronized Behaviour and Constructive Interference

Systemic amplification often arises not because every institution is weak, but because many institutions react similarly.

Consider simultaneous deleveraging.

Institution (i) attempts to reduce risk by selling assets:

$$\Delta a_{i,t} < 0. \tag{10.18}$$

For one institution, this may improve resilience.

If many institutions sell the same asset at the same time, the price falls:

$$\Delta p_t = -\phi \sum_{i=1}^{N} |\Delta a_{i,t}|, \qquad \phi > 0. \tag{10.19}$$

Lower prices generate valuation losses, margin calls, and further sales.

The sequence becomes

$$\text{risk reduction} \to \text{asset sales} \to \text{price decline} \to \text{margin calls} \to \text{further sales}. \tag{10.20}$$

The pathways become constructively aligned.

In amplitude language,

$$P(A) = \sum_{k} |\Psi_k|^2 + \sum_{j \neq k} \Psi_j^* \Psi_k. \tag{10.21}$$

Positive cross terms represent systemic reinforcement.

The resulting collective risk may greatly exceed the sum of institution-level probabilities.

10.4.1 Destructive Interference and Coordinated Stabilization

Systemic interventions can reduce risk by disrupting synchronization.

Examples include:

- staggered liquidity facilities;
- temporary margin relief;
- asset-purchase programmes;
- coordinated communication;
- deposit guarantees;
- resolution stays;
- targeted funding support;
- circuit breakers.

These actions may not eliminate the underlying losses. Their immediate effect may instead be to alter timing and sequence.

A policy-response amplitude may be written as

$$\Psi_{\text{policy}} = R_{\text{policy}} \exp\left(\mathrm{i}\theta_{\text{policy}}\right). \tag{10.22}$$

The combined amplitude is

$$\Psi_{\text{total}} = \Psi_{\text{stress}} + \Psi_{\text{policy}}. \tag{10.23}$$

A timely policy response may produce destructive interference with the adverse sequence.

The policy may succeed not only because it adds resources, but because it prevents institutions from acting simultaneously.

This interpretation gives timing a central macroprudential role.

10.5 The Network Hamiltonian

Let the weighted network adjacency matrix be

$$\mathbf{W} * t = [w * ij, t] \,. \tag{10.24}$$

Define the degree matrix

$$\mathbf{D} * t = \operatorname{diag}\left(d * 1, t, \ldots, d_{N,t}\right), \tag{10.25}$$

with

$$d_{i,t} = \sum_j w_{ij,t}. \tag{10.26}$$

The graph Laplacian is

$$\mathbf{L} * t = \mathbf{D} * t - \mathbf{W}_t. \tag{10.27}$$

A discrete systemic Hamiltonian may be written as

$$\widehat{H} * \mathrm{sys} = -\eta * \mathrm{sys}^2 \mathbf{L} * t + \mathbf{V} * \mathrm{sys}(t), \tag{10.28}$$

where $\mathbf{V}_{\mathrm{sys}}$ is a diagonal or block-structured systemic potential.

The evolution equation is

$$\mathrm{i}\eta_{\mathrm{sys}} \frac{\partial \Psi_{\mathrm{sys}}}{\partial t} = \widehat{H} * \mathrm{sys}\Psi * \mathrm{sys}. \tag{10.29}$$

The Laplacian term governs propagation through the network.

The potential term represents local fragility and protective barriers.

The parameter η_{sys} controls the scale of propagation and coherence.

As elsewhere in the book, these are effective modelling objects rather than physical quantities.

10.5.1 The Dominant Systemic Path

The most dangerous systemic path is not necessarily the route through the weakest institution.

It may pass through:

- a highly connected intermediary;
- a central counterparty;
- a shared collateral market;

- a dominant payment provider;
- a confidence-sensitive funding node;
- a sovereign-financial nexus.

Let γ denote a systemic pathway through institutions, markets, and infrastructures.

The systemic barrier action is

$$S_{\text{sys}}[\gamma] = \int_{\gamma\cap\mathcal{B}*\text{sys}} \sqrt{2\left[V*\text{sys}(\mathbf{Y},t) - E_{\text{sys}}\right]}, \mathrm{d}s_{\mathbf{M}_{\text{sys}}}. \tag{10.30}$$

The dominant path is

$$\gamma^*_{\text{sys}} = \arg, \min *\gamma S*\text{sys}[\gamma]. \tag{10.31}$$

The systemic transmission measure is

$$T_{\text{sys}} \approx \exp\left[-\frac{2S_{\text{sys}}\left[\gamma^*_{\text{sys}}\right]}{\eta_{\text{sys}}}\right]. \tag{10.32}$$

The dominant path may involve several institutions that are not individually the most fragile.

Connectivity, timing, and barrier geometry determine the route.

10.5.2 Central Nodes and Shared Infrastructure

A central node may create both resilience and fragility.

Central clearing can reduce bilateral exposure and improve transparency. Yet it concentrates operational, liquidity, and margin risk.

A shared cloud provider can improve reliability for each institution while creating correlated operational dependence.

A common collateral framework can simplify funding while making many institutions sensitive to the same valuation shock.

Let q_i denote the centrality of node (i). One simple systemic importance score is

$$\mathcal{S}*i = q*i \times \pi_i \times \delta_i, \tag{10.33}$$

where:

- q_i is network centrality;
- π_i is local failure probability;
- δ_i is propagation severity.

However, a minimum-action analysis may identify a node with moderate π_i but very high propagation relevance.

The most important node is therefore not always the node with the highest standalone probability of failure.

10.6 Endogenous and Moving Systemic Barriers

Systemic barriers are endogenous.

They change as institutions react to stress.

Collateral values fall during fire sales. Margin requirements rise as volatility increases. Liquidity facilities become crowded. Political willingness to intervene may weaken. Resolution capacity may be exhausted by repeated failures.

Let the systemic barrier potential be

$$V_{\text{sys}} = V_{\text{sys}}\left(\mathbf{Y}_t, t\right). \tag{10.34}$$

Its evolution may satisfy

$$\frac{\mathrm{d}V_{\text{sys}}}{\mathrm{d}t} = \frac{\partial V_{\text{sys}}}{\partial t} + \nabla V_{\text{sys}}^{\mathsf{T}} \dot{\mathbf{Y}}_t. \tag{10.35}$$

The second term is especially important.

The system's response can alter its own protection.

For example, asset sales intended to improve liquidity may lower collateral values and reduce the barrier for other institutions.

The barrier becomes partly a product of the transition itself.

10.6.1 A Minimal Banking-System Example

Consider three banks:

$$\mathcal{V} = \{B_1, B_2, B_3\}. \tag{10.36}$$

Each bank holds a common asset (A), uses short-term wholesale funding, and maintains bilateral exposures.

The state of bank (i) is

$$\mathbf{x} * i = \left(k * i, l_i, f_i, a_i\right), \tag{10.37}$$

where k_i is capital, l_i liquidity, f_i funding stability, and a_i asset quality.

Suppose each bank individually satisfies its prudential thresholds.

A moderate loss reduces the capital of B_1. Bank B_1 sells asset (A). The sale lowers the market price of (A), reducing the capital of B_2 and B_3. Margin requirements rise. Wholesale lenders withdraw.

The pathways are:

1. direct loss at B_1;
2. common-asset price decline;
3. margin amplification;
4. funding withdrawal;
5. confidence contagion.

Each pathway may be individually subcritical.

Their synchronization may create material systemic transmission.

The total amplitude is

$$\Psi_A = \Psi_{\text{loss}} + \Psi_{\text{fire}} + \Psi_{\text{margin}} + \Psi_{\text{fund}} + \Psi_{\text{conf}}. \tag{10.38}$$

The adverse probability is

$$P(A) = |\Psi_A|^2 . \tag{10.39}$$

The model asks whether the timing and interaction of these pathways add explanatory value beyond a classical network-contagion model.

10.7 Distinguishing Systemic Tunneling from Classical Contagion

Classical contagion includes:

- direct counterparty loss;
- common shocks;
- fire-sale externalities;
- runs;
- network cascades;
- liquidity spirals.

These mechanisms can be represented with balance-sheet networks, agent-based models, threshold cascades, and stochastic processes.

A tunneling representation should not replace them automatically.

It becomes useful only when:

- a collective barrier can be defined;
- each individual pathway appears subcritical;
- material systemic probability nevertheless emerges;
- timing and synchronization materially affect the result;
- the amplitude model improves explanation or prediction.

The distinction is therefore similar to that established in Chapter 6.

Systemic tunneling is not a synonym for contagion. It is a candidate representation of sub-barrier systemic transmission shaped by pathway interaction and phase.

10.7.1 Systemic Coherence

Coherence refers to the persistence of phase relationships among systemic pathways.

A highly coherent system may display synchronized:

- asset allocation;
- risk measurement;
- collateral management;
- funding behaviour;
- liquidity hoarding;
- regulatory response.

Let

$$\theta_{i,t} \tag{10.40}$$

represent the phase of institution (i)'s response.

Here each $\theta_{i,t}$ must use the same prespecified origin and horizon as Equation (10.15); otherwise the cross-institution coherence statistic has no stable interpretation.

A simple coherence measure is

$$\mathcal{C} * t = \left| \frac{1}{N} \sum *i = 1^N \exp\left(\mathrm{i}\theta_{i,t}\right) \right|. \tag{10.41}$$

The value satisfies

$$0 \leq \mathcal{C}_t \leq 1. \tag{10.42}$$

A value near zero indicates dispersed responses. A value near one indicates strong synchronization.

High coherence may be beneficial in coordinated stabilization. It may be dangerous when institutions deleverage, hoard liquidity, or withdraw funding simultaneously.

The sign and consequence of coherence depend on the pathway.

10.8 Stress Testing the Network Geometry

Systemic stress testing should examine not only losses but changes in network structure and barrier integrity.

A systemic scenario may include:

- asset-price decline;
- funding-market closure;
- collateral haircut increase;
- operational outage;
- sovereign downgrade;
- deposit withdrawal;
- infrastructure failure.

The stress test should recompute:

$$\mathbf{W} * t, \quad \mathbf{L} * t, \quad V_{\text{sys}}, \quad \gamma^*_{\text{sys}}, \quad T_{\text{sys}}. \tag{10.43}$$

The key question is not merely whether capital falls below a threshold.

It is whether the geometry changes so that a previously remote adverse region becomes accessible through a lower-action path.

10.8.1 Macroprudential Intervention

The framework suggests four broad forms of systemic intervention.

Raise Barrier Height

Increase usable capital, liquidity, resolution resources, or infrastructure resilience.

Increase Barrier Width

Add time, stages, and institutional separation between local stress and systemic failure.

Examples include longer funding maturities, resolution stays, and diversified clearing arrangements.

Reduce Permeability

Limit concentration, common exposures, shared dependencies, and excessive interconnectedness.

Change Phase

Prevent synchronized harmful behaviour.

Countercyclical buffers, staggered margining, differentiated liquidity access, and coordinated communication can alter timing.

This yields the systemic resilience identity

$$\boxed{\begin{aligned}\text{Systemic Resilience} = {} & \text{Institutional Strength} + \text{Network Diversification} \\ & + \text{Barrier Capacity} + \text{Behavioural De-synchronization} \\ & + \text{Timely Coordinated Intervention.}\end{aligned}} \tag{10.44}$$

10.8.2 Governance and Reporting

A systemic-risk report should include:

1. the current systemic state;
2. the network topology;
3. institution-level fragility;
4. collective barrier capacity;
5. common dependencies;
6. behavioural coherence;
7. dominant systemic pathways;
8. minimum-action path;
9. systemic transmission estimate;
10. classical contagion benchmark;
11. intervention sensitivities;
12. model uncertainty.

The systemic probability may be decomposed as

$$P_{\text{sys}} = P_{\text{local}} + \Delta P_{\text{network}} + \Delta P_{\text{coherence}} + \Delta P_{\text{barrier}} + \varepsilon_{\text{model}}. \tag{10.45}$$

This decomposition separates institution-level fragility from network amplification, synchronization, barrier deterioration, and residual uncertainty.

10.9 What Systemic Tunneling Is Not

Systemic tunneling is not every financial crisis.

It is not ordinary counterparty contagion.

It is not simply a common shock.

It is not any network cascade.

It is not evidence that markets are literal quantum systems.

It is not justified because a complex model fits a historical crisis.

The classification is appropriate only when a collective barrier exists, a Freidlin–Wentzell or comparable classical rare-event and network benchmark has been fitted, phase has a prespecified origin and horizon, all probability outputs are normalized or separately calibrated, and the phase extension adds measurable out-of-sample value. Minimum action, exponential barrier sensitivity, and a dominant path are not sufficient because classical stochastic theory already produces those results.

10.9.1 A Practical Implementation Workflow

A practical implementation may follow ten steps:

1. Define the systemic adverse event.
2. Identify institutions, markets, and infrastructures.
3. Construct institution-level state vectors.
4. Build the exposure and dependency network.
5. Define stable, fragile, and adverse systemic regions.
6. Construct the collective barrier and its capacity constraints.
7. Identify contagion and behavioural pathways.
8. Estimate pathway magnitudes, phases, and coherence.
9. Calculate the dominant systemic path and transmission.
10. Compare policy interventions and classical benchmarks.

The resulting object may be summarized as

$$\boxed{\begin{aligned}\mathfrak{T}_{\text{sys}} = {} & \text{Institutional States} + \text{Network Geometry} \\ & + \text{Collective Barriers} + \text{Pathway Coherence} \\ & + \text{Policy Timing}.\end{aligned}} \tag{10.46}$$

10.10 Chapter Synthesis

This chapter has extended the tunneling framework from individual risks to institutional and systemic risk.

The systemic state is composed of institution-level states, market conditions, infrastructure dependencies, and the network linking them. The resulting state space is a networked manifold rather than a simple collection of independent institutions.

The stable systemic region contains states in which losses can be absorbed, funding remains differentiated, infrastructures continue to operate, and protective mechanisms retain capacity. The fragile region contains systems that remain functional but depend increasingly on concentrated funding, common collateral, central facilities, and synchronized expectations. The adverse region contains market freezes, cascading defaults, fire sales, payment disruption, and collective confidence collapse.

Systemic barriers include capital, liquidity, resolution, clearing, insurance, circuit breakers, infrastructure redundancy, and public support. These protections are collective and often capacity-constrained. They may protect one institution effectively but weaken when many institutions require them simultaneously.

The chapter demonstrated that individual resilience does not imply systemic resilience. Institutions may be individually solvent while sharing the same assets, funding sources, models, providers, and behavioural responses.

Systemic pathways include counterparty loss, funding withdrawal, margin spirals, fire sales, runs, infrastructure failure, sovereign feedback, and confidence contagion. Each pathway may be encoded as a complex amplitude whose phase represents timing and synchronization.

Constructive interference occurs when institutions act in the same direction at the same time. Destructive interference may be created by policy interventions that alter timing, separate reactions, restore confidence, or add liquidity before an adverse sequence becomes coherent.

The dominant systemic path is the route minimizing the integrated barrier action. It may pass through a central intermediary, shared infrastructure, collateral market, or confidence-sensitive funding channel rather than through the weakest standalone institution.

Systemic barriers are endogenous. Asset sales reduce collateral values, margin calls absorb liquidity, and repeated interventions consume policy capacity. The protective structure can therefore deteriorate as a consequence of the crisis response itself.

The model must remain subordinate to classical alternatives. Network contagion, common shocks, fire-sale models, and agent-based simulations remain the baseline. A tunneling representation is justified only when phase, coherence, and sub-barrier transmission add explanatory or predictive value.

The central proposition of Chapter 10 is:

> Systemic resilience depends not only on strengthening individual institutions, but on governing the network geometry, shared dependencies, collective barrier capacity, and synchronized behaviour through which local disturbances can become coherent pathways to systemic failure.

Chapter 11 will turn from application to detection. It will examine how hidden pathways, weakening barriers, phase alignment, and emerging minimum-action routes can be identified before adverse transmission becomes visible.

Part IV

Detection, Measurement, and Governance

Chapter 11

Detecting Hidden Pathways

11.1 From Theory to Empirical Detection

The first ten chapters developed a geometric and amplitude-based framework for understanding risk. They introduced state manifolds, metrics, stochastic mobility, protective barriers, transition pathways, complex amplitudes, phase, interference, barrier action, and tunneling transmission. These concepts were then applied to credit, operational, cyber, legal, regulatory, institutional, and systemic risk.

The present chapter marks a decisive transition.

The question is no longer only:

> How should hidden risk pathways be represented?

It is now:

> How can those pathways, barriers, amplitudes, and transmission mechanisms be identified, estimated, tested, and monitored using real evidence?

This empirical challenge determines whether the framework becomes a practical contribution to risk management or remains an attractive conceptual analogy.

Traditional monitoring frequently begins with thresholds. A borrower breaches a covenant, a liquidity ratio falls below a minimum, a control fails, an incident is detected, or a systemic indicator reaches a critical level. Such indicators remain important, but they often become visible only after the underlying structure has already deteriorated.

The tunneling framework seeks earlier signals.

It asks whether:

- the system is moving toward a fragile region;
- the covariance geometry is rotating toward an adverse boundary;
- the protective barrier is shrinking or becoming unreliable;
- a low-action pathway is emerging;
- several adverse mechanisms are becoming synchronized;

- probability weight is increasingly transmitted toward an adverse region.

The chapter's central empirical proposition is:

> Hidden pathways become empirically detectable when changes in state geometry, barrier integrity, directional mobility, pathway action, and temporal synchronization are measured jointly and shown to improve prediction and intervention beyond credible classical models.

11.1.1 The Empirical Detection Problem

Let the observed system state be

$$\mathbf{x}_{i,t} \in \mathcal{M}, \tag{11.1}$$

where (i) identifies the entity, institution, borrower, system, or network being studied, and $\mathcal{M}$ is the estimated state manifold.

Let

$$S, \qquad F, \qquad A \subset \mathcal{M} \tag{11.2}$$

denote stable, fragile, and adverse regions.

Let

$$\mathcal{B}_{i,t} \subset \mathcal{M} \tag{11.3}$$

be the protective barrier.

A conventional early-warning model estimates

$$P\left(\mathbf{x} * i, t + \Delta \in A \mid \mathcal{I} * i, t\right), \tag{11.4}$$

where $\mathcal{I}_{i,t}$ is the information set available at time (t).

The tunneling-detection problem is broader. It asks whether an adverse region is becoming more accessible even when the current state remains formally outside it.

A general detection object may be written as

$$\mathcal{D} * i, t = f\left(d * A, i, t, I_{\mathcal{B},i,t}, \mathbf{b} * i, t, \mathbf{A} * i, t, S^{*}_{R,i,t}, \mathcal{C} * i, t, T * R, i, t\right), \tag{11.5}$$

where:

- $d_{A,i,t}$ is distance to the adverse region;
- $I_{\mathcal{B},i,t}$ is barrier integrity;

- $\mathbf{b} * i, t$ is drift;
- $\mathbf{A} * i, t$ is covariance or mobility;
- $S^*_{R,i,t}$ is minimum barrier action;
- $\mathcal{C} * i, t$ is pathway coherence;
- (T*R,i,t) is estimated transmission.

No one component is sufficient.

A system may be close to an adverse region but protected by a strong and wide barrier. A barrier may be weakening while the state remains distant. A pathway may have low action but negligible probability weight. High pathway synchronization may be harmless if the synchronized mechanisms are stabilizing rather than adverse.

Detection must therefore combine geometry, protection, movement, pathways, and timing.

11.1.2 Three Empirical Layers

The theoretical objects introduced in the book are not observed directly in complete form. They must be estimated through explicit measurement models.

The empirical architecture should distinguish three layers:

$$\boxed{\begin{aligned}\text{Raw Observations} &\longrightarrow \text{Estimated Structural Objects}\\ &\longrightarrow \text{Risk Interpretation.}\end{aligned}} \tag{11.6}$$

Raw observations may include:

- financial statements;
- transactions;
- credit spreads;
- incident logs;
- control tests;
- collateral values;
- contracts;
- emails;
- regulatory communications;
- network exposures;
- event timestamps.

Estimated structural objects include:

- the state manifold;
- the metric;
- drift and covariance;
- barrier height and width;
- pathway probabilities;
- phase;
- coherence;
- barrier action;
- transmission.

Risk interpretation includes claims that:

- the system is becoming fragile;
- the barrier is narrowing;
- adverse mobility is increasing;
- a hidden path is emerging;
- pathways are constructively aligning;
- intervention is required.

This separation is essential.

A decline in liquidity may be observed. A narrowing liquidity barrier is estimated. A tunneling interpretation is a model-based conclusion.

11.1.3 A Staged Empirical Research Programme

The empirical implementation should not begin with a complete quantum-like model.

Attempting simultaneously to estimate the manifold, metric, barrier, amplitudes, phases, Hamiltonian, scale parameter, and transmission coefficient would create severe identification problems.

A more credible programme proceeds through four stages:

1. classical state geometry;
2. barrier and pathway reconstruction;
3. amplitude and phase extension;

4. operational detection and intervention.

The sequence may be summarized as

$$\boxed{\begin{aligned}\text{Classical Geometry} &\longrightarrow \text{Pathway Structure}\\ &\longrightarrow \text{Amplitude Extension}\\ &\longrightarrow \text{Detection System.}\end{aligned}} \tag{11.7}$$

This ordering ensures that amplitude-based variables extend a functioning empirical model rather than replace one.

11.1.4 Selecting an Empirical Laboratory

A first implementation should use a narrow and clearly defined class of systems.

A suitable empirical laboratory should provide:

1. repeated observations;
2. observable adverse events;
3. measurable protective structures;
4. several plausible transition pathways;
5. sufficient data to support out-of-sample validation.

Possible laboratories include:

- corporate borrowers observed quarterly;
- consumer accounts observed monthly;
- cyber systems observed through event logs;
- standardized operational processes;
- banking institutions connected through an exposure network.

Corporate lending is a particularly useful first application because leverage, liquidity, collateral, refinancing requirements, covenant headroom, ratings, spreads, and default events are conceptually interpretable.

Cyber and operational data may offer more precise event timing, but they often suffer from incomplete incident reporting, changing control architectures, and inconsistent taxonomies.

11.1.5 The Unit of Observation and Prediction Horizon

The empirical design must define both the unit of observation and the adverse transition.

The unit may be:

- a borrower-quarter;
- an account-month;
- a system-hour;
- a process-day;
- a legal matter-month;
- a network-date.

Let the observation record be

$$\mathcal{O} * i, t = (\mathbf{x} * i, t, \mathbf{p} * i, t, \mathbf{e} * i, t, \mathbf{n} * i, t, Y * i, t + \Delta)\,, \tag{11.8}$$

where:

- $\mathbf{x} * i, t$ contains state variables;
- $\mathbf{p} * i, t$ contains protection variables;
- $\mathbf{e} * i, t$ contains event and timing variables;
- $\mathbf{n} * i, t$ contains network or dependency information;
- $Y_{i,t+\Delta}$ indicates an adverse event within horizon Δ.

The prediction horizon must be explicit.

A twelve-month corporate-default model and a six-hour cyber-incident model require different states, pathways, barrier definitions, and phase variables.

11.2 Building a Timestamp-Correct Data Model

The empirical programme should begin with a coherent data model, not with the tunneling equation.

For each entity, the dataset should preserve:

- the date on which each variable became observable;
- the value available at the prediction date;
- subsequent revisions;
- missingness;
- management interventions;
- the exact timing of adverse events;
- censoring and sample-selection rules.

The information set is

$$\mathcal{I} * i, t = \sigma\left(\mathcal{O} * i, s : s \leq t\right). \tag{11.9}$$

Every probability, phase, barrier measure, and pathway estimate must use only information available within $\mathcal{I}_{i,t}$.

This prevents look-ahead bias.

A financial statement published after quarter-end cannot be treated as known at quarter-end. A vulnerability discovered after an incident cannot be inserted into the pre-event model unless a contemporaneous proxy was available.

11.2.1 Constructing the State Vector

For a corporate-credit pilot, define

$$\mathbf{x}_{i,t} = \begin{bmatrix} \ell_{i,t} \\ q_{i,t} \\ m_{i,t} \\ c_{i,t} \\ r_{i,t} \\ h_{i,t} \\ s_{i,t} \end{bmatrix}, \tag{11.10}$$

where:

- $\ell_{i,t}$ is leverage;
- $q_{i,t}$ is liquidity;
- $m_{i,t}$ is operating margin or cash-flow generation;
- $c_{i,t}$ is collateral coverage;
- $r_{i,t}$ is refinancing access;
- $h_{i,t}$ is covenant headroom;
- $s_{i,t}$ is sponsor or lender support.

Variables should be transformed so that directional movement has a consistent interpretation.

A standardized coordinate is

$$z_{j,i,t} = \frac{x_{j,i,t} - \mu_j}{\sigma_j}. \tag{11.11}$$

Some variables require nonlinear transformations. For example,

$$z_{q,i,t} = -\log\left(\frac{q_{i,t}}{q_{\text{safe}}}\right) \tag{11.12}$$

creates a liquidity-stress coordinate that rises as coverage deteriorates.

The design should also preserve metadata on data quality, revisions, source reliability, and reporting delay.

11.2.2 Estimating the State Manifold

The empirical manifold is a representation of feasible states and meaningful neighbourhoods.

Let the normalized state cloud be

$$\mathcal{X} = \{\mathbf{z} * i, t\} * i, t. \tag{11.13}$$

An embedding function

$$\mathcal{E}_{\boldsymbol{\vartheta}} : \mathbb{R}^p \longrightarrow \mathbb{R}^d, \qquad d < p, \tag{11.14}$$

produces latent coordinates

$$\mathbf{y} * i, t = \mathcal{E} * \boldsymbol{\vartheta}\left(\mathbf{z}_{i,t}\right). \tag{11.15}$$

Candidate methods include:

- principal component analysis;
- factor models;
- diffusion maps;
- spectral embeddings;
- locally linear embedding;
- autoencoders;
- variational autoencoders;
- supervised manifold learning.

The embedding should preserve more than visual separation.

A useful manifold should preserve:

- local neighbourhoods;
- temporal continuity;
- economically meaningful directions;

- recoverability of original variables;
- stability across samples.

A combined embedding objective may be

$$\mathcal{L}*\text{embed} = \lambda*N\mathcal{L}*\text{neighbourhood}+\lambda*T\mathcal{L}*\text{temporal}+\lambda*E\mathcal{L}*\text{economic}+\lambda*Y\mathcal{L}_{\text{outcome}}. \tag{11.16}$$

The manifold should not become merely a hidden classifier. Its geometry must remain interpretable.

11.2.3 Machine Learning for Manifold Estimation

An autoencoder defines

$$\mathbf{y}*i,t = \mathcal{E}*\boldsymbol{\vartheta}\left(\mathbf{x}_{i,t}\right), \tag{11.17}$$

and reconstructs

$$\widehat{\mathbf{x}}*i,t = \mathcal{D}*\boldsymbol{\phi}\left(\mathbf{y}_{i,t}\right). \tag{11.18}$$

The reconstruction loss is

$$\mathcal{L}*\text{rec} = \sum*i,t\left|\mathbf{x}*i,t-\widehat{\mathbf{x}}*i,t\right|^2. \tag{11.19}$$

A risk-aware autoencoder may use

$$\mathcal{L}*\text{manifold} = \lambda*\text{rec}\mathcal{L}*\text{rec}+\lambda*\text{temp}\mathcal{L}*\text{temp}+\lambda*\text{risk}\mathcal{L}*\text{risk}+\lambda*\text{mono}\mathcal{L}_{\text{mono}}. \tag{11.20}$$

The monotonicity term can prevent the model from learning economically implausible directions.

For example, increasing leverage while holding other variables constant should not normally move a borrower deeper into the stable region.

11.2.4 Estimating the Metric

The metric defines the meaning of risk distance.

A statistical metric may use inverse local covariance:

$$\mathbf{g}_{\text{stat}}\left(\mathbf{y}\right) = \widehat{\mathbf{A}}^{-1}\left(\mathbf{y}\right). \tag{11.21}$$

A structural metric may use loss, policy, or expert weights:

$$\mathbf{g} * \mathrm{str} = \mathrm{diag}\left(w * 1, \ldots, w_d\right). \tag{11.22}$$

A hybrid metric is

$$\mathbf{g} = \lambda_g \mathbf{g} * \mathrm{stat} + \left(1 - \lambda * g\right) \mathbf{g}_{\mathrm{str}}. \tag{11.23}$$

The metric should be validated by testing whether geometrically close states exhibit higher observed transition frequencies or lower transition costs.

11.2.5 Defining Stable, Fragile, and Adverse Regions

Let

$$Y_{i,t+\Delta} = \begin{cases} 1, & \text{if an adverse event occurs within } \Delta, \\ 0, & \text{otherwise.} \end{cases} \tag{11.24}$$

The adverse event may include default, distressed exchange, liquidity exhaustion, covenant acceleration, severe cyber loss, operational shutdown, legal sanction, or systemic impairment.

An empirical adverse region may be defined as

$$A = \left\{ \mathbf{y} : \widehat{P}\left(Y = 1 \mid \mathbf{y}\right) \geq \tau_A \right\}. \tag{11.25}$$

The fragile region is

$$F = \left\{ \mathbf{y} : \tau_F \leq \widehat{P}(Y = 1 \mid \mathbf{y}) < \tau_A \right\}. \tag{11.26}$$

The stable region is the low-risk remainder.

To reduce circularity, adverse-region boundaries should be estimated on training samples and frozen before out-of-sample validation. Cross-fitting may also be used.

11.3 Estimating Distance to the Adverse Region

The Euclidean distance is

$$d_A^{\mathrm{Euc}}\left(\mathbf{y} * i, t\right) = \min * \mathbf{a} \in A \left|\mathbf{y}_{i,t} - \mathbf{a}\right|. \tag{11.27}$$

A manifold-aware distance is

$$d_A\left(\mathbf{y} * i, t\right) = \inf * \gamma \int_\gamma \sqrt{\dot{\gamma}(u)^{\mathsf{T}} \mathbf{g}\left(\gamma(u)\right) \dot{\gamma}(u)}, \mathrm{d}u. \tag{11.28}$$

If the manifold is represented as a graph, geodesic distance may be estimated through shortest-path algorithms.

Distance is informative but incomplete. Two systems at the same distance may face very different barriers and mobility directions.

11.3.1 Estimating the Classical Law of Motion

The classical model provides the null hypothesis.

In discrete time,

$$\Delta \mathbf{y} * i, t = \mathbf{b}\left(\mathbf{y} * i, t, t\right) \Delta t + \boldsymbol{\varepsilon} * i, t + \mathbf{J} * i, t. \tag{11.29}$$

Candidate methods include:

- dynamic panel models;
- vector autoregressions;
- state-space models;
- hidden Markov models;
- regime-switching systems;
- Gaussian processes;
- neural ordinary differential equations;
- neural stochastic differential equations.

The local covariance is

$$\widehat{\mathbf{A}}\left(\mathbf{y} * i, t, t\right) = \operatorname{Cov}\left(\Delta \mathbf{y} * i, t \mid \mathbf{y}_{i,t}\right). \tag{11.30}$$

The classical model should account for drift, diffusion, jumps, regimes, and hidden states before amplitude-based mechanisms are introduced.

11.3.2 Neural Differential Equations

A neural ordinary differential equation estimates

$$\frac{\mathrm{d}\mathbf{y} * t}{\mathrm{d}t} = \mathbf{b} * \boldsymbol{\vartheta}\left(\mathbf{y}_t, t\right). \tag{11.31}$$

A neural stochastic differential equation estimates

$$\mathrm{d}\mathbf{y} * t = \mathbf{b} * \boldsymbol{\vartheta}\left(\mathbf{y} * t, t\right) \mathrm{d}t + \boldsymbol{\Sigma} * \boldsymbol{\phi}\left(\mathbf{y} * t, t\right) \mathrm{d}\mathbf{W} * t. \tag{11.32}$$

These methods are useful when dynamics are nonlinear, observations are irregular, and mobility is state-dependent.

They should be constrained to preserve positive-semidefinite covariance and plausible trajectories.

11.3.3 Directional Mobility as an Early-Warning Signal

Let

$$\mathbf{n}_{A,i,t} \tag{11.33}$$

be the local direction toward the adverse boundary.

The projected adverse drift is

$$b_{A,i,t} = \mathbf{n} * A, i, t^{\top} \widehat{\mathbf{b}} * i, t. \tag{11.34}$$

The projected adverse variance is

$$a_{A,i,t} = \mathbf{n} * A, i, t^{\top} \widehat{\mathbf{A}} * i, t \mathbf{n}_{A,i,t}. \tag{11.35}$$

A local mobility-warning index is

$$\mathcal{M} * A, i, t = \alpha * b b_{A,i,t} + \alpha_a \sqrt{a_{A,i,t}}. \tag{11.36}$$

Let the dominant covariance eigenvector satisfy

$$\widehat{\mathbf{A}} * i, t \mathbf{v} * 1, i, t = \lambda_{1,i,t} \mathbf{v}_{1,i,t}. \tag{11.37}$$

Its alignment with the adverse direction is

$$\cos \phi_{i,t} = \frac{\mathbf{v} * 1, i, t^{\top} \mathbf{n} * A, i, t}{|\mathbf{v} * 1, i, t| \, |\mathbf{n} * A, i, t|}. \tag{11.38}$$

An increasing value suggests that the principal direction of uncertainty is rotating toward danger.

11.3.4 Constructing the Barrier Empirically

The barrier must be constructed from actual protective mechanisms.

For corporate credit, define a nominal protection score

$$P_{i,t} = w_q q_{i,t} + w_c c_{i,t} + w_h h_{i,t} + w_r r_{i,t} + w_s s_{i,t}. \tag{11.39}$$

Nominal protection must be adjusted for availability and reliability.

For component (j),

$$\widetilde{P} * j, i, t = P * j, i, t A_{j,i,t} R_{j,i,t}, \tag{11.40}$$

where:

- $P_{j,i,t}$ is nominal size;
- $A_{j,i,t}$ is availability under stress;
- $R_{j,i,t}$ is reliability when exercised.

The effective protection score is

$$\widetilde{P} * i, t = \sum * j \widetilde{P}_{j,i,t}. \tag{11.41}$$

The barrier should be estimated as a finite region over which protection materially reduces the probability of entering (A).

11.3.5 Causal Estimation of Barrier Effectiveness

A protection variable is not automatically a causal barrier.

High-risk borrowers may be required to post more collateral. Complex systems may have more controls because they are inherently riskier. Sovereigns receiving support may already be distressed.

The empirical objective is to estimate

$$\frac{\partial P\left(Y_{i,t+\Delta} = 1\right)}{\partial P_{j,i,t}}. \tag{11.42}$$

Possible methods include:

- fixed-effects models;
- matched comparisons;
- instrumental variables;
- difference-in-differences;
- regression discontinuity;
- synthetic controls;
- causal forests;
- doubly robust learners.

A heterogeneous treatment effect may be written as

$$\tau_j(\mathbf{x}) = \mathbb{E}\left[Y(P_j = 1) - Y(P_j = 0) \mid \mathbf{X} = \mathbf{x}\right]. \tag{11.43}$$

This permits barrier strength to vary across states.

A liquidity facility may be effective for a viable borrower but nearly irrelevant for an insolvent one. A cyber control may be protective in one architecture but redundant in another.

11.4 Machine Learning for Barrier Estimation

A nonlinear barrier function may be estimated as

$$\widehat{B} * i, t = g * \boldsymbol{\phi}\left(\mathbf{p} * i, t, \mathbf{x} * i, t, \mathbf{r}_{i,t}\right), \tag{11.44}$$

where $\mathbf{r}_{i,t}$ contains reliability and scenario variables.

Tree-based models, generalized additive models, and gradient boosting can identify interactions such as:

- collateral protecting only when market liquidity remains available;
- covenants becoming effective only before refinancing pressure becomes acute;
- monitoring controls losing effectiveness above a workload threshold;
- official support depending on political conditionality.

Machine learning can therefore help estimate the shape and state dependence of the barrier.

11.4.1 Measuring Barrier Height, Width, and Integrity

Barrier height may be estimated as

$$H_{\mathcal{B},i,t} = \widetilde{P} * i, t - E * i, t, \tag{11.45}$$

where $E_{i,t}$ is an effective transition-capacity or stress measure.

Barrier width has several empirical interpretations.

Temporal width is

$$W_i^{\text{time}} = t_i^A - t_i^F. \tag{11.46}$$

Sequential width is

$$W_i^{\text{seq}} = \min_{\gamma} \left|\gamma \cap \mathcal{B}\right|. \tag{11.47}$$

Geometric width is

$$W_i^{\text{geo}} = \inf_{\gamma} \int_{\gamma \cap \mathcal{B}} \mathrm{d}s_{\mathbf{g}}. \tag{11.48}$$

A composite integrity measure is

$$I_{\mathcal{B},i,t} = \beta_H \widetilde{H} * i, t + \beta * W \widetilde{W} * i, t + \beta * C \widetilde{C} * i, t + \beta * D \widetilde{D}_{i,t}. \tag{11.49}$$

The change in integrity is

$$\Delta I_{\mathcal{B},i,t} = I_{\mathcal{B},i,t} - I_{\mathcal{B},i,t-1}. \tag{11.50}$$

A declining integrity measure may provide an early warning before visible state deterioration.

11.4.2 Estimating Control and Barrier Independence

Let $F_{j,t}$ be the failure indicator for protective layer (j).

A basic dependency matrix is

$$\mathbf{C} * F = \mathrm{Cov}\left(F * 1, t, \ldots, F_{K,t}\right). \tag{11.51}$$

A conditional graphical model may estimate

$$P\left(F_1, \ldots, F_K \mid Z\right), \tag{11.52}$$

where (Z) contains shared infrastructure, personnel, vendors, governance, or market conditions.

If λ_j are the eigenvalues of the dependency-adjusted matrix, the effective number of independent layers may be

$$K_{\text{eff}} = \frac{\left(\sum_{j=1}^{K} \lambda_j\right)^2}{\sum_{j=1}^{K} \lambda_j^2}. \tag{11.53}$$

This turns the qualitative idea of shared dependency into an empirical barrier-width measure.

11.4.3 Pathway Discovery

A pathway is an ordered sequence connecting the current state to the adverse region.

Candidate pathways may be identified using:

- expert-defined causal structures;
- sequence mining;
- process mining;
- event trees;

- temporal Bayesian networks;
- hidden Markov models;
- graph transition models;
- causal discovery;
- sequence-learning methods.

Let an event sequence be

$$\mathcal{E} * i = \{(e * i, 1, t_{i,1}), \ldots, (e_{i,m}, t_{i,m})\}. \tag{11.54}$$

A pathway should satisfy:

- temporal ordering;
- minimum support;
- elevated adverse probability;
- structural plausibility;
- out-of-sample stability.

For corporate credit, one pathway may be

$$\text{margin decline} \rightarrow \text{rating downgrade} \rightarrow \text{spread widening} \rightarrow \text{refinancing failure} \rightarrow \text{default}. \tag{11.55}$$

Another may be

$$\text{collateral decline} \rightarrow \text{covenant breach} \rightarrow \text{lender acceleration} \rightarrow \text{liquidity exhaustion}. \tag{11.56}$$

11.4.4 Sequence Models and Artificial Intelligence

Pathways are inherently sequential.

Methods include:

- hidden Markov models;
- hidden semi-Markov models;
- recurrent neural networks;
- long short-term memory networks;

- temporal convolutional networks;
- transformers;
- temporal point processes.

A sequence model may estimate

$$P\left(e_{t+1} \mid e_{1:t}, \mathbf{x}_t\right). \tag{11.57}$$

Such models can rank likely future pathways or estimate completion probabilities.

They are especially useful for:

- cyber-event logs;
- consumer-payment histories;
- rating and market-event sequences;
- legal and regulatory timelines;
- operational incident chains.

Attention weights may support interpretation, but they should not automatically be treated as causal explanations.

11.4.5 Temporal Point Processes

Let $N_k(t)$ count events of type (k). The conditional intensity is

$$\lambda_k(t) = \lim_{\Delta t \to 0} \frac{P\left(N_k(t+\Delta t) - N_k(t) = 1 \mid \mathcal{H}_t\right)}{\Delta t}. \tag{11.58}$$

A mutually exciting process may be

$$\lambda_k(t) = \mu_k + \sum_j \int_0^t \phi_{jk}(t-s), \mathrm{d}N_j(s). \tag{11.59}$$

This provides an important classical benchmark for timing interaction.

If ordinary event excitation explains the observed sequencing adequately, an amplitude model may add little value.

11.5 Graph Learning and Hidden Paths

Risk systems may be represented as graphs:

$$\mathcal{G} = \left(\mathcal{V}, \mathcal{E}, \mathbf{W}\right). \tag{11.60}$$

Nodes may represent borrowers, institutions, controls, systems, contracts, vendors, or events.

Edges may represent exposures, dependencies, transitions, or shared infrastructure.

Graph methods can estimate:

- shortest paths;
- minimum-resistance routes;
- central nodes;
- bottlenecks;
- communities;
- dependency concentration;
- propagation pathways.

A graph neural network may update node representations through

$$\mathbf{h} * i^{(\ell+1)} = \sigma\left[\mathbf{W} * 0^{(\ell)}\mathbf{h} * i^{(\ell)} + \sum * j \in \mathcal{N}(i)\alpha_{ij}^{(\ell)}\mathbf{W} * 1^{(\ell)}\mathbf{h} * j^{(\ell)}\right]. \tag{11.61}$$

Dynamic graph models can represent changing counterparty, funding, vendor, and infrastructure relationships.

The principal limitation is data completeness. An apparently hidden pathway may reflect a missing network edge rather than an actual tunneling mechanism.

11.5.1 Classical Path Resistance

Let the estimated probability of traversing edge (e) be

$$p_{e,t}. \tag{11.62}$$

Define classical edge resistance as

$$r_{e,t} = -\log p_{e,t}. \tag{11.63}$$

The resistance of path γ is

$$\mathcal{R} * t[\gamma] = \sum * e \in \gamma r_{e,t}. \tag{11.64}$$

The dominant classical path is

$$\gamma^*_{\mathrm{cl},t} = \arg, \min * \gamma \mathcal{R} * t[\gamma]. \tag{11.65}$$

This provides a transparent benchmark for the later barrier-action model.

11.5.2 Estimating Pathway Probabilities

For pathway (k), define

$$p_{k,i,t} = P\left(\gamma_k \text{ completes before } t + \Delta \mid \mathcal{I}_{i,t}\right). \tag{11.66}$$

Possible estimation methods include:

- competing-risk survival models;
- multistate models;
- hidden semi-Markov models;
- sequence transformers;
- temporal point processes;
- dynamic Bayesian networks.

The pathway magnitude is initially anchored in classical probability:

$$R_{k,i,t} = \sqrt{p_{k,i,t}}. \tag{11.67}$$

The amplitude is

$$\Psi_{k,i,t} = R_{k,i,t} \exp\left(\mathrm{i}\theta_{k,i,t}\right). \tag{11.68}$$

11.5.3 Estimating Phase

Phase is the most difficult empirical object in the framework.

It is not observed directly. It is a latent transformation of timing, sequencing, persistence, and synchronization variables.

First construct an effective timing coordinate in the same units as the decision horizon:

$$\tau^{\text{eff}}_{k,i,t} = \boldsymbol{\omega} * k^{\mathsf{T}} \mathbf{z} * k, i, t^{\text{phase}}, \tag{11.69}$$

where the phase-feature vector may contain:

- normalized time to maturity;
- overlap of adverse mechanisms;
- event-order consistency;
- duration of stress;
- detection delay;

- intervention delay;
- network synchronization.

Fix an origin $\tau_{0,i,t}$ and a prespecified period or decision horizon $H_{i,t} > 0$. Phase is then

$$\theta_{k,i,t} = 2\pi \left[\frac{\tau^{\text{eff}}_{k,i,t} - \tau_{0,i,t}}{H_{i,t}} \mod 1 \right]. \tag{11.70}$$

Under this mapping, a delay of $H/2$ corresponds to π. The origin, horizon, feature scaling, and wrapping rule must be fixed from institutional structure or training data before out-of-sample evaluation. If no defensible horizon exists, the model should use ordinary timing features rather than phase.

Regularization may be imposed through

$$\mathcal{L} * \text{phase} = \mathcal{L} * \text{prediction} + \lambda_\theta \sum_k |\boldsymbol{\omega} * k| * 1. \tag{11.71}$$

A phase term should be retained only when it is stable, interpretable, and linked to observable timing evidence.

11.5.4 Pathway Coherence

For (K) pathways, define

$$\mathcal{C} * i,t = \left| \frac{1}{K} \sum *k = 1^K \exp\left(\mathrm{i}\theta_{k,i,t}\right) \right|. \tag{11.72}$$

The index satisfies

$$0 \leq \mathcal{C}_{i,t} \leq 1. \tag{11.73}$$

A value near one indicates strong alignment.

The detection framework should distinguish adverse coherence from stabilizing coherence:

$$\mathcal{C} * i,t^-, \qquad \mathcal{C} * i,t^+. \tag{11.74}$$

High coherence among adverse refinancing, confidence, and collateral pathways is dangerous. High coherence among liquidity support, covenant relief, and sponsor intervention may be stabilizing.

11.6 Estimating Interference

For two pathways, define the raw structural amplitude score

$$Q^{\text{amp}}_{12,i,t} = p_{1,i,t} + p_{2,i,t} + 2\sqrt{p_{1,i,t}p_{2,i,t}}\cos\left(\Delta\theta_{12,i,t}\right). \tag{11.75}$$

The raw cross-term diagnostic is

$$I^{\text{raw}}_{12,i,t} = 2\sqrt{p_{1,i,t}p_{2,i,t}}\cos\left(\Delta\theta_{12,i,t}\right). \tag{11.76}$$

For multiple pathways,

$$Q^{\text{amp}}_{A,i,t} = \left|\sum_{k=1}^{K}\Psi_{k,i,t}\right|^2. \tag{11.77}$$

These raw scores are not probabilities when $R_k = \sqrt{p_k}$ is constructed from marginal pathway probabilities. The bounded structural alignment index is

$$J^{\text{amp}}_{A,i,t} = \frac{Q^{\text{amp}}_{A,i,t}}{\left(\sum_k R_{k,i,t}\right)^2}, \qquad 0 \le J^{\text{amp}}_{A,i,t} \le 1. \tag{11.78}$$

An adverse-event probability must be produced by a normalized full-state amplitude or by a separately calibrated prediction model using J^{amp}_A as an additional feature.

The empirical question is not whether this expression can fit historical outcomes.

The question is whether the phase-dependent structure adds stable value beyond:

- logistic interactions;
- tree ensembles;
- neural networks;
- copulas;
- Bayesian networks;
- point processes;
- sequence models.

Interference is empirically credible only if it predicts timing-dependent effects that strong classical models do not capture equally well.

11.6.1 Estimating the Barrier Action

For edge (e), define:

- $V_{e,i,t}$: barrier resistance;
- $E_{i,t}$: transition capacity;

- $M_{e,i,t}$: movement resistance;
- Δs_e: path length.

The discrete barrier action is

$$S_{R,i,t}[\gamma] = \sum_{e \in \gamma \cap \mathcal{B}} \sqrt{2M_{e,i,t}\left(V_{e,i,t} - E_{i,t}\right)} \Delta s_e. \tag{11.79}$$

The dominant tunneling path is

$$\gamma^*_{R,i,t} = \arg, \min *\gamma S * R, i, t[\gamma]. \tag{11.80}$$

The transmission proxy is

$$T_{R,i,t} = \exp\left[-\frac{2S_{R,i,t}\left[\gamma^*_{R,i,t}\right]}{\eta}\right]. \tag{11.81}$$

The scale parameter may be calibrated through log loss, likelihood, or Brier score rather than arbitrary assignment.

11.6.2 Scaling and Identification of the Action

The action combines quantities with potentially incompatible units.

One empirical strategy is to express components in log-probability units.

Define

$$V_{e,t} = -\log P\left(e \text{ is traversed} \mid \mathcal{I}_t\right). \tag{11.82}$$

Let resistance $M_{e,t}$ be inversely related to observed mobility.

A simpler alternative action score is

$$S_R[\gamma] = \sum_{e \in \gamma} \alpha_1 r_e + \alpha_2 w_e + \alpha_3 d_e, \tag{11.83}$$

where:

- r_e is transition resistance;
- w_e is barrier width;
- d_e is dependency concentration.

The theoretically inspired square-root formulation should be tested against this simpler benchmark.

11.6.3 Artificial Intelligence for Unstructured Evidence

Much of the information required to estimate barriers and pathways is unstructured.

It appears in:

- credit agreements;
- covenant documents;
- legal opinions;
- board minutes;
- audit reports;
- incident reports;
- regulatory communications;
- emails;
- management commentary.

A language model may transform document (d) into structured variables:

$$\widehat{\mathbf{z}} * d = \mathcal{F} * \mathrm{LLM}\,(d; \mathcal{S})\,, \tag{11.84}$$

where $\mathcal{S}$ is a controlled schema.

For a credit agreement, the schema may include:

- covenant thresholds;
- cure periods;
- cross-default clauses;
- collateral rights;
- material-adverse-change provisions;
- waiver mechanisms.

For an operational incident, it may include:

- initiating event;
- failed controls;
- event order;
- detection delay;

- escalation delay;
- recovery time;
- shared dependencies.

Language models can materially improve measurement of barrier quality, pathway order, and intervention timing.

11.6.4 Language Models as Measurement Instruments

A language model should be treated as a measurement instrument.

Its output is

$$\widehat{\mathbf{z}} * d = \mathbf{z} * d + \boldsymbol{\varepsilon}_d^{\mathrm{LLM}}, \tag{11.85}$$

where $\boldsymbol{\varepsilon}_d^{\mathrm{LLM}}$ is extraction error.

Validation should require:

- human-labelled benchmarks;
- field-level accuracy;
- source citations;
- abstention when evidence is insufficient;
- consistency testing;
- human approval for high-stakes fields.

The model should extract evidence, not invent barrier strength.

11.6.5 Retrieval-Augmented Evidence Systems

Let the controlled corpus be

$$\mathcal{K} = \{d_1, \ldots, d_M\}. \tag{11.86}$$

For query (q), retrieval selects

$$\mathcal{R}(q) \subset \mathcal{K}. \tag{11.87}$$

The language model produces

$$\widehat{\mathbf{z}} = \mathcal{F}_{\mathrm{LLM}}\left(q, \mathcal{R}(q)\right). \tag{11.88}$$

This is useful for reconstructing:

- barrier changes;
- covenant amendments;
- intervention timing;
- incident chronology;
- regulatory reinterpretation;
- management response.

Every extracted variable should remain linked to its supporting source.

11.7 AI Agents for Workflow Orchestration

A governed AI agent may assist with:

1. data ingestion;
2. schema validation;
3. document retrieval;
4. variable extraction;
5. pathway updating;
6. barrier-action recomputation;
7. report drafting;
8. exception escalation.

The agent should not possess unrestricted authority to alter production models or approve risk decisions.

The architecture should preserve

$$\text{Data} \to \text{Estimation} \to \text{Validation} \to \text{Decision}. \tag{11.89}$$

Each stage should have permissions, logs, and human accountability.

11.7.1 Generative AI and Synthetic Scenarios

A conditional generative model may produce plausible future trajectories:

$$\widetilde{\mathbf{x}} * t+1 : t+H^{(s)} \sim p * \boldsymbol{\vartheta}\left(\mathbf{x} * t+1 : t+H \mid \mathbf{x} * 1 : t, \mathbf{u}_t, s\right). \tag{11.90}$$

Generative models can explore:

- rare pathway combinations;
- alternative event ordering;
- moving barriers;
- counterfactual interventions;
- network cascades.

Synthetic trajectories should not be treated as observed evidence.

Their purpose is stress testing, falsification, method development, and identification of data gaps.

11.7.2 Reinforcement Learning for Intervention Design

Let the intervention be

$$\mathbf{u}_t \in \mathcal{U}. \tag{11.91}$$

The controlled state transition is

$$\mathbf{x} * t + 1 \sim P\left(\cdot \mid \mathbf{x} * t, \mathbf{u}_t\right). \tag{11.92}$$

An intervention policy may minimize

$$\min_{\pi} \mathbb{E} * \pi \left[\sum * t = 0^H \gamma^t \left(L_t + C(\mathbf{u}_t)\right)\right]. \tag{11.93}$$

Reinforcement learning may help compare actions that:

- raise barrier height;
- widen the barrier;
- reduce connectivity;
- change phase;
- accelerate recovery.

In high-stakes applications, reinforcement learning should remain a simulation and recommendation tool. Conservative policies, hard constraints, and human approval are essential.

11.7.3 Structure-Informed Machine Learning

The theoretical architecture can be used to constrain machine learning.

The model may impose:

- positive-semidefinite covariance;

- normalized probabilities;
- nonnegative barrier widths;
- accounting identities;
- monotonicity;
- continuity of trajectories.

For an amplitude model, define the residual

$$\mathcal{R} * \Psi = \mathrm{i}\eta \frac{\partial \Psi}{\partial t} - \widehat{H} * R\Psi. \tag{11.94}$$

A structure-informed model may minimize

$$\mathcal{L} = \mathcal{L} * \mathrm{data} + \lambda * \Psi \left| \mathcal{R}_{\Psi} \right|^2. \tag{11.95}$$

This does not prove that the system follows physical quantum mechanics. It tests whether the proposed effective equation provides useful regularization.

11.7.4 Hybrid Estimation Architecture

The most promising implementation is likely to be hybrid.

A possible stack is:

1. statistical hazard model for the baseline;
2. supervised autoencoder for the manifold;
3. neural SDE or dynamic panel model for movement;
4. causal learner for barrier effectiveness;
5. sequence model for pathway discovery;
6. graph model for dependencies;
7. language model for document extraction;
8. constrained phase and interference layer;
9. Bayesian or bootstrap uncertainty layer.

The final prediction may combine

$$\widehat{P} * i, t = \mathcal{C} \left(\widehat{P} * i, t^{\mathrm{classical}}, \widehat{T} * i, t^{\mathrm{barrier}}, \widehat{\Delta P} * i, t^{\mathrm{interference}}, \widehat{P}_{i,t}^{\mathrm{network}} \right), \tag{11.96}$$

where $\mathcal{C}$ is a calibrated combination rule.

11.7.5 The Detection Score

Define normalized components:

$$
\begin{aligned}
D_{\text{state},i,t} &= f_1\left(d_{A,i,t}\right), & (11.97)\\
D_{\text{barrier},i,t} &= f_2\left(I_{\mathcal{B},i,t}, \Delta I_{\mathcal{B},i,t}\right), & (11.98)\\
D_{\text{mobility},i,t} &= f_3\left(b_{A,i,t}, a_{A,i,t}, \phi_{i,t}\right), & (11.99)\\
D_{\text{action},i,t} &= f_4\left(S^*_{R,i,t}, \Delta S^*_{R,i,t}\right), & (11.100)\\
D_{\text{phase},i,t} &= f_5\left(\mathcal{C}_{i,t}, \Delta\theta_{i,t}\right), & (11.101)\\
D_{\text{transmission},i,t} &= f_6\left(T_{R,i,t}\right). & (11.102)
\end{aligned}
$$

The composite score is

$$\mathcal{D}_{i,t} = \sum_{j=1}^{6} w_j D_{j,i,t}, \qquad \sum_{j=1}^{6} w_j = 1. \tag{11.103}$$

The score should trigger escalation rather than mechanically classify tunneling.

An escalation rule may be

$$\mathcal{L}_{i,t} = \begin{cases} 0, & \mathcal{D}_{i,t} < \tau_1, \\ 1, & \tau_1 \le \mathcal{D}_{i,t} < \tau_2, \\ 2, & \tau_2 \le \mathcal{D}_{i,t} < \tau_3, \\ 3, & \mathcal{D}_{i,t} \ge \tau_3. \end{cases} \tag{11.104}$$

11.8 Detection as Explanation

The model should explain why risk is rising.

The report should identify:

- which state variable is deteriorating;
- which barrier component is weakening;
- whether covariance is rotating toward danger;
- which pathway has the lowest action;
- which timing relationship creates coherence;
- which intervention would change the geometry.

A useful explanation is structural:

> The warning increased because refinancing access deteriorated, the effective liquidity barrier narrowed, and refinancing and confidence pathways became temporally aligned.

11.8.1 Mapping Detection to Intervention

State deterioration may require exposure reduction, liquidity support, or operational correction.

Barrier deterioration may require collateral, stronger controls, additional redundancy, legal clarification, or recapitalization.

Mobility rotation may require concentration reduction or directional hedging.

Declining path action may require closing the lowest-resistance corridor.

Phase alignment may require changing timing, accelerating detection, delaying harmful actions, or advancing stabilizing support.

Increasing transmission may require escalation, stress testing, and management intervention.

The value of the model lies not only in predicting risk but in revealing how it may be reduced.

11.8.2 Validation Architecture

The framework must be validated rigorously.

Backtesting

Test whether the score rose before historical events.

Out-of-Sample Prediction

Use temporal and entity-based holdout samples.

Relevant metrics include:

- area under the receiver-operating curve;
- precision-recall;
- Brier score;
- log loss;
- calibration;
- lead time;
- false-warning rate.

Ablation

Remove individual components.

For phase,

$$\Delta V_{\text{phase}} = V_{\text{full}} - V_{\text{without phase}}. \tag{11.105}$$

Similar tests should be performed for barrier movement, action, and covariance rotation.

Placebo Tests

Stable periods and unaffected peer groups should not generate persistent warnings.

Mechanism Validation

The predicted dominant pathway should be compared with documented post-event mechanisms.

Intervention Validation

Test whether interventions predicted to change phase, barrier width, or action produced the expected effect.

11.8.3 Comparing Classical and Tunneling Models

Let

$$\widehat{P}_{i,t}^{\text{cl}} \tag{11.106}$$

be the classical prediction, and

$$\widehat{P}_{i,t}^{\text{tun}} \tag{11.107}$$

the tunneling-extended prediction.

The incremental value is

$$\Delta V = V\left(\widehat{P}^{\text{tun}}\right) - V\left(\widehat{P}^{\text{cl}}\right). \tag{11.108}$$

The extension should be retained only if

$$\Delta V > 0 \tag{11.109}$$

out of sample and under relevant regime changes.

A more complex model that fits history but produces unstable phases, poor calibration, or excessive false alarms should be rejected.

11.8.4 Rare Events and Synthetic Experiments

Severe events may be too rare for conventional estimation.

Synthetic simulations can test whether the detection framework incorrectly labels rare classical events as tunneling.

A simulation model may be

$$\mathbf{x} * t + 1 = \mathcal{F}\left(\mathbf{x} * t, \mathbf{u} * t, \boldsymbol{\varepsilon} * t\right). \tag{11.110}$$

The framework should be tested on:

- purely classical systems;
- systems with jumps;
- moving barriers;
- omitted variables;
- timing interactions;
- explicit amplitude-like structures.

A valid method should not diagnose tunneling simply because an event is rare, nonlinear, or surprising.

11.8.5 Uncertainty Quantification

Every output should include uncertainty.

For the detection score,

$$\widehat{\mathcal{D}} * i, t \pm \mathrm{CI} * i, t. \tag{11.111}$$

For the dominant path,

$$P\left(\gamma^* = \gamma_k \mid \mathcal{I}_{i,t}\right). \tag{11.112}$$

Transmission uncertainty may be decomposed as

$$\operatorname{Var}\left(\widehat{T} * R\right) = V * \text{data} + V_{\text{barrier}} + V_{\text{phase}} + V_{\text{model}}. \tag{11.113}$$

This prevents tunneling risk from being presented with unjustified precision.

11.9 AI-Specific Model Risks

Artificial-intelligence methods introduce additional risks:

- training-data leakage;
- label leakage;
- hallucinated extraction;

- unstable embeddings;
- spurious pathway discovery;
- uncalibrated probabilities;
- discriminatory proxies;
- drift;
- weak reproducibility;
- excessive automation.

Controls should include:

- temporal validation;
- independent review;
- source-grounded extraction;
- model cards;
- uncertainty thresholds;
- version control;
- drift monitoring;
- fallback classical models;
- human approval.

AI should not undermine the governance-first orientation of the framework.

11.9.1 A Minimum Viable Empirical Pilot

A first corporate-credit pilot might include:

1. a three-dimensional state manifold using leverage, liquidity, and refinancing access;
2. default or distressed exchange as the adverse event;
3. a barrier score based on liquidity, collateral, and covenant headroom;
4. three pathways: operating deterioration, refinancing failure, and collateral-covenant escalation;
5. pathway probabilities estimated through competing-risk survival models;
6. phase proxies based on maturity timing, spread reaction, and intervention delay;
7. graph-based minimum-action paths;

8. a composite detection score;
9. comparison with a conventional hazard model.

A credible estimation stack might include:

- a gradient-boosted survival model;
- a supervised autoencoder;
- a dynamic panel model or neural SDE;
- a causal forest for barrier effects;
- a competing-risk pathway model;
- a constrained phase model;
- a graph shortest-path algorithm;
- a retrieval-augmented language model for covenant extraction;
- bootstrap or Bayesian uncertainty estimation.

Only after this pilot demonstrates stable incremental value should the implementation expand to continuous amplitude evolution, dynamic Hamiltonians, and richer network structures.

11.9.2 A Progressive Implementation Strategy

The implementation should advance through six levels.

Level One: Interpretable Statistical Baseline

Estimate hazards, states, and protection effects with conventional methods.

Level Two: Nonlinear Machine Learning

Use tree-based and additive models to identify nonlinear risk regions and heterogeneous barriers.

Level Three: Representation and Sequence Learning

Estimate manifolds and pathways through autoencoders, sequence models, and graphs.

Level Four: AI-Assisted Evidence Engineering

Use language models to extract structured variables from documents and incident narratives.

Level Five: Constrained Amplitude Estimation

Estimate phase, interference, action, and transmission under structural restrictions.

Level Six: Decision and Intervention Models

Use causal analysis, simulation, and conservative reinforcement learning to compare interventions.

Progression should require evidence that the previous level is stable and useful.

11.9.3 Governance and Reproducibility

The empirical system should maintain an auditable evidence package containing:

- data definitions;
- timestamps;
- transformations;
- manifold specifications;
- barrier rules;
- pathway definitions;
- phase variables;
- estimation code;
- validation results;
- rejected models;
- intervention assumptions;
- limitations.

Committee reporting should distinguish:

$$\text{Observed Data}, \qquad \text{Estimated Structure}, \qquad \text{Model Interpretation}. \tag{11.114}$$

This distinction is especially important for phase, action, and transmission, which are model-derived rather than directly observed.

11.9.4 What Detection Can Establish

A rising detection score does not prove that an adverse event will occur.

It indicates that the adverse region is becoming more accessible through a combination of:

- state movement;
- weakening protection;
- adverse mobility;

- declining path action;
- pathway synchronization;
- increasing transmission.

A post-event reconstruction cannot prove that a system experienced physical quantum tunneling.

The appropriate conclusion is:

> The evidence is consistent with an increasingly accessible sub-barrier pathway that is represented more effectively by the extended model than by the selected classical benchmarks.

11.9.5 The Minimum Detection Architecture

The empirical framework may be summarized as

$$\mathfrak{D} * i, t = \left(\widehat{\mathcal{M}}, \widehat{\mathbf{g}}, \widehat{S}, \widehat{F}, \widehat{A}, \widehat{\mathbf{b}}, \widehat{\mathbf{A}}, \widehat{\mathcal{B}}, \widehat{I} * \mathcal{B}, \widehat{\Gamma}, \widehat{\Psi}, \widehat{S} * R^{*}, \widehat{T} * R, \mathcal{D}_{i,t}\right). \tag{11.115}$$

The complete empirical logic is

$$\boxed{\begin{aligned} \text{Timestamped Evidence} &\longrightarrow \text{State Geometry} \\ &\longrightarrow \text{Barrier Reconstruction} \longrightarrow \text{Pathway Discovery} \\ &\longrightarrow \text{Phase and Action} \longrightarrow \text{Validated Warning} \\ &\longrightarrow \text{Governed Intervention.} \end{aligned}} \tag{11.116}$$

11.10 Chapter Synthesis

This chapter has transformed the tunneling framework into an empirical research and implementation programme.

Detection begins before formal barrier breach. The objective is to identify structural changes that make an adverse region increasingly accessible while the system may still appear conventionally safe.

The empirical programme begins with timestamp-correct data, a narrow unit of analysis, an observable adverse event, and measurable protective mechanisms.

The state manifold is estimated from feasible configurations. Its metric defines meaningful distance. Stable, fragile, and adverse regions are constructed without contaminating out-of-sample evaluation.

The classical law of motion is estimated first. Drift, covariance, jumps, regimes, and hidden states form the null model.

Directional mobility provides an early warning when drift and covariance rotate toward the adverse boundary.

The protective barrier is constructed from reliability-adjusted controls, liquidity, collateral, covenants, legal rights, infrastructure, or policy capacity. Its causal effectiveness, height, width, continuity, independence, and movement are estimated separately from the state.

Candidate pathways are reconstructed through process mining, sequence models, temporal point processes, graph methods, causal structures, and expert evidence.

Pathway magnitudes remain anchored in classical probabilities. Phase is estimated from observable timing, persistence, sequencing, synchronization, and intervention delay. Interference is accepted only if it adds stable value beyond strong classical interaction models.

Barrier action identifies the most accessible route through the protective structure. Transmission converts that route into a comparable risk indicator.

Machine learning and artificial intelligence can contribute to manifold learning, nonlinear dynamics, barrier estimation, pathway discovery, graph analysis, document extraction, scenario generation, and intervention design. Their role is to estimate difficult empirical objects, not to invent them.

A composite detection score combines state proximity, barrier integrity, directional mobility, path action, phase coherence, and transmission. It supports escalation and intervention rather than mechanically declaring a tunneling event.

The framework must be tested through backtesting, temporal validation, entity validation, ablation, placebo tests, mechanism reconstruction, intervention analysis, and uncertainty quantification.

Its empirical legitimacy depends on incremental value over classical models.

The central proposition of Chapter 11 is:

> Hidden risk pathways can be detected empirically when structured and unstructured evidence is transformed into a coherent model of state geometry, barrier integrity, directional mobility, pathway action, and temporal synchronization—and when that model produces earlier, more accurate, more explainable, and more actionable warnings than credible classical alternatives.

Chapter 12 will move from detection to measurement. It will define tunneling-risk indicators, barrier-action measures, pathway concentration statistics, coherence metrics, transmission estimates, and reporting structures that permit comparison across systems and through time.

Chapter 12

Measuring Tunneling Risk

12.1 From Detection to Measurement

Chapter 11 transformed the tunneling framework into an empirical detection programme.

It showed how raw observations may be converted into estimated structural objects:

$$\text{Data} \longrightarrow \widehat{\mathcal{M}}, \widehat{\mathbf{g}}, \widehat{\mathbf{b}}, \widehat{\mathbf{A}}, \widehat{\mathcal{B}}, \widehat{\Gamma}, \widehat{\boldsymbol{\theta}}, \widehat{S} * R, \widehat{T} * R. \tag{12.1}$$

The purpose of the present chapter is different.

Detection asks:

> Is a hidden adverse pathway becoming increasingly accessible?

Measurement asks:

> How large is that accessibility, what produces it, how uncertain is it, how does it compare with other systems and periods, and what decision should follow?

Chapter 12 therefore does not introduce a separate empirical framework. It continues the one established in Chapter 11.

The estimation pipeline remains:

$$\boxed{\begin{aligned} \text{Timestamped Evidence} &\longrightarrow \text{Estimated Geometry} \\ &\longrightarrow \text{Barrier Reconstruction} \longrightarrow \text{Pathway Estimation} \\ &\longrightarrow \text{Phase and Action} \longrightarrow \text{Detection} \\ &\longrightarrow \text{Measurement} \longrightarrow \text{Decision.} \end{aligned}} \tag{12.2}$$

The empirical objects estimated in Chapter 11 become the inputs to the measurement system developed here.

The chapter's central proposition is:

> Tunneling risk becomes operationally measurable when estimated state geometry, barrier integrity, pathway accessibility, temporal coherence, and transmission are

converted into normalized, uncertainty-aware, decision-linked indicators that remain valid beyond the sample in which they were estimated.

12.1.1 The Measurement Problem

Let the minimum detection architecture from Chapter 11 be

$$\widehat{\mathfrak{D}} * i,t = \left(\widehat{\mathcal{M}}, \widehat{\mathbf{g}}, \widehat{S}, \widehat{F}, \widehat{A}, \widehat{\mathbf{b}}, \widehat{\mathbf{A}}, \widehat{\mathcal{B}}, \widehat{I} * \mathcal{B}, \widehat{\Gamma}, \widehat{\Psi}, \widehat{S} * R^{*}, \widehat{T} * R\right). \tag{12.3}$$

The task of Chapter 12 is to construct a measurement vector

$$\mathbf{M} * i, t^{\mathrm{tun}} = \left(D * i, t, B_{i,t}, M_{i,t}, A_{i,t}, C_{i,t}, I_{i,t}, T_{i,t}, U_{i,t}\right), \tag{12.4}$$

where:

- $D_{i,t}$ measures state distance;
- $B_{i,t}$ measures barrier condition;
- $M_{i,t}$ measures adverse mobility;
- $A_{i,t}$ measures pathway action;
- $C_{i,t}$ measures coherence;
- $I_{i,t}$ measures interference;
- $T_{i,t}$ measures transmission;
- $U_{i,t}$ measures uncertainty.

The measurement system should satisfy six requirements:

1. empirical traceability;
2. comparability through time;
3. comparability across entities;
4. decomposition into interpretable components;
5. uncertainty quantification;
6. direct connection to intervention.

A measurement that cannot be traced back to data is not auditable.

A measure that cannot be compared through time cannot support monitoring.

A score that combines unrelated quantities without transparent normalization cannot support governance.

12.1.2 Measurement Is Not Re-estimation

An important distinction must be maintained.

Chapter 11 estimated structural objects. Chapter 12 measures risk using those estimates.

The estimation problem is

$$\widehat{\mathbf{\Theta}} = \arg, \min *\mathbf{\Theta}\mathcal{L}\left(\mathbf{\Theta}; \mathcal{D} * \mathrm{train}\right), \tag{12.5}$$

where $\mathbf{\Theta}$ contains manifold, barrier, pathway, phase, and transmission parameters.

The measurement problem is

$$\mathbf{M} * i, t^{\mathrm{tun}} = \mathcal{Q}\left(\mathcal{I} * i, t; \widehat{\mathbf{\Theta}}\right), \tag{12.6}$$

where $\mathcal{Q}$ is the fixed measurement function.

The parameters should not be refitted every time a risk score is calculated. Otherwise, changes in the score may reflect model re-estimation rather than genuine changes in risk.

Production measurement should therefore use a versioned parameter set over a defined monitoring period.

12.1.3 The Measurement Window

Every tunneling-risk measure should specify:

- an observation date (t);
- a prediction horizon Δ;
- a reference distribution;
- a calibration version;
- an information cutoff.

Write the measure explicitly as

$$\mathbf{M}_{i,t}^{\mathrm{tun}}\left(\Delta; \mathcal{R}, v\right), \tag{12.7}$$

where $\mathcal{R}$ is the reference population and (v) is the model version.

A one-year credit-risk measure cannot be compared mechanically with a one-month liquidity measure. A systemic-risk transmission measure calibrated during normal conditions may require separate interpretation during crisis.

The horizon and reference class are part of the measure.

12.1.4 State Distance

The first component is distance from the current state to the adverse region.

Using the metric estimated in Chapter 11,

$$d_{A,i,t} = \inf_{\gamma:\mathbf{y}*i,t\to A} \int *\gamma \sqrt{\dot{\gamma}(u)^{\mathsf{T}} \widehat{\mathbf{g}}\left(\gamma(u)\right) \dot{\gamma}(u)}, \mathrm{d}u. \tag{12.8}$$

Distance should be measured not only to the adverse region but also to the fragile region:

$$d_{F,i,t} = \inf_{\gamma:\mathbf{y}*i,t\to F} \int *\gamma \mathrm{d}s_{\widehat{\mathbf{g}}}. \tag{12.9}$$

A normalized proximity score may be

$$D_{i,t} = \exp\left(-\frac{d_{A,i,t}}{\kappa_D}\right), \tag{12.10}$$

where κ_D is calibrated from the empirical distribution of pre-event distances.

The value lies in ((0,1]). Higher values indicate greater proximity.

However, distance alone is not a tunneling measure.

A system may be geometrically close to the adverse region but protected by a strong barrier. Another may be farther away but face a weak, narrow, or highly permeable barrier.

12.1.5 Empirical Calibration of Distance

The scale parameter κ_D should be estimated rather than chosen arbitrarily.

One possibility is

$$\widehat{\kappa} * D = \operatorname{median}\left\{d * A, i, t : Y_{i,t+\Delta} = 1\right\}, \tag{12.11}$$

computed within the training sample.

A more formal calibration may minimize prediction loss:

$$\widehat{\kappa} * D = \arg, \min *\kappa > 0 \sum_{i,t} \mathcal{L}\left[Y_{i,t+\Delta}, \exp\left(-\frac{d_{A,i,t}}{\kappa}\right)\right]. \tag{12.12}$$

Distance should also be reported in percentile form:

$$P_{D,i,t} = \widehat{F} * D\left(D * i, t\right), \tag{12.13}$$

where $\widehat{F}_D$ is the reference empirical distribution.

This permits comparison among entities within the same risk class.

12.1.6 Barrier Measurement

The empirical barrier was reconstructed in Chapter 11 from nominal protection, availability, reliability, width, continuity, and independence.

The barrier measurement vector is

$$\mathbf{B} * i,t = \left(H * i,t, W_{i,t}, C^B_{i,t}, R^B_{i,t}, K^{\text{eff}}_{i,t}, Q^B_{i,t}\right), \tag{12.14}$$

where:

- $H_{i,t}$ is barrier height;
- $W_{i,t}$ is barrier width;
- $C^B_{i,t}$ is continuity;
- $R^B_{i,t}$ is reliability;
- $K^{\text{eff}}_{i,t}$ is the effective number of independent layers;
- $Q^B_{i,t}$ is residual barrier capacity.

A normalized barrier-integrity score may be

$$B^{\text{int}}_{i,t} = \sum_{j=1}^{6} \beta_j Z_{j,i,t}, \qquad \sum_{j=1}^{6} \beta_j = 1, \tag{12.15}$$

where $Z_{j,i,t}$ are normalized barrier components.

Because stronger barriers reduce risk, define barrier weakness as

$$B_{i,t} = 1 - B^{\text{int}}_{i,t}. \tag{12.16}$$

High $B_{i,t}$ therefore indicates weak protection.

12.1.7 Barrier Trend

The current level of barrier integrity is only part of the problem.

The rate of deterioration may be more informative:

$$\dot{B} * i,t^{\text{int}} \approx \frac{B * i,t^{\text{int}} - B^{\text{int}}_{i,t-h}}{h}. \tag{12.17}$$

A barrier-deterioration score may be

$$B^{\text{trend}}_{i,t} = \max\left\{0, -\dot{B}^{\text{int}}_{i,t}\right\}. \tag{12.18}$$

The complete barrier score is

$$B_{i,t}^{\text{total}} = \lambda_B B_{i,t} + (1 - \lambda_B)\, B_{i,t}^{\text{trend}}. \tag{12.19}$$

This distinction permits the model to separate a persistently weak barrier from a strong barrier that is deteriorating rapidly.

12.2 Residual Barrier Capacity

Barrier height should be interpreted relative to the current stress load.

Define residual capacity as

$$Q_{i,t}^{B} = \widetilde{P} * i, t - L * i, t^{\text{stress}}, \tag{12.20}$$

where:

- $\widetilde{P} * i, t$ is reliability-adjusted protection;
- $L * i, t^{\text{stress}}$ is the current or scenario-adjusted load.

The utilization ratio is

$$U_{i,t}^{B} = \frac{L_{i,t}^{\text{stress}}}{\widetilde{P}_{i,t}}. \tag{12.21}$$

A ratio near one indicates that protective capacity is nearly exhausted.

For systemic applications, the denominator should reflect total usable collective capacity, not nominal institutional capacity.

12.2.1 Directional Mobility

The empirical law of motion from Chapter 11 provides drift and covariance.

Let $\mathbf{n}_{A,i,t}$ be the local adverse direction.

Projected drift is

$$b_{A,i,t} = \mathbf{n} * A, i, t^{\mathsf{T}} \widehat{\mathbf{b}} * i, t. \tag{12.22}$$

Projected covariance is

$$a_{A,i,t} = \mathbf{n} * A, i, t^{\mathsf{T}} \widehat{\mathbf{A}} * i, t \mathbf{n}_{A,i,t}. \tag{12.23}$$

A standardized adverse-mobility score is

$$M_{i,t} = \omega_b Z\left(b_{A,i,t}\right) + \omega_a Z\left(\sqrt{a_{A,i,t}}\right) + \omega_\phi Z\left(\cos\phi_{i,t}\right), \tag{12.24}$$

where $\phi_{i,t}$ is the angle between the dominant covariance direction and the adverse boundary.

This measure captures three different phenomena:

- expected movement toward danger;
- increasing uncertainty in the adverse direction;
- rotation of the mobility geometry toward the adverse boundary.

12.2.2 Mobility Acceleration

Risk may increase because adverse mobility is accelerating even when its current level remains moderate.

Define

$$\Delta M_{i,t} = M_{i,t} - M_{i,t-h}. \tag{12.25}$$

A persistent positive trend may be measured by

$$M_{i,t}^{\mathrm{pers}} = \frac{1}{m} \sum_{j=0}^{m-1} \mathbb{I}\left(\Delta M_{i,t-j} > 0\right). \tag{12.26}$$

This is useful when transient noise should be distinguished from sustained rotation toward the adverse region.

12.2.3 Minimum Barrier Action

The central pathway measure is minimum barrier action.

For candidate path γ,

$$S_{R,i,t}[\gamma] = \sum_{e \in \gamma \cap \mathcal{B}} \sqrt{2M_{e,i,t}\left(V_{e,i,t} - E_{i,t}\right)}\Delta s_e. \tag{12.27}$$

The minimum action is

$$S_{R,i,t}^{*} = \min_{\gamma \in \Gamma_{i,t}} S_{R,i,t}[\gamma]. \tag{12.28}$$

Low action indicates an accessible route through the barrier.

To ensure that high values represent greater risk, define the normalized action-risk score as

$$A_{i,t} = \exp\left(-\frac{S_{R,i,t}^{*}}{\kappa_A}\right). \tag{12.29}$$

The parameter κ_A is estimated from historical pre-event action levels.

12.2.4 Action Trend

The direction of action matters.

Define

$$\Delta S^*_{R,i,t} = S^*_{R,i,t} - S^*_{R,i,t-h}. \tag{12.30}$$

A negative value indicates that the dominant route is becoming easier.

An action-compression score is

$$A^{\text{comp}}_{i,t} = \max\left\{0, -\Delta S^*_{R,i,t}\right\}. \tag{12.31}$$

The full action-risk indicator may combine current action and its change:

$$A^{\text{total}}_{i,t} = \lambda_A A_{i,t} + (1-\lambda_A)\, Z\left(A^{\text{comp}}_{i,t}\right). \tag{12.32}$$

12.2.5 Dominant-Path Stability

The identity of the minimum-action pathway may itself be unstable.

Let

$$\gamma^*_{i,t} = \arg, \min *\gamma S * R, i, t[\gamma]. \tag{12.33}$$

Define path persistence over a window of (m) observations:

$$P^{\gamma}_{i,t} = \frac{1}{m}\sum_{j=0}^{m-1} \mathbb{I}\left(\gamma^*_{i,t-j} = \gamma^*_{i,t}\right). \tag{12.34}$$

High persistence suggests a structurally dominant route.

Low persistence may indicate:

- model instability;
- rapidly changing geometry;
- several near-equivalent paths;
- poor pathway identification.

The identity and stability of the dominant path should therefore be reported together.

12.2.6 Pathway Concentration

The probability or amplitude weight may be concentrated in one path or distributed among many.

Let normalized pathway weights be

$$\pi_{k,i,t} = \frac{R_{k,i,t}^2}{\sum_{j=1}^K R_{j,i,t}^2}. \tag{12.35}$$

A Herfindahl-style concentration measure is

$$H_{\Gamma,i,t} = \sum_{k=1}^{K} \pi_{k,i,t}^2. \tag{12.36}$$

The effective number of pathways is

$$K_{\Gamma,i,t}^{\text{eff}} = \frac{1}{H_{\Gamma,i,t}}. \tag{12.37}$$

High concentration means one or two pathways dominate.

Low concentration means vulnerability is dispersed.

This has direct intervention implications:

- concentrated risk may be reduced by targeting the dominant route;
- dispersed risk may require broad barrier redesign.

12.2.7 Pathway Entropy

An alternative measure is pathway entropy:

$$\mathcal{H} * \Gamma, i, t = -\sum *k = 1^K \pi_{k,i,t} \log \pi_{k,i,t}. \tag{12.38}$$

Normalize it as

$$\widetilde{\mathcal{H}} * \Gamma, i, t = \frac{\mathcal{H} * \Gamma, i, t}{\log K}. \tag{12.39}$$

Values near zero indicate concentration.

Values near one indicate dispersion.

Concentration and entropy should not be interpreted as risk levels by themselves. They describe risk architecture.

12.3 Coherence Measurement

Let pathway phases estimated in Chapter 11 be

$$\theta_{k,i,t}. \tag{12.40}$$

Phase is not a free angular parameter. For a timing variable $\tau_{k,i,t}$, define a reference origin $\tau_{0,i,t}$ and a prespecified period or decision horizon $H_{i,t} > 0$:

$$\theta_{k,i,t} = 2\pi \left[\frac{\tau_{k,i,t} - \tau_{0,i,t}}{H_{i,t}} \bmod 1 \right]. \tag{12.41}$$

The global origin does not affect phase differences, but the period does. The horizon must therefore be fixed from institutional timing, contractual cycles, or training data before out-of-sample evaluation. Analysts must report what delay corresponds to π: under this definition it is exactly $H/2$. If no defensible periodic or horizon-based interpretation exists, timing should enter through ordinary lags, splines, kernels, or interaction terms rather than through a phase angle.

Alternative phase constructions may be used, but they must specify the origin, period, wrapping rule, and treatment of multiple events. Phase parameters selected after observing the outcome invalidate the interference test. Validation should compare the phase representation with conventional timing features and should include placebo periods, phase randomization, and destructive-alignment cases.

The amplitude-weighted coherence measure is

$$C_{i,t} = \frac{\left| \sum_{k=1}^{K} R_{k,i,t} e^{\mathrm{i}\theta_{k,i,t}} \right|}{\sum_{k=1}^{K} R_{k,i,t}}. \tag{12.42}$$

This measure lies between zero and one.

A high value means that pathways with material magnitude are phase-aligned.

A separate unweighted coherence measure is

$$C_{i,t}^{\mathrm{unw}} = \left| \frac{1}{K} \sum_{k=1}^{K} e^{\mathrm{i}\theta_{k,i,t}} \right|. \tag{12.43}$$

The weighted version is generally more relevant for risk because it gives greater importance to consequential pathways.

12.3.1 Adverse and Stabilizing Coherence

The pathway set should be divided into adverse and stabilizing pathways.

Let

$$\Gamma^{-} = \{k : \text{pathway } k \text{ increases adverse transmission}\}, \tag{12.44}$$

and

$$\Gamma^{+} = \{k : \text{pathway } k \text{ supports stabilization}\}. \tag{12.45}$$

Then

$$C_{i,t}^{-} = \frac{\left|\sum_{k\in\Gamma^{-}} R_{k,i,t} e^{\mathrm{i}\theta_{k,i,t}}\right|}{\sum_{k\in\Gamma^{-}} R_{k,i,t}}, \tag{12.46}$$

and

$$C_{i,t}^{+} = \frac{\left|\sum_{k\in\Gamma^{+}} R_{k,i,t} e^{\mathrm{i}\theta_{k,i,t}}\right|}{\sum_{k\in\Gamma^{+}} R_{k,i,t}}. \tag{12.47}$$

High adverse coherence is a warning.

High stabilizing coherence may indicate effective and timely intervention.

12.3.2 Structural Amplification Measurement

The classical sum of standalone pathway probabilities is

$$P_{i,t}^{\mathrm{sum}} = \sum_{k=1}^{K} p_{k,i,t}. \tag{12.48}$$

Using $R_{k,i,t} = \sqrt{p_{k,i,t}}$, define the raw structural amplitude score

$$Q_{i,t}^{\mathrm{amp}} = \left|\sum_{k=1}^{K} R_{k,i,t} e^{\mathrm{i}\theta_{k,i,t}}\right|^{2}. \tag{12.49}$$

This quantity is not a probability. It can exceed one, and it is not bounded by the classical union probability. Marginal pathway-completion probabilities do not constitute a normalized orthogonal amplitude basis, particularly when pathways overlap or share the same terminal event.

A bounded structural alignment index is

$$J_{i,t}^{\mathrm{amp}} = \frac{Q_{i,t}^{\mathrm{amp}}}{\left(\sum_{k=1}^{K} R_{k,i,t}\right)^{2}} = C_{i,t}^{2}, \qquad 0 \le J_{i,t}^{\mathrm{amp}} \le 1. \tag{12.50}$$

The raw cross-term diagnostic is

$$I_{i,t}^{\mathrm{raw}} = Q_{i,t}^{\mathrm{amp}} - P_{i,t}^{\mathrm{sum}}, \tag{12.51}$$

but it must also be labelled as a structural score rather than a probability difference.

Positive values indicate constructive interference.

Negative values indicate destructive interference.

To obtain an adverse-event probability, the classical prediction and phase features must enter a separately normalized calibration model, for example

$$P_{i,t}^{\mathrm{cal}} = \operatorname{logit}^{-1}\left(\alpha + \beta \operatorname{logit} P_{i,t}^{\mathrm{cl}} + \gamma J_{i,t}^{\mathrm{amp}} + \boldsymbol{\delta}^{\mathsf{T}} \mathbf{z}_{i,t}\right), \tag{12.52}$$

where all coefficients are estimated on training data and evaluated out of sample. Only a complete amplitude Ψ normalized over the full state space may use the Born rule directly.

12.3.3 Pairwise Interference Matrix

To identify which pathway pairs generate amplification, define

$$\mathbf{I} * i, t = [I * jk, i, t], \tag{12.53}$$

where

$$I_{jk,i,t} = 2R_{j,i,t}R_{k,i,t}\cos\left(\theta_{j,i,t} - \theta_{k,i,t}\right). \tag{12.54}$$

This matrix permits the analyst to identify:

- the strongest amplifying pair;
- the strongest stabilizing pair;
- clusters of mutually aligned pathways;
- pathways whose timing should be interrupted.

The pairwise interference matrix is particularly useful for intervention design because it makes the phase contribution operationally specific.

12.3.4 Transmission Measurement

Transmission is derived from minimum action:

$$T_{R,i,t} = \exp\left[-\frac{2S_{R,i,t}^{*}}{\eta}\right]. \tag{12.55}$$

This measure lies between zero and one when the action and scale are positive.

It should be interpreted as an effective transmission or accessibility indicator, not automatically as a literal physical tunneling probability.

The parameter η should be calibrated in Chapter 11 using out-of-sample prediction.

Chapter 12 uses the calibrated value and monitors the resulting measure.

12.3.5 Transmission Elasticities

The exponential form implies strong sensitivity to barrier geometry.

The elasticity of transmission with respect to action is

$$\frac{\partial \log T_{R,i,t}}{\partial S^*_{R,i,t}} = -\frac{2}{\eta}. \tag{12.56}$$

For barrier component (z),

$$\frac{\partial \log T_{R,i,t}}{\partial z} = -\frac{2}{\eta}\frac{\partial S^*_{R,i,t}}{\partial z}. \tag{12.57}$$

These elasticities are central to intervention analysis.

They show how much transmission may change if the institution:

- raises barrier height;
- increases width;
- removes a dependency;
- changes event timing;
- closes a low-action route.

12.3.6 Risk Decomposition

The total adverse-risk estimate should be decomposed.

Let

$$P^{\mathrm{cl}}_{i,t} \tag{12.58}$$

be the classical benchmark.

Let

$$P^{\mathrm{ext}}_{i,t} \tag{12.59}$$

be the tunneling-extended estimate.

A general decomposition is

$$P^{\mathrm{ext}}_{i,t} = P^{\mathrm{cl}}_{i,t} + \Delta P^{\mathcal{B}}_{i,t} + \Delta P^{\Gamma}_{i,t} + \Delta P^{\mathrm{int}}_{i,t} + \varepsilon^{\mathrm{dec}}_{i,t}, \tag{12.60}$$

where:

- $\Delta P^{\mathcal{B}}$ is the barrier adjustment;
- ΔP^{Γ} is the pathway adjustment;
- ΔP^{int} is the interference adjustment;

- $\varepsilon^{\mathrm{dec}}$ is residual decomposition error.

This decomposition prevents the extended model from becoming a single opaque number.

12.4 Classical Risk Contribution

The classical component includes:

- current state risk;
- drift;
- diffusion;
- jumps;
- observed regime effects;
- ordinary nonlinear interactions.

It should remain the starting point.

The tunneling extension should not absorb risks already explained adequately by the classical model.

12.4.1 Barrier Contribution

The barrier contribution may be estimated as

$$\Delta P_{i,t}^{\mathcal{B}} = \widehat{P}\left(Y = 1 \mid \mathbf{x} * i, t, \widehat{\mathcal{B}} * i, t\right) - \widehat{P}\left(Y = 1 \mid \mathbf{x} * i, t, \mathcal{B} * i, t^{\mathrm{ref}}\right), \tag{12.61}$$

where $\mathcal{B}^{\mathrm{ref}}$ is the reference barrier.

This isolates how much current protection changes risk relative to normal or policy-standard protection.

12.4.2 Pathway Contribution

The pathway contribution measures whether the identified transition structure adds risk beyond state variables alone.

One empirical definition is

$$\Delta P_{i,t}^{\Gamma} = \widehat{P}\left(Y = 1 \mid \mathbf{x} * i, t, \Gamma * i, t\right) - \widehat{P}\left(Y = 1 \mid \mathbf{x}_{i,t}\right). \tag{12.62}$$

This measures the value of sequence, route, and dependency information.

12.4.3 Phase-Extension Contribution

The incremental phase contribution is

$$\Delta P_{i,t}^{\text{phase}} = P_{i,t}^{\text{cal,phase}} - P_{i,t}^{\text{cal,classical}}, \tag{12.63}$$

where both probabilities are normalized and calibrated on the same evaluation sample. The classical pathway model should incorporate nonlinear dependence, conventional timing interactions, and the Freidlin–Wentzell or Kramers-style action benchmark where appropriate.

The phase contribution should be reported only if it survives the validation requirements established in Chapter 11. Raw amplitude cross terms may be shown as structural diagnostics, but never as probability increments.

12.4.4 Measurement Normalization

Raw measures use different units.

Distance is geometric. Barrier height may be monetary or structural. Action is model-dependent. Coherence is dimensionless. Transmission is exponential.

They must not be combined without normalization.

Three normalization methods are useful.

Z-Score Normalization

For variable $X_{i,t}$,

$$Z_{X,i,t} = \frac{X_{i,t} - \mu_{X,\mathcal{R}}}{\sigma_{X,\mathcal{R}}}. \tag{12.64}$$

Percentile Normalization

$$P_{X,i,t} = \widehat{F} * X, \mathcal{R}\left(X * i, t\right). \tag{12.65}$$

Probability Calibration

Map the measure into observed event frequency:

$$\widetilde{X} * i, t = \widehat{P}\left(Y * i, t + \Delta = 1 \mid X_{i,t}\right). \tag{12.66}$$

Percentiles are useful for ranking. Probability calibration is useful for risk interpretation. Z-scores are useful for decomposition and monitoring.

12.4.5 Reference Classes

Normalization must use a meaningful reference class.

Possible reference classes include:

- borrowers within the same rating category;
- systems with similar operational architecture;
- banks within the same business model;
- legal matters within the same jurisdiction;
- sovereigns within comparable currency regimes.

Let the reference class be

$$\mathcal{R}_i. \tag{12.67}$$

Then every normalized measure should be written as

$$Z_{X,i,t}^{(\mathcal{R}_i)}. \tag{12.68}$$

Cross-domain comparison should be conducted through standardized scales, not raw values.

A corporate-credit action score and a cyber action score may both be placed on a zero-to-one scale, but they remain class-specific empirical objects.

12.4.6 Time-Series Normalization

Entity-specific deterioration may be hidden by cross-sectional comparisons.

Define the historical percentile for entity (i):

$$P_{X,i,t}^{\text{hist}} = \widehat{F} * X, i^{\text{hist}} \left(X * i, t \right). \tag{12.69}$$

The system should report both:

- cross-sectional percentile;
- historical percentile.

An entity may remain safer than peers while deteriorating rapidly relative to its own history.

12.4.7 Composite Tunneling-Risk Index

A composite index may combine the principal dimensions.

Define normalized risk-oriented components:

$$\widetilde{\mathbf{M}} * i, t = \left(\widetilde{D} * i, t, \widetilde{B} * i, t, \widetilde{M} * i, t, \widetilde{A} * i, t, \widetilde{C} * i, t^{-}, \widetilde{I} * i, t^{+}, \widetilde{T} * i, t \right), \tag{12.70}$$

where $\widetilde{I}^{+}$ denotes constructive adverse interference.

The composite tunneling-risk index is

$$\mathrm{TRI}*i,t=\sum*j=1^7 w_j\widetilde{M}*j,i,t, \qquad w*j\geq 0, \qquad \sum_{j=1}^{7} w_j=1. \tag{12.71}$$

The weights should be estimated or governed explicitly.

Possible methods include:

- logistic calibration;
- constrained regression;
- Bayesian model averaging;
- gradient boosting with monotonic constraints;
- expert weights subject to validation;
- optimization against economic loss.

12.5 Nonlinear Composite Measures

A linear index may understate interaction.

A nonlinear alternative is

$$\mathrm{TRI}*i,t^{\mathrm{nl}}=1-\prod*j=1^7\left(1-\widetilde{M}*j,i,t\right)^{w*j}. \tag{12.72}$$

This form rises sharply when several components are simultaneously elevated.

Another possibility is a machine-learning combination function:

$$\mathrm{TRI}*i,t^{\mathrm{ML}}=f*\boldsymbol{\omega}\left(\widetilde{\mathbf{M}}_{i,t}\right). \tag{12.73}$$

The model should use monotonicity constraints so that increasing barrier weakness, adverse coherence, or transmission cannot mechanically reduce measured risk without a documented reason.

12.5.1 The Role of Machine Learning in Measurement

Chapter 11 used machine learning primarily to estimate structural objects.

Chapter 12 may use machine learning for:

- normalization;
- nonlinear combination;
- threshold calibration;
- uncertainty estimation;

- regime-conditioned measurement;
- intervention sensitivity.

However, the production score should remain decomposable.

A black-box model that receives all variables and produces one risk score would abandon the structural gains of the framework.

The preferred architecture is:

$$\boxed{\begin{aligned}\text{Interpretable Components} &\longrightarrow \text{Constrained Combination}\\ &\longrightarrow \text{Decision Score.}\end{aligned}} \tag{12.74}$$

12.5.2 Regime-Dependent Measurement

Risk measures may have different meanings across regimes.

Let the estimated regime be

$$Z_t \in \{1, \ldots, R\}. \tag{12.75}$$

A regime-specific measure is

$$\mathrm{TRI} * i, t = \mathrm{TRI}\left(\widetilde{\mathbf{M}} * i, t \mid Z_t\right). \tag{12.76}$$

The reference distribution, weights, and escalation thresholds may change across normal, stressed, and crisis regimes.

Regime dependence should not become an excuse for moving thresholds whenever the model performs poorly. Regime definitions must be specified and validated in advance.

12.5.3 Measurement Uncertainty

Every measured component is estimated with error.

Let

$$\widehat{X} * i, t = X * i, t + \varepsilon_{X,i,t}. \tag{12.77}$$

For transmission,

$$\mathrm{Var}\left(\widehat{T} * R, i, t\right) = V * \text{state} + V_{\text{barrier}} + V_{\text{path}} + V_{\text{phase}} + V_{\text{parameter}} + V_{\text{model}}. \tag{12.78}$$

The measurement system should report:

- point estimate;

- confidence or credible interval;
- data-quality rating;
- model-version stability;
- sensitivity range.

12.5.4 Bootstrap Measurement

One practical uncertainty method is bootstrap resampling.

For bootstrap sample (b), calculate

$$\mathrm{TRI}_{i,t}^{(b)}. \tag{12.79}$$

Then estimate

$$\widehat{\mathrm{Var}}\left(\mathrm{TRI}*i,t\right) = \frac{1}{B-1}\sum*b=1^{B}\left(\mathrm{TRI}*i,t^{(b)} - \overline{\mathrm{TRI}}*i,t\right)^{2}. \tag{12.80}$$

The bootstrap should respect temporal and entity dependence. Ordinary independent resampling may be inappropriate for panel, sequence, or network data.

12.5.5 Bayesian Measurement

A Bayesian approach propagates posterior uncertainty.

Let

$$p\left(\boldsymbol{\Theta} \mid \mathcal{D}\right) \tag{12.81}$$

be the posterior from Chapter 11.

The posterior distribution of the risk index is

$$p\left(\mathrm{TRI}*i,t \mid \mathcal{I}*i,t\right) = \int p\left(\mathrm{TRI}*i,t \mid \boldsymbol{\Theta}, \mathcal{I}*i,t\right) p\left(\boldsymbol{\Theta} \mid \mathcal{D}\right), \mathrm{d}\boldsymbol{\Theta}. \tag{12.82}$$

This is especially useful for phase, action, and dominant-path uncertainty.

12.5.6 Dominant-Path Uncertainty

Instead of reporting only one path, report

$$P\left(\gamma_{i,t}^{*} = \gamma_k \mid \mathcal{I}_{i,t}\right). \tag{12.83}$$

A path may be considered robustly dominant only when its posterior or bootstrap selection probability exceeds a governance threshold.

If several paths have similar probabilities, the intervention strategy should reflect that uncertainty.

12.5.7 Model Confidence Measure

Define a model-confidence score

$$Q_{i,t}^{\text{model}} = f\left(q_{i,t}^{\text{data}}, q_{i,t}^{\text{stability}}, q_{i,t}^{\text{calibration}}, q_{i,t}^{\text{path}}\right). \tag{12.84}$$

The production risk report should show both

$$\text{TRI} * i, t \qquad \text{and} \qquad Q * i, t^{\text{model}}. \tag{12.85}$$

A high risk score with low model confidence requires investigation, not automatic action.

12.6 Calibration to Observed Outcomes

The composite measure should be calibrated to event frequencies.

Let raw score $s_{i,t}$ be mapped through

$$\widehat{p} * i, t^{\text{tun}} = c\left(s * i, t\right), \tag{12.86}$$

where (c) may be estimated through:

- logistic calibration;
- isotonic regression;
- beta calibration;
- spline calibration;
- Bayesian calibration.

Calibration should be assessed through:

- reliability plots;
- Brier score;
- calibration intercept;
- calibration slope;
- expected calibration error.

12.6.1 Lead-Time Measurement

A useful risk measure should provide advance warning.

For event (i) occurring at t_i^*, define first warning time

$$t_i^W = \inf \{t < t_i^* : \mathrm{TRI}_{i,t} \geq \tau\}. \tag{12.87}$$

Lead time is

$$L_i^{\text{lead}} = t_i^* - t_i^W. \tag{12.88}$$

The empirical value of the framework depends not only on prediction accuracy but on whether the warning arrives early enough for intervention.

12.6.2 False-Warning Burden

Let

$$N_{\text{false}} \tag{12.89}$$

be the number of warnings not followed by an adverse event within the horizon.

A false-warning rate is

$$\mathrm{FWR} = \frac{N_{\text{false}}}{N_{\text{warnings}}}. \tag{12.90}$$

An operational measure should balance:

- early lead time;
- event capture;
- false-warning burden;
- intervention cost.

12.6.3 Economic Value of Measurement

Statistical performance is not sufficient.

Define the expected value of using the tunneling measure as

$$V_{\mathrm{TRI}} = \mathbb{E}\left[L^{\text{without}} - L^{\text{with}} - C^{\text{intervention}} - C^{\text{false warning}}\right]. \tag{12.91}$$

The measure has practical value only if

$$V_{\mathrm{TRI}} > 0. \tag{12.92}$$

This requirement links measurement to real risk management.

12.6.4 Intervention Sensitivity

For intervention variable u_j, define

$$\mathcal{S} * j, i, t^{\mathrm{int}} = -\frac{\partial \mathrm{TRI} * i, t}{\partial u_j}. \tag{12.93}$$

A positive value indicates that increasing u_j reduces risk.

Possible intervention variables include:

- liquidity;
- collateral;
- control independence;
- maturity;
- monitoring speed;
- recovery capacity;
- network concentration;
- communication timing.

The most effective intervention is not necessarily the one that changes the current state most. It may be the one that raises action, reduces coherence, or closes the dominant pathway.

12.6.5 Intervention Efficiency

Let intervention (j) have cost C_j.

Define efficiency as

$$E_{j,i,t}^{\mathrm{int}} = \frac{\mathrm{TRI} * i, t^{\mathrm{before}} - \mathrm{TRI} * i, t^{\mathrm{after},j}}{C_j}. \tag{12.94}$$

This permits comparison among interventions that act on different structural dimensions.

For example:

- adding liquidity raises barrier height;
- extending maturity increases width;
- separating systems reduces dependency;
- accelerating response changes phase.

12.6.6 Counterfactual Measurement

A counterfactual measurement system estimates

$$\mathrm{TRI}_{i,t}\,(\mathbf{u}) \tag{12.95}$$

under alternative interventions $\mathbf{u}$.

This can be implemented through:

- structural simulation;
- causal machine learning;
- scenario models;
- digital twins;
- conservative reinforcement learning.

The counterfactual must distinguish prediction from causal effect.

Feature importance is not sufficient to justify intervention.

12.6.7 Risk Appetite and Limits

The composite measure should connect to risk appetite.

Define escalation thresholds

$$0 < \tau_1 < \tau_2 < \tau_3 < 1. \tag{12.96}$$

Then

$$\mathcal{L}_{i,t} = \begin{cases} 0, & \mathrm{TRI}_{i,t} < \tau_1, \\ 1, & \tau_1 \le \mathrm{TRI}_{i,t} < \tau_2, \\ 2, & \tau_2 \le \mathrm{TRI}_{i,t} < \tau_3, \\ 3, & \mathrm{TRI}_{i,t} \ge \tau_3. \end{cases} \tag{12.97}$$

The thresholds should be calibrated to:

- expected loss;
- intervention capacity;
- false-warning tolerance;
- risk appetite;
- governance requirements.

12.7 Persistence-Based Escalation

A single threshold breach may be noisy.

Define persistence:

$$\Pi_{i,t}(\tau, m) = \frac{1}{m} \sum_{j=0}^{m-1} \mathbb{I}\left(\mathrm{TRI}_{i,t-j} \geq \tau\right). \tag{12.98}$$

Escalation may require both:

$$\mathrm{TRI}_{i,t} \geq \tau \tag{12.99}$$

and

$$\Pi_{i,t}(\tau, m) \geq \pi^*. \tag{12.100}$$

Exceptions may be allowed for sudden action compression or transmission spikes.

12.7.1 Rate-of-Change Triggers

Even if the level remains below the limit, rapid deterioration may require escalation.

Define

$$\Delta\mathrm{TRI} * i, t = \mathrm{TRI} * i, t - \mathrm{TRI}_{i,t-h}. \tag{12.101}$$

A rate trigger is

$$\Delta\mathrm{TRI}_{i,t} \geq \delta^*. \tag{12.102}$$

This is particularly relevant when barriers move quickly or pathways become abruptly synchronized.

12.7.2 Component Limits

The composite score should not conceal extreme component values.

Separate limits may be established for:

- barrier weakness;
- adverse coherence;
- action compression;
- transmission;
- model uncertainty.

For example,

$$T_{R,i,t} \geq \tau_T \tag{12.103}$$

may trigger review even if the composite score remains moderate.

12.7.3 The Tunneling-Risk Scorecard

A production scorecard should include:

1. current state location;
2. distance to fragile and adverse regions;
3. barrier height and width;
4. barrier integrity and trend;
5. projected adverse drift;
6. projected adverse variance;
7. dominant path;
8. dominant-path stability;
9. minimum action and change;
10. pathway concentration;
11. adverse coherence;
12. raw phase cross-term and calibrated phase contribution;
13. transmission;
14. composite risk index;
15. uncertainty and model confidence;
16. recommended interventions.

The scorecard should preserve the distinction among:

$$\text{Level}, \qquad \text{Trend}, \qquad \text{Driver}, \qquad \text{Uncertainty}, \qquad \text{Action}. \tag{12.104}$$

12.7.4 A Compact Reporting Table

A practical reporting structure is:

Table 12.1: Compact tunneling-risk reporting scorecard.

Dimension	Current	Trend	Primary Driver	Management Response
State proximity	High	Deteriorating	Liquidity decline	Reduce exposure
Barrier integrity	Moderate	Falling	Collateral reliability	Increase usable protection
Path action	Low	Compressing	Refinancing route	Extend maturity
Adverse coherence	High	Rising	Spread and confidence timing	Accelerate support
Transmission	Elevated	Rising	Combined pathway effect	Escalate and stress test

12.7.5 Observed, Estimated, and Interpreted Measures

Committee reporting should preserve the empirical hierarchy established in Chapter 11.

Table 12.2: Separation of observed evidence, estimated measures, and risk interpretations.

Observed Evidence	Estimated Measure	Risk Interpretation
Debt maturities concentrate	Barrier width declines	Refinancing path shortens
Collateral volatility rises	Barrier reliability declines	Nominal protection overstated
Rating and spread reactions align	Adverse coherence rises	Pathways reinforce
Response time improves	Destructive interference increases	Transmission reduced

This prevents model-derived quantities from being presented as directly observed facts.

12.7.6 Cross-Sectional Comparison

For entities (i) and (j), comparison should use normalized components.

A simple distance between risk profiles is

$$d_M(i,j) = \sqrt{\left(\widetilde{\mathbf{M}} * i,t - \widetilde{\mathbf{M}} * j,t\right)^{\mathsf{T}} \mathbf{W} * M \left(\widetilde{\mathbf{M}} * i,t - \widetilde{\mathbf{M}}_{j,t}\right)}. \tag{12.105}$$

This allows clustering of entities with similar tunneling-risk architectures.

Two entities may have the same total score but different profiles:

- one may have weak barriers and low coherence;
- another may have strong barriers but rapidly compressing action;
- a third may have moderate state risk but high pathway synchronization.

These profiles require different interventions.

12.8 Tunneling-Risk Typology

The measurement vector permits a typology.

12.8.1 Proximity-Dominated Risk

High state proximity, moderate barrier risk, low interference.

12.8.2 Barrier-Dominated Risk

Moderate state risk, severe barrier weakness.

12.8.3 Mobility-Dominated Risk

The state is not yet close, but drift and covariance point toward danger.

12.8.4 Pathway-Dominated Risk

A low-action path is emerging despite moderate aggregate indicators.

12.8.5 Coherence-Dominated Risk

Several individually manageable pathways become synchronized.

12.8.6 Uncertainty-Dominated Risk

The risk score is elevated but parameter and pathway uncertainty are also high.

The typology supports differentiated governance.

12.8.7 Empirical Measurement Across Risk Classes

The same measurement logic applies across applications, but the empirical inputs differ.

Corporate Credit

Measure:

- leverage and liquidity distance;
- collateral and covenant barrier integrity;
- refinancing-path action;
- maturity and confidence phase alignment.

Consumer Credit

Measure:

- debt-service and savings distance;
- income and unused-credit barrier;

- employment and payment-deterioration pathways;
- timing of income shock and credit saturation.

Operational and Cyber Risk

Measure:

- technical and operational state;
- control-layer independence;
- attack or failure-path action;
- event timing, dwell time, and response coherence.

Legal and Conduct Risk

Measure:

- legal-defensibility state;
- contract and evidence barrier quality;
- interpretive-path action;
- timing alignment among ambiguity, conduct, and evidence.

Systemic Risk

Measure:

- network-state distance;
- collective barrier capacity;
- contagion-path action;
- behavioural synchronization;
- shared-infrastructure concentration.

12.8.8 Measurement Invariance and Domain Specificity

A common scale does not imply identical meaning.

The general architecture is invariant:

$$\text{State} + \text{Barrier} + \text{Mobility} + \text{Pathways} + \text{Phase} + \text{Transmission}. \tag{12.106}$$

The empirical definitions are domain-specific.

Therefore, cross-domain comparisons should be restricted to:

- normalized severity;
- percentile;
- trend;
- confidence;
- intervention urgency.

Raw barrier action from legal risk should not be compared numerically with raw action from cyber risk.

12.8.9 Measurement Stability

A production measure must remain stable under reasonable specification changes.

Let $\mathcal{S}$ be a set of admissible specifications.

Define specification variance:

$$V_{i,t}^{\text{spec}} = \mathrm{Var} * s \in \mathcal{S} \left[\mathrm{TRI} * i, t^{(s)} \right]. \tag{12.107}$$

A high value indicates fragile measurement.

The report should identify whether instability originates in:

- manifold construction;
- barrier definition;
- pathway set;
- phase specification;
- calibration.

12.8.10 Model Drift

Let the measurement distribution in the development period be

$$F_{\text{dev}} \tag{12.108}$$

and the current distribution be

$$F_{\text{cur}}. \tag{12.109}$$

Drift may be measured using:

- population stability index;

- Wasserstein distance;
- Kullback–Leibler divergence;
- calibration drift;
- embedding drift.

If model drift exceeds a threshold, the measurement should be flagged before recalibration.

12.8.11 Recalibration Governance

Recalibration should occur according to defined rules.

Possible triggers include:

- calibration failure;
- material data drift;
- structural change in pathways;
- regulatory change;
- sustained performance deterioration.

The recalibrated model should receive a new version identifier.

Historical scores should not be silently rewritten.

12.8.12 Backtesting Measurement Quality

Chapter 11 backtested detection. Chapter 12 backtests the quality of the measures themselves.

The tests should examine:

1. calibration;
2. rank ordering;
3. lead time;
4. stability;
5. decomposition accuracy;
6. intervention relevance.

For component (X), test whether higher values correspond to higher observed adverse frequency:

$$\frac{\partial P\left(Y=1 \mid X\right)}{\partial X} > 0. \tag{12.110}$$

For barrier integrity, the expected direction is reversed.

12.8.13 Component Ablation

The contribution of each component should be tested.

Let the full value function be

$$V_{\text{full}}. \tag{12.111}$$

For component (j),

$$\Delta V_j = V_{\text{full}} - V_{\text{without } j}. \tag{12.112}$$

A component that contributes no stable value should not remain in the production index merely because it is theoretically appealing.

12.9 Measurement Falsification

The framework should be exposed to falsification.

It should fail if:

- low action does not precede adverse events;
- barrier weakness has no protective meaning;
- coherence adds no value beyond classical timing models;
- transmission is poorly calibrated;
- the dominant path does not correspond to actual mechanisms;
- interventions predicted to reduce risk do not do so.

The empirical burden established in Chapter 11 therefore continues into measurement.

12.9.1 A Minimum Viable Measurement System

For the corporate-credit pilot proposed in Chapter 11, the first measurement system could include:

1. adverse-state distance;
2. liquidity-collateral-covenant barrier integrity;
3. projected adverse drift and variance;
4. minimum action across three candidate pathways;
5. pathway concentration;
6. maturity and response coherence;

7. raw phase cross-term and calibrated phase contribution;
8. transmission;
9. composite tunneling-risk index;
10. confidence interval.

The system should compare these measures with:

- conventional default probability;
- rating;
- credit spread;
- classical hazard score.

12.9.2 Illustrative Corporate Measurement

Suppose borrower (i) has:

- moderate distance to the adverse region;
- declining liquidity-barrier integrity;
- projected covariance rotating toward refinancing stress;
- falling action along the refinancing path;
- increasing phase alignment between maturity pressure and confidence withdrawal.

The measurement profile may be

$$\widetilde{\mathbf{M}}_{i,t} = (0.55, 0.78, 0.71, 0.82, 0.76, 0.69, 0.80)\,. \tag{12.113}$$

The conventional default probability may remain moderate because leverage has not yet breached its threshold.

The tunneling-risk index may nevertheless be high because the barrier is weakening and the dominant path is compressing.

The recommended action may be to extend maturities and secure committed liquidity before attempting a broader balance-sheet restructuring.

The example illustrates the purpose of the framework: to identify structural accessibility before ordinary failure indicators become dominant.

12.9.3 Use of Artificial Intelligence in Production Measurement

AI tools may assist with measurement by:

- updating document-derived barrier variables;

- identifying changes in pathway evidence;
- generating score explanations;
- monitoring data and model drift;
- producing counterfactual scenarios;
- drafting committee reports.

A language model may explain a score using retrieved evidence:

$$\mathcal{E} * i, t^{\mathrm{AI}} = \mathcal{F} * \mathrm{LLM}\left(\mathbf{M} * i, t^{\mathrm{tun}}, \mathcal{R} * i, t^{\mathrm{evidence}}\right). \tag{12.114}$$

The explanation must remain grounded in source evidence and approved structural measures.

The language model should not independently alter the score.

12.9.4 AI-Assisted Anomaly Detection

Machine learning may monitor whether current measurement combinations are unusual.

Let the normalized measurement vector be

$$\widetilde{\mathbf{M}}_{i,t}. \tag{12.115}$$

An anomaly score may be

$$A_{i,t}^{\mathrm{anom}} = -\log \widehat{p}\left(\widetilde{\mathbf{M}}_{i,t}\right). \tag{12.116}$$

High anomaly does not imply high risk. It indicates that the current combination is unusual relative to historical observations.

Anomaly detection is therefore a supplementary escalation tool.

12.9.5 Governed Automation

A production workflow may be

1. ingest current data;
2. validate timestamps and quality;
3. update structural inputs;
4. calculate component measures;
5. calculate uncertainty;
6. compare limits and trends;

7. generate evidence-grounded explanation;

8. escalate exceptions for human review.

Automated measurement should remain separate from decision authority.

12.9.6 The Empirical Measurement Architecture

The complete architecture is

$$\mathfrak{M} * i, t^{\mathrm{tun}} = \left(\widehat{\mathfrak{D}} * i, t, \mathbf{M} * i, t^{\mathrm{tun}}, \mathrm{TRI} * i, t, \mathbf{U} * i, t, \mathbf{S} * i, t^{\mathrm{int}}, \mathcal{L}_{i,t}\right), \tag{12.117}$$

where:

- $\widehat{\mathfrak{D}} * i, t$ is the detection architecture from Chapter 11;
- $\mathbf{M} * i, t^{\mathrm{tun}}$ is the component vector;
- $\mathrm{TRI} * i, t$ is the composite tunneling-risk index;
- $\mathbf{U} * i, t$ contains uncertainty measures;
- $\mathbf{S} * i, t^{\mathrm{int}}$ contains intervention sensitivities;
- $\mathcal{L} * i, t$ is the escalation level.

The complete empirical sequence is

$$\boxed{\begin{aligned} \text{Detection Objects} &\longrightarrow \text{Component Measures} \\ &\longrightarrow \text{Normalization} \longrightarrow \text{Risk Decomposition} \\ &\longrightarrow \text{Uncertainty} \longrightarrow \text{Limits and Sensitivities} \\ &\longrightarrow \text{Governed Action.} \end{aligned}} \tag{12.118}$$

12.9.7 What the Measurement Does Not Mean

A high tunneling-risk index does not establish that failure is inevitable.

It does not prove that the underlying institution is a physical quantum system.

It does not replace conventional default probabilities, operational-loss models, legal analysis, or systemic stress tests.

It indicates that the estimated geometry, barriers, pathways, and timing relationships jointly imply elevated adverse accessibility.

The appropriate interpretation is:

> The system exhibits a high and increasing degree of empirically estimated sub-barrier accessibility relative to its reference class and classical benchmark.

12.10 Chapter Synthesis

Chapter 11 developed the empirical machinery required to detect hidden pathways. It showed how to estimate the manifold, metric, classical dynamics, protective barrier, candidate pathways, phase proxies, action, and transmission.

Chapter 12 has continued that empirical programme by converting those estimated objects into measurable risk indicators.

State distance measures proximity to fragile and adverse regions. Barrier measures capture height, width, reliability, continuity, independence, capacity, and deterioration.

Directional mobility measures whether drift and covariance are increasingly aligned with the adverse region.

Minimum action measures the resistance of the most accessible path. Action compression measures whether that path is becoming easier through time.

Pathway concentration and entropy describe whether vulnerability is concentrated or dispersed.

Coherence measures the temporal alignment of material pathways. Interference measures how much their combined effect differs from the sum of their standalone probabilities.

Transmission translates minimum action into an effective accessibility indicator. Elasticities show how transmission responds to changes in barrier geometry and pathway structure.

The total calibrated risk estimate is decomposed into classical, barrier, pathway, and incremental phase contributions. Raw amplitude cross terms remain structural diagnostics and are not probability increments.

Normalization makes the measures comparable through time and across entities within relevant reference classes. Cross-domain comparison is limited to standardized severity, trend, confidence, and urgency.

A composite tunneling-risk index combines interpretable components through a constrained and validated aggregation rule.

Uncertainty is propagated from data, barriers, paths, phase, parameters, and model specification. The system reports model confidence alongside risk.

The measurement architecture is linked directly to escalation thresholds, rate-of-change triggers, risk appetite, intervention sensitivity, and economic value.

Machine learning may support calibration, nonlinear combination, anomaly detection, and counterfactual analysis. Artificial intelligence may assist with evidence extraction and explanation, but it should not replace structural measurement or human accountability.

The chapter's central proposition is:

> The empirical framework becomes operational when the hidden structures estimated in Chapter 11 are transformed into transparent measures of distance, barrier weakness, adverse mobility, path action, coherence, interference, and transmission; normalized against relevant benchmarks; accompanied by uncertainty; and connected directly to limits, interventions, and governance.

Chapter 13 will use these measures as the foundation for stress testing. It will examine how the manifold, barrier, covariance geometry, pathways, phase relationships, and transmission measures change under adverse scenarios and endogenous institutional responses.

Chapter 13

Stress Testing the Geometry

13.1 From Measurement to Stressed Geometry

Chapter 11 developed the empirical architecture required to detect hidden pathways. Chapter 12 transformed those estimated objects into normalized, uncertainty-aware, and decision-linked measures.

The present chapter asks what happens when the environment changes adversely.

Traditional stress testing usually begins with a scenario, shocks selected variables, and calculates losses, capital depletion, liquidity usage, or default rates. This remains necessary, but it is not sufficient for the framework developed in this book.

A tunneling stress test must examine whether stress changes:

- the position of the system;
- the shape of the state manifold;
- the location of fragile and adverse regions;
- the metric governing distance;
- the direction and magnitude of stochastic mobility;
- the height, width, continuity, and reliability of the barrier;
- the set and ordering of available pathways;
- the phase relationships among pathways;
- minimum barrier action;
- effective transmission into the adverse region.

The empirical sequence established in Chapters 11 and 12 therefore becomes

$$\boxed{\begin{aligned}\text{Estimated Objects} &\longrightarrow \text{Measured Risk}\\ &\longrightarrow \text{Scenario Shock} \longrightarrow \text{Stressed Geometry}\\ &\longrightarrow \text{Recomputed Pathways} \longrightarrow \text{Stressed Transmission}\\ &\longrightarrow \text{Intervention Comparison.}\end{aligned}} \tag{13.1}$$

The central proposition of this chapter is:

> A tunneling stress test does not merely estimate how much loss follows from an adverse scenario. It estimates how the scenario changes the geometry of the system, the integrity of its protective barriers, the accessibility of its pathways, and the synchronization of its responses. Any resulting adverse probability must come from the normalized or separately calibrated prediction layer.

13.1.1 The General Stress-Test Object

Let the baseline detection and measurement architecture be

$$\mathfrak{M} * i, t^0 = \left(\widehat{\mathcal{M}} * t, \widehat{\mathbf{g}} * t, \widehat{\mathbf{b}} * t, \widehat{\mathbf{A}} * t, \widehat{\mathcal{B}} * t, \widehat{\Gamma} * t, \widehat{\boldsymbol{\theta}} * t, \widehat{S} * R, t^*, \widehat{T} * R, t, \mathrm{TRI}_t\right). \tag{13.2}$$

Let scenario (s) be described by

$$\mathcal{S}^{(s)} = \left(\mathbf{z}^{(s)}, \boldsymbol{\xi}^{(s)}, \mathbf{u}^{(s)}, \mathcal{R}^{(s)}\right), \tag{13.3}$$

where:

- $\mathbf{z}^{(s)}$ contains exogenous shocks;
- $\boldsymbol{\xi}^{(s)}$ contains endogenous behavioural responses;
- $\mathbf{u}^{(s)}$ contains policy or management interventions;
- $\mathcal{R}^{(s)}$ contains scenario-specific structural rules.

The stressed architecture is

$$\mathfrak{M} * i, t^{(s)} = \mathcal{F} * \text{stress}\left(\mathfrak{M}^0_{i,t}, \mathcal{S}^{(s)}\right). \tag{13.4}$$

The output is not only a stressed loss. It is a complete stressed measurement vector:

$$\mathbf{M} * i, t^{(s)} = \left(D * i, t^{(s)}, B^{(s)}_{i,t}, M^{(s)}_{i,t}, A^{(s)}_{i,t}, C^{(s)}_{i,t}, I^{(s)}_{i,t}, T^{(s)}_{i,t}, U^{(s)}_{i,t}\right). \tag{13.5}$$

13.1.2 Four Types of Stress

The framework distinguishes four broad forms of stress.

State Stress

The scenario moves the system through the existing geometry.

Examples include:

- lower revenue;
- higher interest rates;
- liquidity outflow;
- collateral loss;
- cyber intrusion;
- regulatory sanction;
- funding withdrawal.

Geometric Stress

The scenario changes the manifold, metric, covariance structure, or adverse-region boundary.

Examples include:

- correlation increases;
- market liquidity collapses;
- feasible refinancing states disappear;
- legal interpretation changes;
- network connectivity rises.

Barrier Stress

The scenario changes protective structures.

Examples include:

- collateral becomes less liquid;
- committed funding becomes unavailable;
- controls share a failed dependency;
- resolution capacity is consumed;
- policy support becomes conditional.

Phase Stress

The scenario changes timing, sequencing, overlap, or synchronization.

Examples include:

- several maturities become concentrated;
- margin calls arrive simultaneously;
- cyber detection is delayed;
- institutions deleverage together;
- stabilizing interventions arrive too late.

A complete tunneling stress test should consider all four.

13.1.3 Traditional Stress Testing and the Geometric Extension

A conventional stress test may estimate

$$L_{i,t}^{(s)} = \mathcal{L}\left(\mathbf{x}_{i,t}, \mathbf{z}^{(s)}\right), \tag{13.6}$$

where $L^{(s)}$ is stressed loss.

The geometric extension estimates

$$\Delta\mathfrak{G} * i, t^{(s)} = \mathfrak{G} * i, t^{(s)} - \mathfrak{G}_{i,t}^{0}, \tag{13.7}$$

where

$$\mathfrak{G} * i, t = (\mathcal{M} * t, \mathbf{g} * t, \mathbf{A} * t, \mathcal{B} * t, \Gamma * t, \boldsymbol{\theta}_t) \tag{13.8}$$

is the geometry relevant to risk transmission.

The stressed loss remains important, but the geometric change explains why a particular scenario produces a disproportionate result.

13.1.4 Stressing the State

Let the baseline state be

$$\mathbf{x}_{i,t}^{0}. \tag{13.9}$$

Under scenario (s),

$$\mathbf{x} * i, t^{(s)} = \mathbf{x} * i, t^{0} + \Delta\mathbf{x}_{i,t}^{(s)}. \tag{13.10}$$

For a corporate borrower, a scenario may impose:

$$\ell_{i,t}^{(s)} = \ell_{i,t}^{0} + \Delta\ell^{(s)}, \tag{13.11}$$
$$q_{i,t}^{(s)} = q_{i,t}^{0} - \Delta q^{(s)}, \tag{13.12}$$
$$m_{i,t}^{(s)} = m_{i,t}^{0} - \Delta m^{(s)}, \tag{13.13}$$
$$r_{i,t}^{(s)} = r_{i,t}^{0} - \Delta r^{(s)}. \tag{13.14}$$

The stressed state is then re-embedded:

$$\mathbf{y} * i, t^{(s)} = \widehat{\mathcal{E}} \left(\mathbf{x} * i, t^{(s)} \right). \tag{13.15}$$

The first measurement is stressed distance to the adverse region:

$$d_{A,i,t}^{(s)} = d_{\widehat{\mathbf{g}}^{(s)}} \left(\mathbf{y}_{i,t}^{(s)}, A^{(s)} \right). \tag{13.16}$$

13.1.5 Stress May Move the Adverse Region

The adverse region should not necessarily remain fixed.

A recession may cause states previously classified as fragile to become adverse because refinancing markets close. A regulatory reinterpretation may expand the legally adverse region. A cyber attack may convert a recoverable outage into a critical-service failure if backup capacity is unavailable.

Write

$$A^{(s)} = \mathcal{A} \left(A^{0}, \mathbf{z}^{(s)}, \mathcal{R}^{(s)} \right). \tag{13.17}$$

The total change in distance is then driven by both state movement and boundary movement:

$$\Delta d_{A}^{(s)} = d \left(\mathbf{y}^{(s)}, A^{(s)} \right) - d \left(\mathbf{y}^{0}, A^{0} \right). \tag{13.18}$$

This distinction is fundamental.

The system may appear stable while the boundary of danger moves closer.

13.2 Stressing the Metric

Stress may change the meaning of distance.

Let the baseline metric be

$$\mathbf{g}^{0}(\mathbf{y}). \tag{13.19}$$

Under stress,

$$\mathbf{g}^{(s)}(\mathbf{y}) = \mathbf{g}^{0}(\mathbf{y}) + \Delta\mathbf{g}^{(s)}(\mathbf{y})\,. \tag{13.20}$$

If liquidity becomes scarce, movement along the liquidity dimension may become more consequential. If legal enforceability changes, contractual ambiguity may become a more expensive direction. If a network becomes concentrated, distance along a central dependency may shrink.

The stressed geodesic distance is

$$d_A^{(s)} = \inf_{\gamma} \int_{\gamma} \sqrt{\dot{\gamma}^{\mathsf{T}} \mathbf{g}^{(s)} \dot{\gamma}}, \mathrm{d}u. \tag{13.21}$$

13.2.1 Stressing the Covariance Geometry

Stress frequently increases correlation, reduces diversification, and rotates mobility toward adverse states.

Let

$$\mathbf{A} * t^{0} = \mathbf{\Sigma} * t^{0} \left(\mathbf{\Sigma}_t^{0}\right)^{\mathsf{T}} \tag{13.22}$$

be the baseline covariance rate.

Under stress,

$$\mathbf{A} * t^{(s)} = \mathbf{A} * t^{0} + \Delta\mathbf{A}_t^{(s)}. \tag{13.23}$$

The stress test should measure:

- eigenvalue expansion;
- eigenvector rotation;
- correlation concentration;
- effective-dimension collapse;
- adverse-direction variance.

Let the stressed eigensystem be

$$\mathbf{A} * t^{(s)} \mathbf{v} * k, t^{(s)} = \lambda_{k,t}^{(s)} \mathbf{v}_{k,t}^{(s)}. \tag{13.24}$$

The adverse alignment is

$$\cos\phi_{i,t}^{(s)} = \frac{\left(\mathbf{v} * 1, i, t^{(s)}\right)^{\mathsf{T}} \mathbf{n} * A, i, t^{(s)}}{\left|\mathbf{v} * 1, i, t^{(s)}\right| \left|\mathbf{n} * A, i, t^{(s)}\right|}. \tag{13.25}$$

13.2.2 Effective Dimension Under Stress

Let the eigenvalues of the covariance matrix be $\lambda_1, \ldots, \lambda_d$.

The effective dimension may be measured as

$$d_{\text{eff}} = \frac{\left(\sum_{k=1}^{d} \lambda_k\right)^2}{\sum_{k=1}^{d} \lambda_k^2}. \tag{13.26}$$

A decline in d_{eff} indicates that uncertainty is concentrating into fewer directions.

If those directions align with the adverse boundary, diversification has collapsed precisely where it is needed most.

13.2.3 Stressing the Barrier

The stressed protection component (j) is

$$\widetilde{P} * j, i, t^{(s)} = P * j, i, t^{(s)} A_{j,i,t}^{(s)} R_{j,i,t}^{(s)}, \tag{13.27}$$

where:

- $P_j^{(s)}$ is nominal protection;
- $A_j^{(s)}$ is stressed availability;
- $R_j^{(s)}$ is stressed reliability.

The total stressed protection is

$$\widetilde{P} * i, t^{(s)} = \sum_j \widetilde{P} * j, i, t^{(s)}. \tag{13.28}$$

The barrier-integrity score becomes

$$I_{\mathcal{B},i,t}^{(s)} = f\left(H_{i,t}^{(s)}, W_{i,t}^{(s)}, C_{i,t}^{(s)}, R_{i,t}^{(s)}, K_{\text{eff},i,t}^{(s)}, Q_{i,t}^{(s)}\right). \tag{13.29}$$

13.2.4 Barrier Capacity Under Simultaneous Use

Many barriers are shared or capacity-constrained.

Let total demand on barrier component (j) be

$$L_{j,t}^{(s)} = \sum_{i=1}^{N} l_{j,i,t}^{(s)}. \tag{13.30}$$

Let usable capacity be

$$K_{j,t}^{(s)}. \tag{13.31}$$

The utilization ratio is

$$U_{j,t}^{(s)} = \frac{L_{j,t}^{(s)}}{K_{j,t}^{(s)}}. \tag{13.32}$$

A barrier that appears adequate for one institution may fail under simultaneous use.

Examples include:

- central-bank facilities;
- insurance pools;
- collateral markets;
- backup systems;
- human-response teams;
- resolution capacity.

13.2.5 Stressed Barrier Width

Barrier width may change through time, sequence, and geometry.

Temporal width under scenario (s) is

$$W_i^{\text{time},(s)} = t_i^{A,(s)} - t_i^{F,(s)}. \tag{13.33}$$

Sequential width is

$$W_i^{\text{seq},(s)} = \min_{\gamma} \left| \gamma \cap \mathcal{B}^{(s)} \right|. \tag{13.34}$$

Geometric width is

$$W_i^{\text{geo},(s)} = \inf_{\gamma} \int_{\gamma \cap \mathcal{B}^{(s)}} \mathrm{d}s_{\mathbf{g}^{(s)}}. \tag{13.35}$$

The test should report which type of width is collapsing.

13.3 Pathway Activation Under Stress

The pathway set is scenario-dependent:

$$\Gamma_{i,t}^{(s)} = \mathcal{G}\left(\Gamma_{i,t}^{0}, \mathbf{x} * i, t^{(s)}, \mathcal{B} * i, t^{(s)}, \mathcal{R}^{(s)}\right). \tag{13.36}$$

A pathway may:

- remain inactive;
- become active;
- disappear;
- merge with another route;
- become shorter;
- acquire a higher probability.

For pathway (k), the stressed completion probability is

$$p_{k,i,t}^{(s)} = P\left(\gamma_k \text{ completes before } t + \Delta \mid \mathcal{I}_{i,t}, \mathcal{S}^{(s)}\right). \tag{13.37}$$

13.3.1 Recalculating Classical Path Resistance

The stressed edge probability is

$$p_{e,i,t}^{(s)}. \tag{13.38}$$

The stressed resistance is

$$r_{e,i,t}^{(s)} = -\log p_{e,i,t}^{(s)}. \tag{13.39}$$

The stressed classical path resistance is

$$\mathcal{R} * i, t^{(s)}[\gamma] = \sum *e \in \gamma r_{e,i,t}^{(s)}. \tag{13.40}$$

The dominant classical path becomes

$$\gamma_{\mathrm{cl},i,t}^{*,(s)} = \arg, \min *\gamma \mathcal{R} * i, t^{(s)}[\gamma]. \tag{13.41}$$

This remains the benchmark against which the tunneling path is compared.

13.3.2 Phase Under Stress

Let the baseline phase model from Chapter 11 be

$$\theta_{k,i,t}^{0} = \boldsymbol{\omega} * k^{\mathsf{T}} \mathbf{z} * k, i, t^{\mathrm{phase},0}. \tag{13.42}$$

Under stress,

$$\theta_{k,i,t}^{(s)} = \boldsymbol{\omega} * k^{\mathsf{T}} \mathbf{z} * k, i, t^{\text{phase},(s)}. \tag{13.43}$$

The stressed phase-feature vector may include:

- compressed maturities;
- longer detection delay;
- overlapping incidents;
- synchronized asset sales;
- delayed policy response;
- increased event persistence.

The change in phase is

$$\Delta\theta_{k,i,t}^{(s)} = \theta_{k,i,t}^{(s)} - \theta_{k,i,t}^{0}. \tag{13.44}$$

13.3.3 Stressed Coherence

The amplitude-weighted coherence under scenario (s) is

$$C_{i,t}^{(s)} = \frac{\left| \sum_k R_{k,i,t}^{(s)} e^{\mathrm{i}\theta_{k,i,t}^{(s)}} \right|}{\sum_k R_{k,i,t}^{(s)}}. \tag{13.45}$$

The coherence shock is

$$\Delta C_{i,t}^{(s)} = C_{i,t}^{(s)} - C_{i,t}^{0}. \tag{13.46}$$

A stress scenario may increase risk even if pathway magnitudes change little, provided their timing becomes more aligned.

13.3.4 Stressed Structural Phase Alignment

The stressed raw structural amplitude score is

$$Q_{i,t}^{\text{amp},(s)} = \left| \sum_k R_{k,i,t}^{(s)} e^{\mathrm{i}\theta_{k,i,t}^{(s)}} \right|^2. \tag{13.47}$$

The stressed standalone sum is

$$P_{i,t}^{\text{sum},(s)} = \sum_k p_{k,i,t}^{(s)}. \tag{13.48}$$

The stressed raw cross-term diagnostic is

$$I_{i,t}^{\mathrm{raw},(s)} = Q_{i,t}^{\mathrm{amp},(s)} - P_{i,t}^{\mathrm{sum},(s)}. \tag{13.49}$$

The bounded stressed alignment index is

$$J_{i,t}^{\mathrm{amp},(s)} = \frac{Q_{i,t}^{\mathrm{amp},(s)}}{\left(\sum_k R_{k,i,t}^{(s)}\right)^2}. \tag{13.50}$$

The stress report should compare $J^{\mathrm{amp},(s)}$ with its baseline and pass it through the validated probability-calibration model. Neither $Q^{\mathrm{amp},(s)}$ nor $I^{\mathrm{raw},(s)}$ is itself a stressed probability.

13.3.5 Recalculating Barrier Action

For stressed path γ,

$$S_{R,i,t}^{(s)}[\gamma] = \sum_{e\in\gamma\cap\mathcal{B}^{(s)}} \sqrt{2M_{e,i,t}^{(s)}\left[V_{e,i,t}^{(s)} - E_{i,t}^{(s)}\right]}\Delta s_e^{(s)}. \tag{13.51}$$

The minimum stressed action is

$$S_{R,i,t}^{*,(s)} = \min_{\gamma\in\Gamma_{i,t}^{(s)}} S_{R,i,t}^{(s)}[\gamma]. \tag{13.52}$$

Action compression is

$$\Delta S_{R,i,t}^{*,(s)} = S_{R,i,t}^{*,(s)} - S_{R,i,t}^{*,0}. \tag{13.53}$$

A negative value indicates that stress has made the dominant path more accessible.

13.4 Stressed Transmission

The stressed transmission measure is

$$T_{R,i,t}^{(s)} = \exp\left[-\frac{2S_{R,i,t}^{*,(s)}}{\eta}\right]. \tag{13.54}$$

The absolute transmission shock is

$$\Delta T_{R,i,t}^{(s)} = T_{R,i,t}^{(s)} - T_{R,i,t}^{0}. \tag{13.55}$$

The relative shock is

$$\Delta T^{\mathrm{rel},(s)}_{R,i,t} = \frac{T^{(s)}_{R,i,t} - T^{0}_{R,i,t}}{T^{0}_{R,i,t} + \epsilon}. \tag{13.56}$$

Because transmission is exponential in action, moderate geometric deterioration can produce large changes in accessibility.

13.4.1 The Stressed Tunneling-Risk Index

Let the normalized stressed components be

$$\widetilde{\mathbf{M}} * i, t^{(s)} = \left(\widetilde{D} * i, t^{(s)}, \widetilde{B} * i, t^{(s)}, \widetilde{M} * i, t^{(s)}, \widetilde{A} * i, t^{(s)}, \widetilde{C} * i, t^{(s)}, \widetilde{I} * i, t^{(s)}, \widetilde{T} * i, t^{(s)}\right). \tag{13.57}$$

The stressed composite measure is

$$\mathrm{TRI} * i, t^{(s)} = \sum * j w_j \widetilde{M}^{(s)}_{j,i,t}. \tag{13.58}$$

The stress impact is

$$\Delta\mathrm{TRI} * i, t^{(s)} = \mathrm{TRI} * i, t^{(s)} - \mathrm{TRI}^{0}_{i,t}. \tag{13.59}$$

The decomposition of this change is

$$\Delta\mathrm{TRI}^{(s)} = \Delta\mathrm{TRI}{*}D^{(s)} + \Delta\mathrm{TRI}{*}B^{(s)} + \Delta\mathrm{TRI}{*}M^{(s)} + \Delta\mathrm{TRI}{*}A^{(s)} + \Delta\mathrm{TRI}{*}C^{(s)} + \Delta\mathrm{TRI}{*}I^{(s)} + \Delta\mathrm{TRI}^{(s)}_{T}. \tag{13.60}$$

This makes the scenario mechanism transparent.

13.4.2 Endogenous Institutional Responses

Institutions do not remain passive under stress.

Let management action be

$$\mathbf{u} * i, t = \pi\left(\mathbf{x} * i, t, \mathcal{S}^{(s)}\right). \tag{13.61}$$

Possible responses include:

- selling assets;
- hoarding liquidity;
- cutting credit;
- drawing committed lines;

- enforcing covenants;
- overriding controls;
- raising collateral requirements;
- accelerating incident response.

The stressed state evolves as

$$\mathbf{x} * t + 1^{(s)} = \mathcal{F}\left(\mathbf{x} * t^{(s)}, \mathbf{z} * t^{(s)}, \mathbf{u} * t^{(s)}\right). \tag{13.62}$$

Actions may stabilize the institution while destabilizing the system.

For example, one institution's asset sale may improve its liquidity but reduce collateral values for others.

13.4.3 Feedback Loops

Let system response create an endogenous shock:

$$\mathbf{z} * t + 1^{\text{endo}} = \mathcal{H}\left(\mathbf{u} * 1, t, \ldots, \mathbf{u}_{N,t}\right). \tag{13.63}$$

The next-period state becomes

$$\mathbf{x} * t + 1 = \mathcal{F}\left(\mathbf{x} * t, \mathbf{z} * t^{\text{exo}}, \mathbf{z} * t^{\text{endo}}\right). \tag{13.64}$$

Important feedback loops include:

- fire sales;
- margin spirals;
- deposit runs;
- credit contraction;
- operational overload;
- legal escalation;
- policy crowding.

13.4.4 Dynamic Multi-Period Stress Testing

A one-period shock may miss path-dependent deterioration.

For horizon (H), simulate

$$\left\{\mathbf{x} * t + h^{(s)}, \mathcal{B} * t + h^{(s)}, \Gamma_{t+h}^{(s)}, \boldsymbol{\theta} * t + h^{(s)}\right\} * h = 1^{H}. \tag{13.65}$$

At every horizon, recompute

$$\left(d^{(s)}_{A,t+h}, I^{(s)}_{\mathcal{B},t+h}, S^{*,(s)}_{R,t+h}, C^{(s)}_{t+h}, T^{(s)}_{R,t+h}, \mathrm{TRI}^{(s)}_{t+h}\right). \tag{13.66}$$

This reveals whether the system:

- recovers;
- stabilizes at a fragile state;
- deteriorates gradually;
- experiences delayed action compression;
- crosses into an adverse region.

13.4.5 Scenario Design

Scenarios should be structured around mechanisms, not only variable shocks.

A complete scenario description should specify:

1. initiating shock;
2. affected state variables;
3. covariance response;
4. barrier response;
5. pathway activation;
6. phase and timing assumptions;
7. endogenous behaviours;
8. management and policy interventions.

A scenario schema is

$$\mathcal{S}^{(s)} = [\text{Shock}, \text{Geometry}, \text{Barrier}, \text{Pathways}, \text{Phase}, \text{Response}]. \tag{13.67}$$

13.5 Empirical Scenario Calibration

Scenario magnitudes should be linked to evidence.

Sources include:

- historical episodes;
- current market-implied distributions;

- macroeconomic models;
- expert scenarios;
- supervisory assumptions;
- simulations;
- machine-learning conditional distributions.

For shock variable (z), a historical quantile scenario may be

$$z^{(s)} = \widehat{F}_z^{-1}(\alpha), \tag{13.68}$$

where α is a severe percentile.

A conditional scenario may be

$$z^{(s)} \sim \widehat{P}(z \mid \mathbf{x} * t, Z * t = r), \tag{13.69}$$

where (r) is the current regime.

13.5.1 Historical Stress Testing

Historical scenarios replay observed episodes.

For historical period (h), apply

$$\mathcal{S}^{(h)} = \left(\Delta\mathbf{x}^{(h)}, \Delta\mathbf{A}^{(h)}, \Delta\mathcal{B}^{(h)}, \Delta\Gamma^{(h)}, \Delta\boldsymbol{\theta}^{(h)}\right). \tag{13.70}$$

The advantage is empirical realism.

The limitation is that future stress may combine mechanisms not observed together historically.

13.5.2 Hypothetical and Structural Scenarios

Hypothetical scenarios may combine shocks deliberately.

For example:

$$\mathcal{S}^{\text{corp}} = \{\text{Revenue decline}, \text{Rate increase}, \text{Collateral fall}, \text{Market closure}\}. \tag{13.71}$$

A structural scenario specifies causal relationships rather than a list of independent shocks.

For instance:

$$\text{Revenue decline} \rightarrow \text{Rating downgrade} \rightarrow \text{Spread widening} \rightarrow \text{Refinancing closure}. \tag{13.72}$$

Structural scenarios are particularly appropriate for pathway stress testing.

13.5.3 Machine Learning for Scenario Generation

Machine learning may estimate joint conditional shock distributions.

Let

$$\mathbf{z} * t + 1 : t + H^{(s)} \sim p * \boldsymbol{\vartheta} \left(\mathbf{z} * t + 1 : t + H \mid \mathcal{I} * t \right). \tag{13.73}$$

Possible methods include:

- conditional variational autoencoders;
- generative adversarial networks;
- diffusion models;
- sequence transformers;
- neural stochastic differential equations;
- copula-based machine learning.

The generated scenarios should preserve:

- marginal distributions;
- tail dependence;
- temporal ordering;
- regime consistency;
- economic constraints.

13.5.4 Generative AI for Narrative Scenario Expansion

Large language models may assist in translating a scenario into structured implications.

Given an initiating event, a language model can propose:

- affected controls;
- legal consequences;
- likely operational sequences;
- management responses;
- potential secondary effects.

However, narrative outputs are hypotheses, not calibrated probabilities.

They should be converted into structured scenario variables, reviewed by experts, and tested against data.

13.5.5 Agent-Based Stress Testing

Agent-based models are useful when endogenous behaviour matters.

Let agent (i) follow policy

$$\mathbf{u} * i, t = \pi_i \left(\mathbf{x} * i, t, \mathbf{m} * t, \mathcal{I} * i, t \right), \tag{13.74}$$

where $\mathbf{m}_t$ is the market state.

The aggregate system evolves through

$$\mathbf{m} * t + 1 = \mathcal{H} \left(\mathbf{m} * t, \mathbf{u} * 1, t, \dots, \mathbf{u} * N, t \right). \tag{13.75}$$

The outputs can be mapped back into the tunneling framework by recomputing barriers, pathways, phase, and action at each step.

13.5.6 Graph and Network Stress Testing

Let the baseline network be

$$\mathcal{G} * t^0 = \left(\mathcal{V}, \mathcal{E} * t^0, \mathbf{W}_t^0 \right). \tag{13.76}$$

Under stress,

$$\mathcal{G} * t^{(s)} = \left(\mathcal{V}, \mathcal{E} * t^{(s)}, \mathbf{W}_t^{(s)} \right). \tag{13.77}$$

The network may change because:

- exposures grow;
- funding links disappear;
- common collateral concentration rises;
- infrastructures fail;
- institutions retreat to common counterparties.

The graph Laplacian becomes

$$\mathbf{L} * t^{(s)} = \mathbf{D} * t^{(s)} - \mathbf{W}_t^{(s)}. \tag{13.78}$$

Network centrality, community structure, bottlenecks, and dominant paths should be recomputed.

13.6 Graph Neural Networks in Stress Testing

A trained graph neural network may estimate stressed node states:

$$\mathbf{h} * i, t^{(s)} = \mathcal{G} * \boldsymbol{\vartheta} \left(\mathbf{x} * i, t^{(s)}, \mathcal{E} * t^{(s)}, \mathbf{W}_t^{(s)} \right). \tag{13.79}$$

Graph methods are useful for estimating:

- propagation speed;
- path activation;
- shared dependencies;
- node importance;
- system-wide action compression.

They should be validated against known contagion mechanisms and network data quality.

13.6.1 Reverse Stress Testing

Forward stress testing begins with a scenario and calculates the outcome.

Reverse stress testing begins with a critical outcome and asks which scenario produces it.

Let the critical condition be

$$\mathrm{TRI} * i, t^{(s)} \geq \tau * \mathrm{crit}, \tag{13.80}$$

or

$$T_{R,i,t}^{(s)} \geq \tau_T. \tag{13.81}$$

The reverse-stress problem is

$$\mathcal{S}^* = \arg, \min_{\mathcal{S}} C(\mathcal{S}) \tag{13.82}$$

subject to

$$\mathrm{TRI} * i, t^{(\mathcal{S})} \geq \tau * \mathrm{crit}. \tag{13.83}$$

The function $C(\mathcal{S})$ measures scenario severity or plausibility cost.

13.6.2 Reverse Stress as Minimum Scenario Action

A scenario cost may be

$$C(\mathcal{S}) = \Delta\mathbf{z}^{\top}\boldsymbol{\Omega}^{-1}\Delta\mathbf{z} + \lambda_{\mathcal{B}}C_{\mathcal{B}} + \lambda_{\theta}C_{\theta}. \tag{13.84}$$

The reverse stress test identifies the least severe combination of:

- state shocks;
- barrier erosion;
- covariance rotation;
- phase alignment;
- pathway activation

that creates a critical condition.

This is the geometry of failure.

13.6.3 Optimization Methods for Reverse Stress

Reverse stress testing may use:

- nonlinear programming;
- mixed-integer optimization;
- evolutionary algorithms;
- Bayesian optimization;
- reinforcement learning;
- adversarial machine learning.

The optimization must respect:

- accounting identities;
- causal ordering;
- feasible states;
- scenario plausibility;
- policy constraints.

13.6.4 Adversarial Stress Testing

An adversarial model searches for small input changes that create large risk changes.

The adversarial scenario is

$$\Delta \mathbf{z}^* = \arg, \max *|\Delta \mathbf{z}| \leq \epsilon \mathrm{TRI} \left(\mathbf{x} * t + \Delta \mathbf{z} \right). \tag{13.85}$$

This identifies local fragility and model sensitivity.

Adversarial stress should be treated as a robustness tool, not automatically as a plausible economic forecast.

13.6.5 Intervention Stress Testing

The purpose of stress testing is not merely to display deterioration.

For intervention u_j, recompute

$$\mathrm{TRI}_{i,t}^{(s,u_j)}. \tag{13.86}$$

The risk reduction is

$$\Delta \mathrm{TRI} * j^{\mathrm{red}} = \mathrm{TRI} * i, t^{(s)} - \mathrm{TRI}_{i,t}^{(s,u_j)}. \tag{13.87}$$

The transmission reduction is

$$\Delta T_j^{\mathrm{red}} = T_{R,i,t}^{(s)} - T_{R,i,t}^{(s,u_j)}. \tag{13.88}$$

13.7 Four Intervention Levers

The principal intervention classes are:

13.7.1 Raise Barrier Height

Examples include:

- capital injection;
- additional liquidity;
- collateral enhancement;
- stronger legal protection.

13.7.2 Increase Barrier Width

Examples include:

- maturity extension;
- additional control stages;
- recovery time;
- resolution stays.

13.7.3 Reduce Permeability

Examples include:

- remove shared dependencies;
- reduce concentration;
- isolate critical systems;
- close legal or operational gaps.

13.7.4 Change Phase

Examples include:

- accelerate response;
- stagger margin calls;
- advance liquidity provision;
- delay harmful synchronized actions;
- coordinate communications.

13.7.5 Intervention Efficiency

Let intervention u_j have cost C_j.

Define stressed intervention efficiency as

$$E_j^{(s)} = \frac{\mathrm{TRI}^{(s)} - \mathrm{TRI}^{(s,u_j)}}{C_j}. \tag{13.89}$$

A second measure uses transmission:

$$E_{T,j}^{(s)} = \frac{T_R^{(s)} - T_R^{(s,u_j)}}{C_j}. \tag{13.90}$$

The preferred intervention may be the one that most effectively increases minimum action rather than the one that produces the largest immediate accounting effect.

13.7.6 Intervention Portfolios

Several interventions may interact.

Let intervention portfolio be

$$\mathbf{u} = (u_1, \dots, u_m) . \tag{13.91}$$

Choose

$$\mathbf{u}^* = \arg, \min_{\mathbf{u}} \left[\mathrm{TRI}^{(s,\mathbf{u})} + \lambda_C C(\mathbf{u}) \right] \tag{13.92}$$

subject to operational and governance constraints.

This permits the model to compare complementary interventions, such as liquidity support plus maturity extension.

13.7.7 Reinforcement Learning for Sequential Intervention

A stress episode unfolds over time.

Let state at time (t) be

$$\mathbf{s}_t = (\mathbf{x}_t, \mathcal{B}_t, \Gamma_t, \boldsymbol{\theta}_t) . \tag{13.93}$$

The action is $\mathbf{u}_t$, and the objective is

$$\min_{\pi} \mathbb{E} * \pi \left[\sum *t = 0^H \gamma^t \left(\mathrm{TRI}_t + C(\mathbf{u}_t) \right) \right] . \tag{13.94}$$

Reinforcement learning can compare timing and sequencing of interventions.

In production, it should remain constrained, simulated, and subject to human approval.

13.7.8 Empirical Implementation Workflow

A practical Chapter 13 implementation proceeds in twelve steps.

1. Freeze the Chapter 11 estimation model and Chapter 12 measurement version.
2. Select scenario horizon and reference date.
3. Define exogenous shocks.
4. Specify endogenous responses.
5. Recalculate stressed state variables.
6. Re-estimate or condition the metric and covariance geometry.
7. Reconstruct stressed barriers.

8. Activate and reweight pathways.

9. Recompute phase, coherence, and interference.

10. Recalculate action, transmission, and the tunneling-risk index.

11. Compare management and policy interventions.

12. Report uncertainty, mechanism, and governance implications.

The sequence is

$$\boxed{\begin{aligned} &\text{Baseline Model} \longrightarrow \text{Scenario} \longrightarrow \text{State and Geometry} \\ &\qquad\qquad\quad \longrightarrow \text{Barrier and Pathways} \longrightarrow \text{Phase and Action} \\ &\qquad\qquad\quad \longrightarrow \text{Transmission} \longrightarrow \text{Intervention.} \end{aligned}} \tag{13.95}$$

13.7.9 A Minimum Viable Empirical Stress Test

For the corporate-credit pilot introduced in Chapters 11 and 12, a minimum viable stress test may use:

- leverage;
- liquidity;
- refinancing access;
- collateral;
- covenant headroom;
- three candidate pathways.

A scenario may impose:

- a 20 percent operating cash-flow decline;
- a 300-basis-point funding-cost increase;
- a 25 percent collateral-value decline;
- partial refinancing-market closure.

The implementation would:

1. update the state vector;
2. move the borrower through the estimated manifold;
3. recompute distance to default and distressed restructuring;
4. reduce collateral reliability and refinancing availability;

5. increase adverse-direction covariance;
6. activate the refinancing and covenant pathways;
7. update maturity-overlap phase variables;
8. calculate minimum action and transmission;
9. compare maturity extension, committed liquidity, and collateral enhancement.

13.8 Illustrative Corporate Example

Suppose the baseline borrower has

$$\mathrm{TRI}^0 = 0.42, \qquad T_R^0 = 0.18, \qquad S_R^{*,0} = 1.72. \tag{13.96}$$

Under the stress scenario,

$$\mathrm{TRI}^{(s)} = 0.79, \qquad T_R^{(s)} = 0.61, \qquad S_R^{*,(s)} = 0.49. \tag{13.97}$$

The risk increase may be decomposed into:

- movement toward the adverse region;
- collateral-barrier deterioration;
- covariance rotation toward refinancing stress;
- action compression along the refinancing pathway;
- phase alignment between maturity concentration and confidence withdrawal.

If maturity extension raises action to

$$S_R^{*,(s,u)} = 1.08, \tag{13.98}$$

then transmission falls to

$$T_R^{(s,u)} = \exp\left(-\frac{2(1.08)}{\eta}\right). \tag{13.99}$$

The intervention works principally by increasing barrier width and altering timing.

13.8.1 Operational and Cyber Implementation

For cyber stress testing, the state may include:

- vulnerability;

- credential exposure;
- monitoring quality;
- response readiness;
- dependency concentration.

A scenario may impose:

- credential compromise;
- zero-day exploitation;
- monitoring delay;
- shared-cloud outage.

The model then recomputes:

- attack-path activation;
- control-layer reliability;
- dwell-time phase;
- recovery-path width;
- minimum action into critical service failure.

13.8.2 Legal and Regulatory Implementation

For legal stress testing, the scenario may include:

- adverse precedent;
- new evidence;
- regulatory reinterpretation;
- whistleblower disclosure;
- enforcement escalation.

The barrier changes through:

- weaker contractual interpretation;
- reduced evidentiary strength;
- lower credibility of prior conduct;
- higher enforcement intensity.

The model should recompute interpretive pathways, evidence timing, legal-barrier action, and adverse coherence.

13.8.3 Systemic Implementation

For systemic stress, the input should include:

- institution-level balance sheets;
- funding links;
- common asset holdings;
- collateral flows;
- infrastructure dependencies;
- behavioural rules.

The scenario may impose:

- asset-price shock;
- funding-market closure;
- margin increase;
- deposit withdrawal;
- infrastructure outage.

The model recomputes:

- network topology;
- collective barrier capacity;
- systemic coherence;
- dominant contagion path;
- aggregate action;
- systemic transmission.

13.8.4 Validation of the Stress-Testing Framework

The stressed model must be validated.

Historical Replay

Apply the model to pre-crisis data and replay the observed scenario.

Test whether it identifies the documented pathway and timing.

Outcome Validation

Compare stressed predictions with realized:

- defaults;
- losses;
- outages;
- sanctions;
- liquidity usage;
- contagion.

Mechanism Validation

Verify whether the predicted minimum-action path corresponds to the actual transmission mechanism.

Intervention Validation

Where interventions occurred, test whether the model correctly predicts their effect on action and transmission.

Sensitivity Validation

Examine whether results are robust to reasonable changes in scenario assumptions.

13.8.5 Scenario Uncertainty

A scenario is not one deterministic path.

Let scenario parameters be

$$\boldsymbol{\zeta}^{(s)} \sim p\left(\boldsymbol{\zeta} \mid s\right). \tag{13.100}$$

The stressed risk distribution is

$$p\left(\mathrm{TRI}^{(s)}\right) = \int p\left(\mathrm{TRI} \mid \boldsymbol{\zeta}\right) p\left(\boldsymbol{\zeta} \mid s\right), \mathrm{d}\boldsymbol{\zeta}. \tag{13.101}$$

The report should show:

- median stressed risk;
- confidence or credible intervals;
- tail percentiles;
- dominant-path probabilities;

- intervention-effect uncertainty.

13.9 Model Uncertainty

Scenario uncertainty and model uncertainty are distinct.

A useful decomposition is

$$\mathrm{Var}\left(\mathrm{TRI}^{(s)}\right) = V_{\mathrm{scenario}} + V_{\mathrm{state}} + V_{\mathrm{barrier}} + V_{\mathrm{path}} + V_{\mathrm{phase}} + V_{\mathrm{model}}. \tag{13.102}$$

The institution should know whether uncertainty arises because the scenario is uncertain or because the model cannot identify the dominant pathway reliably.

13.9.1 AI-Specific Stress Testing

AI tools used in Chapters 11 and 12 should themselves be stressed.

Relevant tests include:

- data drift;
- missing-variable stress;
- document-extraction error;
- adversarial inputs;
- model-regime breakdown;
- pathway-instability stress;
- hallucination and grounding failure.

An AI-assisted risk system has its own barrier architecture and should be governed accordingly.

13.9.2 Governance of Stress Testing

A stress-test report should contain:

1. scenario definition;
2. empirical calibration;
3. baseline state and risk measures;
4. stressed state and geometry;
5. barrier deterioration;
6. covariance rotation;
7. pathway activation;

8. phase and interference effects;
9. minimum-action path;
10. transmission and tunneling-risk change;
11. intervention comparison;
12. uncertainty;
13. limitations.

The report should clearly distinguish:

$$\text{Scenario Assumption}, \qquad \text{Model Estimate}, \qquad \text{Management Judgment}. \tag{13.103}$$

13.9.3 Stress-Test Scorecard

A practical scorecard may contain:

Table 13.1: Baseline and stressed tunneling-risk scorecard.

Dimension	**Baseline**	**Stress**	**Main Driver**	**Preferred Intervention**
State distance	Moderate	High risk	Liquidity decline	Reduce exposure
Barrier integrity	Strong	Weak	Collateral reliability	Add usable protection
Adverse mobility	Low	High	Covariance rotation	Hedge concentration
Minimum action	High	Low	Refinancing route	Extend maturity
Coherence	Moderate	High	Maturity synchronization	Advance liquidity support
Transmission	Low	Elevated	Action compression	Escalate intervention

13.9.4 What Stress Testing Can Establish

A stress test does not predict that the scenario will occur.

It does not prove that the most adverse modeled pathway will be realized.

It does not establish physical quantum behaviour.

It shows how the estimated risk architecture responds under specified assumptions.

The appropriate interpretation is:

> Under the stated scenario and behavioural assumptions, the estimated geometry becomes less protective, the dominant pathway becomes more accessible, and effective adverse transmission rises materially.

13.9.5 The Complete Stress-Testing Architecture

The complete empirical stress object is

$$\mathfrak{S} * i, t = \left(\mathfrak{M} * i, t^0, \mathcal{S}^{(s)}, \mathfrak{M} * i, t^{(s)}, \Delta\mathbf{M} * i, t^{(s)}, \Delta\mathrm{TRI} * i, t^{(s)}, \mathbf{u}^*, \mathbf{U} * i, t^{(s)}\right), \qquad (13.104)$$

where:

- $\mathfrak{M}^0$ is the baseline architecture;
- $\mathcal{S}^{(s)}$ is the scenario;
- $\mathfrak{M}^{(s)}$ is the stressed architecture;
- $\Delta\mathbf{M}^{(s)}$ is the change in measured components;
- $\Delta\mathrm{TRI}^{(s)}$ is the composite risk change;
- $\mathbf{u}^*$ is the preferred intervention;
- $\mathbf{U}^{(s)}$ is uncertainty.

13.10 Chapter Synthesis

Chapter 11 showed how hidden pathways may be detected empirically. Chapter 12 converted those estimated objects into measurable, normalized, and decision-ready indicators.

Chapter 13 has extended the empirical programme into stress testing.

A tunneling stress test changes more than the value of the state variables. It may alter the state manifold, the metric, the adverse-region boundary, covariance geometry, barrier structure, pathway set, phase relationships, and transmission mechanism.

Stress may move the system toward the barrier while simultaneously moving or weakening the barrier itself.

The covariance geometry may expand, rotate, and collapse into fewer dimensions. Diversification may disappear as adverse mobility becomes concentrated.

Protective barriers may lose height, width, reliability, continuity, and independence. Shared barriers may become capacity-constrained when many institutions use them simultaneously.

Previously inactive pathways may become dominant. Classical path resistance and tunneling action must be recalculated under every scenario.

Phase relationships may change because adverse events become more synchronized, persistent, or delayed. Constructive interference may therefore increase even when standalone pathway probabilities rise only moderately.

The exponential relationship between action and transmission creates nonlinear stressed outcomes. Modest action compression may cause a material increase in adverse accessibility.

Institutional behaviour is endogenous. Management actions and policy responses may stabilize one entity while amplifying system-wide stress.

Dynamic, multi-period stress testing is required to represent feedback, path dependence, delayed failure, and recovery.

Scenarios should be calibrated through historical evidence, market distributions, structural models, expert judgment, machine learning, and simulation. Generative AI may expand scenarios, but narrative outputs must be converted into governed, reviewable assumptions.

Reverse stress testing identifies the least severe combination of state movement, barrier erosion, covariance rotation, pathway activation, and phase alignment that produces a critical risk level.

Intervention stress testing compares actions according to how they raise barrier height, increase width, reduce permeability, or change phase. The preferred intervention is the one that produces the greatest robust reduction in transmission and tunneling risk relative to cost.

The empirical legitimacy of the framework depends on historical replay, outcome validation, mechanism validation, intervention testing, sensitivity analysis, and uncertainty quantification.

The central proposition of Chapter 13 is:

> Stress testing becomes structurally informative when it evaluates not only how much a system loses, but how adverse scenarios reshape the geometry of risk, weaken protective barriers, redirect mobility, activate hidden pathways, synchronize adverse mechanisms, compress minimum action, and alter the effectiveness of intervention.

Chapter 14 will move from stressed diagnosis to governance. It will examine how institutions can deliberately design, monitor, and control barrier permeability, pathway accessibility, phase alignment, and intervention authority within a formal risk-management architecture.

Chapter 14

Governing Barrier Permeability

14.1 From Stress Diagnosis to Governed Intervention

Chapter 11 developed an empirical architecture for detecting hidden pathways. Chapter 12 converted the estimated geometry, barriers, pathways, phase relationships, action, and transmission into measurable risk indicators. Chapter 13 subjected those objects to forward, reverse, dynamic, network, and intervention stress tests.

The present chapter completes the governance sequence.

Detection establishes that a hidden pathway may be emerging. Measurement establishes its magnitude and principal drivers. Stress testing shows how the pathway behaves under adverse conditions. Governance determines who must act, which intervention is permissible, when it must occur, and how its effectiveness will be verified.

The empirical progression is therefore

$$\boxed{\begin{aligned} &\text{Detection} \longrightarrow \text{Measurement} \longrightarrow \text{Stress Testing} \\ &\qquad\qquad \longrightarrow \text{Limits} \longrightarrow \text{Authority} \longrightarrow \text{Intervention} \\ &\qquad\qquad \longrightarrow \text{Verification} \longrightarrow \text{Accountability.} \end{aligned}} \tag{14.1}$$

Governance must address more than whether a protective structure formally exists. It must determine whether the structure remains available, reliable, independent, sufficiently wide, and capable of interrupting the dominant pathway under realistic stress.

The central proposition of this chapter is:

> Effective risk governance means deliberately controlling the permeability of the institution's protective architecture by assigning ownership, limits, intervention authority, timing, testing, evidence requirements, and accountability to the barriers and pathways through which adverse transitions may occur.

14.1.1 Barrier Permeability as a Governance Variable

A barrier separates a stable or fragile region from an adverse region. Its effectiveness depends on more than nominal size.

A barrier may be:

- high but narrow;
- wide but unreliable;
- continuous in policy but fragmented in operation;
- composed of several controls that share one dependency;
- strong in ordinary conditions but weak under simultaneous stress;
- available in principle but activated too late.

Barrier permeability should therefore be treated as an explicit governance variable.

Let the empirically measured barrier characteristics be

$$\mathbf{B} * i, t = \left(H * i, t, W_{i,t}, C_{i,t}, R_{i,t}, K_{i,t}^{\text{eff}}, Q_{i,t}, \tau_{i,t}^{\text{resp}}\right), \tag{14.2}$$

where:

- $H_{i,t}$ is height;
- $W_{i,t}$ is width;
- $C_{i,t}$ is continuity;
- $R_{i,t}$ is reliability;
- $K_{i,t}^{\text{eff}}$ is the effective number of independent layers;
- $Q_{i,t}$ is residual capacity;
- $\tau_{i,t}^{\text{resp}}$ is response time.

A general permeability function is

$$\Pi_{i,t} = \pi\left(\mathbf{B} * i, t, \mathbf{M} * i, t, \Gamma_{i,t}, \boldsymbol{\theta}_{i,t}\right), \tag{14.3}$$

where $\mathbf{M} * i, t$ is mobility, $\Gamma * i, t$ is the pathway set, and $\boldsymbol{\theta}_{i,t}$ contains phase relationships.

High permeability means that the adverse region is accessible through one or more low-resistance routes despite the existence of nominal protections.

14.1.2 An Empirical Permeability Measure

A practical permeability indicator should remain connected to the estimates developed in Chapters 11 and 12.

One formulation is

$$\Pi_{i,t}^{\text{emp}} = \omega_B \widetilde{B} * i, t^{\text{weak}} + \omega * A\widetilde{A} * i, t^{\text{risk}} + \omega * C\widetilde{C} * i, t^{-} + \omega * T\widetilde{T}_{R,i,t}, \tag{14.4}$$

where:

- $\widetilde{B}^{\text{weak}}$ is normalized barrier weakness;
- $\widetilde{A}^{\text{risk}}$ is normalized low-action risk;
- $\widetilde{C}^{-}$ is adverse coherence;
- $\widetilde{T}_R$ is normalized transmission.

The weights satisfy

$$\omega_j \geq 0, \qquad \sum_j \omega_j = 1. \tag{14.5}$$

A nonlinear alternative is

$$\Pi_{i,t}^{\text{nl}} = 1 - \prod_j (1 - X_{j,i,t})^{\omega_j}, \tag{14.6}$$

where X_j are normalized risk-oriented components.

The nonlinear form captures the possibility that several moderate weaknesses become dangerous when they occur together.

14.1.3 Governance Does Not Begin with the Score

The governance process should not begin only after the permeability score becomes elevated.

It begins when the institution defines:

- which adverse transitions matter;
- which barriers are intended to prevent them;
- which pathways could bypass those barriers;
- who owns each barrier;
- which indicators show deterioration;
- which interventions are authorized;
- how quickly intervention must occur.

The governance architecture is therefore prospective.

It should be designed before the adverse pathway becomes active.

14.1.4 The Barrier Register

Every institution should maintain a barrier register.

For barrier (j), define the record

$$\mathcal{R}_j^B = (O_j, E_j, A_j, R_j, D_j, L_j, T_j, V_j)\,, \tag{14.7}$$

where:

- O_j is the owner;
- E_j is the adverse event the barrier addresses;
- A_j is activation authority;
- R_j is expected reliability;
- D_j is dependency information;
- L_j is the associated limit;
- T_j is the testing schedule;
- V_j is the evidence and model version.

The barrier register should include both formal protections and operational protections.

Examples include:

- capital buffers;
- liquidity facilities;
- collateral;
- covenants;
- legal rights;
- cyber controls;
- backup systems;
- escalation processes;
- resolution plans;
- central-bank facilities.

14.1.5 The Pathway Register

A complementary pathway register should document the known routes to adverse outcomes.

For pathway (k),

$$\mathcal{R}_k^{\Gamma} = (\gamma_k, P_k, B_k, O_k, I_k, \tau_k, S_k, V_k)\,, \tag{14.8}$$

where:

- γ_k is the ordered pathway;
- P_k is estimated completion probability;
- B_k is the barrier segment crossed;
- O_k is the pathway owner;
- I_k is the intervention set;
- τ_k is the intervention window;
- S_k is current action;
- V_k is evidence and model version.

The pathway register should be updated when Chapter 11 detection identifies a new route or Chapter 13 stress testing activates a previously dormant path.

14.1.6 Ownership of Barriers

A barrier without an owner is not a governed barrier.

Ownership should include responsibility for:

- design;
- capacity;
- maintenance;
- testing;
- monitoring;
- escalation;
- remediation;
- evidence quality.

The barrier owner should not necessarily be the business unit that benefits from the barrier.

For example:

- treasury may own liquidity capacity;

- legal may own enforceability;
- operations may own recovery;
- cybersecurity may own detection and containment;
- the board may own aggregate risk appetite.

14.2 Ownership of Pathways

The dominant pathway often crosses organizational boundaries.

A refinancing pathway may involve the business, treasury, legal, credit, and investor relations. A cyber pathway may involve identity, infrastructure, monitoring, and incident response. A conduct pathway may involve sales, compliance, legal, and governance.

A pathway owner should therefore coordinate across barrier owners.

The pathway owner's task is to ensure that:

- pathway evidence is complete;
- the sequence is monitored;
- intervention authority is clear;
- timing is coordinated;
- no transition falls between organizational mandates.

14.2.1 A Responsibility Matrix

For barrier (j) and pathway (k), define a responsibility matrix

$$\mathbf{R}^{\mathrm{gov}} = [r_{jk}] \,, \tag{14.9}$$

where r_{jk} records whether the function is:

- responsible;
- accountable;
- consulted;
- informed.

The matrix should cover:

- barrier operation;
- pathway monitoring;
- intervention approval;

- model validation;
- stress testing;
- post-event review.

14.2.2 Translating Risk Appetite into Structural Limits

Traditional risk appetite frequently uses exposure, capital, liquidity, concentration, and loss limits.

The tunneling framework adds structural limits.

Let the governance-limit vector be

$$\boldsymbol{\tau}^{\text{gov}} = (\tau_B, \tau_S, \tau_C, \tau_I, \tau_T, \tau_\Pi, \tau_U), \tag{14.10}$$

where:

- τ_B is the barrier-weakness limit;
- τ_S is the minimum-action floor;
- τ_C is the adverse-coherence limit;
- τ_I is the constructive-interference limit;
- τ_T is the transmission limit;
- τ_Π is the permeability limit;
- τ_U is the uncertainty limit.

A breach occurs if

$$B_{i,t}^{\text{weak}} \geq \tau_B, \tag{14.11}$$

or

$$S_{R,i,t}^{*} \leq \tau_S, \tag{14.12}$$

or

$$T_{R,i,t} \geq \tau_T. \tag{14.13}$$

14.2.3 Empirical Calibration of Governance Limits

Limits should not be selected arbitrarily.

They may be calibrated from:

- historical pre-event distributions;
- observed intervention capacity;
- loss tolerance;
- false-warning cost;
- lead-time requirements;
- supervisory standards;
- reverse stress tests.

For component (X), a historical limit may be

$$\tau_X = \widehat{F}^{-1}_{X,\mathrm{pre-event}}(\alpha), \tag{14.14}$$

where α is a selected severe percentile.

A decision-based limit may minimize

$$\tau^*_X = \arg, \min *\tau \left[C * \mathrm{miss}(\tau) + C_{\mathrm{false}}(\tau) + C_{\mathrm{intervention}}(\tau)\right]. \tag{14.15}$$

14.2.4 Level, Trend, and Acceleration Limits

Governance should not rely only on current levels.

For variable $X_{i,t}$, monitor:

$$X_{i,t}, \qquad \Delta X_{i,t}, \qquad \Delta^2 X_{i,t}. \tag{14.16}$$

A rapid decline in action may require escalation even when the current level remains above the formal floor.

Define action-compression trigger

$$-\Delta S^*_{R,i,t} \geq \tau_{\Delta S}. \tag{14.17}$$

Define permeability-acceleration trigger

$$\Delta^2 \Pi_{i,t} \geq \tau_{\Delta^2 \Pi}. \tag{14.18}$$

14.2.5 Persistence-Based Escalation

Transient noise should be distinguished from sustained deterioration.

Define persistence above threshold τ_X:

$$\mathcal{P} * X, i, t = \frac{1}{m} \sum * h = 0^{m-1} \mathbb{I} \left(X_{i,t-h} \geq \tau_X \right). \tag{14.19}$$

A sustained breach occurs when

$$\mathcal{P}_{X,i,t} \geq \pi^*. \tag{14.20}$$

Sudden action compression or transmission spikes may override the persistence requirement.

14.2.6 Risk Appetite as a Permeability Budget

The institution may define an aggregate permeability budget.

Let exposure (i) have importance weight w_i.

Aggregate permeability is

$$\Pi_t^{\text{agg}} = \sum_{i=1}^{N} w_i \Pi_{i,t}. \tag{14.21}$$

The risk-appetite condition is

$$\Pi_t^{\text{agg}} \leq \Pi_{\max}. \tag{14.22}$$

Concentration should also be considered.

If permeability is concentrated in a few critical entities or pathways, the aggregate average may be misleading.

14.3 Permeability Concentration

Let normalized permeability contributions be

$$\rho_{i,t} = \frac{w_i \Pi_{i,t}}{\sum_j w_j \Pi_{j,t}}. \tag{14.23}$$

Concentration is

$$H_{\Pi,t} = \sum_i \rho_{i,t}^2. \tag{14.24}$$

High concentration suggests that targeted intervention may substantially reduce aggregate risk.

Low concentration suggests broad structural redesign.

14.3.1 Precommitted Intervention Authority

An institution should not wait for crisis conditions to determine who may act.

Intervention authority should be defined in advance for:

- adding liquidity;
- increasing collateral;
- reducing exposure;
- tightening limits;
- suspending activity;
- isolating systems;
- extending maturities;
- escalating legal review;
- activating recovery plans;
- communicating with regulators.

Let intervention u_j require authority level a_j.

The authority map is

$$\mathcal{A} : u_j \longmapsto a_j. \tag{14.25}$$

14.3.2 Intervention Triggers

An intervention rule may be written as

$$u_{j,i,t} = \begin{cases} 1, & \text{if } \mathcal{G} * j \left(\mathbf{M} * i, t^{\text{tun}}, \mathbf{U}_{i,t}\right) = 1, \\ 0, & \text{otherwise}, \end{cases} \tag{14.26}$$

where $\mathcal{G}_j$ is the governance trigger for intervention (j).

For example, a maturity-extension trigger may depend on:

- refinancing-path action;
- maturity concentration;
- liquidity-barrier weakness;
- confidence-path coherence;
- model confidence.

14.3.3 Four Governance Levers

The intervention set can be organized around four structural levers.

Raise Barrier Height

Examples include:

- add capital;
- secure liquidity;
- improve collateral;
- strengthen legal enforceability;
- increase recovery capacity.

Increase Barrier Width

Examples include:

- extend maturity;
- add control stages;
- create review intervals;
- build recovery time;
- introduce resolution stays.

Reduce Permeability

Examples include:

- remove shared dependencies;
- reduce concentration;
- isolate critical systems;
- close contractual gaps;
- diversify support mechanisms.

Change Phase

Examples include:

- accelerate detection;
- advance stabilizing liquidity;
- stagger margin calls;
- delay synchronized asset sales;
- coordinate communication.

14.3.4 Timing as a Governance Variable

A correct intervention delivered too late may fail.

Let adverse-path completion time be

$$\tau^{A}_{\gamma,i,t}. \tag{14.27}$$

Let intervention-response time be

$$\tau^{\text{resp}}_{u_j,i,t}. \tag{14.28}$$

The intervention is timely only if

$$\tau^{\text{resp}}_{u_j,i,t} < \tau^{A}_{\gamma,i,t}. \tag{14.29}$$

A safety margin is

$$\Delta\tau^{\text{safe}}_{j,i,t} = \tau^{A}_{\gamma,i,t} - \tau^{\text{resp}}_{u_j,i,t}. \tag{14.30}$$

Negative values indicate that the intervention process is structurally too slow.

14.3.5 Empirical Estimation of Response Time

Response-time distributions should be estimated from:

- historical incidents;
- approval logs;
- treasury drawdowns;
- recovery exercises;
- legal escalation records;
- crisis simulations.

Let observed response times be

$$\left\{\tau^{\text{obs}}_{u_j,n}\right\}_{n=1}^{N_j}. \tag{14.31}$$

Estimate the response-time distribution

$$\tau^{\text{resp}}_{u_j} \sim \widehat{F}_{u_j}. \tag{14.32}$$

Governance should use a conservative percentile rather than the historical mean.

14.3.6 Service-Level Agreements for Intervention

For intervention u_j, define a maximum response time

$$\tau_{u_j}^{\max}. \tag{14.33}$$

The compliance indicator is

$$C_{u_j,t}^{\mathrm{SLA}} = \mathbb{I}\left(\tau_{u_j,t}^{\mathrm{actual}} \le \tau_{u_j}^{\max}\right). \tag{14.34}$$

Repeated failure to meet intervention timing should reduce the reliability adjustment applied to the barrier.

14.4 Intervention Selection

The preferred intervention should be chosen according to expected reduction in tunneling risk, cost, feasibility, timing, and uncertainty.

For intervention u_j, define expected risk reduction

$$\Delta\mathrm{TRI} * j, i, t = \mathrm{TRI} * i, t^{\mathrm{before}} - \mathbb{E}\left[\mathrm{TRI}_{i,t}^{\mathrm{after},j}\right]. \tag{14.35}$$

Intervention efficiency is

$$E_{j,i,t} = \frac{\Delta\mathrm{TRI}_{j,i,t}}{C_j}. \tag{14.36}$$

A broader decision score is

$$V_{j,i,t} = \alpha\Delta\mathrm{TRI} * j, i, t + \beta\Delta T * j, i, t + \gamma\Delta S_{j,i,t}^{*} - \delta C_j - \zeta U_{j,i,t}, \tag{14.37}$$

where $U_{j,i,t}$ is intervention uncertainty.

14.4.1 Intervention Portfolios

Several interventions may complement or offset one another.

Let the portfolio be

$$\mathbf{u} = (u_1, \dots, u_m). \tag{14.38}$$

The governed portfolio problem is

$$\mathbf{u}^{*} = \arg, \min_{\mathbf{u}} \left[\mathbb{E}\left(\mathrm{TRI}^{\mathrm{after}}(\mathbf{u})\right) + \lambda_C C(\mathbf{u}) + \lambda_U U(\mathbf{u})\right], \tag{14.39}$$

subject to:

- authority constraints;
- liquidity constraints;
- legal constraints;
- operational capacity;
- timing requirements;
- risk-appetite limits.

14.4.2 Empirical Evaluation of Intervention Effectiveness

The effectiveness of an intervention should be estimated, not assumed.

Possible designs include:

- matched historical cases;
- difference-in-differences;
- regression discontinuity;
- instrumental variables;
- causal forests;
- structural simulation;
- controlled exercises.

The causal effect is

$$\tau_j(\mathbf{x}) = \mathbb{E}\left[Y(u_j = 1) - Y(u_j = 0) \mid \mathbf{X} = \mathbf{x}\right]. \tag{14.40}$$

The governance system should distinguish a predictive intervention from a causally effective intervention.

14.4.3 Barrier Assurance

Barrier governance requires independent assurance that protections operate as intended.

Assurance methods include:

- control testing;
- collateral revaluation;
- liquidity-draw tests;
- legal enforceability reviews;
- cyber penetration tests;

- recovery exercises;
- resolution simulations;
- third-party audits.

Let test result for barrier (j) be

$$Z_{j,n}^{\text{test}}. \tag{14.41}$$

Estimated barrier reliability is

$$\widehat{R} * j, t = \mathbb{E}\left[Z * j^{\text{test}} \mid \mathcal{I}_t\right]. \tag{14.42}$$

14.4.4 Testing Barrier Independence

Nominally separate controls may fail together.

Let failure indicators be

$$F_{1,t}, \ldots, F_{K,t}. \tag{14.43}$$

Estimate conditional dependence through

$$P\left(F_1, \ldots, F_K \mid Z\right), \tag{14.44}$$

where (Z) includes shared people, vendors, data, infrastructure, and governance.

The effective number of independent barriers is

$$K_{\text{eff}} = \frac{\left(\sum_j \lambda_j\right)^2}{\sum_j \lambda_j^2}, \tag{14.45}$$

where λ_j are dependency-adjusted eigenvalues.

A decline in K_{eff} should trigger barrier redesign.

14.4.5 Barrier Testing Frequency

Testing frequency should depend on:

- criticality;
- permeability;
- deterioration speed;
- uncertainty;

- dependency concentration;
- recent intervention.

Let testing interval be

$$\Delta t_j^{\text{test}} = f\left(\Pi_j, \dot{\Pi} * j, U_j, K * \text{eff}, j\right). \tag{14.46}$$

Higher permeability and uncertainty should produce more frequent testing.

14.4.6 Reverse Stress Testing as a Governance Tool

Chapter 13 used reverse stress testing to identify the least severe scenario producing a critical risk condition.

Governance should use that scenario to define:

- contingency plans;
- precommitted authorities;
- minimum barrier standards;
- scenario-specific limits;
- escalation paths.

Let the critical scenario be

$$\mathcal{S}^* = \arg, \min_{\mathcal{S}} C(\mathcal{S}) \tag{14.47}$$

subject to

$$\Pi^{(\mathcal{S})} \geq \tau_\Pi. \tag{14.48}$$

The institution should demonstrate that available interventions prevent or materially delay that scenario.

14.5 Red-Team Governance

Red-team exercises should attempt to:

- identify hidden barrier gaps;
- activate unmodeled pathways;
- exploit shared dependencies;
- create timing conflicts;

- delay intervention;
- challenge model assumptions.

The red team should not merely test whether a control can be bypassed. It should test whether several ordinary weaknesses can align into a low-action route.

14.5.1 Post-Event Reconstruction

After an incident, near miss, or material intervention, the institution should reconstruct:

- the observed state trajectory;
- barrier performance;
- pathway sequence;
- intervention timing;
- action compression;
- coherence;
- model errors.

Let predicted dominant path be

$$\widehat{\gamma}_t^* \tag{14.49}$$

and observed path be

$$\gamma_{\text{obs}}. \tag{14.50}$$

Pathway validation should assess

$$d_\Gamma \left(\widehat{\gamma} * t^*, \gamma * \text{obs}\right). \tag{14.51}$$

14.5.2 Learning from Near Misses

Near misses are particularly valuable because the adverse pathway became accessible but was interrupted.

A near-miss analysis should identify whether success resulted from:

- higher barrier capacity;
- greater width;
- lower permeability;
- destructive interference;

- timely intervention;
- good fortune.

The final category is important. A system should not treat a favorable outcome as evidence of barrier effectiveness without mechanism analysis.

14.5.3 Model Governance Across the Empirical Chain

The governance perimeter must include the full empirical architecture.

It should cover:

1. raw data;
2. timestamp integrity;
3. state-variable definitions;
4. manifold estimation;
5. metric construction;
6. drift and covariance estimation;
7. barrier modeling;
8. pathway discovery;
9. phase proxies;
10. action calculation;
11. transmission calibration;
12. composite measurement;
13. stress testing;
14. intervention modeling.

Oversight of the final score alone is insufficient.

14.5.4 Model Inventory

Each model component should have an inventory record

$$\mathcal{R}_m^{\text{model}} = (O_m, P_m, D_m, M_m, V_m, U_m, L_m, R_m)\,, \tag{14.52}$$

where:

- O_m is owner;

- P_m is purpose;
- D_m is data;
- M_m is method;
- V_m is validation status;
- U_m is uncertainty;
- L_m is limitations;
- R_m is review date.

14.5.5 Model Validation

Validation should include:

- conceptual soundness;
- data quality;
- out-of-sample performance;
- calibration;
- stability;
- benchmark comparison;
- sensitivity;
- implementation testing;
- decision impact.

The tunneling extension should remain only if it provides stable incremental value beyond credible classical models.

14.5.6 Governance of Phase Estimates

Phase requires particularly strong governance because it is latent and potentially weakly identified.

The institution should document:

- observable timing proxies;
- parameter constraints;
- regularization;
- stability across samples;

- ablation results;
- comparison with classical timing models.

A phase contribution should not drive intervention if model confidence is below a defined threshold.

14.6 Governance of Artificial Intelligence Tools

Machine learning and artificial intelligence may support:

- manifold learning;
- nonlinear dynamics;
- barrier estimation;
- pathway discovery;
- document extraction;
- scenario generation;
- intervention analysis;
- report drafting.

AI tools introduce risks including:

- data leakage;
- hallucinated extraction;
- unstable embeddings;
- spurious pathways;
- uncalibrated outputs;
- automation bias;
- weak reproducibility.

14.6.1 AI Control Architecture

An AI-assisted workflow should maintain separation among:

$$\text{Evidence Extraction} \to \text{Model Estimation} \to \text{Validation} \to \text{Decision}. \tag{14.53}$$

The system should include:

- controlled data access;

- source-grounded outputs;
- human review;
- confidence thresholds;
- version control;
- audit logs;
- fallback classical processes.

14.6.2 Language Models as Controlled Measurement Tools

If a language model extracts barrier or pathway variables from documents, every value should be linked to source evidence.

For extracted variable z_d,

$$\widehat{z}_d^{\mathrm{LLM}} = z_d + \varepsilon_d^{\mathrm{LLM}}. \tag{14.54}$$

Governance should require:

- field-level validation;
- citation to source passages;
- abstention;
- human approval for material fields;
- monitoring of extraction drift.

14.6.3 Agentic AI and Intervention Authority

An AI agent may update scores, retrieve evidence, and propose interventions. It should not independently execute high-impact actions unless explicitly authorized within a tightly constrained policy.

Let the agent's permitted action set be

$$\mathcal{U} * \mathrm{AI} \subset \mathcal{U} * \mathrm{institution}. \tag{14.55}$$

High-impact actions should remain outside $\mathcal{U}_{\mathrm{AI}}$ or require human approval.

14.6.4 Three Lines of Governance

The governance structure may be organized through three lines.

First Line

Owns the business process, operates controls, monitors barriers, and initiates interventions.

Second Line

Defines methodology, limits, risk appetite, challenge, and escalation standards.

Third Line

Provides independent assurance over data, models, barriers, authority, and governance effectiveness.

The tunneling framework should not blur these responsibilities.

14.6.5 Committee Architecture

Relevant committees may include:

- business risk committee;
- credit committee;
- operational-risk committee;
- model-risk committee;
- asset-liability committee;
- crisis-management committee;
- board risk committee.

Each committee should receive only the information necessary for its mandate, while preserving consistent definitions across governance levels.

14.6.6 The Management Scorecard

A management scorecard should contain:

1. current permeability;
2. barrier integrity and trend;
3. minimum action and compression;
4. dominant pathway;
5. pathway concentration;
6. adverse coherence;
7. transmission;
8. model confidence;
9. breached limits;

10. accountable owner;
11. available interventions;
12. response deadline.

14.7 The Board Dashboard

The board should receive a more strategic view.

The dashboard should show:

- aggregate permeability;
- permeability concentration;
- critical barriers;
- dominant enterprise pathways;
- stress-test results;
- reverse-stress vulnerabilities;
- intervention readiness;
- model uncertainty;
- overdue remediation.

The board's question is not merely:

> How much risk do we currently have?

It is:

> Through which routes can risk travel, how permeable are our protections, and are we prepared to act before those routes become irreversible?

14.7.1 Observed, Estimated, and Governed Information

Reporting should preserve the empirical hierarchy.

Table 14.1: Separation of observed evidence, estimated structure, governance interpretation, and required action.

Observed Evidence	**Estimated Structure**	**Governance Interpretation**	**Required Action**
Liquidity falls	Barrier height declines	Capacity approaching limit	Secure committed funding
Response delays rise	Adverse phase alignment increases	Intervention may arrive too late	Shorten approval chain
Controls share vendor	Effective layers decline	Barrier independence overstated	Add independent fallback
Path action compresses	Dominant route becomes accessible	Escalation threshold breached	Close or widen pathway

14.7.2 Empirical Governance Workflow

A practical implementation can follow fourteen steps.

1. Define adverse regions and material transition types.
2. Identify the protective barriers associated with each transition.
3. Populate the barrier register.
4. Populate the pathway register.
5. Assign barrier and pathway owners.
6. Freeze validated empirical definitions from Chapters 11 and 12.
7. Calibrate limits using historical, economic, and reverse-stress evidence.
8. Define intervention authority and response-time standards.
9. Calculate current permeability and component measures.
10. Run forward and reverse stress tests.
11. Compare available intervention portfolios.
12. Escalate breaches according to level, trend, and persistence.
13. Verify intervention effects.
14. Update models, controls, and governance records.

The workflow is

$$\boxed{\begin{aligned}\text{Map Barriers and Paths} &\longrightarrow \text{Assign Ownership} \longrightarrow \text{Calibrate Limits}\\ &\longrightarrow \text{Preauthorize Actions} \longrightarrow \text{Monitor Permeability}\\ &\longrightarrow \text{Intervene} \longrightarrow \text{Verify} \longrightarrow \text{Learn.}\end{aligned}} \tag{14.56}$$

14.7.3 A Minimum Viable Governance Pilot

The corporate-credit pilot developed in Chapters 11–13 can be extended into a minimum viable governance system.

The pilot may include:

- one adverse event: default or distressed restructuring;
- three pathways: operating deterioration, refinancing failure, and collateral-covenant escalation;
- three main barriers: liquidity, collateral, and covenant headroom;
- one composite permeability measure;
- action, coherence, and transmission limits;
- three interventions: maturity extension, committed liquidity, and collateral enhancement.

14.7.4 Pilot Data Requirements

The governance pilot requires:

- quarterly financials;
- maturity schedules;
- collateral values;
- covenant terms;
- refinancing indicators;
- approval timelines;
- intervention records;
- default and restructuring outcomes.

Unstructured covenant documents may be processed through a controlled retrieval-augmented language-model system, with human approval.

14.7.5 Pilot Limit Calibration

Suppose historical analysis shows that adverse events become materially more frequent when:

$$S_R^* < 0.75, \tag{14.57}$$

or

$$T_R > 0.55, \tag{14.58}$$

or

$$C^- > 0.70. \tag{14.59}$$

These become provisional governance limits, subject to validation and false-warning analysis.

14.7.6 Pilot Intervention Rules

A possible intervention rule is:

- if liquidity-barrier weakness breaches its limit, secure committed liquidity;
- if refinancing action falls below its floor, extend maturity;
- if collateral-covenant action compresses, increase collateral or renegotiate covenants;
- if adverse coherence rises sharply, accelerate sponsor and lender coordination.

Each intervention should have:

- an owner;
- authority;
- deadline;
- expected effect;
- cost;
- verification metric.

14.8 Illustrative Governance Decision

Suppose borrower (i) has:

$$\Pi_{i,t} = 0.74, \qquad S_{R,i,t}^* = 0.52, \qquad C_{i,t}^- = 0.79, \qquad T_{R,i,t} = 0.63. \tag{14.60}$$

The dominant path is refinancing failure.

The relevant barriers are liquidity and maturity width.

The approved intervention portfolio may include:

- maturity extension;
- committed liquidity;
- accelerated lender coordination.

After intervention, the model estimates:

$$S^{*,\text{after}}_{R,i,t} = 1.12, \qquad C^{-,\text{after}}_{i,t} = 0.46, \qquad T^{\text{after}}_{R,i,t} = 0.27. \tag{14.61}$$

The governance process should then verify whether actual liquidity, maturity, and coordination changes occurred as assumed.

14.8.1 Operational and Cyber Governance

For operational and cyber risk, the barrier register may include:

- access controls;
- segregation of duties;
- monitoring;
- incident response;
- backups;
- recovery systems;
- vendor redundancy.

The governance system should monitor:

- control independence;
- detection delay;
- dwell time;
- response capacity;
- shared-vendor concentration;
- dominant attack or failure path.

14.8.2 Legal and Conduct Governance

For legal, regulatory, and conduct risk, barriers may include:

- contractual clarity;
- evidence quality;

- disclosure;
- review;
- role separation;
- remediation;
- escalation authority.

Governance should assign ownership for:

- interpretive ambiguity;
- evidence preservation;
- conduct consistency;
- regulatory communication;
- remediation timing.

14.8.3 Systemic Governance

For systemic risk, no single institution may own the complete barrier.

Governance may involve:

- supervisors;
- central banks;
- clearing houses;
- infrastructure operators;
- resolution authorities;
- market participants.

Systemic barrier governance should address:

- collective liquidity capacity;
- shared collateral;
- infrastructure concentration;
- policy timing;
- synchronized behaviour;
- cross-authority coordination.

14.8.4 Cross-Institutional Intervention Protocols

Because systemic pathways cross institutions, intervention protocols should specify:

- information sharing;
- activation authority;
- sequencing;
- communication;
- burden allocation;
- confidentiality;
- post-crisis accountability.

The absence of coordination may itself become a low-action systemic pathway.

14.8.5 Governance Metrics

Governance effectiveness should be measured.

Possible metrics include:

- percentage of barriers with assigned owners;
- percentage of critical pathways with tested interventions;
- intervention response-time compliance;
- barrier-test success rate;
- effective-layer count;
- remediation aging;
- limit-breach duration;
- model-confidence coverage;
- post-intervention risk reduction.

14.8.6 Barrier Ownership Coverage

Define

$$C_{\text{owner}} = \frac{N_{\text{barriers with owner}}}{N_{\text{critical barriers}}}. \tag{14.62}$$

The target should be one for all critical barriers.

14.9 Intervention Readiness

For intervention (j), define readiness as

$$R_j^{\text{int}} = f\left(A_j, C_j, \tau_j, T_j, D_j\right), \tag{14.63}$$

where:

- A_j is authority clarity;
- C_j is capacity;
- τ_j is response speed;
- T_j is testing status;
- D_j is dependency independence.

14.9.1 Governance Effectiveness Index

A composite governance index may be

$$G_t = \alpha_1 C_{\text{owner},t} + \alpha_2 C_{\text{test},t} + \alpha_3 C_{\text{SLA},t} + \alpha_4 R_{\text{int},t} + \alpha_5 C_{\text{remediation},t}. \tag{14.64}$$

The index should not substitute for component reporting.

14.9.2 Governance Under Uncertainty

The institution should distinguish:

- high risk with high confidence;
- high risk with low confidence;
- low risk with high confidence;
- low risk with low confidence.

High risk with high confidence supports decisive intervention.

High risk with low confidence supports immediate investigation and precautionary measures.

Low risk with low confidence may require data improvement and conservative limits.

14.9.3 The Precautionary Principle

Where model confidence is low but the potential adverse consequence is severe, governance may use a precautionary threshold.

Let expected loss under uncertainty be

$$\mathbb{E}\left[L \mid \mathcal{I}_t\right]. \tag{14.65}$$

A precautionary intervention may be justified when

$$\mathbb{E}[L] + \lambda_U \, \mathrm{SD}(L) > \tau_L. \tag{14.66}$$

14.9.4 Governance Failure Modes

Common governance failures include:

- barrier existence mistaken for barrier effectiveness;
- fragmented ownership;
- limits based only on current levels;
- slow approval chains;
- shared dependencies ignored;
- model outputs accepted without challenge;
- intervention effects assumed rather than verified;
- false precision;
- historical success confused with resilience;
- AI automation without accountability.

14.9.5 What Governing Permeability Is Not

Governing permeability is not an instruction to eliminate all risk.

It is not a demand for infinitely high barriers.

It is not a replacement for business judgment.

It is not a claim that every adverse event is tunneling.

It is not a justification for opaque automated intervention.

It is a structured way to govern how adverse transitions can occur and how the institution can prevent, delay, redirect, or interrupt them.

14.9.6 The Complete Governance Architecture

The complete governance object is

$$\mathfrak{G} * i, t^{\mathrm{gov}} = \left(\mathfrak{D} * i, t, \mathfrak{M} * i, t, \mathfrak{S} * i, t, \mathcal{R}^{B}, \mathcal{R}^{\Gamma}, \boldsymbol{\tau}^{\mathrm{gov}}, \mathcal{A}, \mathcal{U}, \mathcal{V}, \mathcal{Q}\right), \tag{14.67}$$

where:

- $\mathfrak{D}$ is the detection architecture;
- $\mathfrak{M}$ is the measurement architecture;
- $\mathfrak{S}$ is the stress-testing architecture;
- $\mathcal{R}^B$ is the barrier register;
- $\mathcal{R}^\Gamma$ is the pathway register;
- $\boldsymbol{\tau}^{\text{gov}}$ is the limit vector;
- $\mathcal{A}$ is the authority map;
- $\mathcal{U}$ is the intervention set;
- $\mathcal{V}$ is the validation architecture;
- $\mathcal{Q}$ is the accountability and reporting structure.

14.10 Chapter Synthesis

Chapter 11 established how hidden pathways may be detected empirically. Chapter 12 converted those estimated objects into transparent measures. Chapter 13 tested how the geometry, barriers, pathways, phase, action, and transmission change under stress.

Chapter 14 has converted those analytical objects into a formal governance architecture.

The central object is barrier permeability: the effective accessibility of an adverse region through nominally protective structures.

Permeability depends on barrier height, width, continuity, reliability, independence, residual capacity, pathway action, adverse coherence, transmission, and intervention timing.

Every critical barrier should appear in a barrier register with a named owner, purpose, capacity, dependencies, limits, testing schedule, and evidence base.

Every material pathway should appear in a pathway register with an ordered sequence, estimated probability, crossed barrier, intervention set, action, timing window, and owner.

Risk appetite should be translated into structural limits for barrier weakness, minimum action, adverse coherence, interference, transmission, permeability, and model uncertainty.

Limits should be calibrated empirically through historical pre-event distributions, intervention economics, false-warning costs, lead-time requirements, and reverse stress testing.

Governance should monitor levels, trends, accelerations, persistence, and concentration.

Intervention authority should be precommitted. The institution should know in advance who may raise capital, add liquidity, increase collateral, extend maturity, isolate systems, suspend activity, or escalate legal review.

The intervention architecture is organized around four levers: raise barrier height, increase width, reduce permeability, and change phase.

Timing is a governance variable. A correct intervention delivered after pathway completion is ineffective. Response-time distributions and service-level standards should therefore be estimated from actual institutional evidence.

Intervention effectiveness should be assessed causally where possible and verified after implementation. Feature importance and intuitive plausibility are not sufficient.

Barrier assurance should test reliability, capacity, and independence through exercises, audits, red-team analysis, recovery tests, liquidity draws, and legal review.

Reverse stress testing should be used to determine whether available interventions can prevent or delay the least severe scenario that generates critical permeability.

Model governance must cover the full empirical chain: data, manifold, metric, dynamics, barrier, pathways, phase, action, transmission, stress testing, and intervention.

Artificial-intelligence tools may support estimation, evidence extraction, scenario design, and reporting, but they require source grounding, validation, access controls, auditability, and human accountability.

Management reporting should identify barrier integrity, dominant pathways, action compression, coherence, transmission, ownership, breached limits, interventions, deadlines, and uncertainty.

The board should receive an enterprise-level view of aggregate permeability, concentration, critical barriers, reverse-stress vulnerabilities, intervention readiness, and overdue remediation.

The chapter's central proposition is:

> Risk becomes governable when the institution moves beyond measuring the probability of adverse outcomes and begins explicitly governing the barriers, pathways, timing relationships, authorities, and interventions that determine whether those outcomes remain accessible.

Chapter 15 will integrate the conceptual, mathematical, empirical, and governance layers into a unified framework for tunneling and risk management.

Part V

Toward a General Theory of Geometric Risk

Chapter 15

A Unified Framework for Geometric Risk

15.1 The Purpose of Unification

The preceding chapters developed a sequence of ideas rather than a single universal model. They began with probability, state spaces, covariance, barriers, and stochastic movement. They then introduced the mathematics of tunneling, separated physical theory from risk interpretation, applied the resulting language across risk domains, and converted the conceptual framework into detection, measurement, stress testing, and governance. The purpose of this final chapter is to state clearly what survives that sequence.

The unified framework does not claim that financial, operational, legal, or institutional systems are quantum systems. Nor does it claim that minimum action, exponential barrier sensitivity, or dominant transition paths are uniquely quantum-like. Classical large-deviation theory, Freidlin–Wentzell quasipotentials, Kramers escape, network contagion, nonlinear hazards, and regime-switching dynamics already generate many of those results. A responsible synthesis must therefore place them inside the classical baseline.

What remains distinctive is narrower. The framework organizes risk around location, movement, protection, pathway, timing, and intervention. It permits an optional phase representation when timing has a defensible origin and horizon and when phase-dependent features add stable value beyond classical timing interactions. It requires raw amplitude combinations to be treated as structural scores unless a complete normalized state amplitude has been specified. The resulting theory is deliberately falsifiable: complexity is retained only when it improves evidence or decisions.

The central unified object is

$$\mathfrak{U}_{i,t} = \Big(\mathcal{M}_t, \mathbf{x}_{i,t}, \mathbf{g}_t, \mathcal{D}_t^{\mathrm{cl}}, \mathcal{B}_t, \Gamma_{i,t}, I_{i,t}^{*}, \mathbf{z}_{i,t}^{\mathrm{phase}}, \mathbf{u}_{i,t}, \mathbf{U}_{i,t}\Big), \tag{15.1}$$

where the components represent the state geometry, classical dynamics, protective barrier, pathways, minimum classical action, optional phase evidence, interventions, and uncertainty. The framework is unified because these components answer a common set of questions across domains, not because every domain must use identical variables or equations.

15.2 Layer One: State, Geometry, and Adverse Regions

Every application begins by defining the unit of analysis, the decision horizon, and the adverse outcome. A state vector should contain the smallest sufficiently informative set of variables needed to model future evolution at that horizon. Its coordinates may be financial ratios, operational capacities, network dependencies, legal evidence, behavioural indicators, or institutional resources. Observations must be timestamp-correct: a variable belongs to the state at time t only if it was available to the decision-maker at time t.

Feasible states form a structured set $\mathcal{M}_t$. Accounting identities, contractual restrictions, technology architecture, regulation, and behaviour may make that set curved, constrained, or regime-dependent. The metric $\mathbf{g}_t$ defines meaningful distance. A metric is useful only if shorter distance is empirically associated with earlier transition, greater intervention urgency, or stronger warning after controlling for ordinary risk forecasts.

The state space is partitioned into stable, fragile, adverse, and recovery regions. These regions must have observable entry criteria. Fragility is not simply a lower grade of adversity; it is the area in which the system still functions but increasingly depends on favourable timing, depleted capacity, exceptions, or concentrated support. Recovery is distinct because the route out of adversity may differ from the route into it. Boundaries may move when markets close, regulations change, infrastructure fails, or service tolerances tighten.

Geometry is therefore conditional on time and purpose. A state can become riskier because it moves, because an adverse boundary approaches, because the metric changes, or because feasible recovery states disappear. These mechanisms should be reported separately. A single probability may reflect all of them, but governance requires knowing which one changed.

15.3 Layer Two: Classical Dynamics and Rare-Event Action

The classical law of motion is the mandatory baseline. It may include drift, diffusion, jumps, latent regimes, feedback, network propagation, and intervention:

$$\mathrm{d}\mathbf{x}_t = \mathbf{b}(\mathbf{x}_t, t)\,\mathrm{d}t + \sqrt{\varepsilon}\boldsymbol{\sigma}(\mathbf{x}_t, t)\,\mathrm{d}\mathbf{W}_t + \mathrm{d}\mathbf{J}_t. \tag{15.2}$$

For continuous small-noise dynamics, the relevant rare-event benchmark is a Freidlin–Wentzell action. With $\mathbf{a} = \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}}$, a path ϕ over $[0, T]$ has action

$$I_{0T}[\phi] = \frac{1}{2}\int_0^T \left(\dot{\phi} - \mathbf{b}(\phi, t)\right)^{\mathsf{T}} \mathbf{a}(\phi, t)^{-1} \left(\dot{\phi} - \mathbf{b}(\phi, t)\right)\,\mathrm{d}t. \tag{15.3}$$

The dominant classical rare-event route minimizes this action subject to the initial and adverse boundary conditions. The quasipotential and Kramers-type escape approximations imply exponential sensitivity to the minimized action or barrier height. Thus a baseline accessibility indicator may be written schematically as

$$T_{i,t}^{(0)} = \exp\left(-\frac{I_{i,t}^{*}}{\varepsilon_{i,t}}\right). \tag{15.4}$$

This expression already yields a minimum-action path, nonlinear barrier sensitivity, and large accessibility changes after modest action compression. These are classical conclusions. A WKB-shaped barrier integral can remain a useful geometric proxy, but it must be compared with the drift-relative diffusion action and cannot be presented as evidence of a quantum-like mechanism [Freidlin and Wentzell, 2012, Kramers, 1940, E et al., 2004, Heymann and Vanden-Eijnden, 2008].

The strong classical null also includes competing explanations: omitted variables, barrier erosion, hidden gaps, regime change, common shocks, nonlinear interactions, network cascades, and model misspecification. The extended model bears the burden of improving on this full benchmark rather than on a deliberately weak comparison.

15.4 Layer Three: Barriers, Capacity, and Pathways

A barrier is measurable protection separating the current or fragile region from an adverse region. It may consist of capital, liquidity, collateral, covenants, legal rights, operational controls, recovery systems, resolution capacity, infrastructure redundancy, or public support. The barrier is characterized by height, width, continuity, availability, reliability, independence, response time, and residual capacity.

Nominal protection is not effective protection. A control unavailable during peak demand, collateral that cannot be realized, or a facility dependent on the same failing infrastructure does not provide the resistance implied by its stated amount. Barrier components must therefore be adjusted for availability, reliability, and shared dependency. Simultaneous stress is especially important because several entities may consume the same limited protection at once.

Pathways are ordered sequences connecting the current state to the adverse boundary. They should be reconstructed from process evidence, historical events, near misses, causal analysis, network structure, documents, and simulation. Each stage requires an observable indicator, a defensible ordering, and an identified interruption point. Pathways that are merely retrospective narratives should not enter the model.

The dominant pathway is determined by classical action or another validated resistance measure. Its value is operational: it identifies where protection is weakest after accounting for the entire route. A low-action path may contain no dramatic isolated defect. Several moderate weaknesses can form a short corridor when their dependencies and timing align. Intervention should target that corridor rather than the largest headline exposure.

15.5 Layer Four: The Optional Phase Extension

Phase is the only component intended to represent information not already captured by classical minimum action and pathway probability. It is therefore optional and tightly constrained. Let $\tau^{\mathrm{eff}}_{k,i,t}$ be an effective pathway timing coordinate, $\tau_{0,i,t}$ a fixed origin, and $H_{i,t} > 0$ a prespecified institutional period or decision horizon. Define

$$\theta_{k,i,t} = 2\pi \left[\frac{\tau^{\mathrm{eff}}_{k,i,t} - \tau_{0,i,t}}{H_{i,t}} \bmod 1 \right]. \tag{15.5}$$

This definition answers the essential interpretive question: a delay of $H/2$ corresponds to π.

The global origin cancels from phase differences, but the period does not. The horizon, timing transformation, and wrapping rule must be fixed before evaluation. If no defensible period or horizon exists, the model must use conventional timing variables rather than phase.

With pathway magnitudes $R_k = \sqrt{p_k}$, the raw combination

$$Q_{i,t}^{\mathrm{amp}} = \left| \sum_k R_{k,i,t} e^{\mathrm{i}\theta_{k,i,t}} \right|^2 \tag{15.6}$$

is a structural score, not a probability. Marginal pathway probabilities do not form a normalized orthogonal basis, and overlapping routes may share the same terminal event. A bounded alignment feature is

$$J_{i,t}^{\mathrm{amp}} = \frac{Q_{i,t}^{\mathrm{amp}}}{\left(\sum_k R_{k,i,t}\right)^2} = C_{i,t}^2. \tag{15.7}$$

An adverse-event probability must come from a normalized full-state amplitude or, more practically, from a separately calibrated model that adds J^{amp} and pairwise phase features to the strong classical prediction. Constructive and destructive cases must both be tested. A phase model that only produces alarming coherence is not credible.

15.6 Identification, Falsification, and Model Comparison

The unified framework is empirical only when its components can fail. State variables can omit mechanisms. Metrics can misrepresent transition difficulty. Barriers can be nominal rather than causal. Pathways can be unstable. Classical actions can be misspecified. Phase horizons can be arbitrary. Calibration can deteriorate across regimes. Each component therefore requires a rejection criterion.

Model comparison should be nested. The first model estimates ordinary adverse-event probability. The second adds state geometry and classical rare-event action. The third adds barrier and pathway structure. The fourth, optional model adds anchored phase features. Incremental performance should be measured through out-of-sample likelihood, calibration, proper scoring rules, discrimination, lead time, stability, and intervention usefulness. Phase randomization, horizon perturbation, placebo timing, and ablation tests should determine whether the phase result is genuine or merely flexible interaction fitting.

The phase extension fails if its horizon changes materially across samples without institutional justification, if destructive alignment is never observed, if classical timing interactions reproduce its performance, if the raw amplitude score is required to exceed probability bounds, or if intervention-induced timing changes do not affect predictions in the expected direction. Minimum action and exponential sensitivity remain useful after such failure, but they remain classical results.

Uncertainty must accompany every layer. Data uncertainty, state uncertainty, barrier reliability, path instability, action estimation, phase choice, parameter error, and model specification should be propagated or reported separately. A low-risk estimate with low confidence may require investigation rather than reassurance.

15.7 Measurement and Stress Testing

Operational measurement should preserve components before constructing a composite score. Distance describes proximity. Directional mobility describes whether drift and covariance point toward adversity. Barrier measures describe effective protection and residual capacity. Classical minimum action identifies the easiest rare-event route. Transmission $T^{(0)}$ summarizes classical accessibility. Phase coherence and J^{amp} describe timing alignment without claiming to be probabilities. Model confidence describes the reliability of the complete estimate.

A calibrated prediction may combine these quantities:

$$P_{i,t}^{\text{cal}} = g\left(P_{i,t}^{\text{cl}}, d_{A,i,t}, I_{i,t}^{*}, T_{i,t}^{(0)}, J_{i,t}^{\text{amp}}, \mathbf{z}_{i,t}, \mathbf{U}_{i,t}\right), \tag{15.8}$$

where g is constrained, trained, and evaluated out of sample. No raw score should be relabelled as probability merely because it lies near the unit interval in one example.

Stress testing changes more than state variables. A scenario may move the state, alter the metric, expand the adverse region, rotate covariance, collapse effective dimension, reduce barrier reliability, activate pathways, compress action, or change timing. The stress test should recompute the classical action before applying any phase extension. Reverse stress testing should identify the least severe combination of state movement, barrier deterioration, and timing change that breaches a governed threshold.

15.8 Governance and Intervention

The framework becomes useful only when analysis is connected to authority. Every critical barrier requires an owner, purpose, capacity, dependency record, limit, testing schedule, evidence base, and activation rule. Every material pathway requires an ordered definition, observable stages, action estimate, intervention window, owner, and model version. A pathway crossing organizational boundaries needs coordination beyond ordinary functional ownership.

Governance should monitor levels, trends, acceleration, persistence, concentration, and uncertainty. A rapidly compressing classical action may justify escalation even before a probability limit is breached. A high phase-alignment feature with weak horizon justification should trigger model challenge rather than automatic intervention. Reports must distinguish observation, model estimate, scenario assumption, and management judgment.

Interventions operate through four structural levers. They may raise barrier height, increase barrier width, reduce permeability, or alter timing. The preferred action is the one that robustly reduces calibrated risk and classical accessibility relative to cost, capacity, delay, and side effects. Post-intervention testing should verify the predicted mechanism. If maturity extension was expected to increase path action, the measured action should rise. If control separation was expected to reduce shared dependency, barrier independence should improve. If timing intervention was expected to reduce phase alignment, the anchored phase feature should fall.

15.9 Domain Translation and Research Agenda

The same questions apply across domains, but the empirical objects differ. Credit risk uses leverage, liquidity, refinancing, collateral, covenants, and support. Operational and cyber risk use control reliability, detection, dependency, recovery, workload, and human response. Legal and regulatory risk use obligations, evidence, conduct, interpretation, jurisdiction, remediation, and enforcement. Systemic risk uses networks, common assets, funding, infrastructure, collective capacity, and coordinated behaviour.

Future research should focus on comparisons rather than analogies. Classical quasipotential and Kramers models should be estimated directly against WKB-shaped proxies. Metrics should be validated against observed transition time and intervention difficulty. Pathways should be evaluated prospectively. Phase horizons should be registered before testing. Destructive alignment should be sought deliberately rather than treated as an inconvenient exception. Calibrated probabilities should be checked against proper scoring rules and probability bounds.

Artificial intelligence may assist with evidence extraction, path discovery, scenario generation, and model monitoring, but it must remain grounded in source records and governed by access control, validation, and human accountability. Generative fluency is not evidence of causal structure.

15.10 Chapter Synthesis

The unified framework begins with a simple proposition: risk depends on how adverse outcomes become accessible, not only on their estimated frequency. State spaces locate the system. Metrics define meaningful distance. Classical dynamics govern ordinary movement. Large-deviation action identifies rare but classically possible paths. Barriers represent effective protection. Pathways reveal ordered mechanisms and interruption points. Stress testing deforms these objects. Governance assigns authority to change them.

The conceptual correction is equally important. Exponential action, dominant paths, and barrier sensitivity are not sufficient evidence for a tunneling extension. Freidlin–Wentzell and Kramers theory already provide those results within classical stochastic dynamics. They should be estimated as the strong null.

Phase remains a possible extension only when it has a fixed origin, a defensible period, and a prespecified timing transformation. Raw amplitude sums constructed from marginal probabilities are structural scores, not probabilities. Bounded alignment features may enter a calibrated model, but only normalized full-state amplitudes may use the Born rule directly. Both constructive and destructive alignment must occur in validation data if the representation is to remain credible.

The resulting framework is smaller, stronger, and more testable. It does not require quantum language to justify geometry, barriers, or minimum action. It retains quantum-like phase only as an empirical hypothesis about timing relationships that classical models may or may not already capture.

The final proposition of the book is therefore:

> Geometric risk management is credible when state, movement, protection, pathways, timing, and intervention are measured within a strong classical rare-event framework,

> and when any phase-based extension is normalized, falsifiable, and demonstrably useful beyond that classical baseline.

The framework should be judged by that standard. Where it improves warning, explanation, and intervention, it should be retained. Where classical models perform as well, the simpler account should prevail.

Part VI

Technical Appendices

Chapter A

Mathematical and Empirical Reference Architecture

A.1 Purpose of the Appendix

This appendix consolidates the principal mathematical, empirical, and governance objects developed throughout the book. Its purpose is to provide a single reference point for readers implementing, testing, or extending the tunneling-risk framework.

The framework combines several layers:

1. the state of the system;
2. the geometry of feasible states;
3. the law of motion;
4. protective barriers;
5. transition pathways;
6. amplitude, phase, and interference;
7. barrier action and transmission;
8. empirical detection and measurement;
9. stress testing;
10. governed intervention.

The complete architecture may be summarized as

$$\boxed{\begin{aligned}&\text{Evidence} \longrightarrow \text{State Geometry} \longrightarrow \text{Dynamics} \longrightarrow \text{Barriers}\\ &\qquad\qquad \longrightarrow \text{Pathways} \longrightarrow \text{Phase and Interference} \longrightarrow \text{Action and Transmission}\\ &\qquad\qquad \longrightarrow \text{Measurement} \longrightarrow \text{Stress Testing} \longrightarrow \text{Governed Intervention.}\end{aligned}} \tag{A.1}$$

This appendix does not replace the conceptual discussions in the chapters. It provides a compact implementation-oriented map of the complete system.

A.2 General Notation

Symbol	Meaning
(t)	Time index
(i)	Entity, borrower, institution, process, system, or network unit
$\mathbf{x} * i, t$	Observed state vector
$\mathbf{y} * i, t$	Embedded or latent state
$\mathcal{M}$	State manifold
$\mathbf{g}$	Metric tensor defining distance
$\mathbf{b}$	Drift vector
$\mathbf{\Sigma}$	Volatility or diffusion loading matrix
$\mathbf{A} = \mathbf{\Sigma}\mathbf{\Sigma}^\top$	Covariance-rate or mobility tensor
$\mathbf{J}_t$	Jump component
(S,F,A)	Stable, fragile, and adverse regions
$\mathcal{B}$	Protective barrier
V_R	Risk potential
Γ	Set of candidate pathways
γ	One pathway
Ψ	Complex amplitude
(R)	Amplitude magnitude
θ	Phase
$S_R[\gamma]$	Barrier action along pathway γ
S_R^*	Minimum barrier action
T_R	Effective transmission measure
$\mathcal{C}$	Pathway coherence
Π	Barrier permeability
TRI	Composite tunneling-risk index

A.3 The State and Its Law of Motion

The general continuous-time state dynamics are

$$\mathrm{d}\mathbf{x} * t = \mathbf{b}\left(\mathbf{x} * t, t\right) \mathrm{d}t + \mathbf{\Sigma}\left(\mathbf{x} * t, t\right) \mathrm{d}\mathbf{W} * t + \mathrm{d}\mathbf{J}_t. \tag{A.2}$$

The components have distinct interpretations:

- $\mathbf{b}$ represents systematic movement;
- $\mathbf{\Sigma}\mathrm{d}\mathbf{W}$ represents continuous stochastic movement;
- $\mathrm{d}\mathbf{J}$ represents discontinuous shocks.

The instantaneous covariance rate is

$$\mathbf{A}\left(\mathbf{x} * t, t\right) = \mathbf{\Sigma}\left(\mathbf{x} * t, t\right) \mathbf{\Sigma}\left(\mathbf{x}_t, t\right)^{\mathsf{T}}. \tag{A.3}$$

The diffusion tensor is

$$\mathbf{D} = \frac{1}{2}\mathbf{A}. \tag{A.4}$$

In discrete time, a practical approximation is

$$\Delta\mathbf{x} * i, t = \mathbf{b}\left(\mathbf{x} * i, t, t\right) \Delta t + \boldsymbol{\varepsilon} * i, t + \mathbf{J} * i, t. \tag{A.5}$$

A.4 State-Space Construction

Observed variables may be high-dimensional. The framework therefore permits an embedding

$$\mathcal{E}_{\boldsymbol{\vartheta}} : \mathbb{R}^p \longrightarrow \mathbb{R}^d, \qquad d < p, \tag{A.6}$$

with

$$\mathbf{y} * i, t = \mathcal{E} * \boldsymbol{\vartheta}\left(\mathbf{x}_{i,t}\right). \tag{A.7}$$

Candidate methods include:

- principal component analysis;
- factor models;
- diffusion maps;
- spectral embedding;
- autoencoders;
- variational autoencoders;
- supervised manifold learning.

A risk-aware embedding objective may be

$$\mathcal{L} * \text{embed} = \lambda_N \mathcal{L} * \text{neighbourhood} + \lambda_T \mathcal{L} * \text{temporal} + \lambda_E \mathcal{L} * \text{economic} + \lambda_Y \mathcal{L}_{\text{outcome}}. \tag{A.8}$$

The embedding should preserve local neighbourhoods, temporal trajectories, and economically meaningful directions.

A.5 The Metric and Risk Distance

The metric tensor $\mathbf{g}$ determines the meaning of distance on the manifold.

The length of path γ is

$$L[\gamma] = \int_\gamma \sqrt{\dot{\gamma}(u)^\mathsf{T} \mathbf{g} \left(\gamma(u)\right) \dot{\gamma}(u)}, \mathrm{d}u. \tag{A.9}$$

The geodesic distance between two states is

$$d_\mathbf{g} \left(\mathbf{x} * 1, \mathbf{x} * 2\right) = \inf_{\gamma:\mathbf{x}*1 \to \mathbf{x}*2} L[\gamma]. \tag{A.10}$$

A statistical metric may be

$$\mathbf{g}_{\text{stat}} = \mathbf{A}^{-1}, \tag{A.11}$$

while a structural metric may be

$$\mathbf{g}_{\text{str}} = \operatorname{diag}\left(w_1, \ldots, w_d\right). \tag{A.12}$$

A hybrid metric is

$$\mathbf{g} = \lambda_g \mathbf{g} * \text{stat} + \left(1 - \lambda_g\right) \mathbf{g} * \text{str}. \tag{A.13}$$

A.6 Risk Regions

The manifold is partitioned into stable, fragile, and adverse regions:

$$\mathcal{M} = S \cup F \cup A. \tag{A.14}$$

For empirical implementation, define an adverse indicator

$$Y_{i,t+\Delta} = \begin{cases} 1, & \text{if an adverse event occurs within } \Delta, \\ 0, & \text{otherwise.} \end{cases} \tag{A.15}$$

The adverse region may be estimated as

$$A = \left\{ \mathbf{y} : \widehat{P}\left(Y = 1 \mid \mathbf{y}\right) \geq \tau_A \right\}. \tag{A.16}$$

Distance to the adverse region is

$$d_A \left(\mathbf{y} * i, t\right) = \inf * \mathbf{a} \in A d_\mathbf{g} \left(\mathbf{y}_{i,t}, \mathbf{a}\right). \tag{A.17}$$

A.7 Protective Barriers

A barrier is a finite region that resists movement into the adverse region.

A generic protection function may be written as

$$B\left(\mathbf{x} * t\right) = P\left(\mathbf{x} * t\right) - L\left(\mathbf{x}_t\right), \tag{A.18}$$

where (P) is effective protection and (L) is the current stress load.

The boundary is

$$B(\mathbf{x}) = 0. \tag{A.19}$$

Nominal protection component (j) should be adjusted for availability and reliability:

$$\widetilde{P} * j, i, t = P * j, i, t A_{j,i,t} R_{j,i,t}. \tag{A.20}$$

Total effective protection is

$$\widetilde{P} * i, t = \sum_j \widetilde{P} * j, i, t. \tag{A.21}$$

Residual barrier capacity is

$$Q^B_{i,t} = \widetilde{P} * i, t - L * i, t^{\text{stress}}. \tag{A.22}$$

Barrier utilization is

$$U^B_{i,t} = \frac{L^{\text{stress}}_{i,t}}{\widetilde{P}_{i,t}}. \tag{A.23}$$

A.8 Barrier Height, Width, Continuity, and Independence

Barrier height may be represented as

$$H_{\mathcal{B},i,t} = \widetilde{P} * i, t - E * i, t, \tag{A.24}$$

where $E_{i,t}$ is effective transition capacity.

Temporal width is

$$W^{\text{time}}_i = t^A_i - t^F_i. \tag{A.25}$$

Sequential width is

$$W_i^{\text{seq}} = \min_{\gamma} |\gamma \cap \mathcal{B}|. \tag{A.26}$$

Geometric width is

$$W_i^{\text{geo}} = \inf_{\gamma} \int_{\gamma \cap \mathcal{B}} \mathrm{d}s_{\mathbf{g}}. \tag{A.27}$$

If λ_j are eigenvalues of the dependency-adjusted control matrix, the effective number of independent layers is

$$K_{\text{eff}} = \frac{\left(\sum_j \lambda_j\right)^2}{\sum_j \lambda_j^2}. \tag{A.28}$$

A composite barrier-integrity measure is

$$I_{\mathcal{B},i,t} = \beta_H \widetilde{H} * i,t + \beta_W \widetilde{W} * i,t + \beta_C \widetilde{C} * i,t + \beta_R \widetilde{R} * i,t + \beta_K \widetilde{K} * i,t^{\text{eff}} + \beta_Q \widetilde{Q} * i,t. \tag{A.29}$$

A.9 Directional Mobility

Let $\mathbf{n}_{A,i,t}$ be the local direction toward the adverse region.

Projected drift is

$$b_{A,i,t} = \mathbf{n} * A,i,t^{\mathsf{T}} \mathbf{b} * i,t. \tag{A.30}$$

Projected variance is

$$a_{A,i,t} = \mathbf{n} * A,i,t^{\mathsf{T}} \mathbf{A} * i,t \mathbf{n}_{A,i,t}. \tag{A.31}$$

Let the dominant eigenvector satisfy

$$\mathbf{A} * i,t \mathbf{v} * 1,i,t = \lambda_{1,i,t} \mathbf{v}_{1,i,t}. \tag{A.32}$$

Its alignment with danger is

$$\cos \phi_{i,t} = \frac{\mathbf{v} * 1,i,t^{\mathsf{T}} \mathbf{n} * A,i,t}{|\mathbf{v} * 1,i,t|\,|\mathbf{n} * A,i,t|}. \tag{A.33}$$

A mobility-warning index is

$$M_{i,t} = \omega_b Z(b_{A,i,t}) + \omega_a Z(\sqrt{a_{A,i,t}}) + \omega_\phi Z(\cos \phi_{i,t}). \tag{A.34}$$

A.10 Pathway Representation

A pathway is an ordered sequence connecting the current state to the adverse region.

Let

$$\Gamma_{i,t} = \{\gamma_{1,i,t}, \ldots, \gamma_{K,i,t}\} \tag{A.35}$$

be the candidate set.

For edge (e), define transition probability

$$p_{e,i,t}. \tag{A.36}$$

Classical edge resistance is

$$r_{e,i,t} = -\log p_{e,i,t}. \tag{A.37}$$

The resistance of path γ is

$$\mathcal{R} * i, t[\gamma] = \sum * e \in \gamma r_{e,i,t}. \tag{A.38}$$

The classical dominant pathway is

$$\gamma^*_{\mathrm{cl},i,t} = \arg, \min * \gamma \mathcal{R} * i, t[\gamma]. \tag{A.39}$$

A.11 Amplitude and Phase

The probability density is related to a complex amplitude:

$$\Psi = Re^{\mathrm{i}\theta}, \qquad \rho = |\Psi|^2 = R^2. \tag{A.40}$$

For pathway (k),

$$\Psi_{k,i,t} = R_{k,i,t} e^{\mathrm{i}\theta_{k,i,t}}. \tag{A.41}$$

A disciplined empirical starting point is

$$R_{k,i,t} = \sqrt{p_{k,i,t}}. \tag{A.42}$$

Phase must be anchored to an observable timing scale. Let $\tau^{\mathrm{eff}}_{k,i,t}$ be an estimated timing coordinate, τ_0 a prespecified origin, and H a prespecified period or decision horizon. Then

$$\theta_{k,i,t} = 2\pi \left[\frac{\tau^{\mathrm{eff}}_{k,i,t} - \tau_0}{H} \bmod 1 \right], \tag{A.43}$$

where $\tau^{\mathrm{eff}}_{k,i,t} = \boldsymbol{\omega}_k^{\mathsf{T}} \mathbf{z}^{\mathrm{phase}}_{k,i,t}$. A delay of $H/2$ therefore corresponds to a phase difference of π. If no defensible origin and horizon can be specified before evaluation, the timing variables should enter a classical model directly rather than being converted into phase.

where the phase-feature vector may include:

- time to maturity;
- event overlap;
- persistence;
- detection delay;
- response delay;
- synchronization.

A.12 Interference

For two pathways,

$$|\Psi_1 + \Psi_2|^2 = R_1^2 + R_2^2 + 2R_1R_2 \cos(\theta_1 - \theta_2). \tag{A.44}$$

The raw pairwise structural-alignment contribution is

$$I^{\mathrm{raw}}_{12} = 2R_1R_2 \cos(\theta_1 - \theta_2). \tag{A.45}$$

For multiple pathways, the raw structural amplitude score is

$$Q^{\mathrm{amp}}_A = \left| \sum_{k=1}^{K} \Psi_k \right|^2. \tag{A.46}$$

The classical standalone sum is

$$P^{\mathrm{sum}}_A = \sum_{k=1}^{K} p_k. \tag{A.47}$$

The bounded structural alignment index is

$$J^{\mathrm{amp}}_A = \frac{Q^{\mathrm{amp}}_A}{\left(\sum_k R_k\right)^2}. \tag{A.48}$$

The index satisfies $0 \leq J_A^{\text{amp}} \leq 1$. The raw cross term $Q_A^{\text{amp}} - P_A^{\text{sum}}$ is a structural diagnostic, not a probability difference. Probability requires full-state normalization or a separately calibrated prediction model.

A.13 Coherence

The unweighted coherence index is

$$\mathcal{C} = \left| \frac{1}{K} \sum_{k=1}^{K} e^{\mathrm{i}\theta_k} \right|. \tag{A.49}$$

The amplitude-weighted coherence index is

$$\mathcal{C}^{\mathrm{w}} = \frac{\left| \sum_{k=1}^{K} R_k e^{\mathrm{i}\theta_k} \right|}{\sum_{k=1}^{K} R_k}. \tag{A.50}$$

Adverse and stabilizing coherence should be measured separately.

A.14 Barrier Action

The multidimensional continuous action is

$$S_R[\gamma] = \int_{\gamma\cap\mathcal{B}} \sqrt{2\left[V_R(\mathbf{x}, t) - E_R\right], \mathrm{d}s_{\mathbf{M}}}. \tag{A.51}$$

A discrete empirical approximation is

$$S_{R,i,t}[\gamma] = \sum_{e\in\gamma\cap\mathcal{B}} \sqrt{2M_{e,i,t}\left(V_{e,i,t} - E_{i,t}\right)}\Delta s_e. \tag{A.52}$$

The minimum-action pathway is

$$\gamma^*_{R,i,t} = \arg, \min *\gamma S * R, i, t[\gamma]. \tag{A.53}$$

The minimum action is

$$S^*_{R,i,t} = S_{R,i,t}\left[\gamma^*_{R,i,t}\right]. \tag{A.54}$$

A.15 Transmission

The transmission measure is

$$T_{R,i,t} = \exp\left[-\frac{2S^*_{R,i,t}}{\eta}\right]. \tag{A.55}$$

The parameter η is an empirical scale parameter.

It may be estimated by minimizing prediction loss:

$$\widehat{\eta} = \arg, \min *\eta > 0 \sum *i, t\mathcal{L}\left[Y_{i,t+\Delta}, T_{R,i,t}(\eta)\right]. \tag{A.56}$$

The transmission measure should be interpreted as effective adverse accessibility, not as proof of literal physical tunneling.

A.16 Pathway Concentration

Let pathway weights be

$$\pi_{k,i,t} = \frac{R_{k,i,t}^2}{\sum_j R_{j,i,t}^2}. \tag{A.57}$$

The pathway concentration index is

$$H_{\Gamma,i,t} = \sum_k \pi_{k,i,t}^2. \tag{A.58}$$

The effective number of pathways is

$$K_{\Gamma,i,t}^{\text{eff}} = \frac{1}{H_{\Gamma,i,t}}. \tag{A.59}$$

Pathway entropy is

$$\mathcal{H} * \Gamma, i, t = -\sum_k \pi * k, i, t \log \pi_{k,i,t}. \tag{A.60}$$

A.17 Composite Detection and Measurement

A general detection vector is

$$\mathbf{M} * i, t^{\text{tun}} = \left(D * i, t, B_{i,t}, M_{i,t}, A_{i,t}, C_{i,t}, I_{i,t}, T_{i,t}, U_{i,t}\right). \tag{A.61}$$

The components represent:

- state proximity;
- barrier weakness;
- adverse mobility;
- action risk;
- coherence;

- interference;
- transmission;
- uncertainty.

A linear composite tunneling-risk index is

$$\mathrm{TRI} * i, t = \sum_j w_j \widetilde{M} * j, i, t, \qquad \sum_j w_j = 1. \tag{A.62}$$

A nonlinear alternative is

$$\mathrm{TRI} * i, t^{\mathrm{nl}} = 1 - \prod_j \left(1 - \widetilde{M} * j, i, t\right)^{w_j}. \tag{A.63}$$

A.18 Normalization

Z-score normalization is

$$Z_{X,i,t} = \frac{X_{i,t} - \mu_{X,\mathcal{R}}}{\sigma_{X,\mathcal{R}}}. \tag{A.64}$$

Percentile normalization is

$$P_{X,i,t} = \widehat{F} * X, \mathcal{R}\left(X * i, t\right). \tag{A.65}$$

Probability calibration is

$$\widetilde{X} * i, t = \widehat{P}\left(Y * i, t + \Delta = 1 \mid X_{i,t}\right). \tag{A.66}$$

The reference class $\mathcal{R}$ should be economically meaningful.

A.19 Stress Testing

A scenario is

$$\mathcal{S}^{(s)} = \left(\mathbf{z}^{(s)}, \boldsymbol{\xi}^{(s)}, \mathbf{u}^{(s)}, \mathcal{R}^{(s)}\right), \tag{A.67}$$

where:

- $\mathbf{z}^{(s)}$ contains exogenous shocks;
- $\boldsymbol{\xi}^{(s)}$ contains endogenous responses;
- $\mathbf{u}^{(s)}$ contains interventions;

- $\mathcal{R}^{(s)}$ contains scenario rules.

The stressed architecture is

$$\mathfrak{M} * i, t^{(s)} = \mathcal{F} * \text{stress}\left(\mathfrak{M}^0_{i,t}, \mathcal{S}^{(s)}\right). \tag{A.68}$$

The stressed transmission is

$$T^{(s)}_{R,i,t} = \exp\left[-\frac{2S^{*,(s)}_{R,i,t}}{\eta}\right]. \tag{A.69}$$

The stressed risk change is

$$\Delta\text{TRI} * i, t^{(s)} = \text{TRI} * i, t^{(s)} - \text{TRI}^0_{i,t}. \tag{A.70}$$

A.20 Reverse Stress Testing

Reverse stress testing identifies the least severe scenario that produces a critical condition.

The problem is

$$\mathcal{S}^* = \arg, \min_{\mathcal{S}} C(\mathcal{S}) \tag{A.71}$$

subject to

$$\text{TRI}^{(\mathcal{S})} \geq \tau_{\text{crit}}, \tag{A.72}$$

or

$$T^{(\mathcal{S})}_R \geq \tau_T. \tag{A.73}$$

Possible optimization methods include:

- nonlinear programming;
- mixed-integer optimization;
- evolutionary search;
- Bayesian optimization;
- adversarial machine learning;
- reinforcement learning.

A.21 Barrier Permeability

An empirical permeability measure may be

$$\Pi_{i,t} = \omega_B \widetilde{B} * i, t^{\text{weak}} + \omega_A \widetilde{A} * i, t^{\text{risk}} + \omega_C \widetilde{C} * i, t^{-} + \omega_T \widetilde{T} * R, i, t. \tag{A.74}$$

A nonlinear permeability measure is

$$\Pi_{i,t}^{\text{nl}} = 1 - \prod_j \left(1 - X_{j,i,t}\right)^{\omega_j}. \tag{A.75}$$

Aggregate permeability is

$$\Pi_t^{\text{agg}} = \sum_i w_i \Pi_{i,t}. \tag{A.76}$$

A.22 Governance Limits

The governance-limit vector is

$$\boldsymbol{\tau}^{\text{gov}} = \left(\tau_B, \tau_S, \tau_C, \tau_I, \tau_T, \tau_\Pi, \tau_U\right). \tag{A.77}$$

Limits may be calibrated from historical pre-event distributions:

$$\tau_X = \widehat{F}_{X,\text{pre}-\text{event}}^{-1}\left(\alpha\right). \tag{A.78}$$

A decision-based threshold is

$$\tau_X^* = \arg, \min * \tau \left[C * \text{miss}(\tau) + C_{\text{false}}(\tau) + C_{\text{intervention}}(\tau)\right]. \tag{A.79}$$

A.23 Intervention Timing

Let path-completion time be

$$\tau_\gamma^A. \tag{A.80}$$

Let response time be

$$\tau_u^{\text{resp}}. \tag{A.81}$$

The intervention is timely when

$$\tau_u^{\text{resp}} < \tau_\gamma^A. \tag{A.82}$$

The safety margin is

$$\Delta\tau^{\mathrm{safe}} = \tau_{\gamma}^{A} - \tau_{u}^{\mathrm{resp}}. \tag{A.83}$$

A.24 Intervention Selection

Expected tunneling-risk reduction is

$$\Delta\mathrm{TRI}_j = \mathrm{TRI}^{\mathrm{before}} - \mathbb{E}\left[\mathrm{TRI}^{\mathrm{after},j}\right]. \tag{A.84}$$

Intervention efficiency is

$$E_j = \frac{\Delta\mathrm{TRI}_j}{C_j}. \tag{A.85}$$

A broader intervention-value function is

$$V_j = \alpha\Delta\mathrm{TRI}_j + \beta\Delta T_j + \gamma\Delta S_j^{*} - \delta C_j - \zeta U_j. \tag{A.86}$$

A.25 Estimation Methods by Structural Object

Object	Candidate Methods	Validation Question
State manifold	PCA, factor models, diffusion maps, autoencoders	Are neighbourhoods, trajectories, and economic directions preserved?
Metric	Inverse covariance, structural weights, metric learning	Do short distances correspond to accessible transitions?
Drift and covariance	Dynamic panels, state-space models, Gaussian processes, neural SDEs	Are future movement and uncertainty calibrated?
Barrier effectiveness	Panel models, causal forests, doubly robust learners	Does protection causally reduce adverse transition?
Pathways	Process mining, HMMs, transformers, temporal point processes	Are sequences stable, observable, and predictive?
Dependencies	Bayesian networks, graph models, graph neural networks	Are shared exposures and propagation routes identified?
Phase	Constrained latent models, timing regressions, sequence embeddings	Does timing add stable value beyond classical interaction?
Action	Shortest paths, transition-path methods, constrained optimization	Does falling action precede adverse events?
Transmission	Likelihood calibration, isotonic calibration, Bayesian calibration	Is estimated accessibility calibrated out of sample?
Intervention	Causal inference, simulation, reinforcement learning	Would the proposed action improve outcomes?

A.26 Artificial Intelligence Tools

Artificial intelligence may support:

- representation learning;
- nonlinear dynamics;
- pathway discovery;
- graph analysis;
- document extraction;
- scenario generation;

- intervention comparison;
- report drafting.

For document (d), an extraction model may produce

$$\widehat{\mathbf{z}} * d = \mathcal{F} * \mathrm{LLM}\left(d; \mathcal{S}\right), \tag{A.87}$$

where $\mathcal{S}$ is a controlled schema.

The output should be treated as a noisy measurement:

$$\widehat{\mathbf{z}}_d = \mathbf{z}_d + \boldsymbol{\varepsilon}_d^{\mathrm{LLM}}. \tag{A.88}$$

High-stakes extraction requires source citation, validation, abstention, and human approval.

A.27 Validation Standards

The framework should be tested through:

1. backtesting;
2. temporal out-of-sample validation;
3. entity holdout validation;
4. regime validation;
5. ablation;
6. placebo testing;
7. mechanism validation;
8. intervention validation;
9. robustness analysis;
10. uncertainty quantification.

Let the classical model produce

$$\widehat{P}^{\mathrm{cl}}, \tag{A.89}$$

and the extended model produce

$$\widehat{P}^{\mathrm{tun}}. \tag{A.90}$$

Incremental value is

$$\Delta V = V\left(\widehat{P}^{\mathrm{tun}}\right) - V\left(\widehat{P}^{\mathrm{cl}}\right). \tag{A.91}$$

The tunneling extension should be retained only where

$$\Delta V > 0 \tag{A.92}$$

out of sample and with acceptable stability.

A.28 Uncertainty Quantification

For estimated transmission,

$$\mathrm{Var}\left(\widehat{T} * R\right) = V * \mathrm{data} + V_{\mathrm{barrier}} + V_{\mathrm{path}} + V_{\mathrm{phase}} + V_{\mathrm{parameter}} + V_{\mathrm{model}}. \tag{A.93}$$

For the dominant pathway, report

$$P\left(\gamma^* = \gamma_k \mid \mathcal{I}_t\right). \tag{A.94}$$

For the composite score, report

$$\widehat{\mathrm{TRI}} * i, t \pm \mathrm{CI} * i, t. \tag{A.95}$$

A.29 Minimum Data Architecture

A minimum empirical dataset should contain:

- entity identifier;
- observation timestamp;
- data-availability timestamp;
- state variables;
- protection variables;
- event and sequence variables;
- network relationships;
- interventions;
- adverse outcomes;
- missingness and revision flags.

The observation record is

$$\mathcal{O} * i, t = \left(\mathbf{x} * i, t, \mathbf{p} * i, t, \mathbf{e} * i, t, \mathbf{n} * i, t, \mathbf{u} * i, t, Y_{i,t+\Delta}\right). \tag{A.96}$$

A.30 Minimum Implementation Workflow

A practical implementation may follow fifteen steps:

1. define the adverse event;
2. define the unit and prediction horizon;
3. construct timestamp-correct data;
4. define and normalize state variables;
5. estimate the manifold;
6. estimate the metric;
7. define stable, fragile, and adverse regions;
8. estimate classical dynamics;
9. reconstruct barriers;
10. discover and estimate pathways;
11. estimate phase proxies;
12. calculate action and transmission;
13. construct normalized measures;
14. stress test and compare interventions;
15. validate and govern the system.

The implementation sequence is

$$\boxed{\begin{aligned}&\text{Define} \longrightarrow \text{Observe} \longrightarrow \text{Estimate} \longrightarrow \text{Measure}\\ &\qquad \longrightarrow \text{Stress} \longrightarrow \text{Intervene} \longrightarrow \text{Validate} \longrightarrow \text{Learn.}\end{aligned}} \tag{A.97}$$

A.31 Minimum Reporting Architecture

A production report should include:

1. current state;
2. distance to fragile and adverse regions;

3. barrier integrity;
4. projected adverse mobility;
5. dominant classical pathway;
6. dominant minimum-action pathway;
7. pathway concentration;
8. coherence and interference;
9. transmission;
10. composite tunneling-risk index;
11. uncertainty;
12. breached limits;
13. intervention options;
14. accountable owner;
15. required response time.

The report should separate:

$$\text{Observed Evidence}, \qquad \text{Estimated Structure}, \qquad \text{Risk Interpretation}, \qquad \text{Governance Action}. \tag{A.98}$$

A.32 Interpretive Guardrails

The following guardrails apply throughout the framework:

1. A rare event is not automatically tunneling.
2. A jump is not automatically tunneling.
3. A hidden variable is not automatically a hidden pathway.
4. Barrier failure caused by a visible gap is a classical failure.
5. Complex amplitudes are modelling objects, not claims of physical quantum behaviour.
6. Phase must be tied to observable timing or sequencing variables.
7. Interference must be tested against strong classical nonlinear benchmarks.
8. Transmission must be calibrated empirically.
9. Model uncertainty must be reported.

10. High-stakes intervention requires accountable human governance.

A.33 Final Reference Proposition

The entire framework can be condensed into one operational statement:

> A system becomes vulnerable when its current state, movement geometry, protective barriers, transition pathways, and timing relationships combine to make an adverse region accessible. It becomes governable when those components can be estimated, measured, stressed, assigned to accountable owners, and altered through timely intervention.

The mathematical architecture is

$$\boxed{\mathfrak{T} * i, t = \left(\mathcal{M}, \mathbf{g}, \mathbf{x} * i, t, \mathbf{b} * i, t, \mathbf{A} * i, t, \mathcal{B} * i, t, \Gamma * i, t, \Psi_{i,t}, S^{*}_{R,i,t}, T_{R,i,t}, \mathrm{TRI} * i, t, \Pi * i, t, \mathcal{U}^{*}_{i,t}\right)}, \tag{A.99}$$

where $\mathcal{U}^{*}_{i,t}$ is the governed intervention set.

The practical architecture is

$$\boxed{\text{Understand how risk can travel, measure how accessible the route has become, and act before the transitio}} \tag{A.100}$$

Chapter B

Empirical Implementation Blueprint

B.1 Purpose of the Appendix

Appendix A consolidated the mathematical and conceptual architecture of the tunneling-risk framework. The present appendix translates that architecture into a practical empirical implementation.

Its purpose is to answer five operational questions:

1. What data are required?
2. How should the data be organized?
3. Which model estimates each theoretical object?
4. How should the components be validated and combined?
5. How should the resulting system be deployed and governed?

The implementation principle is deliberately sequential:

$$\boxed{\begin{aligned} &\text{Evidence} \longrightarrow \text{Classical Model} \longrightarrow \text{State Geometry} \\ &\qquad\quad\ \longrightarrow \text{Barrier Estimation} \longrightarrow \text{Pathway Reconstruction} \\ &\qquad\quad\ \longrightarrow \text{Phase and Action} \longrightarrow \text{Measurement} \\ &\qquad\quad\ \longrightarrow \text{Stress Testing} \longrightarrow \text{Governed Intervention.} \end{aligned}} \tag{B.1}$$

The tunneling extension should be introduced only after a credible classical benchmark has been constructed.

B.2 The Minimum Empirical Research Question

Every implementation should begin with a narrowly defined research question.

A complete question identifies:

- the unit of analysis;

- the adverse event;
- the prediction horizon;
- the protective barrier;
- the candidate pathways;
- the intended decision.

For example:

> For a portfolio of corporate borrowers observed quarterly, can changes in refinancing-barrier integrity, pathway action, and maturity-related phase alignment improve twelve-month prediction of default or distressed restructuring beyond a conventional hazard model?

This question is preferable to the broader and less testable question:

> Can quantum tunneling predict credit risk?

The empirical programme should test specific structural claims rather than an unrestricted analogy.

B.3 Unit of Analysis

Let the unit be indexed by (i), and time by (t).

Examples include:

Application	Unit	Adverse Outcome
Corporate credit	Borrower-quarter	Default, distressed exchange, liquidity exhaustion
Consumer credit	Account-month	Serious delinquency, charge-off, restructuring
Operational risk	Process-day or system-hour	Service failure, control breach, material loss
Cyber risk	Asset-event sequence	Intrusion, data loss, critical outage
Legal risk	Matter-month	Adverse ruling, sanction, unenforceability
Systemic risk	Institution-network date	Market freeze, contagion, payment disruption

The prediction horizon should be fixed before model development.

Let

$$\Delta \tag{B.2}$$

denote the horizon.

The adverse indicator is

$$Y_{i,t+\Delta} = \begin{cases} 1, & \text{if the adverse event occurs within } \Delta, \\ 0, & \text{otherwise.} \end{cases} \tag{B.3}$$

B.4 Timestamp-Correct Data Architecture

A production-quality implementation requires two dates for every variable:

1. the date to which the observation refers;
2. the date on which the information became available.

Let

$$t_{i,j}^{\text{obs}} \tag{B.4}$$

be the observation date of variable (j), and

$$t_{i,j}^{\text{avail}} \tag{B.5}$$

its availability date.

A variable may enter the information set at time (t) only if

$$t_{i,j}^{\text{avail}} \le t. \tag{B.6}$$

The information set is

$$\mathcal{I} * i, t = \sigma \left\{ x * i, j, s : t_{i,j,s}^{\text{avail}} \le t \right\}. \tag{B.7}$$

This rule prevents look-ahead bias.

B.5 Core Data Tables

A minimum data architecture should contain the following linked tables.

B.5.1 Entity Table

Field	Description
entity_id	Unique entity identifier
entity_type	Borrower, account, system, institution, matter
sector	Industry or operational class
jurisdiction	Relevant legal or regulatory jurisdiction
reference_class	Peer group for normalization
start_date	Entry into the observed sample
end_date	Exit, closure, maturity, or censoring date

B.5.2 State-Observation Table

Field	Description
entity_id	Entity identifier
observation_date	Economic reference date
availability_date	Date available to the model
variable_name	State-variable name
raw_value	Observed value
normalized_value	Transformed model input
source	Data source
quality_flag	Missing, estimated, revised, or verified

B.5.3 Protection Table

Field	Description
protection_id	Unique barrier-component identifier
protection_type	Liquidity, collateral, covenant, control, legal right
nominal_capacity	Stated protection amount or score
availability	Availability under the relevant scenario
reliability	Probability of functioning when required
dependency_group	Shared dependency identifier
activation_time	Expected time to activate
owner	Responsible institutional function

B.5.4 Event Table

Field	Description
event_id	Unique event identifier
event_type	Downgrade, breach, outage, withdrawal, intervention
event_time	Timestamp of event occurrence
detection_time	Timestamp when event became known
severity	Observed or estimated severity
source	Evidence source
pathway_candidate	Candidate pathway membership

B.5.5 Network Table

Field	Description
source_node	Originating node
target_node	Receiving node
edge_type	Exposure, funding, collateral, vendor, control
weight	Strength or amount of dependency
start_time	Beginning of relationship
end_time	End of relationship
confidence	Confidence in edge existence or weight

B.5.6 Outcome Table

Field	Description
entity_id	Entity identifier
outcome_type	Default, outage, sanction, contagion
outcome_date	Date of adverse event
loss	Observed economic or operational loss
resolution_date	Date of recovery or closure
observed_path	Post-event reconstructed pathway

B.6 Data-Quality Controls

The empirical system should calculate a data-quality score

$$Q_{i,t}^{\text{data}} = f\left(C_{i,t}, T_{i,t}, A_{i,t}, R_{i,t}\right), \tag{B.8}$$

where:

- $C_{i,t}$ is completeness;
- $T_{i,t}$ is timeliness;
- $A_{i,t}$ is accuracy;

- $R_{i,t}$ is source reliability.

A simple weighted score is

$$Q_{i,t}^{\text{data}} = w_C C_{i,t} + w_T T_{i,t} + w_A A_{i,t} + w_R R_{i,t}. \tag{B.9}$$

Low-quality observations should produce larger uncertainty or model abstention.

B.7 Missing-Data Strategy

Missing values should be classified according to mechanism:

- missing completely at random;
- missing conditionally at random;
- missing because of deterioration;
- missing because the variable is not applicable;
- missing because the source failed.

The missingness indicator should be retained:

$$m_{j,i,t} = \mathbb{I}\left(x_{j,i,t} \text{ is missing}\right). \tag{B.10}$$

Missingness may itself contain risk information.

Imputation methods may include:

- last observation carried forward;
- sector median;
- multiple imputation;
- Kalman smoothing;
- model-based imputation;
- learned embeddings with missingness masks.

All imputed fields should be marked.

B.8 Data Transformation

Variables should be transformed to improve comparability and consistency.

A standardized variable is

$$z_{j,i,t} = \frac{x_{j,i,t} - \mu_{j,\text{train}}}{\sigma_{j,\text{train}}}. \tag{B.11}$$

Robust scaling may use

$$z^{\text{rob}}_{j,i,t} = \frac{x_{j,i,t} - \text{median}(x_j)}{\text{IQR}(x_j)}. \tag{B.12}$$

Directional transformations should ensure that higher values consistently represent either greater risk or greater protection.

B.9 Train, Validation, and Test Design

The data should be split through time.

Let:

- $\mathcal{D}_{\text{train}}$ is the estimation sample;
- $\mathcal{D}_{\text{valid}}$ is the model-selection sample;
- $\mathcal{D}_{\text{test}}$ is the untouched evaluation sample.

The ordering must satisfy

$$t_{\text{train}} < t_{\text{valid}} < t_{\text{test}}. \tag{B.13}$$

Random splitting should generally be avoided for time-dependent risk systems.

Entity holdout may also be required:

$$\mathcal{I} * \text{train} \cap \mathcal{I} * \text{test} = \varnothing, \tag{B.14}$$

where $\mathcal{I}$ denotes entity sets.

B.10 Module 1: Classical Baseline

The first model should estimate ordinary adverse-event risk.

A logistic baseline is

$$P\left(Y_{i,t+\Delta} = 1 \mid \mathbf{x} * i, t\right) = \Lambda\left(\alpha + \boldsymbol{\beta}^\mathsf{T}\mathbf{x} * i, t\right). \tag{B.15}$$

A survival baseline may be

$$h_i(t) = h_0(t)\exp\left(\boldsymbol{\beta}^\mathsf{T}\mathbf{x}_{i,t}\right). \tag{B.16}$$

Stronger benchmarks may include:

- gradient-boosted trees;
- random forests;
- generalized additive models;
- deep survival models;
- regime-switching hazard models.

The classical model provides

$$\widehat{P}_{i,t}^{\mathrm{cl}}. \tag{B.17}$$

B.11 Module 2: State-Manifold Estimation

Let

$$\mathbf{y} * i, t = \mathcal{E} * \boldsymbol{\vartheta}\left(\mathbf{x}_{i,t}\right) \tag{B.18}$$

be the latent state.

A supervised autoencoder may minimize

$$\begin{aligned} \mathcal{L} * \text{manifold} = \lambda * \text{rec}\left|\mathbf{x} * i, t - \widehat{\mathbf{x}} * i, t\right|^2 \\ + \lambda_{\text{temp}}\left|\mathbf{y} * i, t - \mathbf{y} * i, t - 1\right|^2 + \lambda_Y \mathcal{L}\left(Y_{i,t+\Delta}, \widehat{Y}_{i,t}\right). \end{aligned} \tag{B.19}$$

The manifold should be evaluated using:

- reconstruction error;
- neighbourhood preservation;
- trajectory smoothness;
- economic monotonicity;
- stability across samples.

B.12 Module 3: Metric Estimation

A local covariance metric is

$$\widehat{\mathbf{g}} * i, t = \left(\widehat{\mathbf{A}} * i, t + \lambda_g \mathbf{I}\right)^{-1}, \tag{B.20}$$

where $\lambda_g > 0$ stabilizes inversion.

A learned metric may minimize

$$\mathcal{L} * \text{metric} = \sum *(i,j) \in \mathcal{P} \left[d_{\mathbf{g}} \left(\mathbf{y}_i, \mathbf{y} * j \right) - c * ij \right]^2 , \tag{B.21}$$

where c_{ij} is an empirical transition cost.

B.13 Module 4: Dynamics and Mobility

A discrete state model is

$$\Delta \mathbf{y} * i, t = \mathbf{b} * \boldsymbol{\theta} \left(\mathbf{y} * i, t \right) + \boldsymbol{\varepsilon} * i, t. \tag{B.22}$$

Estimate

$$\widehat{\mathbf{b}} * i, t = \mathbb{E} \left[\Delta \mathbf{y} * i, t \mid \mathbf{y}_{i,t} \right], \tag{B.23}$$

and

$$\widehat{\mathbf{A}} * i, t = \operatorname{Cov} \left(\Delta \mathbf{y} * i, t \mid \mathbf{y}_{i,t} \right). \tag{B.24}$$

Candidate estimators include:

- local linear regression;
- state-space models;
- Gaussian processes;
- neural stochastic differential equations;
- regime-dependent covariance models.

B.14 Module 5: Barrier Estimation

Effective protection is

$$\widetilde{P} * j, i, t = P * j, i, t A_{j,i,t} R_{j,i,t}. \tag{B.25}$$

Barrier effectiveness should be estimated through the change in adverse probability associated with protection.

A causal effect is

$$\tau_j(\mathbf{x}) = \mathbb{E} \left[Y(P_j = 1) - Y(P_j = 0) \mid \mathbf{X} = \mathbf{x} \right]. \tag{B.26}$$

Possible estimators include:

- panel regression;
- propensity-score weighting;
- causal forests;
- doubly robust learners;
- regression discontinuity;
- difference-in-differences.

Barrier reliability may be updated through observed tests:

$$\widehat{R} * j, t = \frac{a_j + n * j, t^{\text{success}}}{a_j + b_j + n_{j,t}^{\text{tests}}}, \tag{B.27}$$

using a beta-binomial Bayesian update.

B.15 Module 6: Pathway Discovery

The event history for entity (i) is

$$\mathcal{E} * i = \left[(e * i, 1, t_{i,1}), \ldots, (e_{i,m}, t_{i,m})\right]. \tag{B.28}$$

Candidate pathway-discovery methods include:

- process mining;
- frequent-sequence mining;
- hidden Markov models;
- temporal Bayesian networks;
- transformers;
- Hawkes processes;
- expert-defined causal maps.

A candidate pathway should satisfy:

1. temporal consistency;
2. minimum empirical support;
3. elevated outcome incidence;
4. interpretability;

5. out-of-sample persistence.

B.16 Module 7: Pathway-Probability Estimation

For pathway (k), estimate

$$p_{k,i,t} = P\left(\gamma_k \text{ completes before } t + \Delta \mid \mathcal{I}_{i,t}\right). \tag{B.29}$$

A competing-risk model may use

$$h_{k,i}(t) = h_{0,k}(t) \exp\left(\boldsymbol{\beta} * k^{\mathsf{T}} \mathbf{x} * i, t\right). \tag{B.30}$$

The estimated cumulative incidence provides $p_{k,i,t}$.

The amplitude magnitude is initialized as

$$R_{k,i,t} = \sqrt{p_{k,i,t}}. \tag{B.31}$$

B.17 Module 8: Phase Estimation

Phase should be tied to measurable timing features.

Define an effective timing coordinate

$$\tau^{\text{eff}}_{k,i,t} = \boldsymbol{\omega} * k^{\mathsf{T}} \mathbf{z} * k, i, t^{\text{phase}}, \tag{B.32}$$

where the feature vector may contain:

- time to maturity;
- event overlap;
- stress duration;
- detection latency;
- intervention delay;
- synchronization with other entities.

A constrained objective is

$$\widehat{\boldsymbol{\omega}} = \arg, \min * \boldsymbol{\omega} \left[\mathcal{L} * \text{outcome} + \lambda_1 \left|\boldsymbol{\omega}\right| * 1 + \lambda_2 \mathcal{L} * \text{stability}\right]. \tag{B.33}$$

Fix an origin $\tau_{0,i,t}$ and a prespecified horizon $H_{i,t} > 0$. Phase is

$$\theta_{k,i,t} = 2\pi \left[\frac{\tau^{\text{eff}}_{k,i,t} - \tau_{0,i,t}}{H_{i,t}} \bmod 1 \right]. \tag{B.34}$$

The origin, horizon, scaling, and wrapping rule must be frozen before evaluation. Without a defensible horizon, use ordinary timing covariates rather than phase.

B.18 Module 9: Coherence and Interference

Weighted coherence is

$$\mathcal{C} * i, t = \frac{\left| \sum_k R * k, i, t e^{\mathrm{i}\theta_{k,i,t}} \right|}{\sum_k R_{k,i,t}}. \tag{B.35}$$

The raw structural amplitude score is

$$Q^{\text{amp}}_{i,t} = \left| \sum_k R_{k,i,t} e^{\mathrm{i}\theta_{k,i,t}} \right|^2. \tag{B.36}$$

The classical standalone sum is

$$P^{\text{sum}}_{i,t} = \sum_k p_{k,i,t}. \tag{B.37}$$

The bounded structural alignment index is

$$J^{\text{amp}}_{i,t} = \frac{Q^{\text{amp}}_{i,t}}{\left(\sum_k R_{k,i,t}\right)^2}. \tag{B.38}$$

The raw cross term $Q^{\text{amp}}_{i,t} - P^{\text{sum}}_{i,t}$ may be retained as a diagnostic, but neither quantity is an adverse-event probability. A calibrated probability model must be estimated separately and accepted only after comparison with strong nonlinear classical and minimum-action benchmarks.

B.19 Module 10: Action Estimation

For edge (e), define:

- transition resistance $V_{e,i,t}$;
- movement resistance $M_{e,i,t}$;
- effective transition capacity $E_{i,t}$;
- path length Δs_e.

A discrete action is

$$S_{R,i,t}[\gamma] = \sum_{e\in\gamma\cap\mathcal{B}} \sqrt{2M_{e,i,t}\left(V_{e,i,t} - E_{i,t}\right)}\Delta s_e. \tag{B.39}$$

The minimum-action path is

$$\gamma^*_{i,t} = \arg, \min *\gamma S * R, i, t[\gamma]. \tag{B.40}$$

The minimum action is

$$S^*_{R,i,t} = S_{R,i,t}\left[\gamma^*_{i,t}\right]. \tag{B.41}$$

B.20 Module 11: Transmission Calibration

Transmission is

$$T_{R,i,t} = \exp\left[-\frac{2S^*_{R,i,t}}{\eta}\right]. \tag{B.42}$$

Calibrate η by minimizing out-of-sample loss:

$$\widehat{\eta} = \arg, \min *\eta > 0 \sum *(i,t) \in \mathcal{D} * \mathrm{valid}\mathcal{L}\left(Y * i, t+\Delta, T_{R,i,t}(\eta)\right). \tag{B.43}$$

Possible losses include:

- binary cross-entropy;
- Brier loss;
- survival likelihood;
- economically weighted loss.

B.21 Module 12: Composite Measurement

The normalized component vector is

$$\widetilde{\mathbf{M}} * i, t = \left(\widetilde{D} * i, t, \widetilde{B} * i, t, \widetilde{M} * i, t, \widetilde{A} * i, t, \widetilde{C} * i, t, \widetilde{I} * i, t, \widetilde{T} * i, t\right). \tag{B.44}$$

A linear composite index is

$$\mathrm{TRI} * i, t = \sum_j w_j \widetilde{M} * j, i, t. \tag{B.45}$$

A nonlinear index is

$$\mathrm{TRI} * i, t^{\mathrm{nl}} = 1 - \prod_{j} \left(1 - \widetilde{M} * j, i, t\right)^{w_j}. \tag{B.46}$$

The combination rule should preserve monotonicity and component transparency.

B.22 Model-Comparison Ladder

The empirical framework should be tested through a progressive ladder.

Level	Model	Purpose
0	Naive frequency or rating benchmark	Establish minimum predictive standard
1	Classical logistic or hazard model	Estimate conventional risk
2	Nonlinear machine-learning model	Capture nonlinear interactions
3	Geometric model	Add manifold, distance, and mobility
4	Barrier and pathway model	Add protection and route structure
5	Phase and interference model	Add timing-dependent interaction
6	Full tunneling-risk system	Add action, transmission, stress, governance

Each higher level should demonstrate incremental value over the preceding one.

B.23 Validation Metrics

Classification performance may include:

- area under the receiver-operating curve;
- area under the precision-recall curve;
- Brier score;
- log loss;
- calibration intercept;
- calibration slope.

Early-warning performance may include:

- median lead time;
- event-capture rate;
- false-warning rate;
- warning persistence;

- intervention time available.

Structural validation may include:

- dominant-path accuracy;
- barrier-effect consistency;
- action compression before events;
- phase stability;
- intervention-effect accuracy.

B.24 Incremental-Value Tests

Let

$$V_m \tag{B.47}$$

be the value of model (m).

For model levels (m) and (m-1), define

$$\Delta V_m = V_m - V_{m-1}. \tag{B.48}$$

The extended component should be retained only if:

$$\Delta V_m > 0 \tag{B.49}$$

and the improvement is stable across time, entities, and regimes.

B.25 Ablation Tests

For component (j), define

$$\Delta V_j^{\text{abl}} = V_{\text{full}} - V_{\text{without } j}. \tag{B.50}$$

Ablation should be conducted for:

- state geometry;
- barrier integrity;
- mobility rotation;
- pathway structure;

- phase;
- action;
- interference;
- transmission.

B.26 Placebo and Falsification Tests

The model should be tested on situations where tunneling should not be detected.

Examples include:

- purely classical simulated diffusion;
- isolated jumps;
- transparent barrier gaps;
- randomly permuted event timing;
- stable systems without adverse outcomes.

If the model continues to identify strong interference or tunneling transmission under these conditions, it is likely overfitted.

B.27 Stress-Testing Implementation

A scenario record should contain:

Field	**Description**
`scenario_id`	Unique scenario identifier
`scenario_type`	Historical, hypothetical, reverse, adversarial
`horizon`	Scenario duration
`state_shocks`	Changes to state variables
`covariance_shocks`	Changes to mobility geometry
`barrier_shocks`	Changes to capacity and reliability
`pathway_changes`	Activated, removed, or reweighted paths
`phase_changes`	Timing and synchronization assumptions
`responses`	Management and policy actions

For every scenario, recompute:

$$\left(\mathbf{y}^{(s)}, \mathbf{g}^{(s)}, \mathbf{A}^{(s)}, \mathcal{B}^{(s)}, \Gamma^{(s)}, \boldsymbol{\theta}^{(s)}, S_R^{*,(s)}, T_R^{(s)}, \mathrm{TRI}^{(s)}\right). \tag{B.51}$$

B.28 Reverse Stress Testing

The reverse-stress objective is

$$\mathcal{S}^* = \arg, \min_{\mathcal{S}} C(\mathcal{S}) \tag{B.52}$$

subject to

$$\mathrm{TRI}^{(\mathcal{S})} \geq \tau_{\mathrm{crit}}. \tag{B.53}$$

The output should identify:

- minimum state shock;
- minimum barrier erosion;
- minimum covariance rotation;
- required phase alignment;
- activated pathway;
- intervention capable of preventing the condition.

B.29 Intervention Evaluation

For intervention u_j, estimate

$$\Delta\mathrm{TRI} * j, i, t = \mathrm{TRI} * i, t^{\mathrm{before}} - \mathrm{TRI}_{i,t}^{\mathrm{after},j}. \tag{B.54}$$

Intervention efficiency is

$$E_{j,i,t} = \frac{\Delta\mathrm{TRI} * j, i, t}{C * j, i, t}. \tag{B.55}$$

The decision system should also consider timing:

$$\tau_{u_j}^{\mathrm{resp}} < \tau_{\gamma}^{A}. \tag{B.56}$$

An effective intervention that cannot be delivered before pathway completion is not operationally effective.

B.30 Uncertainty Propagation

Let the parameter vector be

$$\boldsymbol{\Theta} = \left(\boldsymbol{\Theta} * \mathcal{M}, \boldsymbol{\Theta} * \mathbf{g}, \boldsymbol{\Theta} * \mathcal{B}, \boldsymbol{\Theta} * \Gamma, \boldsymbol{\Theta}_{\theta}, \eta\right). \tag{B.57}$$

The posterior or bootstrap distribution should propagate through the measurement system:

$$\mathrm{TRI} * i, t^{(b)} = \mathcal{Q}\left(\mathcal{I} * i, t; \boldsymbol{\Theta}^{(b)}\right). \tag{B.58}$$

Report:

- point estimate;
- confidence interval;
- dominant-path probability;
- model-confidence score;
- sensitivity range.

B.31 Model-Confidence Score

A model-confidence measure may be

$$Q_{i,t}^{\text{model}} = w_D Q_{i,t}^{\text{data}} + w_S Q_{i,t}^{\text{stability}} + w_C Q_{i,t}^{\text{calibration}} + w_P Q_{i,t}^{\text{path}} + w_U Q_{i,t}^{\text{uncertainty}}. \tag{B.59}$$

Risk and confidence should be reported together.

B.32 Use of Machine Learning

Machine learning may be used for:

- nonlinear baseline risk;
- manifold estimation;
- state-dependent covariance;
- heterogeneous barrier effects;
- pathway discovery;
- graph propagation;
- anomaly detection;
- counterfactual intervention.

The selected method should match the structural object being estimated.

A black-box model should not replace the decomposed architecture.

B.33 Use of Large Language Models

Language models may extract structured variables from:

- contracts;
- credit agreements;
- legal opinions;
- incident reports;
- board minutes;
- regulatory correspondence.

For field (j), the extracted value is

$$\widehat{z} * j, d^{\mathrm{LLM}} = z * j, d + \varepsilon_{j,d}^{\mathrm{LLM}}. \tag{B.60}$$

The extraction pipeline should require:

- controlled schema;
- retrieval from an approved corpus;
- source passage;
- confidence score;
- human review for material fields;
- abstention when evidence is insufficient.

B.34 AI-Agent Workflow

A governed AI agent may perform the following sequence:

1. ingest data;
2. validate formats and timestamps;
3. retrieve relevant documents;
4. extract structured evidence;
5. update state and barrier variables;
6. run approved models;
7. calculate risk components;
8. draft an explanation;
9. escalate exceptions.

The agent should not independently:

- modify model parameters;
- approve material overrides;
- execute high-impact interventions;
- suppress uncertainty;
- change governance limits.

B.35 Production System Architecture

A practical technical architecture may contain six layers.

B.35.1 Layer 1: Data Ingestion

Sources:

- databases;
- market feeds;
- document stores;
- operational logs;
- network systems.

B.35.2 Layer 2: Feature and Evidence Store

Contains:

- versioned features;
- source links;
- timestamps;
- missingness flags;
- model-ready datasets.

B.35.3 Layer 3: Estimation Services

Contains separate services for:

- classical risk;
- manifold and metric;
- dynamics;
- barriers;

- pathways;
- phase and action.

B.35.4 Layer 4: Measurement and Stress Engine

Calculates:

- component measures;
- tunneling-risk index;
- uncertainty;
- forward stress;
- reverse stress;
- interventions.

B.35.5 Layer 5: Governance Engine

Applies:

- limits;
- escalation rules;
- authority maps;
- response-time standards;
- model-confidence requirements.

B.35.6 Layer 6: Reporting and Audit

Produces:

- management scorecards;
- board dashboards;
- model logs;
- evidence packages;
- intervention records.

B.36 Model Versioning

Each production result should identify:

$$v = (v_{\text{data}}, v_{\text{features}}, v_{\text{model}}, v_{\text{calibration}}, v_{\text{limits}}) . \tag{B.61}$$

Historical results should not be silently rewritten after recalibration.

B.37 Monitoring and Drift

Monitor:

- input drift;
- embedding drift;
- calibration drift;
- pathway drift;
- phase drift;
- intervention-effect drift.

For distributions F_{dev} and F_{cur}, a drift measure may be

$$D_{\text{W}} = W_1 (F_{\text{dev}}, F_{\text{cur}}) , \tag{B.62}$$

where W_1 is the Wasserstein distance.

Drift above a threshold should trigger investigation and possibly recalibration.

B.38 Minimum Governance Records

The production system should maintain:

1. data dictionary;
2. feature registry;
3. model inventory;
4. barrier register;
5. pathway register;
6. scenario library;
7. intervention library;
8. validation reports;
9. limit approvals;
10. incident and near-miss reviews.

B.39 Corporate-Credit Pilot

The remainder of this appendix presents a minimum viable corporate-credit implementation.

B.39.1 Objective

Estimate whether the tunneling-risk framework improves twelve-month prediction of default or distressed restructuring.

B.39.2 Unit

$$i = \text{corporate borrower}, \qquad t = \text{quarter}. \tag{B.63}$$

B.39.3 State Vector

Define

$$\mathbf{x} * i, t = (\ell * i, t, q_{i,t}, m_{i,t}, c_{i,t}, r_{i,t}, h_{i,t}, s_{i,t})\,, \tag{B.64}$$

where:

- ℓ: leverage;
- q: liquidity;
- m: cash-flow margin;
- c: collateral coverage;
- r: refinancing access;
- h: covenant headroom;
- s: sponsor or lender support.

B.39.4 Adverse Outcome

$$Y_{i,t+4} = 1 \tag{B.65}$$

if default, distressed exchange, or material restructuring occurs during the following four quarters.

B.39.5 Classical Baseline

Estimate:

$$h_i(t) = h_0(t) \exp\left(\boldsymbol{\beta}^{\mathsf{T}} \mathbf{x}_{i,t}\right). \tag{B.66}$$

Compare with a gradient-boosted survival model.

B.39.6 Manifold

Use a supervised autoencoder to construct

$$\mathbf{y}_{i,t} \in \mathbb{R}^3. \tag{B.67}$$

The three latent dimensions should be interpreted through feature sensitivity and nearest-neighbour analysis.

B.39.7 Barrier

The effective protection score is

$$\widetilde{P} * i, t = w_q \widetilde{q} * i, t + w_c \widetilde{c} * i, t + w_h \widetilde{h} * i, t + w_s \widetilde{s}_{i,t}. \tag{B.68}$$

Each component is adjusted for availability and reliability.

B.39.8 Pathways

Use three initial pathways.

Operating Path

$$\text{margin decline} \to \text{cash-flow pressure} \to \text{liquidity depletion} \to \text{default}. \tag{B.69}$$

Refinancing Path

$$\text{spread widening} \to \text{market-access decline} \to \text{maturity wall} \to \text{restructuring}. \tag{B.70}$$

Collateral-Covenant Path

$$\text{collateral decline} \to \text{covenant pressure} \to \text{lender acceleration} \to \text{liquidity exhaustion}. \tag{B.71}$$

B.39.9 Phase Variables

For the refinancing path, define

$$\theta_{\text{ref},i,t} = \omega_1 \tau_{i,t}^{\text{maturity}} + \omega_2 \tau_{i,t}^{\text{spread}} + \omega_3 \tau_{i,t}^{\text{support}}. \tag{B.72}$$

For the collateral-covenant path,

$$\theta_{\text{cov},i,t} = \omega_4 \tau_{i,t}^{\text{valuation}} + \omega_5 \tau_{i,t}^{\text{breach}} + \omega_6 \tau_{i,t}^{\text{waiver}}. \tag{B.73}$$

B.39.10 Action

Calculate action for every pathway and identify

$$\gamma^*_{i,t} = \arg, \min *\gamma S * R, i, t[\gamma]. \tag{B.74}$$

B.39.11 Measurement

Construct:

$$\text{TRI} * i, t = w_D D * i, t + w_B B_{i,t} + w_M M_{i,t} + w_A A_{i,t} + w_C C_{i,t} + w_I I_{i,t} + w_T T_{i,t}. \tag{B.75}$$

B.39.12 Validation

The pilot succeeds only if it demonstrates:

1. improved out-of-sample calibration;
2. positive incremental predictive value;
3. useful lead time;
4. stable dominant-path identification;
5. interpretable intervention recommendations.

B.40 Illustrative Production Record

A production output for borrower (i) may contain:

Field	Illustrative Value
Classical twelve-month probability	(8.4
Distance percentile	(71)
Barrier-weakness percentile	(82)
Adverse-mobility percentile	(76)
Minimum action	(0.61)
Dominant pathway	Refinancing failure
Adverse coherence	(0.74)
Interference contribution	(+3.2) percentage points
Transmission	(0.58)
Composite tunneling-risk index	(0.77)
Model confidence	(0.81)
Recommended intervention	Maturity extension plus committed liquidity
Required response window	Thirty days

B.41 Decision Logic

A simple decision rule may be:

$$\text{Escalate} = \mathbb{I}\left(\text{TRI} * i, t \geq \tau * \text{TRI}\right) \vee \mathbb{I}\left(S^*_{R,i,t} \leq \tau_S\right) \vee \mathbb{I}\left(\Delta S^*_{R,i,t} \leq -\tau_{\Delta S}\right). \tag{B.76}$$

The intervention should be selected using:

$$u^* = \arg, \max_u \frac{\mathbb{E}\left[\text{TRI}^{\text{before}} - \text{TRI}^{\text{after},u}\right]}{C(u)}. \tag{B.77}$$

B.42 Implementation Checklist

Before deployment, confirm that:

1. the adverse outcome is clearly defined;
2. timestamps are correct;
3. the classical benchmark is credible;
4. the manifold is stable and interpretable;
5. the metric has empirical meaning;
6. barrier variables have protective relevance;
7. candidate pathways are observable;
8. phase variables are measurable;
9. action units are coherent;
10. transmission is calibrated;
11. uncertainty is propagated;
12. limits are empirically supported;
13. interventions are authorized;
14. model versions are controlled;
15. reporting distinguishes observation from interpretation.

B.43 Minimum Acceptance Criteria

The empirical system should not move into production unless:

1. the classical model performs adequately;
2. the geometric model adds stable value;
3. the barrier model is interpretable;

4. pathways reproduce observed mechanisms;
5. phase survives ablation and timing benchmarks;
6. action compression precedes adverse events;
7. transmission is calibrated;
8. false warnings remain operationally manageable;
9. interventions can be delivered in time;
10. model governance approves the complete chain.

B.44 Final Implementation Proposition

The empirical implementation can be summarized as

$$\boxed{\begin{array}{l}\text{A tunneling-risk system is credible only when:}\\ \text{its state geometry is estimated from timestamp-correct data,}\\ \text{its barriers correspond to measurable protection,}\\ \text{its pathways are observable and reproducible,}\\ \text{its phases are grounded in timing evidence,}\\ \text{its action and transmission are calibrated,}\\ \text{and its outputs improve real decisions.}\end{array}} \tag{B.78}$$

The practical objective is not to create the most complex possible model. It is to build the simplest empirically defensible system capable of revealing how risk can travel, how protection can fail, and which intervention can interrupt the transition before it becomes irreversible.

Chapter C

Worked Applications and Illustrative Calculations

C.1 Purpose of the Appendix

Appendix A consolidated the mathematical architecture of tunneling and risk management. Appendix B translated that architecture into an empirical implementation blueprint. The present appendix demonstrates how the framework may be applied in practice through a series of worked examples.

The objective is not to present fully calibrated production models. The numerical values are illustrative. They are chosen to show how the principal objects of the framework connect:

$$\boxed{\begin{aligned}&\text{State} \longrightarrow \text{Geometry} \longrightarrow \text{Barrier}\\ &\qquad \longrightarrow \text{Pathways} \longrightarrow \text{Phase} \longrightarrow \text{Action}\\ &\qquad \longrightarrow \text{Transmission} \longrightarrow \text{Intervention.}\end{aligned}} \tag{C.1}$$

The appendix contains four worked applications:

1. corporate credit;
2. operational and cyber risk;
3. legal and regulatory risk;
4. institutional and systemic risk.

Each example follows the same sequence:

1. define the adverse event;
2. construct the state;
3. identify the protective barrier;
4. define candidate pathways;

5. estimate pathway magnitudes;
6. construct phase proxies;
7. calculate interference;
8. calculate minimum action and transmission;
9. compare interventions;
10. identify governance implications.

C.2 Worked Example 1: Corporate Credit

C.2.1 Problem Definition

Consider a corporate borrower with a twelve-month prediction horizon.

The adverse event is defined as:

$$A_{\text{corp}} = \{\text{default}, \text{distressed exchange}, \text{material restructuring}\}\,. \tag{C.2}$$

The borrower is currently operating normally, but it faces:

- weakening cash flow;
- a large maturity concentration;
- declining collateral values;
- uncertain sponsor support.

C.2.2 State Vector

Define the state as

$$\mathbf{x}_t = (\ell_t, q_t, m_t, c_t, r_t, h_t, s_t)\,, \tag{C.3}$$

where:

- ℓ_t is leverage;
- q_t is liquidity;
- m_t is operating margin;
- c_t is collateral coverage;
- r_t is refinancing access;
- h_t is covenant headroom;

- s_t is sponsor support.

Suppose the normalized state is

$$\mathbf{x}_t = (0.68, 0.41, 0.46, 0.55, 0.38, 0.44, 0.52)\,, \tag{C.4}$$

where larger values represent greater strength except for leverage, which has been risk-oriented so that a larger value represents greater stress.

C.2.3 Distance to the Adverse Region

Suppose the estimated geodesic distance to the adverse region is

$$d_A = 1.35. \tag{C.5}$$

Using a calibrated distance scale

$$\kappa_D = 1.20, \tag{C.6}$$

the proximity score is

$$D = \exp\left(-\frac{1.35}{1.20}\right) = 0.325. \tag{C.7}$$

The borrower is not yet extremely close to the adverse region.

C.2.4 Protective Barrier

Assume the principal protections are:

- cash liquidity;
- collateral;
- covenant headroom;
- sponsor support.

Nominal protection is

$$P = w_q q + w_c c + w_h h + w_s s. \tag{C.8}$$

Let the weights be

$$(w_q, w_c, w_h, w_s) = (0.30, 0.25, 0.25, 0.20)\,. \tag{C.9}$$

Let availability and reliability be:

Protection	**Nominal**	**Availability**	**Reliability**
Liquidity	(0.41)	(0.95)	(0.90)
Collateral	(0.55)	(0.80)	(0.75)
Covenant headroom	(0.44)	(0.90)	(0.85)
Sponsor support	(0.52)	(0.60)	(0.55)

Effective protection is therefore

$$\begin{aligned}\widetilde{P} &= 0.30\,(0.41 \times 0.95 \times 0.90) \\ &\quad + 0.25\,(0.55 \times 0.80 \times 0.75) \\ &\quad + 0.25\,(0.44 \times 0.90 \times 0.85) \\ &\quad + 0.20\,(0.52 \times 0.60 \times 0.55) \\ &= 0.3061. \end{aligned} \tag{C.10}$$

Suppose the current stress load is

$$L^{\text{stress}} = 0.29. \tag{C.11}$$

Residual capacity is

$$Q^B = 0.3061 - 0.290 = 0.0161. \tag{C.12}$$

Barrier utilization is

$$U^B = \frac{0.290}{0.3061} = 0.947. \tag{C.13}$$

The borrower remains marginally protected, but approximately (94.7%) of effective capacity is already utilized. The remaining buffer is therefore materially smaller than the nominal protection figures suggest.

C.2.5 Candidate Pathways

Assume three pathways.

Operating Deterioration

$$\gamma_1 : \text{margin decline} \rightarrow \text{cash-flow pressure} \rightarrow \text{liquidity depletion} \rightarrow \text{default}. \tag{C.14}$$

Refinancing Failure

$$\gamma_2 : \text{spread widening} \rightarrow \text{market-access loss} \rightarrow \text{maturity wall} \rightarrow \text{restructuring}. \tag{C.15}$$

Collateral-Covenant Escalation

$$\gamma_3 : \text{collateral decline} \to \text{covenant breach} \to \text{acceleration} \to \text{liquidity exhaustion}. \tag{C.16}$$

Suppose the estimated pathway-completion probabilities are

$$p_1 = 0.07, \qquad p_2 = 0.11, \qquad p_3 = 0.06. \tag{C.17}$$

The corresponding magnitudes are

$$R_1 = \sqrt{0.07} = 0.265, \tag{C.18}$$

$$R_2 = \sqrt{0.11} = 0.332, \tag{C.19}$$

and

$$R_3 = \sqrt{0.06} = 0.245. \tag{C.20}$$

C.2.6 Phase Construction

Fix a twelve-month decision horizon $H = 12$ months and take the assessment date as the phase origin. The phase transformation is

$$\theta_k = 2\pi \frac{\tau_k}{H} \pmod{2\pi}, \tag{C.21}$$

so a delay of six months corresponds to π. The following phases therefore correspond to pathway timing offsets of approximately (1.15), (1.49), and (1.80) months. The horizon and origin must be fixed before observing the outcome.

Suppose the phase estimates are

$$\theta_1 = 0.60, \qquad \theta_2 = 0.78, \qquad \theta_3 = 0.94. \tag{C.22}$$

The phases are relatively close. The pathways are therefore temporally aligned.

The weighted coherence is

$$\mathcal{C} = \frac{\left| R_1 e^{\mathrm{i}\theta_1} + R_2 e^{\mathrm{i}\theta_2} + R_3 e^{\mathrm{i}\theta_3} \right|}{R_1 + R_2 + R_3}. \tag{C.23}$$

Substituting the values gives approximately

$$\mathcal{C} \approx 0.99. \tag{C.24}$$

This indicates strong alignment among the adverse pathways.

C.2.7 Structural Amplification Calculation

The standalone sum is

$$P^{\text{sum}} = 0.07 + 0.11 + 0.06 = 0.24. \tag{C.25}$$

The raw structural amplitude score is

$$Q^{\text{amp}} = \left| R_1 e^{\mathrm{i}\theta_1} + R_2 e^{\mathrm{i}\theta_2} + R_3 e^{\mathrm{i}\theta_3} \right|^2. \tag{C.26}$$

Using the values above,

$$Q^{\text{amp}} \approx 0.695. \tag{C.27}$$

This value is not a probability and must not be compared with (0.24) as though both were estimates of the same event probability. The normalized structural alignment index is

$$J^{\text{amp}} = \frac{0.695}{(0.265 + 0.332 + 0.245)^2} = 0.983. \tag{C.28}$$

This is a strong constructive-alignment result. It is a bounded structural feature, not an adverse-event probability. A probability must be obtained from a normalized state amplitude or from a separately calibrated model that uses J^{amp} alongside the classical prediction.

To demonstrate destructive alignment, suppose an intervention delays the refinancing pathway by half the decision horizon. Then

$$\theta_2' = 0.78 + \pi = 3.922. \tag{C.29}$$

Keeping the pathway magnitudes unchanged gives

$$\mathcal{C}' \approx 0.203, \qquad Q^{\text{amp}\prime} \approx 0.029, \qquad J^{\text{amp}\prime} \approx 0.041. \tag{C.30}$$

The same standalone pathway probabilities now produce a reassuring structural signal because their timing is opposed. This counterexample makes the phase claim falsifiable: if estimated timing changes do not alter the out-of-sample structural feature or calibrated probability as predicted, the phase extension should be rejected.

C.2.8 Barrier Action

Suppose the path actions are

$$S_R[\gamma_1] = 1.28, \tag{C.31}$$

$$S_R[\gamma_2] = 0.74, \tag{C.32}$$

and

$$S_R[\gamma_3] = 1.05. \tag{C.33}$$

The dominant path is therefore

$$\gamma^* = \gamma_2, \tag{C.34}$$

with

$$S_R^* = 0.74. \tag{C.35}$$

Suppose the calibrated transmission scale is

$$\eta = 1.40. \tag{C.36}$$

Transmission is

$$T_R = \exp\left(-\frac{2(0.74)}{1.40}\right) = 0.347. \tag{C.37}$$

C.2.9 Composite Risk Measure

Suppose the normalized component scores are:

Component	**Score**
State proximity	(0.33)
Barrier weakness	(0.78)
Adverse mobility	(0.69)
Action risk	(0.71)
Adverse coherence	(0.99)
Structural phase amplification	(0.86)
Transmission	(0.35)

With equal weights,

$$\mathrm{TRI} = \frac{0.33 + 0.78 + 0.69 + 0.71 + 0.99 + 0.86 + 0.35}{7} = 0.673. \tag{C.38}$$

The borrower is not extremely close to default, but the combination of barrier weakness, low refinancing action, and phase alignment produces an elevated structural warning.

C.2.10 Intervention Comparison

Consider three interventions.

Intervention	New Action	New Transmission	Cost
Committed liquidity	(0.96)	(0.254)	(0.18)
Maturity extension	(1.20)	(0.180)	(0.22)
Additional collateral	(0.91)	(0.273)	(0.16)

Intervention efficiency based on transmission reduction is

$$E_{T,j} = \frac{T_R^{\text{before}} - T_R^{\text{after},j}}{C_j}. \tag{C.39}$$

For committed liquidity,

$$E_{T,\text{liq}} = \frac{0.347 - 0.254}{0.18} = 0.517. \tag{C.40}$$

For maturity extension,

$$E_{T,\text{mat}} = \frac{0.347 - 0.180}{0.22} = 0.759. \tag{C.41}$$

For collateral enhancement,

$$E_{T,\text{coll}} = \frac{0.347 - 0.273}{0.16} = 0.463. \tag{C.42}$$

Maturity extension is the most efficient intervention because it increases barrier width and changes the timing of the refinancing pathway.

C.3 Worked Example 2: Operational and Cyber Risk

C.3.1 Problem Definition

Consider a cloud-based payment system.

The adverse event is

$$A_{\text{cyber}} = \{\text{critical service outage}, \text{material data compromise}, \text{fraudulent payment execution}\}. \tag{C.43}$$

The system has strong nominal controls but depends heavily on a shared identity provider and one cloud region.

C.3.2 State Vector

Define

$$\mathbf{x}_t = (v_t, c_t, m_t, r_t, d_t, h_t, u_t)\,, \tag{C.44}$$

where:

- v_t is vulnerability;
- c_t is control effectiveness;
- m_t is monitoring quality;
- r_t is recovery readiness;
- d_t is dependency concentration;
- h_t is human reliability;
- u_t is workload or utilization.

Suppose the risk-oriented normalized state is

$$\mathbf{x}_t = (0.62, 0.35, 0.40, 0.38, 0.81, 0.44, 0.73)\,. \tag{C.45}$$

C.3.3 Barrier Architecture

The principal barriers are:

- multifactor authentication;
- privilege segregation;
- anomaly detection;
- incident response;
- backup and recovery;
- infrastructure redundancy.

Suppose nominally there are six layers, but several share the same identity and cloud infrastructure.

The dependency-adjusted effective number of layers is

$$K_{\text{eff}} = 2.7. \tag{C.46}$$

The institution therefore has six documented controls but fewer than three effectively independent protective layers.

C.3.4 Candidate Pathways

Assume four pathways.

Credential Path

$$\gamma_1 : \text{credential compromise} \to \text{account access} \to \text{privilege escalation} \to \text{fraud}. \tag{C.47}$$

Software Exploit Path

$$\gamma_2 : \text{vulnerability exploitation} \to \text{lateral movement} \to \text{data access} \to \text{data loss}. \tag{C.48}$$

Cloud Dependency Path

$$\gamma_3 : \text{regional cloud outage} \to \text{identity failure} \to \text{service interruption} \to \text{critical outage}. \tag{C.49}$$

Response Failure Path

$$\gamma_4 : \text{alert} \to \text{classification delay} \to \text{containment delay} \to \text{recovery failure}. \tag{C.50}$$

Suppose

$$(p_1, p_2, p_3, p_4) = (0.05, 0.04, 0.09, 0.07)\,. \tag{C.51}$$

C.3.5 Phase and Timing

Suppose the phases are

$$(\theta_1, \theta_2, \theta_3, \theta_4) = (1.20, 1.36, 1.31, 1.48)\,. \tag{C.52}$$

The closeness of the phase values reflects:

- common maintenance timing;
- shared identity infrastructure;
- high workload;
- delayed escalation.

Weighted coherence is high:

$$\mathcal{C}_{\text{cyber}} \approx 0.99. \tag{C.53}$$

C.3.6 Action

Suppose the estimated actions are

$$(S_1, S_2, S_3, S_4) = (1.05, 1.20, 0.58, 0.82)\,. \tag{C.54}$$

The cloud-dependency pathway has the lowest action.

The dominant path is therefore

$$\gamma^* = \gamma_3. \tag{C.55}$$

For

$$\eta = 1.10, \tag{C.56}$$

transmission is

$$T_R = \exp\left(-\frac{2(0.58)}{1.10}\right) = 0.348. \tag{C.57}$$

C.3.7 Interventions

Consider:

1. adding a second independent identity provider;
2. reducing incident-response delay;
3. adding a second cloud region;
4. increasing monitoring sensitivity.

A second cloud region and independent identity service change both barrier independence and pathway width.

Suppose this raises the cloud-path action from

$$0.58 \tag{C.58}$$

to

$$1.42. \tag{C.59}$$

Transmission becomes

$$T_R^{\text{after}} = \exp\left(-\frac{2(1.42)}{1.10}\right) = 0.076. \tag{C.60}$$

The intervention succeeds because it removes a shared dependency and materially increases effective barrier width.

C.3.8 Governance Interpretation

The principal governance finding is not that the institution lacks controls.

It is that the controls are less independent than their nominal count suggests.

The appropriate report is:

> The critical operational pathway is driven by concentrated identity and cloud dependencies. The nominal six-layer control architecture provides approximately 2.7 independent layers. Adding independent identity and regional recovery capacity substantially raises path action and reduces modeled transmission.

C.4 Worked Example 3: Legal and Regulatory Risk

C.4.1 Problem Definition

Consider a financial institution offering a product under a contract whose disclosure language is legally defensible but potentially ambiguous.

The adverse region is

$$A_{\text{legal}} = \{\text{adverse ruling}, \text{regulatory sanction}, \text{unenforceability}, \text{material restitution}\}\,. \tag{C.61}$$

C.4.2 State Vector

Define

$$\mathbf{x}_t = (a_t, e_t, d_t, c_t, p_t, r_t)\,, \tag{C.62}$$

where:

- a_t is contractual ambiguity;
- e_t is evidence quality;
- d_t is disclosure quality;
- c_t is conduct consistency;
- p_t is precedent strength;
- r_t is remediation readiness.

Suppose the normalized risk-oriented state is

$$\mathbf{x}_t = (0.72, 0.46, 0.49, 0.58, 0.61, 0.37)\,. \tag{C.63}$$

C.4.3 Protective Barrier

The legal barrier consists of:

- contractual wording;
- documented disclosure;
- evidence preservation;
- internal legal review;
- remediation procedures;
- prior consistent conduct.

Suppose the nominal legal-protection score is

$$P_{\text{legal}}^{\text{nom}} = 0.68. \tag{C.64}$$

However, reliability is reduced because:

- customer communications differ from the contract;
- evidence is distributed across systems;
- prior remediation was inconsistent.

The effective legal protection is

$$\widetilde{P}_{\text{legal}} = 0.68 \times 0.72 = 0.490. \tag{C.65}$$

C.4.4 Candidate Pathways

Assume three pathways.

Interpretive Path

$$\gamma_1 : \text{contract ambiguity} \to \text{adverse interpretation} \to \text{unenforceability}. \tag{C.66}$$

Conduct-Evidence Path

$$\gamma_2 : \text{inconsistent conduct} \to \text{adverse evidence} \to \text{regulatory sanction}. \tag{C.67}$$

Disclosure-Remediation Path

$$\gamma_3 : \text{weak disclosure} \to \text{customer harm} \to \text{late remediation} \to \text{restitution}. \tag{C.68}$$

Suppose

$$(p_1, p_2, p_3) = (0.08, 0.12, 0.09) . \tag{C.69}$$

C.4.5 Phase: A Destructive-Alignment Case

Suppose the institution receives the same three signals, but an early remediation programme separates their effective decision times across a six-month governance horizon:

- a regulator request;
- an internal whistleblower report;
- several customer complaints

The phase origin is the first signal and the period is fixed at six months, so a three-month separation corresponds to π.

The phases are

$$(\theta_1, \theta_2, \theta_3) = (2.10, 4.20, 6.30) . \tag{C.70}$$

Using magnitudes $R_k = \sqrt{p_k}$, the weighted coherence is approximately

$$\mathcal{C}_{\text{legal}} \approx 0.058. \tag{C.71}$$

This indicates destructive timing alignment rather than synchronization. The low value is a reassuring result: the interpretive, conduct, and remediation pathways do not reinforce one another even though their standalone probabilities are unchanged.

C.4.6 Action and Transmission

Suppose

$$(S_1, S_2, S_3) = (1.12, 0.69, 0.83) . \tag{C.72}$$

The dominant path is the conduct-evidence pathway.

Let

$$\eta = 1.25. \tag{C.73}$$

Then

$$T_R = \exp\left(-\frac{2(0.69)}{1.25}\right) = 0.332. \tag{C.74}$$

C.4.7 Interventions

Possible interventions include:

1. preserve and centralize evidence;
2. suspend disputed practices;
3. issue corrective disclosure;
4. begin customer remediation;
5. engage the regulator promptly.

An early remediation programme may:

- raise barrier height through evidence and corrective action;
- increase width by inserting review and remediation stages;
- change phase by bringing stabilizing action forward.

Suppose the intervention raises minimum action to

$$S_R^{*,\text{after}} = 1.31. \tag{C.75}$$

Transmission becomes

$$T_R^{\text{after}} = \exp\left(-\frac{2(1.31)}{1.25}\right) = 0.123. \tag{C.76}$$

C.4.8 Governance Interpretation

The key governance lesson is that legal protection is not defined only by contractual language.

Observed conduct, evidence quality, and response timing may create a lower-action pathway than the formal interpretive path.

The correct conclusion is:

> The principal legal vulnerability is not contractual wording alone. It is the interaction of inconsistent conduct, dispersed evidence, and delayed remediation. Early evidence preservation and corrective action materially increase barrier action and reduce adverse transmission.

C.5 Worked Example 4: Institutional and Systemic Risk

C.5.1 Problem Definition

Consider a system of three banks:

$$B_1, \qquad B_2, \qquad B_3. \tag{C.77}$$

All three:

- hold a common bond;
- rely on wholesale funding;
- post collateral to a common central counterparty;
- use the same payment infrastructure.

The adverse region is

$$A_{\text{sys}} = \{\text{market freeze}, \text{fire-sale spiral}, \text{funding seizure}, \text{payment disruption}\}. \tag{C.78}$$

C.5.2 Institutional States

Let the institutional state be

$$\mathbf{x}_i = (k_i, l_i, f_i, a_i), \tag{C.79}$$

where:

- k_i is capital;
- l_i is liquidity;
- f_i is funding stability;
- a_i is asset quality.

Suppose:

$$\mathbf{x}_1 = (0.62, 0.48, 0.44, 0.58), \tag{C.80}$$

$$\mathbf{x}_2 = (0.71, 0.57, 0.51, 0.64), \tag{C.81}$$

and

$$\mathbf{x}_3 = (0.76, 0.61, 0.55, 0.68). \tag{C.82}$$

Each institution appears individually viable.

C.5.3 Network

Let the weighted exposure matrix be

$$\mathbf{W} = \begin{bmatrix} 0 & 0.18 & 0.10 \\ 0.14 & 0 & 0.16 \\ 0.08 & 0.12 & 0 \end{bmatrix}. \tag{C.83}$$

The common-asset concentration is

$$H_{\text{asset}} = 0.74. \tag{C.84}$$

Infrastructure concentration is

$$H_{\text{infra}} = 0.82. \tag{C.85}$$

The system is therefore highly dependent on shared assets and infrastructure.

C.5.4 Systemic Barriers

The collective barriers are:

- institution-level capital;
- liquidity buffers;
- central-bank liquidity;
- central-counterparty default resources;
- payment-system resilience.

Suppose nominal collective capacity is

$$P_{\text{sys}}^{\text{nom}} = 1.00. \tag{C.86}$$

Under simultaneous stress, availability falls to

$$A_{\text{sys}} = 0.68, \tag{C.87}$$

and reliability falls to

$$R_{\text{sys}} = 0.72. \tag{C.88}$$

Effective protection is

$$\widetilde{P}_{\text{sys}} = 1.00 \times 0.68 \times 0.72 = 0.490. \tag{C.89}$$

The apparent system-wide barrier is therefore approximately half as strong as its nominal value.

C.5.5 Systemic Pathways

Assume five pathways:

1. direct counterparty loss;
2. common-asset fire sale;
3. margin spiral;
4. wholesale-funding withdrawal;
5. payment-infrastructure disruption.

Suppose their probabilities are

$$(p_1, p_2, p_3, p_4, p_5) = (0.04, 0.10, 0.08, 0.09, 0.03)\,. \tag{C.90}$$

C.5.6 Behavioural Synchronization

Suppose the three institutions follow similar risk rules and respond to stress by selling the common bond.

Let their response phases be

$$(\theta_{B_1}, \theta_{B_2}, \theta_{B_3}) = (0.45, 0.51, 0.56)\,. \tag{C.91}$$

Systemic coherence is

$$\mathcal{C} * \text{sys} = \left| \frac{1}{3} \sum *i = 1^3 e^{\mathrm{i}\theta_{B_i}} \right| \approx 0.999. \tag{C.92}$$

The institutions are responding almost simultaneously.

C.5.7 Systemic Action

Suppose the path actions are:

Pathway	**Action**
Direct counterparty loss	(1.32)
Common-asset fire sale	(0.54)
Margin spiral	(0.68)
Funding withdrawal	(0.72)
Payment disruption	(1.18)

The common-asset fire-sale pathway has minimum action:

$$S^*_{\text{sys}} = 0.54. \tag{C.93}$$

Let

$$\eta_{\text{sys}} = 1.30. \tag{C.94}$$

Systemic transmission is

$$T_{\text{sys}} = \exp\left(-\frac{2(0.54)}{1.30}\right) = 0.436. \tag{C.95}$$

C.5.8 Policy Interventions

Consider:

1. temporary asset-purchase facility;
2. staggered margin requirements;
3. central-bank liquidity;
4. coordinated communication.

The asset-purchase facility raises the action of the fire-sale pathway.

The staggered-margin policy changes phase and reduces synchronization.

Suppose the combined intervention produces:

$$S^{*,\text{after}}_{\text{sys}} = 1.16, \tag{C.96}$$

and reduces coherence to

$$\mathcal{C}^{\text{after}}_{\text{sys}} = 0.61. \tag{C.97}$$

Transmission becomes

$$T^{\text{after}}_{\text{sys}} = \exp\left(-\frac{2(1.16)}{1.30}\right) = 0.168. \tag{C.98}$$

C.5.9 Governance Interpretation

The individually strongest institution is not necessarily the most important object.

The decisive vulnerability is the common-asset pathway combined with synchronized risk reduction.

The appropriate systemic conclusion is:

> The principal systemic vulnerability arises from correlated fire sales and shared market infrastructure rather than from isolated institutional insolvency. Policy is most effective when it both raises the resistance of the common-asset pathway and reduces behavioural synchronization.

C.6 Cross-Case Comparison

The four cases use the same general architecture but different empirical objects.

Domain	Dominant Barrier	Dominant Path	Phase Variable	Intervention
Corporate credit	Liquidity and maturity	Refinancing failure	Maturity overlap	Extend maturity
Cyber	Independent infrastructure	Cloud dependency	Detection and outage timing	Add independent recovery
Legal	Evidence and remediation	Conduct-evidence route	Complaint and regulator timing	Early remediation
Systemic	Collective liquidity and market capacity	Common-asset fire sale	Synchronized deleveraging	Asset support and de-synchronization

C.7 Illustrative Calculation Template

For any application, the analyst may use the following generic template.

C.7.1 Step 1: Define State

$$\mathbf{x} * i, t = (x * 1, i, t, \ldots, x_{p,i,t}) \,. \tag{C.99}$$

C.7.2 Step 2: Estimate Distance

$$d_{A,i,t} = d_{\mathbf{g}} \left(\mathbf{x}_{i,t}, A \right) . \tag{C.100}$$

C.7.3 Step 3: Estimate Effective Protection

$$\widetilde{P} * i, t = \sum_j w_j P * j, i, t A_{j,i,t} R_{j,i,t}. \tag{C.101}$$

C.7.4 Step 4: Define Pathways

$$\Gamma_{i,t} = \{\gamma_1, \ldots, \gamma_K\} \,. \tag{C.102}$$

C.7.5 Step 5: Estimate Magnitudes

$$R_{k,i,t} = \sqrt{p_{k,i,t}}. \tag{C.103}$$

C.7.6 Step 6: Estimate Phases

$$\theta_{k,i,t} = \boldsymbol{\omega} * k^{\mathsf{T}} \mathbf{z} * k, i, t^{\text{phase}}. \tag{C.104}$$

C.7.7 Step 7: Calculate Coherence

$$\mathcal{C} * i, t = \frac{\left|\sum_k R * k, i, t e^{\mathrm{i}\theta_{k,i,t}}\right|}{\sum_k R_{k,i,t}}. \tag{C.105}$$

C.7.8 Step 8: Calculate Interference

$$I_{i,t} = \left|\sum_k R_{k,i,t} e^{\mathrm{i}\theta_{k,i,t}}\right|^2 - \sum_k p_{k,i,t}. \tag{C.106}$$

C.7.9 Step 9: Calculate Minimum Action

$$S^*_{R,i,t} = \min_\gamma S_{R,i,t}[\gamma]. \tag{C.107}$$

C.7.10 Step 10: Calculate Transmission

$$T_{R,i,t} = \exp\left(-\frac{2S^*_{R,i,t}}{\eta}\right). \tag{C.108}$$

C.7.11 Step 11: Calculate Composite Risk

$$\mathrm{TRI} * i, t = \sum_j w_j \widetilde{M} * j, i, t. \tag{C.109}$$

C.7.12 Step 12: Compare Interventions

$$u^* = \arg, \max_u \frac{\mathrm{TRI}^{\text{before}} - \mathrm{TRI}^{\text{after},u}}{C(u)}. \tag{C.110}$$

C.8 Interpretation Protocol

Every worked calculation should be interpreted through five questions.

C.8.1 What Is Observed?

Examples include:

- declining liquidity;
- shared cloud infrastructure;
- inconsistent conduct;
- synchronized asset sales.

C.8.2 What Is Estimated?

Examples include:

- barrier integrity;

- pathway probability;
- phase;
- minimum action;
- transmission.

C.8.3 What Is the Classical Explanation?

The analyst should test:

- drift;
- diffusion;
- jumps;
- threshold breach;
- common shock;
- visible barrier gap.

C.8.4 What Does the Tunneling Extension Add?

It should add one or more of:

- hidden-path identification;
- barrier-action measurement;
- timing interaction;
- interference;
- intervention insight.

C.8.5 What Action Follows?

The action should:

- raise barrier height;
- increase width;
- reduce permeability;
- change phase;
- improve evidence.

C.9 Numerical Guardrails

The illustrative calculations in this appendix should not be transferred directly into production. Production use requires:

1. empirically estimated scales;
2. calibrated probabilities;
3. validated pathway definitions;
4. stable phase parameters;
5. benchmark comparison;
6. uncertainty intervals;
7. model governance.

Particular caution is required when:

- interference is very large;
- transmission changes sharply;
- the dominant pathway is unstable;
- the effective barrier depends heavily on expert scoring;
- the sample contains few adverse events.

C.10 Suggested Computational Outputs

A research notebook supporting these examples should generate:

1. state-space visualization;
2. stable, fragile, and adverse regions;
3. barrier surface;
4. candidate-path graph;
5. pathway probability table;
6. phase and coherence plot;
7. pairwise-interference matrix;
8. minimum-action path;
9. transmission curve;

10. intervention comparison;
11. uncertainty intervals.

C.11 Suggested Notebook Structure

A minimum notebook sequence is:

Notebook	**Purpose**
NB01	Load and validate timestamp-correct data
NB02	Construct normalized state variables
NB03	Estimate and visualize the manifold
NB04	Estimate drift, covariance, and adverse mobility
NB05	Construct and test protective barriers
NB06	Discover and validate pathways
NB07	Estimate pathway probabilities
NB08	Estimate phase and coherence
NB09	Calculate action and transmission
NB10	Construct the tunneling-risk index
NB11	Run forward and reverse stress tests
NB12	Compare interventions and generate reports

C.12 Final Worked-Application Proposition

The examples demonstrate that the same formal architecture can be used across very different domains.

The state variables change. The barriers change. The pathways change. The meaning of phase changes. The intervention authority changes.

The underlying logic remains constant:

$$\boxed{\begin{array}{l}\text{Risk becomes structurally visible when:}\\ \text{the current state is located,}\\ \text{the effective barrier is measured,}\\ \text{the relevant pathways are reconstructed,}\\ \text{their timing relationships are estimated,}\\ \text{minimum action and transmission are calculated,}\\ \text{and interventions are evaluated by how they alter that structure.}\end{array}} \tag{C.111}$$

The practical objective is not to force every risk problem into the same numerical model. It is to use a common structural language for understanding how adverse outcomes become accessible and how timely intervention can make them less so.

Closing Note

The governing idea of this research programme is simple: probability estimates how often adverse outcomes may occur, while geometry explains how those outcomes become reachable. The apparent simplicity of that statement should not obscure its consequences. Once risk is viewed as accessibility within a structured space, the analysis can no longer end with a score, a probability, or a loss distribution. It must also ask where the system is located, how its environment is changing, which directions are becoming easier to traverse, what protects the system, and whether an unexpected route can bypass or penetrate that protection.

This perspective does not replace probability. It gives probability an architecture. A risk estimate is produced by a system occupying a state, moving under a law of motion, responding to shocks and interventions, and facing boundaries that separate ordinary operation from fragility and adversity. The same numerical probability can arise from slow deterioration, concentrated exposure, a weakening barrier, a short pathway, synchronized mechanisms, or profound uncertainty. These structures matter because they imply different warning indicators and different actions. A model that distinguishes them can be more useful even when its headline probability is unchanged.

The state-space representation was the first step. A borrower, institution, operating process, technological platform, legal position, or network was represented through coordinates sufficient to describe its relevant condition. Feasible combinations formed a manifold shaped by accounting identities, contracts, regulation, infrastructure, behaviour, and institutional capacity. A metric assigned meaning to movement. Curvature captured the fact that the consequence of a change depends on where it occurs. A small liquidity decline can be harmless in a well-capitalized, diversified borrower and decisive near a maturity concentration. A control weakness can be tolerable when recovery is independent and dangerous when several protections share one dependency.

Covariance then became more than a table of statistical relationships. It described directional mobility. Eigenvalues identified the magnitude of uncertainty along principal directions; eigenvectors showed where that uncertainty pointed. The important question was not simply whether volatility increased, but whether mobility rotated toward an adverse boundary. Under stress, effective dimension could collapse as previously diversified movements became concentrated into one harmful direction. This interpretation connected classical probability to geometry without pretending that covariance alone supplied a causal mechanism.

Protective barriers formed the next layer. Capital, liquidity, collateral, covenants, legal rights, operational controls, recovery capacity, resolution regimes, and public facilities were treated not as binary objects but as structured protections. Their effectiveness depended on height, width, continuity, reliability, availability, independence, and residual capacity. A barrier could be high but thin, wide but unreliable, formally continuous but operationally fragmented, or strong for one path and weak for another. The distinction between nominal and effective protection became fundamental. A control that exists only in policy, a facility unavailable under simultaneous demand, or collateral that cannot be realized when needed does not provide the resistance implied by its name.

Pathways made the framework dynamic and actionable. Adverse outcomes are rarely reached by one undifferentiated movement. They emerge through ordered sequences: margin decline, cash-flow pressure, liquidity depletion, and default; vulnerability, privilege escalation, delayed detection, and data loss; ambiguous obligation, inconsistent conduct, regulatory attention, and sanction; asset sales, price decline, margin calls, and further sales. Reconstructing these routes reveals interruption points. It also reveals that the most important path is not necessarily the one containing the largest isolated weakness. Several moderate vulnerabilities can form a low-resistance corridor when their timing and dependencies align.

The concepts of action and transmission provided a way to compare such routes. Action accumulated resistance along a path rather than measuring only the highest barrier point. The minimum-action path identified the route offering the greatest effective accessibility. Transmission translated that resistance into a nonlinear indicator. The exponential form mattered: modest compression of barrier width or action could produce a material increase in accessibility. These conclusions are not uniquely quantum-like. Freidlin–Wentzell quasipotentials, Kramers escape, and classical minimum-action methods already generate exponential rare-event sensitivity and dominant paths. They therefore form the mandatory baseline. The recurring lesson remains useful but classical: resilience depends on how protection is distributed across plausible routes and how much time remains for response.

Phase and interference introduced the most distinctive part of the tunneling analogy. Phase represented observable timing, sequencing, persistence, synchronization, and delay. When pathways aligned, their aggregate effect could exceed the sum of their standalone contributions; when intervention disrupted their timing, the combined effect could fall. This representation was never intended to establish physical quantum behaviour in institutions. Its claim was narrower: complex amplitudes may provide a disciplined model of relational information that ordinary additive probabilities omit. That claim remains conditional on empirical grounding. Phase must be estimated rather than narrated, and interference must outperform strong classical interaction models before it can be accepted.

For this reason, the classical null hypothesis remained central. Drift, diffusion, jumps, hidden regimes, omitted variables, network cascades, feedback, moving barriers, quasipotential escape, and classical minimum-action paths can explain surprising transitions without tunneling. A visible gap in protection is bypass, not penetration. A barrier that disappeared before the event cannot support a sub-barrier classification. An implausible prediction from a misspecified model is evidence against the model, not evidence for exotic dynamics. The tunneling extension bears the burden of proof because complexity should never be used to protect an idea from contradiction. Its incremental claim is limited to anchored, normalized phase structure that outperforms classical timing interactions.

The empirical chapters converted these principles into a research design. Evidence had to be timestamp-correct. States, metrics, boundaries, dynamics, barriers, paths, phase, action, and transmission had to be estimated as distinct objects. Classical models were fitted first. Candidate paths were reconstructed through process evidence, sequence methods, networks, causal structures, simulation, documents, and expert challenge. Validation required temporal and entity holdouts, historical replay, ablation, placebo tests, sensitivity analysis, intervention evidence, and uncertainty quantification. The appropriate conclusion was never that a physical tunneling event had been observed. It was that an extended structural model explained or predicted accessibility better than specified alternatives.

Measurement made the architecture usable. Distance, barrier weakness, adverse mobility, minimum action, coherence, interference, transmission, and model confidence could be reported separately and, where justified, combined into a tunneling-risk index. The component view remained more important than the composite. A decision-maker needed to know why the measure increased, which evidence drove the change, which pathway dominated, and whether uncertainty was expanding. A high score indicated estimated accessibility, not inevitability. A low score with weak evidence required caution rather than reassurance.

Stress testing extended the analysis from current measurement to adverse deformation. A scenario could move the state, expand the adverse region, alter the metric, rotate covariance, reduce effective dimension, weaken barrier components, activate new pathways, synchronize responses, and compress minimum action. The framework therefore distinguished state stress, geometric stress, barrier stress, and phase stress. This broader view explained why apparently moderate shocks can generate disproportionate outcomes: the scenario changes not only the system's position but also the terrain, protection, and available routes.

Governance completed the analytical sequence. A barrier without an owner, evidence base, capacity assessment, testing schedule, and activation authority was not a governed barrier. A pathway crossing organizational boundaries required coordination that ordinary functional ownership might not provide. Structural limits had to address barrier weakness, action, coherence, transmission, permeability, and uncertainty, while escalation rules had to consider levels, trends, acceleration, persistence, and concentration. Intervention authority had to be established before the available response window closed.

The four intervention levers provided a practical summary. Institutions could raise barrier height, increase barrier width, reduce permeability, or change phase. Capital, liquidity, collateral, control reliability, legal evidence, redundancy, and public support could increase resistance. Longer maturities, additional review stages, independent recovery routes, and resolution stays could create distance and time. Dependency reduction, diversification, segmentation, and capacity expansion could close low-resistance corridors. Earlier support, staggered obligations, coordinated communication, or delayed harmful actions could alter synchronization. The preferred intervention was not the one that looked strongest in isolation, but the one that robustly reduced adverse accessibility relative to cost, delay, capacity, and side effects.

The cross-domain applications demonstrated both the unity and the limits of the framework. Credit risk emphasized liquidity, refinancing, collateral, covenants, and confidence. Operational and cyber risk emphasized control layers, detection, shared infrastructure, recovery, and human response. Legal, regulatory, and conduct risk emphasized evidence, interpretation, behaviour, jurisdiction, and remediation. Systemic risk emphasized networks, common assets, collective barriers, infrastructure concentration, and synchronized behaviour. The formal language remained consistent, but the variables, evidence, materiality thresholds, and intervention authorities changed. A common architecture did not imply a single universal calibration.

Artificial intelligence occupies an important but subordinate role in this programme. It can extract evidence from documents, identify candidate paths, summarize incidents, generate scenarios, detect nonlinear patterns, and support reporting. It can also hallucinate sources, confuse observation with inference, amplify data bias, and create false confidence through fluent explanation. An AI-assisted risk system therefore has its own barrier architecture: grounding, validation, access control, versioning, monitoring, human review, and accountability. The use of advanced tools does not relax evidentiary standards. It makes those standards more important.

Several research questions remain open. Metrics must be calibrated so that geometric distance predicts real transition difficulty. Barrier action must be normalized across paths, entities, and time. Phase proxies must demonstrate stability and incremental value. Dynamic manifolds must accommodate structural change without turning every regime shift into an after-the-fact explanation. Network and agent-based models must be compared carefully with amplitude formulations. Intervention effects must be estimated causally where possible. Rare adverse events require methods that use near misses, partial transitions, simulations, and expert evidence without manufacturing precision. These are not peripheral technical matters; they determine whether the framework becomes science, disciplined practice, or merely metaphor.

The ethical dimension also deserves attention. Better detection of hidden pathways can improve resilience, but it can also justify intrusive monitoring, opaque limits, automated exclusion, or excessive conservatism. Structural models may encode institutional assumptions about which states are acceptable and which transitions are adverse. Those choices should be visible and contestable. Individuals and organizations affected by model-driven decisions need appropriate explanation, review, and accountability. The goal is not to construct infinitely high barriers or eliminate uncertainty. It is to govern risk proportionately while preserving legitimate activity, adaptability, and recovery.

The ultimate value of this framework will not be determined by the elegance of its equations. It will be determined by whether institutions detect consequential fragility earlier, understand why protection may fail, intervene before pathways become irreversible, and learn from outcomes without rewriting the past. A successful implementation should produce fewer surprises that were visible but unconnected, fewer controls that existed but could not function, and fewer interventions that arrived after their effective window had closed.

The closing proposition is therefore both analytical and practical. Risk is shaped by location, movement, protection, pathway, timing, and response. Probability summarizes the resulting uncertainty; geometry organizes the conditions that produce it. Tunneling directs attention to the possibility that nominally remote outcomes may remain accessible through finite, changing, and imperfect barriers. Used carefully, this perspective does not make risk mysterious. It makes the hidden structure of risk available for measurement, challenge, and intervention.

The work ends where a new empirical programme begins. Each application must define its state, construct its geometry, test its classical explanations, measure its barriers, reconstruct its pathways, estimate its timing, and evaluate its interventions. The framework should be retained only where it improves evidence and decisions. That is the standard by which the ideas in this book should be judged, revised, or rejected. The final question is not whether risk resembles physics. It is whether a geometric account helps us see, in time to act, how failure becomes reachable.

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